RM0090
Reference manual

STM32F405/415, STM32F407/417, STM32F427/437 and STM32F429/439 advanced Arm®-based 32-bit MCUs

Introduction

This reference manual targets application developers. It provides complete information on how to use the STM32F405xx/07xx, STM32F415xx/17xx, STM32F42xxx and STM32F43xxx microcontroller memory and peripherals.

The STM32F405xx/07xx, STM32F415xx/17xx, STM32F42xxx and STM32F43xxx constitute a family of microcontrollers with different memory sizes, packages and peripherals.

For ordering information, mechanical and electrical device characteristics please refer to the datasheets.

For information on the Arm® Cortex®-M4 with FPU core, please refer to the *Cortex®*-M4 with FPU *Technical Reference Manual*.

The STM32F405xx/07xx, STM32F415xx/17xx, STM32F42xxx and STM32F43xxx microcontrollers include ST state-of-the-art patented technology.

Related documents

Available from STMicroelectronics web site (*www.st.com*):

- STM32F40x and STM32F41x datasheets
- STM32F42x and STM32F43x datasheets
- STM32F40x and STM32F41x errata sheets
- STM32F42x and STM32F43x errata sheets
- For information on the Arm® *Cortex®*-M4 with FPU*, refer* to the *STM32F3xx/F4xxx Cortex®*-M4 with FPU *programming manual (PM0214).*

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1 Documentation conventions

The STM32F405xx/07xx, STM32F415xx/17xx, STM32F42xxx and STM32F43xxx devices have an Arm®[(a)] Cortex®-M4 with FPU core.

1.1 List of abbreviations for registers

The following abbreviations are used in register descriptions:

read/write (rw)	Software can read and write to these bits.
read-only (r)	Software can only read these bits.
write-only (w)	Software can only write to this bit. Reading the bit returns the reset value.
read/clear (rc_w1)	Software can read as well as clear this bit by writing 1. Writing ‘0’ has no effect on the bit value.
read/clear (rc_w0)	Software can read as well as clear this bit by writing 0. Writing ‘1’ has no effect on the bit value.
read/clear by read (rc_r)	Software can read this bit. Reading this bit automatically clears it to ‘0’. Writing ‘0’ has no effect on the bit value.
read/set (rs)	Software can read as well as set this bit. Writing ‘0’ has no effect on the bit value.
read-only write trigger (rt_w)	Software can read this bit. Writing ‘0’ or ‘1’ triggers an event but has no effect on the bit value.
toggle (t)	Software can only toggle this bit by writing ‘1’. Writing ‘0’ has no effect.
Reserved (Res.)	Reserved bit, must be kept at reset value.

a. Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere.

1.2 Glossary

This section gives a brief definition of acronyms and abbreviations used in this document:

- The CPU core integrates two debug ports:
  - JTAG debug port (JTAG-DP) provides a 5-pin standard interface based on the Joint Test Action Group (JTAG) protocol.
  - SWD debug port (SWD-DP) provides a 2-pin (clock and data) interface based on the Serial Wire Debug (SWD) protocol.

    For both the JTAG and SWD protocols, please refer to the Cortex®-M4 with FPU Technical Reference Manual
- Word: data/instruction of 32-bit length.
- Half word: data/instruction of 16-bit length.
- Byte: data of 8-bit length.
- Double word: data of 64-bit length.
- IAP (in-application programming): IAP is the ability to reprogram the flash memory of a microcontroller while the user program is running.
- ICP (in-circuit programming): ICP is the ability to program the flash memory of a microcontroller using the JTAG protocol, the SWD protocol or the bootloader while the device is mounted on the user application board.
- I-Code: this bus connects the Instruction bus of the CPU core to the Flash instruction interface. Prefetch is performed on this bus.
- D-Code: this bus connects the D-Code bus (literal load and debug access) of the CPU to the Flash data interface.
- Option bytes: product configuration bits stored in the flash memory.
- OBL: option byte loader.
- AHB: advanced high-performance bus.
- CPU: refers to the Cortex®-M4 with FPU core.

1.3 Peripheral availability

For peripheral availability and number across all STM32F405xx/07xx and STM32F415xx/17xx sales types, please refer to the STM32F405xx/07xx and STM32F415xx/17xx datasheets.

For peripheral availability and number across all STM32F42xxx and STM32F43xxx sales types, please refer to the STM32F42xxx and STM32F43xxx datasheets.

2 Memory and bus architecture

2.1 System architecture

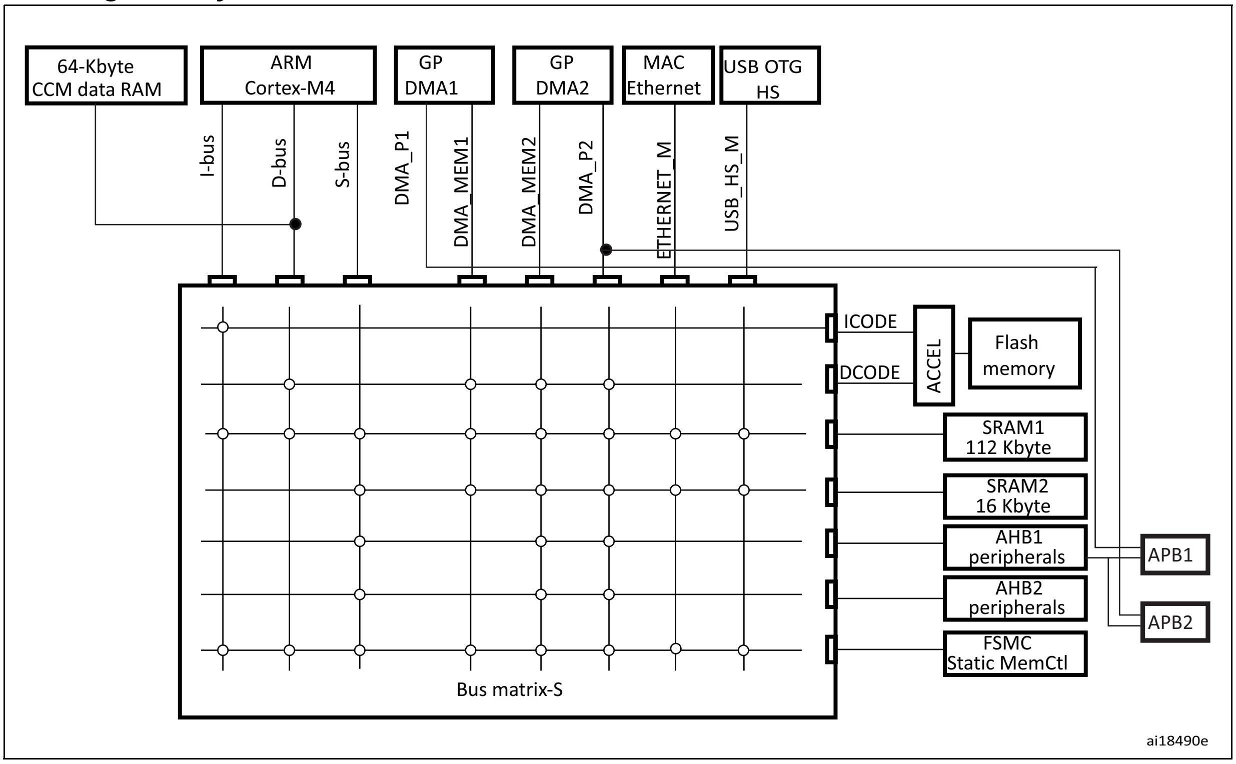
In STM32F405xx/07xx and STM32F415xx/17xx, the main system consists of 32-bit multilayer AHB bus matrix that interconnects:

The main system consists of 32-bit multilayer AHB bus matrix that interconnects:

- Eight masters:
  - Cortex®-M4 with FPU core I-bus, D-bus and S-bus
  - DMA1 memory bus
  - DMA2 memory bus
  - DMA2 peripheral bus
  - Ethernet DMA bus
  - USB OTG HS DMA bus
- Seven slaves:
  - Internal flash memory ICode bus
  - Internal flash memory DCode bus
  - Main internal SRAM1 (112 KB)
  - Auxiliary internal SRAM2 (16 KB)
  - AHB1 peripherals including AHB to APB bridges and APB peripherals
  - AHB2 peripherals
  - FSMC

The bus matrix provides access from a master to a slave, enabling concurrent access and efficient operation even when several high-speed peripherals work simultaneously. The 64-Kbyte CCM (core coupled memory) data RAM is not part of the bus matrix and can be accessed only through the CPU. This architecture is shown in *Figure 1*.

Figure 1. System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices

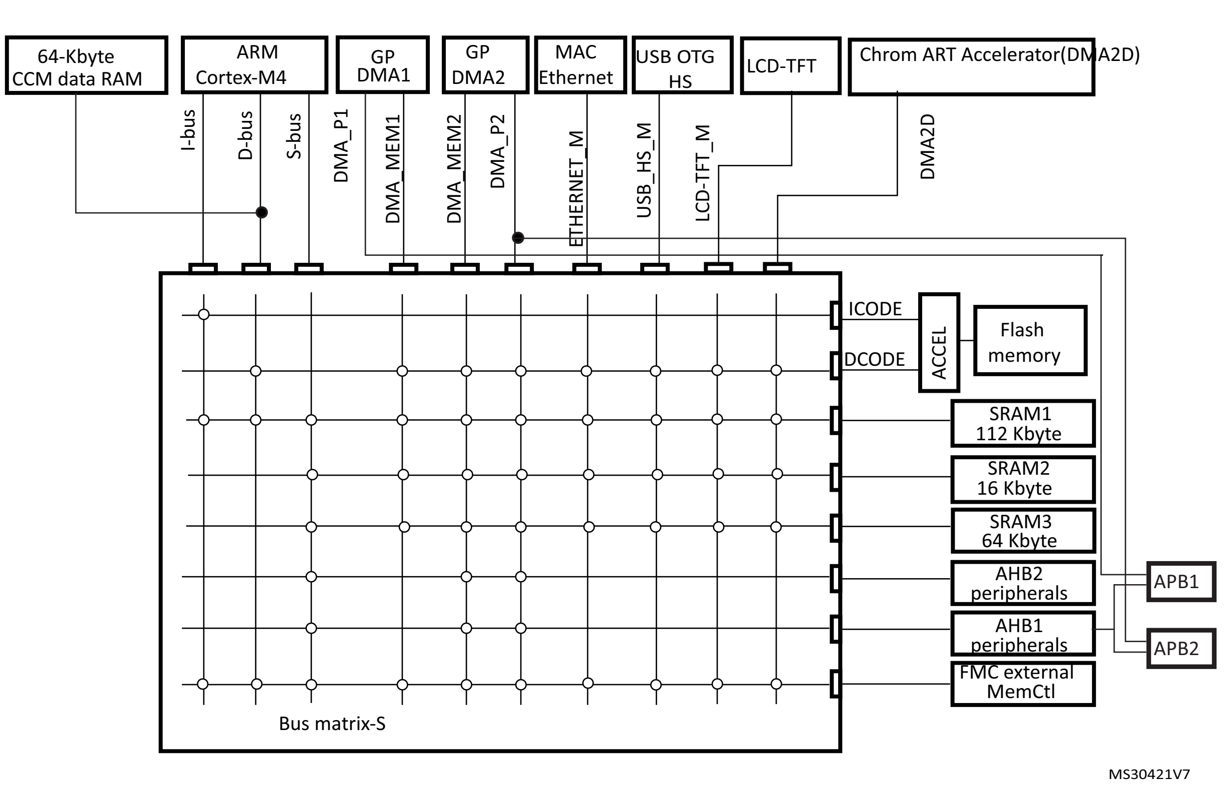


In the STM32F42xx and STM32F43xx devices, the main system consists of 32-bit multilayer AHB bus matrix that interconnects:

- Ten masters:
  - Cortex®-M4 with FPU core I-bus, D-bus and S-bus
  - DMA1 memory bus
  - DMA2 memory bus
  - DMA2 peripheral bus
  - Ethernet DMA bus
  - USB OTG HS DMA bus
  - LCD Controller DMA-bus
  - DMA2D (Chrom-Art Accelerator™) memory bus
- Eight slaves:
  - Internal flash memory ICode bus
  - Internal flash memory DCode bus
  - Main internal SRAM1 (112 KB)
  - Auxiliary internal SRAM2 (16 KB)
  - Auxiliary internal SRAM3 (64 KB)
  - AHB1peripherals including AHB to APB bridges and APB peripherals
  - AHB2 peripherals
  - FMC

The bus matrix provides access from a master to a slave, enabling concurrent access and efficient operation even when several high-speed peripherals work simultaneously. The 64-Kbyte CCM (core coupled memory) data RAM is not part of the bus matrix and can be accessed only through the CPU. This architecture is shown in *Figure 2*.

Figure 2. System architecture for STM32F42xxx and STM32F43xxx devices



2.1.1 I-bus

This bus connects the Instruction bus of the Cortex®-M4 with FPU core to the BusMatrix. This bus is used by the core to fetch instructions. The target of this bus is a memory containing code (internal flash memory/SRAM or external memories through the FSMC/FMC).

2.1.2 D-bus

This bus connects the databus of the Cortex®-M4 with FPU to the 64-Kbyte CCM data RAM to the BusMatrix. This bus is used by the core for literal load and debug access. The target of this bus is a memory containing code or data (internal flash memory or external memories through the FSMC/FMC).

2.1.3 S-bus

This bus connects the system bus of the Cortex®-M4 with FPU core to a BusMatrix. This bus is used to access data located in a peripheral or in SRAM. Instructions may also be fetched on this bus (less efficient than ICode). The targets of this bus are the internal SRAM1, SRAM2 and SRAM3, the AHB1 peripherals including the APB peripherals, the AHB2 peripherals and the external memories through the FSMC/FMC.

2.1.4 DMA memory bus

This bus connects the DMA memory bus master interface to the BusMatrix. It is used by the DMA to perform transfer to/from memories. The targets of this bus are data memories: internal SRAMs (SRAM1, SRAM2 and SRAM3) and external memories through the FSMC/FMC.

2.1.5 DMA peripheral bus

This bus connects the DMA peripheral master bus interface to the BusMatrix. This bus is used by the DMA to access AHB peripherals or to perform memory-to-memory transfers. The targets of this bus are the AHB and APB peripherals plus data memories: internal SRAMs (SRAM1, SRAM2 and SRAM3) and external memories through the FSMC/FMC.

2.1.6 Ethernet DMA bus

This bus connects the Ethernet DMA master interface to the BusMatrix. This bus is used by the Ethernet DMA to load/store data to a memory. The targets of this bus are data memories: internal SRAMs (SRAM1, SRAM2, SRAM3), internal flash memory, and external memories through the FSMC/FMC.

2.1.7 USB OTG HS DMA bus

This bus connects the USB OTG HS DMA master interface to the BusMatrix. This bus is used by the USB OTG DMA to load/store data to a memory. The targets of this bus are data memories: internal SRAMs (SRAM1, SRAM2, SRAM3), internal flash memory, and external memories through the FSMC/FMC.

2.1.8 LCD-TFT controller DMA bus

This bus connects the LCD controller DMA master interface to the BusMatrix. It is used by the LCD-TFT DMA to load/store data to a memory. The targets of this bus are data memories: internal SRAMs (SRAM1, SRAM2, SRAM3), external memories through FMC, and internal flash memory.

2.1.9 DMA2D bus

This bus connect the DMA2D master interface to the BusMatrix. This bus is used by the DMA2D graphic Accelerator to load/store data to a memory. The targets of this bus are data memories: internal SRAMs (SRAM1, SRAM2, SRAM3), external memories through FMC, and internal flash memory.

2.1.10 BusMatrix

The BusMatrix manages the access arbitration between masters. The arbitration uses a round-robin algorithm.

2.1.11 AHB/APB bridges (APB)

The two AHB/APB bridges, APB1 and APB2, provide full synchronous connections between the AHB and the two APB buses, allowing flexible selection of the peripheral frequency.

Refer to the device datasheets for more details on APB1 and APB2 maximum frequencies, and to *Table 1* for the address mapping of AHB and APB peripherals.

After each device reset, all peripheral clocks are disabled (except for the SRAM and flash memory interface). Before using a peripheral you have to enable its clock in the RCC_AHBxENR or RCC_APBxENR register.

Note: When a 16- or an 8-bit access is performed on an APB register, the access is transformed into a 32-bit access: the bridge duplicates the 16- or 8-bit data to feed the 32-bit vector.

2.2 Memory organization

Program memory, data memory, registers and I/O ports are organized within the same linear 4 Gbyte address space.

The bytes are coded in memory in little endian format. The lowest numbered byte in a word is considered the word’s least significant byte and the highest numbered byte, the word’s most significant.

For the detailed mapping of peripheral registers, please refer to the related chapters.

The addressable memory space is divided into 8 main blocks, each of 512 MB.

All the memory areas that are not allocated to on-chip memories and peripherals are considered “Reserved”). Refer to the memory map figure in the product datasheet.

2.3 Memory map

See the datasheet corresponding to your device for a comprehensive diagram of the memory map. *Table 1* gives the boundary addresses of the peripherals available in all STM32F4xx devices.

Table 1. STM32F4xx register boundary addresses

Boundary address	Peripheral	Bus	Register map
0xA000 0000 - 0xA000 0FFF	FSMC control register (STM32F405xx/07xx and STM32F415xx/17xx)/ FMC control register (STM32F42xxx and STM32F43xxx)	AHB3	*Section 36.6.9: FSMC register map on page 1603* *Section 37.8: FMC register map on page 1683*

Table 1. STM32F4xx register boundary addresses (continued)

Boundary address	Peripheral	Bus	Register map
0x5006 0800 - 0x5006 0BFF	RNG	AHB2	Section 24.4.4: RNG register map on page 774
0x5006 0400 - 0x5006 07FF	HASH		Section 25.4.9: HASH register map on page 798
0x5006 0000 - 0x5006 03FF	CRYP		Section 23.6.13: CRYP register map on page 766
0x5005 0000 - 0x5005 03FF	DCMI		Section 15.8.12: DCMI register map on page 481
0x5000 0000 - 0x5003 FFFF	USB OTG FS		Section 34.16.6: OTG_FS register map on page 1329
0x4004 0000 - 0x4007 FFFF	USB OTG HS	AHB1	Section 35.12.6: OTG_HS register map on page 1475
0x4002 B000 - 0x4002 BBFF	DMA2D		Section 11.5: DMA2D registers on page 355
0x4002 8000 - 0x4002 93FF	ETHERNET MAC		Section 33.8.5: Ethernet register maps on page 1239
0x4002 6400 - 0x4002 67FF	DMA2		Section 10.5.11: DMA register map on page 338
0x4002 6000 - 0x4002 63FF	DMA1		
0x4002 4000 - 0x4002 4FFF	BKPSRAM		-
0x4002 3C00 - 0x4002 3FFF	Flash interface register		Section 3.9: Flash interface registers
0x4002 3800 - 0x4002 3BFF	RCC		Section 7.3.24: RCC register map on page 267
0x4002 3000 - 0x4002 33FF	CRC		Section 4.4.4: CRC register map on page 116
0x4002 2800 - 0x4002 2BFF	GPIOK		Section 8.4.11: GPIO register map on page 290
0x4002 2400 - 0x4002 27FF	GPIOJ		
0x4002 2000 - 0x4002 23FF	GPIOI		Section 8.4.11: GPIO register map on page 290
0x4002 1C00 - 0x4002 1FFF	GPIOH		
0x4002 1800 - 0x4002 1BFF	GPIOG		
0x4002 1400 - 0x4002 17FF	GPIOF		
0x4002 1000 - 0x4002 13FF	GPIOE		
0x4002 0C00 - 0x4002 0FFF	GPIOD		
0x4002 0800 - 0x4002 0BFF	GPIOC		
0x4002 0400 - 0x4002 07FF	GPIOB		
0x4002 0000 - 0x4002 03FF	GPIOA		
0x4001 6800 - 0x4001 6BFF	LCD-TFT	APB2	Section 16.7.26: LTDC register map on page 515
0x4001 5800 - 0x4001 5BFF	SAI1		Section 29.17.9: SAI register map on page 966
0x4001 5400 - 0x4001 57FF	SPI6	APB2	Section 28.5.10: SPI register map on page 928
0x4001 5000 - 0x4001 53FF	SPI5		

Table 1. STM32F4xx register boundary addresses (continued)

Boundary address	Peripheral	Bus	Register map
0x4001 4800 - 0x4001 4BFF	TIM11	APB2	*Section 19.5.12: TIM10/11/13/14 register map on page 697*
0x4001 4400 - 0x4001 47FF	TIM10		
0x4001 4000 - 0x4001 43FF	TIM9		*Section 19.4.13: TIM9/12 register map on page 687*
0x4001 3C00 - 0x4001 3FFF	EXTI		*Section 12.3.7: EXTI register map on page 390*
0x4001 3800 - 0x4001 3BFF	SYSCFG		*Section 9.2.8: SYSCFG register maps for STM32F405xx/07xx and STM32F415xx/17xx on page 297* and *Section 9.3.8: SYSCFG register maps for STM32F42xxx and STM32F43xxx on page 304*
0x4001 3400 - 0x4001 37FF	SPI4	APB2	*Section 28.5.10: SPI register map on page 928*
0x4001 3000 - 0x4001 33FF	SPI1	APB2	*Section 28.5.10: SPI register map on page 928*
0x4001 2C00 - 0x4001 2FFF	SDIO		*Section 31.9.16: SDIO register map on page 1077*
0x4001 2000 - 0x4001 23FF	ADC1 - ADC2 - ADC3		*Section 13.13.18: ADC register map on page 433*
0x4001 1400 - 0x4001 17FF	USART6		*Section 30.6.8: USART register map on page 1021*
0x4001 1000 - 0x4001 13FF	USART1		
0x4001 0400 - 0x4001 07FF	TIM8		*Section 17.4.21: TIM1 and TIM8 register map on page 590*
0x4001 0000 - 0x4001 03FF	TIM1		
0x4000 7C00 - 0x4000 7FFF	UART8	APB1	*Section 30.6.8: USART register map on page 1021*
0x4000 7800 - 0x4000 7BFF	UART7		

Table 1. STM32F4xx register boundary addresses (continued)

Boundary address	Peripheral	Bus	Register map
0x4000 7400 - 0x4000 77FF	DAC	APB1	Section 14.5.15: DAC register map on page 456
0x4000 7000 - 0x4000 73FF	PWR		Section 5.6: PWR register map on page 151
0x4000 6800 - 0x4000 6BFF	CAN2		Section 32.9.5: bxCAN register map on page 1121
0x4000 6400 - 0x4000 67FF	CAN1		
0x4000 5C00 - 0x4000 5FFF	I2C3		Section 27.6.11: I2C register map on page 875
0x4000 5800 - 0x4000 5BFF	I2C2		
0x4000 5400 - 0x4000 57FF	I2C1		
0x4000 5000 - 0x4000 53FF	UART5		Section 30.6.8: USART register map on page 1021
0x4000 4C00 - 0x4000 4FFF	UART4		
0x4000 4800 - 0x4000 4BFF	USART3		
0x4000 4400 - 0x4000 47FF	USART2		
0x4000 4000 - 0x4000 43FF	I2S3ext		Section 28.5.10: SPI register map on page 928
0x4000 3C00 - 0x4000 3FFF	SPI3 / I2S3		
0x4000 3800 - 0x4000 3BFF	SPI2 / I2S2		
0x4000 3400 - 0x4000 37FF	I2S2ext		
0x4000 3000 - 0x4000 33FF	IWDG		Section 21.4.5: IWDG register map on page 715
0x4000 2C00 - 0x4000 2FFF	WWDG		Section 22.6.4: WWDG register map on page 722
0x4000 2800 - 0x4000 2BFF	RTC & BKP Registers		Section 26.6.21: RTC register map on page 839
0x4000 2000 - 0x4000 23FF	TIM14		Section 19.5.12: TIM10/11/13/14 register map on page 697
0x4000 1C00 - 0x4000 1FFF	TIM13		
0x4000 1800 - 0x4000 1BFF	TIM12		Section 19.4.13: TIM9/12 register map on page 687
0x4000 1400 - 0x4000 17FF	TIM7		Section 20.4.9: TIM6 and TIM7 register map on page 710
0x4000 1000 - 0x4000 13FF	TIM6		
0x4000 0C00 - 0x4000 0FFF	TIM5		Section 18.4.21: TIMx register map on page 651
0x4000 0800 - 0x4000 0BFF	TIM4		
0x4000 0400 - 0x4000 07FF	TIM3		
0x4000 0000 - 0x4000 03FF	TIM2		

2.3.1 Embedded SRAM

The STM32F405xx/07xx and STM32F415xx/17xx feature 4 Kbytes of backup SRAM (see *Section 5.1.2: Battery backup domain*) plus 192 Kbytes of system SRAM.

The STM32F42xxx and STM32F43xxx feature 4 Kbytes of backup SRAM (see *Section 5.1.2: Battery backup domain*) plus 256 Kbytes of system SRAM.

The embedded SRAM can be accessed as bytes, half-words (16 bits) or full words (32 bits). Read and write operations are performed at CPU speed with 0 wait state. The embedded SRAM is divided into up to three blocks:

- SRAM1 and SRAM2 mapped at address 0x2000 0000 and accessible by all AHB masters.
- SRAM3 (available on STM32F42xxx and STM32F43xxx) mapped at address 0x2002 0000 and accessible by all AHB masters.
- CCM (core coupled memory) mapped at address 0x1000 0000 and accessible only by the CPU through the D-bus.

The AHB masters support concurrent SRAM accesses (from the Ethernet or the USB OTG HS): for instance, the Ethernet MAC can read/write from/to SRAM2 while the CPU is reading/writing from/to SRAM1 or SRAM3.

The CPU can access the SRAM1 through the System bus or through the I-Code/D-Code buses when boot from SRAM is selected or when physical remap is selected (*Section 9.2.1: SYSCFG memory remap register (SYSCFG_MEMRMP)* in the SYSCFG controller). To get the max performance on SRAM execution, physical remap should be selected (boot or software selection).

2.3.2 Flash memory overview

The flash memory interface manages CPU AHB I-Code and D-Code accesses to the flash memory. It implements the erase and program flash memory operations and the read and write protection mechanisms. It accelerates code execution with a system of instruction prefetch and cache lines.

The flash memory is organized as follows:

- A main memory block divided into sectors.
- System memory from which the device boots in System memory boot mode
- 512 OTP (one-time programmable) bytes for user data.
- Option bytes to configure read and write protection, BOR level, watchdog software/hardware and reset when the device is in Standby or Stop mode.

Refer to *Section 3: Embedded flash memory interface* for more details.

2.3.3 Bit banding

The Cortex®-M4 with FPU memory map includes two bit-band regions. These regions map each word in an alias region of memory to a bit in a bit-band region of memory. Writing to a word in the alias region has the same effect as a read-modify-write operation on the targeted bit in the bit-band region.

In the STM32F4xx devices both the peripheral registers and the SRAM are mapped to a bit-band region, so that single bit-band write and read operations are allowed. The operations are only available for Cortex®-M4 with FPU accesses, and not from other bus masters (e.g. DMA).

A mapping formula shows how to reference each word in the alias region to a corresponding bit in the bit-band region. The mapping formula is:

bit_word_addr = *bit_band_base* + (*byte_offset* x 32) + (*bit_number* × 4)

where:

- *bit_word_addr* is the address of the word in the alias memory region that maps to the targeted bit
- *bit_band_base* is the starting address of the alias region
- *byte_offset* is the number of the byte in the bit-band region that contains the targeted bit
- *bit_number* is the bit position (0-7) of the targeted bit

Example

The following example shows how to map bit 2 of the byte located at SRAM address 0x20000300 to the alias region:

0x22006008 = 0x22000000 + (0x300*32) + (2*4)

Writing to address 0x22006008 has the same effect as a read-modify-write operation on bit 2 of the byte at SRAM address 0x20000300.

Reading address 0x22006008 returns the value (0x01 or 0x00) of bit 2 of the byte at SRAM address 0x20000300 (0x01: bit set; 0x00: bit reset).

For more information on bit-banding, please refer to the *Cortex*®-M4 with FPU *programming manual* (see *Related documents on page 1*).

2.4 Boot configuration

Due to its fixed memory map, the code area starts from address 0x0000 0000 (accessed through the ICode/DCode buses) while the data area (SRAM) starts from address 0x2000 0000 (accessed through the system bus). The Cortex®-M4 with FPU CPU always fetches the reset vector on the ICode bus, which implies to have the boot space available only in the code area (typically, flash memory). STM32F4xx microcontrollers implement a special mechanism to be able to boot from other memories (like the internal SRAM).

In the STM32F4xx, three different boot modes can be selected through the BOOT[1:0] pins as shown in *Table 2*.

Table 2. Boot modes

Boot mode selection pins		Boot mode	Aliasing
BOOT1	BOOT0		
x	0	Main flash memory	Main flash memory is selected as the boot space
0	1	System memory	System memory is selected as the boot space
1	1	Embedded SRAM	Embedded SRAM is selected as the boot space

The values on the BOOT pins are latched on the 4th rising edge of SYSCLK after a reset. It is up to the user to set the BOOT1 and BOOT0 pins after reset to select the required boot mode.

BOOT0 is a dedicated pin while BOOT1 is shared with a GPIO pin. Once BOOT1 has been sampled, the corresponding GPIO pin is free and can be used for other purposes.

The BOOT pins are also resampled when the device exits the Standby mode. Consequently, they must be kept in the required Boot mode configuration when the device is in the Standby mode. After this startup delay is over, the CPU fetches the top-of-stack value from address 0x0000 0000, then starts code execution from the boot memory starting from 0x0000 0004.

Note: When the device boots from SRAM, in the application initialization code, you have to relocate the vector table in SRAM using the NVIC exception table and the offset register.

In STM32F42xxx and STM32F43xxx devices, when booting from the main flash memory, the application software can either boot from bank 1 or from bank 2. By default, boot from bank 1 is selected.

To select boot from flash memory bank 2, set the BFB2 bit in the user option bytes. When this bit is set and the boot pins are in the boot from main flash memory configuration, the device boots from system memory, and the boot loader jumps to execute the user application programmed in flash memory bank 2. For further details, please refer to AN2606.

Embedded bootloader

The embedded bootloader mode is used to reprogram the flash memory using one of the following serial interfaces:

- USART1 (PA9/PA10)
- USART3 (PB10/11 and PC10/11)
- CAN2 (PB5/13)
- USB OTG FS (PA11/12) in Device mode (DFU: device firmware upgrade).

The USART peripherals operate at the internal 16 MHz oscillator (HSI) frequency, while the CAN and USB OTG FS require an external clock (HSE) multiple of 1 MHz (ranging from 4 to 26 MHz).

The embedded bootloader code is located in system memory. It is programmed by ST during production. For additional information, refer to application note AN2606.

Physical remap in STM32F405xx/07xx and STM32F415xx/17xx

Once the boot pins are selected, the application software can modify the memory accessible in the code area (in this way the code can be executed through the ICode bus in place of the System bus). This modification is performed by programming the *Section 9.2.1: SYSCFG memory remap register (SYSCFG_MEMRMP)* in the SYSCFG controller.

The following memories can thus be remapped:

- Main flash memory
- System memory
- Embedded SRAM1 (112 KB)
- FSMC bank 1 (NOR/PSRAM 1 and 2)

Table 3. Memory mapping vs. Boot mode/physical remap in STM32F405xx/07xx and STM32F415xx/17xx

Addresses	Boot/Remap in main flash memory	Boot/Remap in embedded SRAM	Boot/Remap in System memory	Remap in FSMC
0x2001 C000 - 0x2001 FFFF	SRAM2 (16 KB)	SRAM2 (16 KB)	SRAM2 (16 KB)	SRAM2 (16 KB)
0x2000 0000 - 0x2001 BFFF	SRAM1 (112 KB)	SRAM1 (112 KB)	SRAM1 (112 KB)	SRAM1 (112 KB)
0x1FFF 0000 - 0x1FFF 77FF	System memory	System memory	System memory	System memory
0x0810 0000 - 0x0FFF FFFF	Reserved	Reserved	Reserved	Reserved
0x0800 0000 - 0x080F FFFF	Flash memory	Flash memory	Flash memory	Flash memory
0x0400 0000 - 0x07FF FFFF	Reserved	Reserved	Reserved	FSMC bank 1 NOR/PSRAM 2 (128 MB Aliased)
0x0000 0000 - 0x000F FFFF(1)(2)	Flash (1 MB) Aliased	SRAM1 (112 KB) Aliased	System memory (30 KB) Aliased	FSMC bank 1 NOR/PSRAM 1 (128 MB Aliased)

1. When the FSMC is remapped at address 0x0000 0000, only the first two regions of bank 1 memory controller (bank 1 NOR/PSRAM 1 and NOR/PSRAM 2) can be remapped. In remap mode, the CPU can access the external memory via ICode bus instead of System bus which boosts up the performance.
2. Even when aliased in the boot memory space, the related memory is still accessible at its original memory space.

Physical remap in STM32F42xxx and STM32F43xxx

Once the boot pins are selected, the application software can modify the memory accessible in the code area (in this way the code can be executed through the ICode bus in place of the System bus). This modification is performed by programming the *Section 9.2.1: SYSCFG memory remap register (SYSCFG_MEMRMP)* in the SYSCFG controller.

The following memories can thus be remapped:

- Main flash memory
- System memory
- Embedded SRAM1 (112 KB)
- FMC bank 1 (NOR/PSRAM 1 and 2)
- FMC SDRAM bank 1

Table 4. Memory mapping vs. Boot mode/physical remap in STM32F42xxx and STM32F43xxx

Addresses	Boot/Remap in main flash memory	Boot/Remap in embedded SRAM	Boot/Remap in System memory	Remap in FMC
0x2002 0000 - 0x2002 FFFF	SRAM3 (64 KB)	SRAM3 (64 KB)	SRAM3 (64 KB)	SRAM3 (64 KB)
0x2001 C000 - 0x2001 FFFF	SRAM2 (16 KB)	SRAM2 (16 KB)	SRAM2 (16 KB)	SRAM2 (16 KB)
0x2000 0000 - 0x2001 BFFF	SRAM1 (112 KB)	SRAM1 (112 KB)	SRAM1 (112 KB)	SRAM1 (112 KB)
0x1FFF 0000 - 0x1FFF 77FF	System memory	System memory	System memory	System memory
0x0810 0000 - 0x0FFF FFFF	Reserved	Reserved	Reserved	Reserved
0x0800 0000 - 0x081F FFFF	Flash memory	Flash memory	Flash memory	Flash memory

Table 4. Memory mapping vs. Boot mode/physical remap in STM32F42xxx and STM32F43xxx (continued)

Addresses	Boot/Remap in main flash memory	Boot/Remap in embedded SRAM	Boot/Remap in System memory	Remap in FMC
0x0400 0000 - 0x07FF FFFF	Reserved	Reserved	Reserved	FMC bank 1 NOR/PSRAM 2 (128 MB Aliased)
0x0000 0000 - 0x001F FFFF(1)(2)	Flash (2 MB) Aliased	SRAM1 (112 KB) Aliased	System memory (30 KB) Aliased	FMC bank 1 NOR/PSRAM 1 (128 MB Aliased) or FMC SDRAM bank 1 (128 MB Aliased)

1. When the FMC is remapped at address 0x0000 0000, only the first two regions of bank 1 memory controller (bank 1 NOR/PSRAM 1 and NOR/PSRAM 2) or SDRAM bank 1 can be remapped. In remap mode, the CPU can access the external memory via ICode bus instead of System bus which boosts up the performance.
2. Even when aliased in the boot memory space, the related memory is still accessible at its original memory space.

3 Embedded flash memory interface

3.1 Introduction

The flash memory interface manages CPU AHB I-Code and D-Code accesses to the flash memory. It implements the erase and program flash memory operations and the read and write protection mechanisms.

The flash memory interface accelerates code execution with a system of instruction prefetch and cache lines.

3.2 Main features

- Flash memory read operations
- Flash memory program/erase operations
- Read / write protections
- Prefetch on I-Code
- 64 cache lines of 128 bits on I-Code
- 8 cache lines of 128 bits on D-Code

Figure 3 shows the flash memory interface connection inside the system architecture.

Figure 3. Flash memory interface connection inside system architecture (STM32F405xx/07xx and STM32F415xx/17xx)

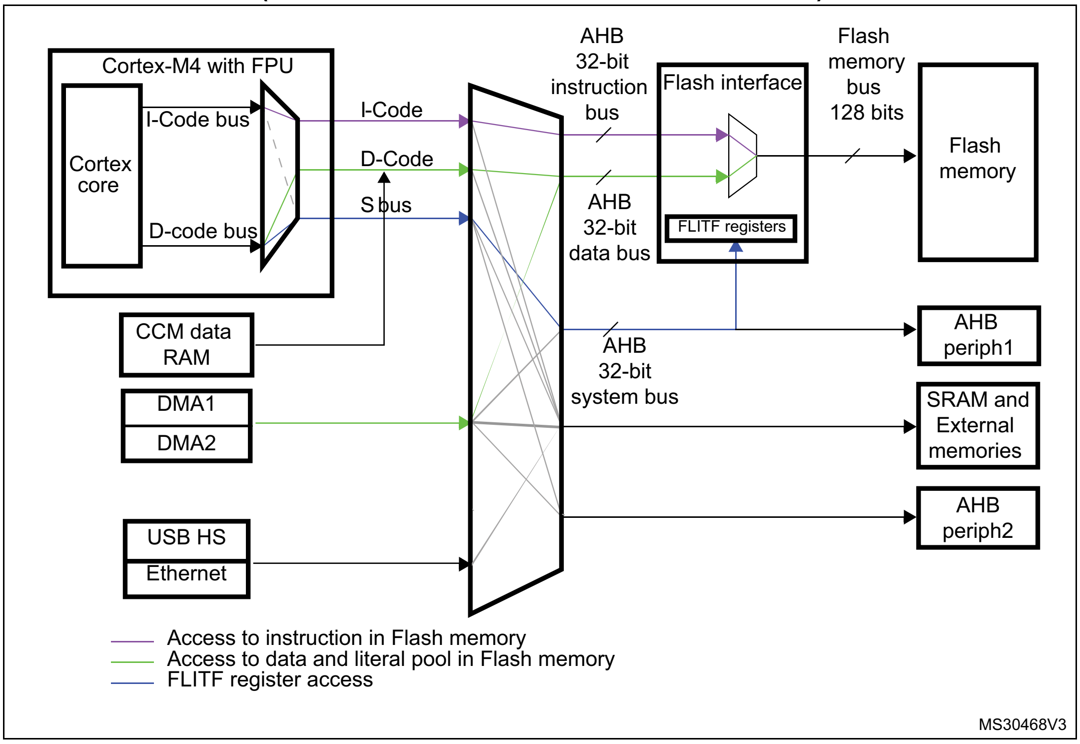
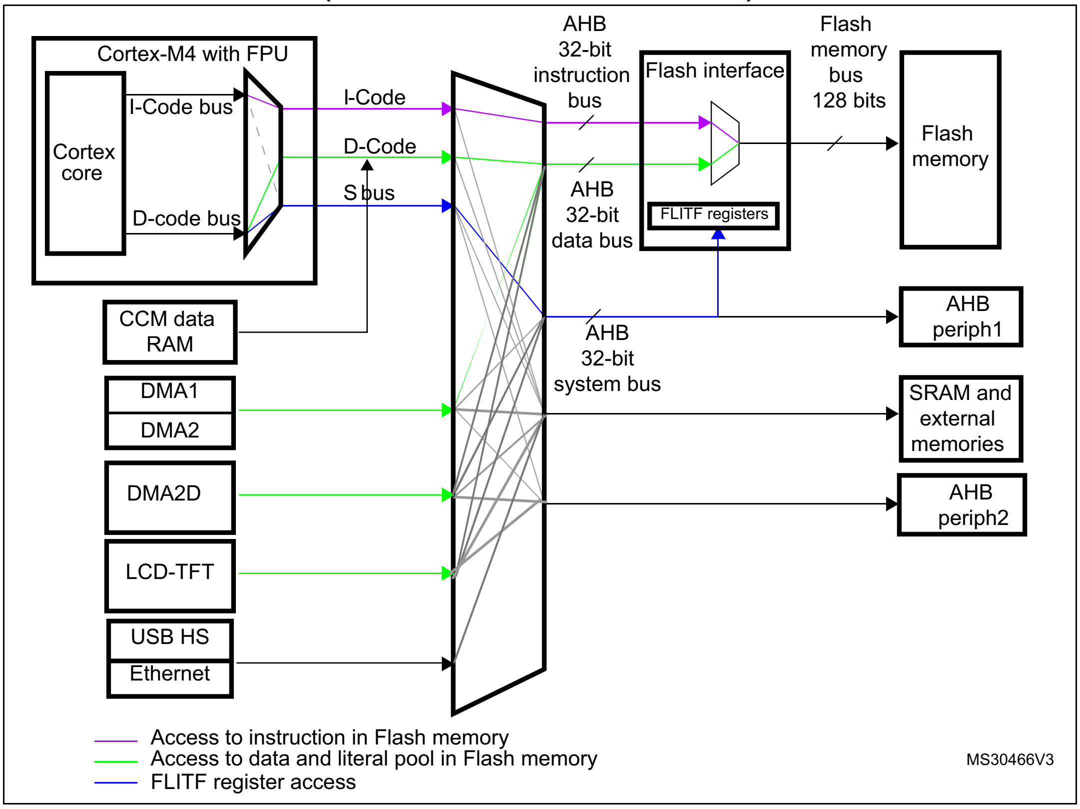


Figure 4. Flash memory interface connection inside system architecture (STM32F42xxx and STM32F43xxx)



3.3 Embedded flash memory in STM32F405xx/07xx and STM32F415xx/17xx

The flash memory has the following main features:

- Capacity up to 1 Mbyte
- 128 bits wide data read
- Byte, half-word, word and double word write
- Sector and mass erase
- Memory organization

  The flash memory is organized as follows:
  - A main memory block divided into 4 sectors of 16 Kbytes, 1 sector of 64 Kbytes, and 7 sectors of 128 Kbytes
  - System memory from which the device boots in System memory boot mode
  - 512 OTP (one-time programmable) bytes for user data

    The OTP area contains 16 additional bytes used to lock the corresponding OTP data block.
  - Option bytes to configure read and write protection, BOR level, watchdog software/hardware and reset when the device is in Standby or Stop mode.
- Low-power modes (for details refer to the Power control (PWR) section of the reference manual)

Table 5. Flash module organization (STM32F40x and STM32F41x)

Block	Name	Block base addresses	Size
Main memory	Sector 0	0x0800 0000 - 0x0800 3FFF	16 Kbytes
	Sector 1	0x0800 4000 - 0x0800 7FFF	16 Kbytes
	Sector 2	0x0800 8000 - 0x0800 BFFF	16 Kbytes
	Sector 3	0x0800 C000 - 0x0800 FFFF	16 Kbytes
	Sector 4	0x0801 0000 - 0x0801 FFFF	64 Kbytes
	Sector 5	0x0802 0000 - 0x0803 FFFF	128 Kbytes
	Sector 6	0x0804 0000 - 0x0805 FFFF	128 Kbytes
	. . .	. . .	. . .
	Sector 11	0x080E 0000 - 0x080F FFFF	128 Kbytes
System memory		0x1FFF 0000 - 0x1FFF 77FF	30 Kbytes
OTP area		0x1FFF 7800 - 0x1FFF 7A0F	528 bytes
Option bytes		0x1FFF C000 - 0x1FFF C00F	16 bytes

3.4 Embedded flash memory in STM32F42xxx and STM32F43xxx

The flash memory has the following main features:

- Capacity up to 2 Mbyte with dual bank architecture supporting read-while-write capability (RWW)
- 128 bits wide data read
- Byte, half-word, word and double word write
- Sector, bank, and mass erase (both banks)
- Dual bank memory organization

  The dual bank organization is available only on 1 Mbyte and 2 Mbyte devices.

  The flash memory is organized as follows:

  – For each bank, a main memory block (1 Mbyte) divided into 4 sectors of 16 Kbytes, 1 sector of 64 Kbytes, and 7 sectors of 128 Kbytes

  – System memory from which the device boots in System memory boot mode

  – 512 OTP (one-time programmable) bytes for user data

    The OTP area contains 16 additional bytes used to lock the corresponding OTP data block.

  – Option bytes to configure read and write protection, BOR level, watchdog, dual bank boot mode, dual bank feature, software/hardware and reset when the device is in Standby or Stop mode.
- Dual bank organization on 1 Mbyte devices

  The dual bank feature on 1 Mbyte devices is enabled by setting the DB1M option bit.

  To obtain a dual bank flash memory, the last 512 Kbytes of the single bank (sectors [8:11]) are re-structured in the same way as the first 512 Kbytes.

  The sector numbering of dual bank memory organization is different from the single bank: the single bank memory contains 12 sectors whereas the dual bank memory contains 16 sectors (see *Table 7: 1 Mbyte flash memory single bank vs dual bank organization (STM32F42xxx and STM32F43xxx)*).

  For erase operation, the right sector numbering must be considered according the DB1M option bit.

  – When the DB1M bit is reset, the erase operation must be performed on the default sector number.

  – When the DB1M bit is set, to perform an erase operation on bank 2, the sector number must be programmed (sector number from 12 to 19). Refer to FLASH_CR register for SNB (Sector number) configuration.

Refer to *Table 8: 1 Mbyte single bank flash memory organization (STM32F42xxx and STM32F43xxx)* and *Table 9: 1 Mbyte dual bank flash memory organization (STM32F42xxx and STM32F43xxx)* for details on 1 Mbyte single bank and 1 Mbyte dual bank organizations.

Table 6. Flash module - 2 Mbyte dual bank organization (STM32F42xxx and STM32F43xxx)

Block	Bank	Name	Block base addresses	Size
Main memory	Bank 1	Sector 0	0x0800 0000 - 0x0800 3FFF	16 Kbytes
		Sector 1	0x0800 4000 - 0x0800 7FFF	16 Kbytes
		Sector 2	0x0800 8000 - 0x0800 BFFF	16 Kbytes
		Sector 3	0x0800 C000 - 0x0800 FFFF	16 Kbyte
		Sector 4	0x0801 0000 - 0x0801 FFFF	64 Kbytes
		Sector 5	0x0802 0000 - 0x0803 FFFF	128 Kbytes
		Sector 6	0x0804 0000 - 0x0805 FFFF	128 Kbytes
		.	-	-
		.	-	-
		-	-	-
		Sector 11	0x080E 0000 - 0x080F FFFF	128 Kbytes
	Bank 2	Sector 12	0x0810 0000 - 0x0810 3FFF	16 Kbytes
		Sector 13	0x0810 4000 - 0x0810 7FFF	16 Kbytes
		Sector 14	0x0810 8000 - 0x0810 BFFF	16 Kbytes
		Sector 15	0x0810 C000 - 0x0810 FFFF	16 Kbytes
		Sector 16	0x0811 0000 - 0x0811 FFFF	64 Kbytes
		Sector 17	0x0812 0000 - 0x0813 FFFF	128 Kbytes
		Sector 18	0x0814 0000 - 0x0815 FFFF	128 Kbytes
		.	.	.
		.	.	.
		.	.	.
		Sector 23	0x081E 0000 - 0x081F FFFF	128 Kbytes
System memory			0x1FFF 0000 - 0x1FFF 77FF	30 Kbytes
OTP			0x1FFF 7800 - 0x1FFF 7A0F	528 bytes
Option bytes	Bank 1		0x1FFF C000 - 0x1FFF C00F	16 bytes
	Bank 2		0x1FFE C000 - 0x1FFE C00F	16 bytes

Table 7. 1 Mbyte flash memory single bank vs dual bank organization (STM32F42xxx and STM32F43xxx)

1 Mbyte single bank flash memory (default)			1 Mbyte dual bank flash memory		
DB1M=0			DB1M=1		
Main memory	Sector number	Sector size	Main memory	Sector number	Sector size
1MB	Sector 0	16 Kbytes	Bank 1 512KB	Sector 0	16 Kbytes
	Sector 1	16 Kbytes		Sector 1	16 Kbytes
	Sector 2	16 Kbytes		Sector 2	16 Kbytes
	Sector 3	16 Kbytes		Sector 3	16 Kbytes
	Sector 4	64 Kbytes		Sector 4	64 Kbytes
	Sector 5	128 Kbytes		Sector 5	128 Kbytes
	Sector 6	128 Kbytes		Sector 6	128 Kbytes
	Sector 7	128 Kbytes		Sector 7	128 Kbytes
	Sector 8	128 Kbytes	Bank 2 512KB	Sector 12	16 Kbytes
	Sector 9	128 Kbytes		Sector 13	16 Kbytes
	Sector 10	128 Kbytes		Sector 14	16 Kbytes
	Sector 11	128 Kbytes		Sector 15	16 Kbytes
	-	-		Sector 16	64 Kbytes
	-	-		Sector 17	128 Kbytes
	-	-		Sector 18	128 Kbytes
	-	-		Sector 19	128 Kbytes

Table 8. 1 Mbyte single bank flash memory organization (STM32F42xxx and STM32F43xxx)

Block	Bank	Name	Block base addresses	Size
Main memory	Single bank	Sector 0	0x0800 0000 - 0x0800 3FFF	16 Kbytes
		Sector 1	0x0800 4000 - 0x0800 7FFF	16 Kbytes
		Sector 2	0x0800 8000 - 0x0800 BFFF	16 Kbytes
		Sector 3	0x0800 C000 - 0x0800 FFFF	16 Kbytes
		Sector 4	0x0801 0000 - 0x0801 FFFF	64 Kbytes
		Sector 5	0x0802 0000 - 0x0803 FFFF	128 Kbytes
		Sector 6	0x0804 0000 - 0x0805 FFFF	128 Kbytes
		Sector 7	0x0806 0000 - 0x0807 FFFF	128 Kbytes
		Sector 8	0x0808 0000 - 0x0809 FFFF	128 Kbytes
		Sector 9	0x080A 0000 - 0x080B FFFF	128 Kbytes
		Sector 10	0x080C 0000 - 0x080D FFFF	128 Kbytes
		Sector 11	0x080E 0000 - 0x080F FFFF	128 Kbytes

Table 8. 1 Mbyte single bank flash memory organization (STM32F42xxx and STM32F43xxx) (continued)

Block	Bank	Name	Block base addresses	Size
System memory			0x1FFF 0000 - 0x1FFFF 77FF	30 Kbytes
OTP			0x1FFF 7800 - 0x1FFF 7A0F	528 bytes
Option bytes			0x1FFF C000 - 0x1FFF C00F	16 bytes
			0x1FFE C000 - 0x1FFE C00F	16 bytes

Table 9. 1 Mbyte dual bank flash memory organization (STM32F42xxx and STM32F43xxx)

Block		Name	Block base addresses	Size
Main memory	Bank 1	Sector 0	0x0800 0000 - 0x0800 3FFF	16 Kbytes
		Sector 1	0x0800 4000 - 0x0800 7FFF	16 Kbytes
		Sector 2	0x0800 8000 - 0x0800 BFFF	16 Kbytes
		Sector 3	0x0800 C000 - 0x0800 FFFF	16 Kbyte
		Sector 4	0x0801 0000 - 0x0801 FFFF	64 Kbytes
		Sector 5	0x0802 0000 - 0x0803 FFFF	128 Kbytes
		Sector 6	0x0804 0000 - 0x0805 FFFF	128 Kbytes
		Sector 7	0x0806 0000 - 0x0807 FFFF	128 Kbytes
	Bank 2	Sector 12	0x0808 0000 - 0x0808 3FFF	16 Kbytes
		Sector 13	0x0808 4000 - 0x0808 7FFF	16 Kbytes
		Sector 14	0x0808 0000 - 0x0808 BFFF	16 Kbytes
		Sector 15	0x0808 C000 - 0x0808 FFFF	16 Kbytes
		Sector 16	0x0809 0000 - 0x0809 FFFF	64 Kbytes
		Sector 17	0x080A 0000 - 0x080B FFFF	128 Kbytes
		Sector 18	0x080C 0000 - 0x080D FFFF	128 Kbytes
		Sector 19	0x080E 0000 - 0x080F FFFF	128 Kbytes
System memory			0x1FFF 0000 - 0x1FFF 77FF	30 Kbytes
OTP			0x1FFF 7800 - 0x1FFF 7A0F	528 bytes
Option bytes	Bank 1		0x1FFF C000 - 0x1FFF C00F	16 bytes
	Bank 2		0x1FFE C000 - 0x1FFE C00F	16 bytes

Table 10. 512 Kbyte single bank flash memory organization (STM32F42xxx and STM32F43xxx)

Block	Bank	Name	Block base addresses	Size
Main memory	Single Bank	Sector 0	0x0800 0000 - 0x0800 3FFF	16 Kbytes
		Sector 1	0x0800 4000 - 0x0800 7FFF	16 Kbytes
		Sector 2	0x0800 8000 - 0x0800 BFFF	16 Kbytes
		Sector 3	0x0800 C000 - 0x0800 FFFF	16 Kbytes
		Sector 4	0x0801 0000 - 0x0801 FFFF	64 Kbytes
		Sector 5	0x0802 0000 - 0x0803 FFFF	128 Kbytes
		Sector 6	0x0804 0000 - 0x0805 FFFF	128 Kbytes
		Sector 7	0x0806 0000 - 0x0807 FFFF	128 Kbytes
System memory			0x1FFF 0000 - 0x1FFFF 77FF	30 Kbytes
OTP			0x1FFF 7800 - 0x1FFF 7A0F	528 bytes
Option bytes			0x1FFF C000 - 0x1FFF C00F	16 bytes
			0x1FFE C000 - 0x1FFE C00F	16 bytes

3.5 Read interface

3.5.1 Relation between CPU clock frequency and flash memory read time

To correctly read data from flash memory, the number of wait states (LATENCY) must be correctly programmed in the Flash access control register (FLASH_ACR) according to the frequency of the CPU clock (HCLK) and the supply voltage of the device.

The prefetch buffer must be disabled when the supply voltage is below 2.1 V. The correspondence between wait states and CPU clock frequency is given in *Table 11* and *Table 12*.

Note: On STM32F405xx/07xx and STM32F415xx/17xx devices:

- when VOS = '0', the maximum value of f_{HCLK} = 144 MHz.

- when VOS = '1', the maximum value of f_{HCLK} = 168 MHz.

On STM32F42xxx and STM32F43xxx devices:

- when VOS[1:0] = '0x01', the maximum value of f_{HCLK} is 120 MHz.

- when VOS[1:0] = '0x10', the maximum value of f_{HCLK} is 144 MHz. It can be extended to 168 MHz by activating the over-drive mode.

- when VOS[1:0] = '0x11, the maximum value of f_{HCLK} is 168 MHz. It can be extended to 180 MHz by activating the over-drive mode.

- The over-drive mode is not available when V_{DD} ranges from 1.8 to 2.1 V.

Refer to Section 5.1.4: Voltage regulator for STM32F42xxx and STM32F43xxx for details on how to activate the over-drive mode.

Table 11. Number of wait states according to CPU clock (HCLK) frequency (STM32F405xx/07xx and STM32F415xx/17xx)

Wait states (WS) (LATENCY)	HCLK (MHz)			
	Voltage range 2.7 V - 3.6 V	Voltage range 2.4 V - 2.7 V	Voltage range 2.1 V - 2.4 V	Voltage range 1.8 V - 2.1 V Prefetch OFF
0 WS (1 CPU cycle)	0 < HCLK ≤30	0 < HCLK ≤24	0 < HCLK ≤22	0 < HCLK ≤20
1 WS (2 CPU cycles)	30 < HCLK ≤60	24 < HCLK ≤48	22 < HCLK ≤44	20 <HCLK ≤40
2 WS (3 CPU cycles)	60 < HCLK ≤90	48 < HCLK ≤72	44 < HCLK ≤66	40 < HCLK ≤60
3 WS (4 CPU cycles)	90 < HCLK ≤120	72 < HCLK ≤96	66 < HCLK ≤88	60 < HCLK ≤80
4 WS (5 CPU cycles)	120 < HCLK ≤150	96 < HCLK ≤120	88 < HCLK ≤110	80 < HCLK ≤100
5 WS (6 CPU cycles)	150 < HCLK ≤168	120 < HCLK ≤144	110 < HCLK ≤132	100 < HCLK ≤120
6 WS (7 CPU cycles)	-	144 < HCLK ≤168	132 < HCLK ≤154	120 < HCLK ≤140
7 WS (8 CPU cycles)	-	-	154 < HCLK ≤168	140 < HCLK ≤160

Table 12. Number of wait states according to CPU clock (HCLK) frequency (STM32F42xxx and STM32F43xxx)

Wait states (WS) (LATENCY)	HCLK (MHz)			
	Voltage range 2.7 V - 3.6 V	Voltage range 2.4 V - 2.7 V	Voltage range 2.1 V - 2.4 V	Voltage range 1.8 V - 2.1 V Prefetch OFF
0 WS (1 CPU cycle)	0 <HCLK ≤30	0 <HCLK ≤24	0 <HCLK ≤22	0 < HCLK ≤20
1 WS (2 CPU cycles)	30 <HCLK ≤60	24 < HCLK ≤48	22 <HCLK ≤44	20 <HCLK ≤40
2 WS (3 CPU cycles)	60 <HCLK ≤90	48 < HCLK ≤72	44 < HCLK ≤66	40 < HCLK ≤60
3 WS (4 CPU cycles)	90 <HCLK ≤120	72 < HCLK ≤96	66 <HCLK ≤88	60 < HCLK ≤80
4 WS (5 CPU cycles)	120 <HCLK ≤150	96 < HCLK ≤120	88 < HCLK ≤110	80 < HCLK ≤100
5 WS (6 CPU cycles)	150 <HCLK ≤180	120 <HCLK ≤144	110 < HCLK ≤132	100 < HCLK ≤120
6 WS (7 CPU cycles)	-	144 <HCLK ≤168	132 < HCLK ≤154	120 < HCLK ≤140
7 WS (8 CPU cycles)	-	168 <HCLK ≤180	154 <HCLK ≤176	140 < HCLK ≤160
8 WS (9 CPU cycles)	-	-	176 <HCLK ≤180	160 < HCLK ≤168

After reset, the CPU clock frequency is 16 MHz and 0 wait state (WS) is configured in the FLASH_ACR register.

It is highly recommended to use the following software sequences to tune the number of wait states needed to access the flash memory with the CPU frequency.

Increasing the CPU frequency

1. Program the new number of wait states to the LATENCY bits in the FLASH_ACR register
2. Check that the new number of wait states is taken into account to access the flash memory by reading the FLASH_ACR register
3. Modify the CPU clock source by writing the SW bits in the RCC_CFGR register
4. If needed, modify the CPU clock prescaler by writing the HPRE bits in RCC_CFGR
5. Check that the new CPU clock source or/and the new CPU clock prescaler value is/are taken into account by reading the clock source status (SWS bits) or/and the AHB prescaler value (HPRE bits), respectively, in the RCC_CFGR register.

Decreasing the CPU frequency

1. Modify the CPU clock source by writing the SW bits in the RCC_CFGR register
2. If needed, modify the CPU clock prescaler by writing the HPRE bits in RCC_CFGR
3. Check that the new CPU clock source or/and the new CPU clock prescaler value is/are taken into account by reading the clock source status (SWS bits) or/and the AHB prescaler value (HPRE bits), respectively, in the RCC_CFGR register
4. Program the new number of wait states to the LATENCY bits in FLASH_ACR
5. Check that the new number of wait states is used to access the flash memory by reading the FLASH_ACR register

Note: A change in CPU clock configuration or wait state (WS) configuration may not be effective straight away. To make sure that the current CPU clock frequency is the one you have configured, you can check the AHB prescaler factor and clock source status values. To make sure that the number of WS you have programmed is effective, you can read the FLASH_ACR register.

3.5.2 Adaptive real-time memory accelerator (ART Accelerator™)

The proprietary Adaptive real-time (ART) memory accelerator is optimized for STM32 industry-standard Arm® Cortex®-M4 with FPU processors. It balances the inherent performance advantage of the Arm® Cortex®-M4 with FPU over flash memory technologies, which normally requires the processor to wait for the flash memory at higher operating frequencies.

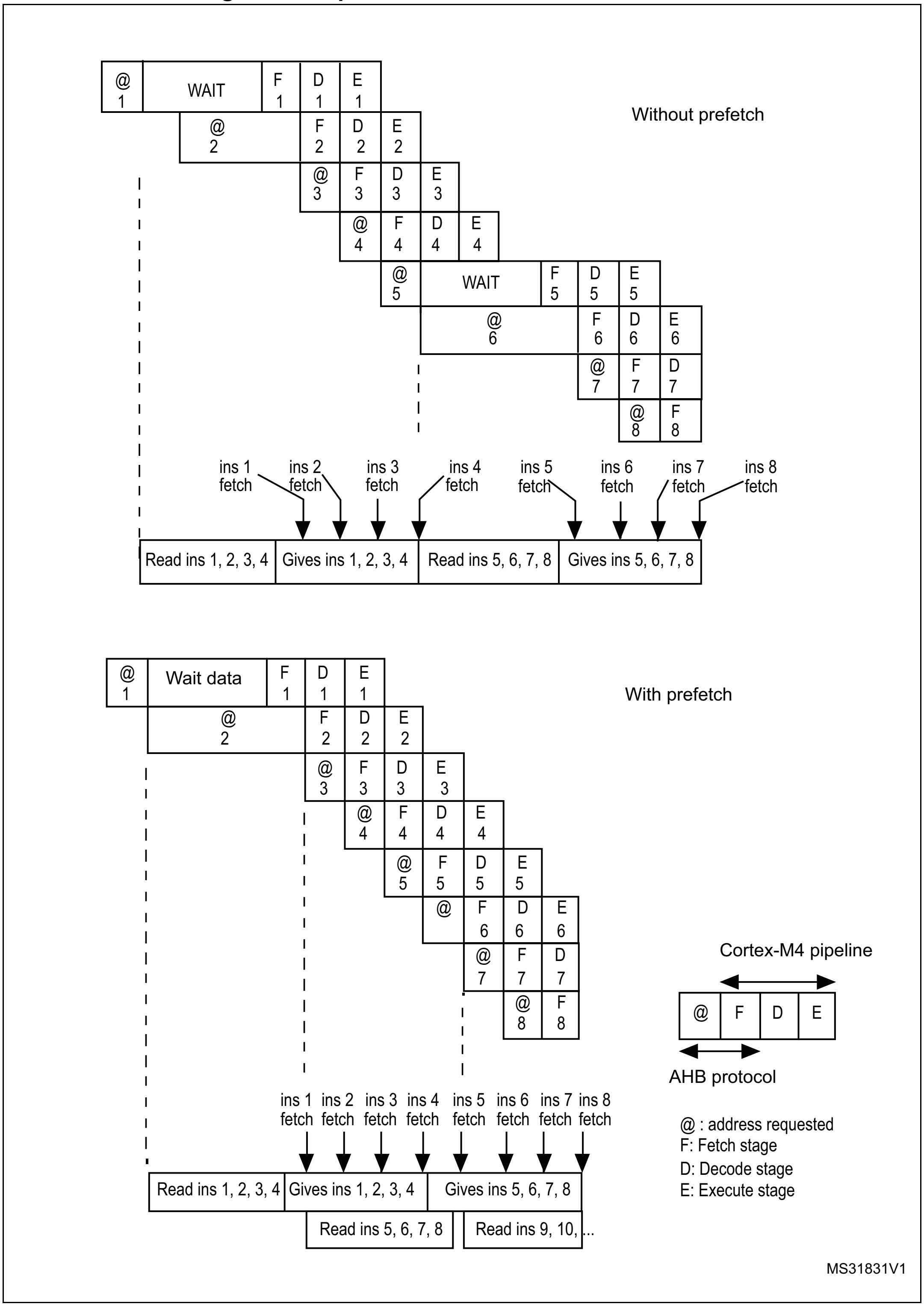
To release the processor full performance, the accelerator implements an instruction prefetch queue and branch cache which increases program execution speed from the 128-bit flash memory. Based on CoreMark benchmark, the performance achieved thanks to the ART accelerator is equivalent to 0 wait state program execution from flash memory at a CPU frequency up to 180 MHz.

Instruction prefetch

Each flash memory read operation provides 128 bits from either four instructions of 32 bits or 8 instructions of 16 bits according to the program launched. So, in case of sequential code, at least four CPU cycles are needed to execute the previous read instruction line. Prefetch on the I-Code bus can be used to read the next sequential instruction line from the flash memory while the current instruction line is being requested by the CPU. Prefetch is enabled by setting the PRFTEN bit in the FLASH_ACR register. This feature is useful if at least one wait state is needed to access the flash memory.

Figure 5 shows the execution of sequential 32-bit instructions with and without prefetch when 3 WSs are needed to access the flash memory.

Figure 5. Sequential 32-bit instruction execution



When the code is not sequential (branch), the instruction may not be present in the currently used instruction line or in the prefetched instruction line. In this case (miss), the penalty in terms of number of cycles is at least equal to the number of wait states.

Instruction cache memory

To limit the time lost due to jumps, it is possible to retain 64 lines of 128 bits in an instruction cache memory. This feature can be enabled by setting the instruction cache enable (ICEN) bit in the FLASH_ACR register. Each time a miss occurs (requested data not present in the currently used instruction line, in the prefetched instruction line or in the instruction cache memory), the line read is copied into the instruction cache memory. If some data contained in the instruction cache memory are requested by the CPU, they are provided without inserting any delay. Once all the instruction cache memory lines have been filled, the LRU (least recently used) policy is used to determine the line to replace in the instruction memory cache. This feature is particularly useful in case of code containing loops.

Data management

Literal pools are fetched from flash memory through the D-Code bus during the execution stage of the CPU pipeline. The CPU pipeline is consequently stalled until the requested literal pool is provided. To limit the time lost due to literal pools, accesses through the AHB databus D-Code have priority over accesses through the AHB instruction bus I-Code.

If some literal pools are frequently used, the data cache memory can be enabled by setting the data cache enable (DCEN) bit in the FLASH_ACR register. This feature works like the instruction cache memory, but the retained data size is limited to 8 rows of 128 bits.

Note: Data in user configuration sector are not cacheable.

3.6 Erase and program operations

For any flash memory program operation (erase or program), the CPU clock frequency (HCLK) must be at least 1 MHz. The contents of the flash memory are not guaranteed if a device reset occurs during a flash memory operation.

Any attempt to read the flash memory on STM32F4xx while it is being written or erased, causes the bus to stall. Read operations are processed correctly once the program operation has completed. This means that code or data fetches cannot be performed while a write/erase operation is ongoing.

On STM32F42xxx and STM32F43xxx devices, two banks are available allowing read operation from one bank while a write/erase operation is performed to the other bank.

3.6.1 Unlocking the Flash control register

After reset, write is not allowed in the Flash control register (FLASH_CR) to protect the flash memory against possible unwanted operations due, for example, to electric disturbances. The following sequence is used to unlock this register:

1. Write KEY1 = 0x45670123 in the Flash key register (FLASH_KEYR)
2. Write KEY2 = 0xCDEF89AB in the Flash key register (FLASH_KEYR)

Any wrong sequence returns a bus error and lock up the FLASH_CR register until the next reset.

The FLASH_CR register can be locked again by software by setting the LOCK bit in the FLASH_CR register.

Note: The FLASH_CR register is not accessible in write mode when the BSY bit in the FLASH_SR register is set. Any attempt to write to it with the BSY bit set causes the AHB bus to stall until the BSY bit is cleared.

3.6.2 Program/erase parallelism

The parallelism size is configured through the PSIZE field in the FLASH_CR register. It represents the number of bytes to be programmed each time a write operation occurs to the flash memory. PSIZE is limited by the supply voltage and by whether the external V_{PP} supply is used or not. It must therefore be correctly configured in the FLASH_CR register before any programming/erasing operation.

A flash memory erase operation can only be performed by sector, bank or for the whole flash memory (mass erase). The erase time depends on PSIZE programmed value. For more details on the erase time, refer to the electrical characteristics section of the device datasheet.

Table 13 provides the correct maximum PSIZE values.

Table 13. Maximum program/erase parallelism

	Voltage range 2.7 - 3.6 V with External V_{PP}	Voltage range 2.7 - 3.6 V	Voltage range 2.4 - 2.7 V	Voltage range 2.1 - 2.4 V	Voltage range 1.8 V - 2.1 V
Max. parallelism size	x64	x32	x16		x8
PSIZE(1:0)	11	10	01		00

Note: Any program or erase operation started with inconsistent program parallelism/voltage range settings may lead to unpredicted results. Even if a subsequent read operation indicates that the logical value was effectively written to the memory, this value may not be retained.

To use V_{PP}, an external high-voltage supply (between 8 and 9 V) must be applied to the V_{PP} pad. The external supply must be able to sustain this voltage range even if the DC consumption exceeds 10 mA. It is advised to limit the use of VPP to initial programming on the factory line. The V_{PP} supply must not be applied for more than an hour, otherwise the flash memory might be damaged.

3.6.3 Erase

The flash memory erase operation can be performed at sector level or on the whole flash memory (Mass Erase). Mass Erase does not affect the OTP sector or the configuration sector.

Sector Erase

To erase a sector, follow the procedure below:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register
2. Set the SER bit and select the sector out of the 12 sectors (for STM32F405xx/07xx and STM32F415xx/17xx) and out of 24 (for STM32F42xxx and STM32F43xxx) in the main memory block you wish to erase (SNB) in the FLASH_CR register
3. Set the STRT bit in the FLASH_CR register
4. Wait for the BSY bit to be cleared

Bank erase in STM32F42xxx and STM32F43xxx devices

To erase bank 1 or bank 2, follow the procedure below:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register
2. Set MER or MER1 bit accordingly in the FLASH_CR register
3. Set the STRT bit in the FLASH_CR register
4. Wait for the BSY bit to be reset.

Mass Erase

To perform Mass Erase, the following sequence is recommended:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register
2. Set the MER bit in the FLASH_CR register (on STM32F405xx/07xx and STM32F415xx/17xx devices)
3. Set both the MER and MER1 bits in the FLASH_CR register (on STM32F42xxx and STM32F43xxx devices).
4. Set the STRT bit in the FLASH_CR register
5. Wait for the BSY bit to be cleared

Note: If MERx and SER bits are both set in the FLASH_CR register, mass erase is performed.

If both MERx and SER bits are reset and the STRT bit is set, an unpredictable behavior may occur without generating any error flag. This condition should be forbidden.

3.6.4 Programming

Standard programming

The flash memory programming sequence is as follows:

1. Check that no main flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register.
2. Set the PG bit in the FLASH_CR register
3. Perform the data write operation(s) to the desired memory address (inside main memory block or OTP area):
   – Byte access in case of x8 parallelism
   – Half-word access in case of x16 parallelism
   – Word access in case of x32 parallelism
   – Double word access in case of x64 parallelism
4. Wait for the BSY bit to be cleared.

Note: Successive write operations are possible without the need of an erase operation when changing bits from '1' to '0'. Writing '1' requires a flash memory erase operation.

If an erase and a program operation are requested simultaneously, the erase operation is performed first.

Programming errors

It is not allowed to program data to the flash memory that would cross the 128-bit row boundary. In such a case, the write operation is not performed and a program alignment error flag (PGAERR) is set in the FLASH_SR register.

The write access type (byte, half-word, word or double word) must correspond to the type of parallelism chosen (x8, x16, x32 or x64). If not, the write operation is not performed and a program parallelism error flag (PGPERR) is set in the FLASH_SR register.

If the standard programming sequence is not respected (for example, if there is an attempt to write to a flash memory address when the PG bit is not set), the operation is aborted and a program sequence error flag (PGSERR) is set in the FLASH_SR register.

Programming and caches

If a flash memory write access concerns some data in the data cache, the Flash write access modifies the data in the flash memory and the data in the cache.

If an erase operation in flash memory also concerns data in the data or instruction cache, you have to make sure that these data are rewritten before they are accessed during code execution. If this cannot be done safely, it is recommended to flush the caches by setting the DCRST and ICRST bits in the FLASH_CR register.

Note: The I/D cache should be flushed only when it is disabled (I/DCEN = 0).

3.6.5 Read-while-write (RWW)

In STM32F42xxx and STM32F43xxx devices, the flash memory is divided into two banks allowing read-while-write operations. This feature allows to perform a read operation from one bank while an erase or program operation is performed to the other bank.

Note: Write-while-write operations are not allowed. As an exampled, It is not possible to perform an erase operation on one bank while programming the other one.

Read from bank 1 while erasing bank 2

While executing a program code from bank 1, it is possible to perform an erase operation on bank 2 (and vice versa). Follow the procedure below:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register (BSY is active when erase/program operation is on going to bank 1 or bank 2)
2. Set MER or MER1 bit in the FLASH_CR register
3. Set the STRT bit in the FLASH_CR register
4. Wait for the BSY bit to be reset (or use the EOP interrupt).

Read from bank 1 while programming bank 2

While executing a program code (over the I-Code bus) from bank 1,it is possible to perform an program operation to the bank 2 (and vice versa). Follow the procedure below:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register (BSY is active when erase/program operation is on going on bank 1 or bank 2)
2. Set the PG bit in the FLASH_CR register
3. Perform the data write operation(s) to the desired memory address inside main memory block or OTP area
4. Wait for the BSY bit to be reset.

3.6.6 Interrupts

Setting the end of operation interrupt enable bit (EOPIE) in the FLASH_CR register enables interrupt generation when an erase or program operation ends, that is when the busy bit (BSY) in the FLASH_SR register is cleared (operation completed, correctly or not). In this case, the end of operation (EOP) bit in the FLASH_SR register is set.

If an error occurs during a program, an erase, or a read operation request, one of the following error flags is set in the FLASH_SR register:

- PGAERR, PGPERR, PGSERR (Program error flags)
- WRPERR (Protection error flag)
- RDERR (Read protection error flag) for STM32F42xxx and STM32F43xxx devices only.

In this case, if the error interrupt enable bit (ERRIE) is set in the FLASH_CR register, an interrupt is generated and the operation error bit (OPERR) is set in the FLASH_SR register.

Note: If several successive errors are detected (for example, in case of DMA transfer to the flash memory), the error flags cannot be cleared until the end of the successive write requests.

Table 14. Flash interrupt request

Interrupt event	Event flag	Enable control bit
End of operation	EOP	EOPIE
Write protection error	WRPERR	ERRIE
Programming error	PGAERR, PGPERR, PGSERR	ERRIE
Read protection error	RDERR	ERRIE

3.7 Option bytes

3.7.1 Description of user option bytes

The option bytes are configured by the end user depending on the application requirements. *Table 15* shows the organization of these bytes inside the user configuration sector.

Table 15. Option byte organization

Address	[63:16]	[15:0]
0x1FFF C000	Reserved	ROP & user option bytes (**RDP** & **USER**)
0x1FFF C008	Reserved	SPRMOD and Write protection **nWRP bits for sectors 0 to 11**

Table 15. Option byte organization (continued)

Address	[63:16]	[15:0]
0x1FFE C000	Reserved	Reserved
0x1FFE C008	Reserved	SPRMOD and Write protection **nWRP bits for sectors 12 to 23**

Table 16. Description of the option bytes (STM32F405xx/07xx and STM32F415xx/17xx)

Option bytes (word, address 0x1FFF C000)	
RDP: *Read protection option byte.* The read protection is used to protect the software code stored in flash memory.	
Bits 15:8	0xAA: Level 0, no protection 0xCC: Level 2, chip protection (debug and boot from RAM features disabled) Others: Level 1, read protection of memories (debug features limited)
USER: User option byte This byte is used to configure the following features: – Select the watchdog event: Hardware or software – Reset event when entering the Stop mode – Reset event when entering the Standby mode	
Bit 7	**nRST_STDBY** 0: Reset generated when entering the Standby mode 1: No reset generated
Bit 6	**nRST_STOP** 0: Reset generated when entering the Stop mode 1: No reset generated
Bit 5	**WDG_SW** 0: Hardware independent watchdog 1: Software independent watchdog
Bit 4	0x1: Not used
Bits 3:2	**BOR_LEV:** BOR reset Level These bits contain the supply level threshold that activates/releases the reset. They can be written to program a new BOR level value into flash memory. 00: BOR Level 3 (VBOR3), brownout threshold level 3 01: BOR Level 2 (VBOR2), brownout threshold level 2 10: BOR Level 1 (VBOR1), brownout threshold level 1 11: BOR off, POR/PDR reset threshold level is applied *Note: For full details on BOR characteristics, refer to the "Electrical characteristics" section of the product datasheet.*
Bits 1:0	0x3: Not used

Table 16. Description of the option bytes (STM32F405xx/07xx and STM32F415xx/17xx) (continued)

Option bytes (word, address 0x1FFF C008)	
Bits 15:12	0xF: Not used
nWRP: *Flash memory write protection option bytes* Sectors 0 to 11 can be write protected.	
Bits 11:0	**nWRPi** 0: Write protection active on selected sector 1: Write protection not active on selected sector

Table 17. Description of the option bytes (STM32F42xxx and STM32F43xxx)

Option bytes (word, address 0x1FFF C000)	
RDP: *Read protection option byte.* The read protection is used to protect the software code stored in flash memory.	
Bit 15:8	0xAA: Level 0, no protection 0xCC: Level 2, chip protection (debug and boot from RAM features disabled) Others: Level 1, read protection of memories (debug features limited)
USER: User option byte This byte is used to configure the following features: Select the watchdog event: Hardware or software Reset event when entering the Stop mode Reset event when entering the Standby mode	
Bit 7	**nRST_STDBY** 0: Reset generated when entering the Standby mode 1: No reset generated
Bit 6	**nRST_STOP** 0: Reset generated when entering the Stop mode 1: No reset generated
Bit 5	**WDG_SW** 0: Hardware independent watchdog 1: Software independent watchdog
Bit 4	**BFB2**: Dual bank boot 0: Boot from flash memory bank 1 or system memory depending on boot pin state (Default). 1: Boot always from system memory (Dual bank boot mode).

Table 17. Description of the option bytes (STM32F42xxx and STM32F43xxx) (continued)

Bits 3:2	**BOR_LEV:** BOR reset Level These bits contain the supply level threshold that activates/releases the reset. They can be written to program a new BOR level value into flash memory. 00: BOR Level 3 (VBOR3), brownout threshold level 3 01: BOR Level 2 (VBOR2), brownout threshold level 2 10: BOR Level 1 (VBOR1), brownout threshold level 1 11: BOR off, POR/PDR reset threshold level is applied *Note: For full details on BOR characteristics, refer to the “Electrical characteristics” section of the product datasheet.*
Bits 1:0	0x3: Not used
Option bytes (word, address 0x1FFF C008)	
Bit 15	**SPRMOD**: Selection of protection mode of nWPRi bits 0: nWPRi bits used for sector i write protection (Default) 1: nWPRi bits used for sector i PCROP protection (Sector)
Bit 14	**DB1M**: Dual bank 1 Mbyte flash memory devices 0: 1 Mbyte single flash memory (contiguous addresses in bank 1) 1: 1 Mbyte dual bank flash memory. The flash memory is organized as two banks of 512 Kbytes each (see *Table 7: 1 Mbyte flash memory single bank vs dual bank organization (STM32F42xxx and STM32F43xxx)* and *Table 9: 1 Mbyte dual bank flash memory organization (STM32F42xxx and STM32F43xxx)*). To perform an erase operation, the right sector must be programmed (see *Table 7* for information on the sector numbering scheme).
Bits 13:12	0x3: not used
nWRP: Flash memory write protection option bytes for bank 1. Sectors 0 to 11 can be write protected.	
Bits 11:0	**nWRPi:** If SPRMOD is reset (default value): 0: Write protection active on sector i. 1: Write protection not active on sector i. If SPRMOD is set (active): 0: PCROP protection not active on sector i. 1: PCROP protection active on sector i.
Option bytes (word, address 0x1FFE C000)	
Bit 15:0	0xFFFF: not used

Table 17. Description of the option bytes (STM32F42xxx and STM32F43xxx) (continued)

Option bytes (word, address 0x1FFE C008)	
Bit 15:12	0xF: not used
nWRP: Flash memory write protection option bytes for bank 2. Sectors 12 to 23 can be write protected.	
Bits 11: 0	nWRPi: If SPRMOD is reset (default value): 0: Write protection active on sector i. 1: Write protection not active on sector i. If SPRMOD is set (active): 0: PCROP protection not active on sector i. 1: PCROP protection active on sector i.

3.7.2 Programming user option bytes

To run any operation on this sector, the option lock bit (OPTLOCK) in the Flash option control register (FLASH_OPTCR) must be cleared. To be allowed to clear this bit, you have to perform the following sequence:

1. Write OPTKEY1 = 0x0819 2A3B in the Flash option key register (FLASH_OPTKEYR)
2. Write OPTKEY2 = 0x4C5D 6E7F in the Flash option key register (FLASH_OPTKEYR)

The user option bytes can be protected against unwanted erase/program operations by setting the OPTLOCK bit by software.

Modifying user option bytes on STM32F405xx/07xx and STM32F415xx/17xx

To modify the user option value, follow the sequence below:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register
2. Write the desired option value in the FLASH_OPTCR register.
3. Set the option start bit (OPTSTRT) in the FLASH_OPTCR register
4. Wait for the BSY bit to be cleared.

Note: The value of an option is automatically modified by first erasing the user configuration sector and then programming all the option bytes with the values contained in the FLASH_OPTCR register.

Modifying user option bytes on STM32F42xxx and STM32F43xxx

The user option bytes for bank 1 and bank 2 cannot be modified independently. They must be updated concurrently.

To modify the user option byte value, follow the sequence below:

1. Check that no flash memory operation is ongoing by checking the BSY bit in the FLASH_SR register
2. Write the bank 2 option byte value in the FLASH_OPTCR1 register
3. Write the bank 1 option byte value in the FLASH_OPTCR register.
4. Set the option start bit (OPTSTRT) in the FLASH_OPTCR register
5. Wait for the BSY bit to be cleared

Note: The value of an option byte is automatically modified by first erasing the user configuration sector (bank 1 and 2) and then programming all the option bytes with the values contained in the FLASH_OPTCR and FLASH_OPTCR1 registers.

3.7.3 Read protection (RDP)

The user area in the flash memory can be protected against read operations by an entrusted code. Three read protection levels are defined:

- Level 0: no read protection

  When the read protection level is set to Level 0 by writing 0xAA into the read protection option byte (RDP), all read/write operations (if no write protection is set) from/to the flash memory or the backup SRAM are possible in all boot configurations (Flash user boot, debug or boot from RAM).

- Level 1: read protection enabled

  It is the default read protection level after option byte erase. The read protection Level 1 is activated by writing any value (except for 0xAA and 0xCC used to set Level 0 and Level 2, respectively) into the RDP option byte. When the read protection Level 1 is set:

  – No access (read, erase, program) to flash memory or backup SRAM can be performed while the debug feature is connected or while booting from RAM or system memory bootloader. A bus error is generated in case of read request.
  – When booting from flash memory, accesses (read, erase, program) to flash memory and backup SRAM from user code are allowed.

  When Level 1 is active, programming the protection option byte (RDP) to Level 0 causes the flash memory and the backup SRAM to be mass-erased. As a result the user code area is cleared before the read protection is removed. The mass erase only erases the user code area. The other option bytes including write protections remain unchanged from before the mass-erase operation. The OTP area is not affected by mass erase and remains unchanged. Mass erase is performed only when Level 1 is active and Level 0 requested. When the protection level is increased (0->1, 1->2, 0->2) there is no mass erase.

- Level 2: debug/chip read protection disabled

  The read protection Level 2 is activated by writing 0xCC to the RDP option byte. When the read protection Level 2 is set:

  – All protections provided by Level 1 are active.
  – Booting from RAM or system memory bootloader is no more allowed.
  – JTAG, SWV (single-wire viewer), ETM, and boundary scan are disabled.
  – User option bytes can no longer be changed.
  – When booting from flash memory, accesses (read, erase and program) to flash memory and backup SRAM from user code are allowed.

  Memory read protection Level 2 is an irreversible operation. When Level 2 is activated, the level of protection cannot be decreased to Level 0 or Level 1.

Note: If the read protection is set while the debugger is still connected (or has been connected since the last power-on) through JTAG/SWD, apply a POR (power-on reset) instead of a system reset.

If the read protection is programmed by software (executing from SRAM), perform a POR to reload the option byte and clear the detected intrusion. This can be done with a transition Standby mode followed by a wake-up.

The JTAG port is permanently disabled when Level 2 is active (acting as a JTAG fuse). As a consequence, boundary scan cannot be performed. STMicroelectronics is not able to perform analysis on defective parts on which the Level 2 protection has been set.

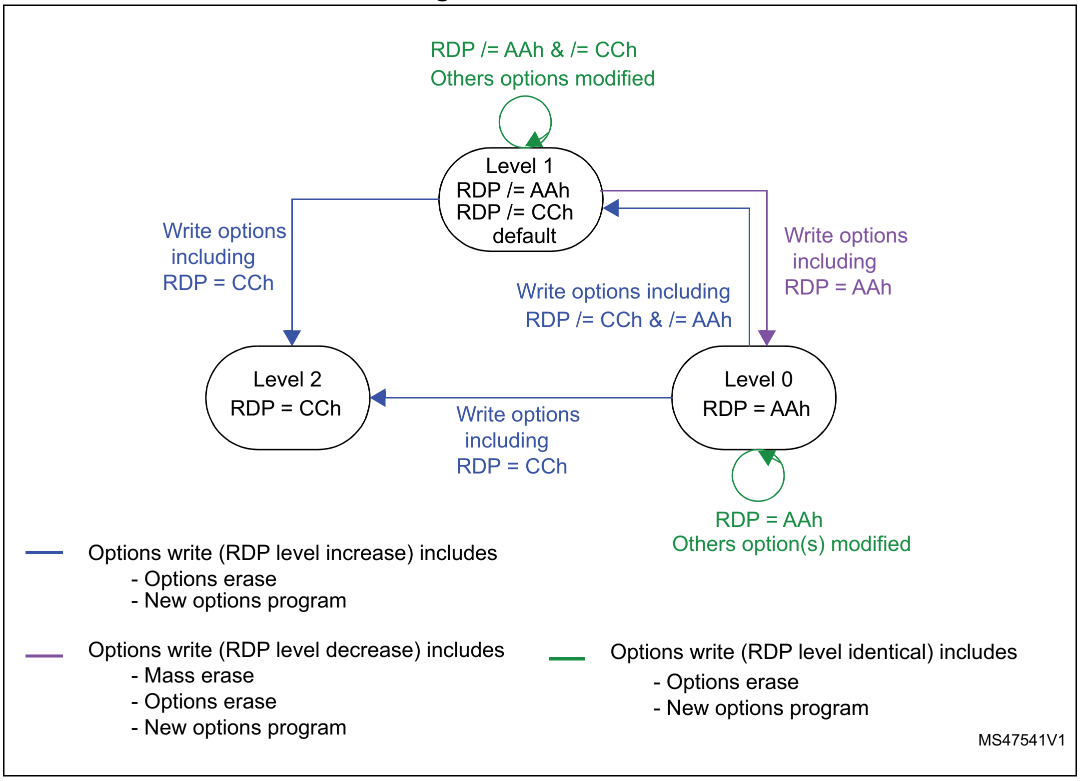
Table 18. Access versus read protection level

Memory area	Protection Level	Debug features, Boot from RAM or from System memory bootloader			Booting from flash memory		
		Read	Write	Erase	Read	Write	Erase
Main Flash Memory and Backup SRAM	Level 1	NO		NO(1)	YES		
	Level 2	NO			YES		
Option Bytes	Level 1	YES			YES		
	Level 2	NO			NO		
OTP	Level 1	NO		NA	YES		NA
	Level 2	NO		NA	YES		NA

1. The main flash memory and backup SRAM are only erased when the RDP changes from level 1 to 0. The OTP area remains unchanged.

Figure 6 shows how to go from one RDP level to another.

Figure 6. RDP levels



3.7.4 Write protections

Up to 24 user sectors in flash memory can be protected against unwanted write operations due to loss of program counter contexts. When the non-write protection nWRPi bit (0 ≤ i ≤11) in the FLASH_OPTCR or FLASH_OPTCR1 registers is low, the corresponding sector cannot be erased or programmed. Consequently, a mass erase cannot be performed if one of the sectors is write-protected.

If an erase/program operation to a write-protected part of the flash memory is attempted (sector protected by write protection bit, OTP part locked or part of the flash memory that can never be written like the ICP), the write protection error flag (WRPERR) is set in the FLASH_SR register.

On STM32F42xxx and STM32F43xxx devices, when the PCROP mode is set, the active level of nWRPi is high, and the corresponding sector i is write protected when nWRPi is high. A PCROP sector is automatically write protected.

Note: When the memory read protection level is selected (RDP level = 1), it is not possible to program or erase flash memory sector i if the CPU debug features are connected (JTAG or single wire) or boot code is being executed from RAM, even if nWRPi = 1.

Write protection error flag

If an erase/program operation to a write protected area of the flash memory is performed, the Write Protection Error flag (WRPERR) is set in the FLASH_SR register.

If an erase operation is requested, the WRPERR bit is set when:

- Mass, bank, sector erase are configured (MER or MER/MER1 and SER = 1)
- A sector erase is requested and the Sector Number SNB field is not valid
- A mass erase is requested while at least one of the user sector is write protected by option bit (MER or MER/MER1 = 1 and nWRPi = 0 with $0 \leq i \leq 11$ bits in the FLASH_OPTCRx register
- A sector erase is requested on a write protected sector. (SER = 1, SNB = i and nWRPi = 0 with $0 \leq i \leq 11$ bits in the FLASH_OPTCRx register)
- The flash memory is readout protected and an intrusion is detected.

If a program operation is requested, the WRPERR bit is set when:

- A write operation is performed on system memory or on the reserved part of the user specific sector.
- A write operation is performed to the user configuration sector
- A write operation is performed on a sector write protected by option bit.
- A write operation is requested on an OTP area which is already locked
- The flash memory is read protected and an intrusion is detected.

3.7.5 Proprietary code readout protection (PCROP)

The proprietary readout protection (PCROP) is available only on STM32F42xxx and STM32F43xxx devices.

Flash memory user sectors (0 to 23) can be protected against D-bus read accesses by using the proprietary readout protection (PCROP).

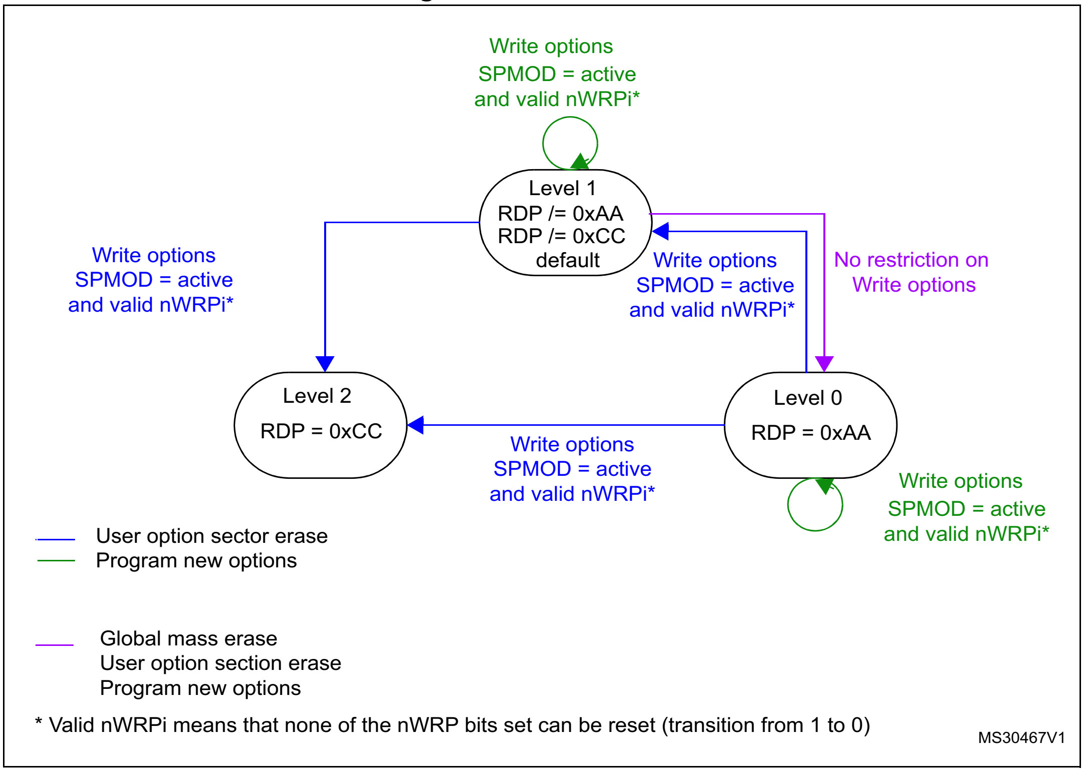
The PCROP protection is selected as follows, through the SPRMOD option bit in the FLASH_OPTCR register:

- SPRMOD = 0: nWRPi control the write protection of respective user sectors
- SPRMOD = 1: nWRPi control the read and write protection (PCROP) of respective user sectors.

When a sector is readout protected (PCROP mode activated), it can only be accessed for code fetch through ICODE Bus on Flash interface:

- Any read access performed through the D-bus triggers a RDERR flag error.
- Any program/erase operation on a PCROPed sector triggers a WRPERR flag error.

Figure 7. PCROP levels



The deactivation of the SPRMOD and/or the unprotection of PCROPed user sectors can only occur when, at the same time, the RDP level changes from 1 to 0. If this condition is not respected, the user option byte modification is cancelled and the write error WRPERR flag is set. The modification of the users option bytes (BOR_LEV, RST_STDBY, ..) is allowed since none of the active nWRPi bits is reset and SPRMOD is kept active.

Note: The active value of nWRPi bits is inverted when PCROP mode is active (SPRMOD =1).

If SPRMOD = 1 and nWRPi =1, then user sector i of bank 1, respectively bank 2 is read/write protected (PCROP).

3.8 One-time programmable bytes

Table 19 shows the organization of the one-time programmable (OTP) part of the OTP area.

Table 19. OTP area organization

Block	[128:96]	[95:64]	[63:32]	[31:0]	Address byte 0
0	OTP0	OTP0	OTP0	OTP0	0x1FFF 7800
	OTP0	OTP0	OTP0	OTP0	0x1FFF 7810
1	OTP1	OTP1	OTP1	OTP1	0x1FFF 7820
	OTP1	OTP1	OTP1	OTP1	0x1FFF 7830
. . .			. . .		. . .
15	OTP15	OTP15	OTP15	OTP15	0x1FFF 79E0
	OTP15	OTP15	OTP15	OTP15	0x1FFF 79F0
Lock block	LOCKB15 ... LOCKB12	LOCKB11 ... LOCKB8	LOCKB7 ... LOCKB4	LOCKB3 ... LOCKB0	0x1FFF 7A00

The OTP area is divided into 16 OTP data blocks of 32 bytes and one lock OTP block of 16 bytes. The OTP data and lock blocks cannot be erased. The lock block contains 16 bytes LOCKBi ($0 \leq i \leq 15$) to lock the corresponding OTP data block (blocks 0 to 15). Each OTP data block can be programmed until the value 0x00 is programmed in the corresponding OTP lock byte. The lock bytes must only contain 0x00 and 0xFF values, otherwise the OTP bytes might not be taken into account correctly.

3.9 Flash interface registers

3.9.1 Flash access control register (FLASH_ACR) for STM32F405xx/07xx and STM32F415xx/17xx

The Flash access control register is used to enable/disable the acceleration features and control the flash memory access time according to CPU frequency.

Address offset: 0x00
Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DCRST	ICRST	DCEN	ICEN	PRFTEN	Reserved					LATENCY[2:0]		
			rw	w	rw	rw	rw						rw	rw	rw

Bits 31:13 Reserved, must be kept cleared.

Bit 12 **DCRST:** Data cache reset
0: Data cache is not reset
1: Data cache is reset
This bit can be written only when the D cache is disabled.

Bit 11 **ICRST:** Instruction cache reset
0: Instruction cache is not reset
1: Instruction cache is reset
This bit can be written only when the I cache is disabled.

Bit 10 **DCEN:** Data cache enable
0: Data cache is disabled
1: Data cache is enabled

Bit 9 **ICEN:** Instruction cache enable
0: Instruction cache is disabled
1: Instruction cache is enabled

Bit 8 **PRFTEN:** Prefetch enable
0: Prefetch is disabled
1: Prefetch is enabled

Bits 7:3 Reserved, must be kept cleared.

Bits 2:0 **LATENCY[2:0]:** Latency
These bits represent the ratio of the CPU clock period to the flash memory access time.
000: Zero wait state
001: One wait state
010: Two wait states
011: Three wait states
100: Four wait states
101: Five wait states
110: Six wait states
111: Seven wait states

3.9.2 Flash access control register (FLASH_ACR) for STM32F42xxx and STM32F43xxx

The Flash access control register is used to enable/disable the acceleration features and control the flash memory access time according to CPU frequency.

Address offset: 0x00
Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved			DCRST	ICRST	DCEN	ICEN	PRFTEN	Reserved				LATENCY[3:0]			
			rw	w	rw	rw	rw					rw	rw	rw	rw

Bits 31:13 Reserved, must be kept cleared.

Bit 12 **DCRST:** Data cache reset
0: Data cache is not reset
1: Data cache is reset
This bit can be written only when the D cache is disabled.

Bit 11 **ICRST:** Instruction cache reset
0: Instruction cache is not reset
1: Instruction cache is reset
This bit can be written only when the I cache is disabled.

Bit 10 **DCEN:** Data cache enable
0: Data cache is disabled
1: Data cache is enabled

Bit 9 **ICEN:** Instruction cache enable
0: Instruction cache is disabled
1: Instruction cache is enabled

Bit 8 **PRFTEN:** Prefetch enable
0: Prefetch is disabled
1: Prefetch is enabled

Bits 7:4 Reserved, must be kept cleared.

Bits 3:0 **LATENCY[3:0]:** Latency
These bits represent the ratio of the CPU clock period to the flash memory access time.
0000: Zero wait state
0001: One wait state
0010: Two wait states
...
1110: Fourteen wait states
1111: Fifteen wait states

3.9.3 Flash key register (FLASH_KEYR)

The Flash key register is used to allow access to the Flash control register and so, to allow program and erase operations.

Address offset: 0x04
Reset value: 0x0000 0000

Access: no wait state, word access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEY[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **FKEYR[31:0]**: FPEC key

The following values must be programmed consecutively to unlock the FLASH_CR register and allow programming/erasing it:

a) KEY1 = 0x45670123

b) KEY2 = 0xCDEF89AB

3.9.4 Flash option key register (FLASH_OPTKEYR)

The Flash option key register is used to allow program and erase operations in the user configuration sector.

Address offset: 0x08
Reset value: 0x0000 0000

Access: no wait state, word access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OPTKEYR[31:16															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OPTKEYR[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **OPTKEYR[31:0]**: Option byte key

The following values must be programmed consecutively to unlock the FLASH_OPTCR register and allow programming it:

a) OPTKEY1 = 0x08192A3B

b) OPTKEY2 = 0x4C5D6E7F

3.9.5 Flash status register (FLASH_SR) for STM32F405xx/07xx and STM32F415xx/17xx

The Flash status register gives information on ongoing program and erase operations.

Address offset: 0x0C
Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td></tr>
<tr><td colspan="15" rowspan="2">Reserved</td><td>BSY</td></tr>
<tr><td>r</td></tr>
<tr><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="8" rowspan="2">Reserved</td><td>PGSERR</td><td>PGPERR</td><td>PGAERR</td><td>WRPERR</td><td colspan="2" rowspan="2">Reserved</td><td>OPERR</td><td>EOP</td></tr>
<tr><td>rc_w1</td><td>rc_w1</td><td>rc_w1</td><td>rc_w1</td><td>rc_w1</td><td>rc_w1</td></tr>
</table>

Bits 31:17 Reserved, must be kept cleared.

Bit 16 **BSY:** Busy
This bit indicates that a flash memory operation is in progress. It is set at the beginning of a flash memory operation and cleared when the operation finishes or an error occurs.
0: no flash memory operation ongoing
1: Flash memory operation ongoing

Bits 15:8 Reserved, must be kept cleared.

Bit 7 **PGSERR:** Programming sequence error
Set by hardware when a write access to the flash memory is performed by the code while the control register has not been correctly configured.
Cleared by writing 1.

Bit 6 **PGPERR:** Programming parallelism error
Set by hardware when the size of the access (byte, half-word, word, double word) during the program sequence does not correspond to the parallelism configuration PSIZE (x8, x16, x32, x64).
Cleared by writing 1.

Bit 5 **PGAERR:** Programming alignment error
Set by hardware when the data to program cannot be contained in the same 128-bit flash memory row.
Cleared by writing 1.

Bit 4 **WRPERR:** Write protection error
Set by hardware when an address to be erased/programmed belongs to a write-protected part of the flash memory.
Cleared by writing 1.

Bits 3:2 Reserved, must be kept cleared.

Bit 1 **OPERR:** Operation error
Set by hardware when a flash operation (programming / erase /read) request is detected and can not be run because of parallelism, alignment, or write protection error. This bit is set only if error interrupts are enabled (ERRIE = 1).

Bit 0 **EOP:** End of operation
Set by hardware when one or more flash memory operations (program/erase) has/have completed successfully. It is set only if the end of operation interrupts are enabled (EOPIE = 1).
Cleared by writing a 1.

3.9.6 Flash status register (FLASH_SR) for STM32F42xxx and STM32F43xxx

The Flash status register gives information on ongoing program and erase operations.

Address offset: 0x0C

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															BSY
															r

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							RDERR	PGSERR	PGPERR	PGAERR	WRPERR	Reserved		OPERR	EOP
							rc_w1	rc_w1	rc_w1	rc_w1	rc_w1			rc_w1	rc_w1

Bits 31:17 Reserved, must be kept cleared.

Bit 16 **BSY:** Busy
This bit indicates that a flash memory operation is in progress to/from one bank. It is set at the beginning of a flash memory operation and cleared when the operation finishes or an error occurs.
0: no flash memory operation ongoing
1: Flash memory operation ongoing

Bits 15:9 Reserved, must be kept cleared.

Bit 8 **RDERR:** Proprietary readout protection (PCROP) error
Set by hardware when a read access through the D-bus is performed to an address belonging to a proprietary readout protected Flash sector.
Cleared by writing 1.

Bit 7 **PGSERR:** Programming sequence error
Set by hardware when a write access to the flash memory is performed by the code while the control register has not been correctly configured.
Cleared by writing 1.

Bit 6 **PGPERR:** Programming parallelism error

Set by hardware when the size of the access (byte, half-word, word, double word) during the program sequence does not correspond to the parallelism configuration PSIZE (x8, x16, x32, x64).
Cleared by writing 1.

Bit 5 **PGAERR:** Programming alignment error

Set by hardware when the data to program cannot be contained in the same 128-bit flash memory row.
Cleared by writing 1.

Bit 4 **WRPERR:** Write protection error

Set by hardware when an address to be erased/programmed belongs to a write-protected part of the flash memory.
Cleared by writing 1.

Bits 3:2 Reserved, must be kept cleared.

Bit 1 **OPERR:** Operation error

Set by hardware when a flash operation (programming/erase/read) request is detected and can not be run because of parallelism, alignment, write or read (PCROP) protection error. This bit is set only if error interrupts are enabled (ERRIE = 1).

Bit 0 **EOP:** End of operation

Set by hardware when one or more flash memory operations (program/erase) has/have completed successfully. It is set only if the end of operation interrupts are enabled (EOPIE = 1).
Cleared by writing a 1.

3.9.7 Flash control register (FLASH_CR) for STM32F405xx/07xx and STM32F415xx/17xx

The Flash control register is used to configure and start flash memory operations.

Address offset: 0x10

Reset value: 0x8000 0000

Access: no wait state when no flash memory operation is ongoing, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCK	Reserved					ERRIE	EOPIE	Reserved							STRT
rs						rw	rw								rs

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						PSIZE[1:0]		Res.	SNB[3:0]				MER	SER	PG
						rw	rw		rw	rw	rw	rw	rw	rw	rw

Bit 31 **LOCK:** Lock
Write to 1 only. When it is set, this bit indicates that the FLASH_CR register is locked. It is cleared by hardware after detecting the unlock sequence.
In the event of an unsuccessful unlock operation, this bit remains set until the next reset.

Bits 30:26 Reserved, must be kept cleared.

Bit 25 **ERRIE:** Error interrupt enable
This bit enables the interrupt generation when the OPERR bit in the FLASH_SR register is set to 1.
0: Error interrupt generation disabled
1: Error interrupt generation enabled

Bit 24 **EOPIE:** End of operation interrupt enable
This bit enables the interrupt generation when the EOP bit in the FLASH_SR register goes to 1.
0: Interrupt generation disabled
1: Interrupt generation enabled

Bits 23:17 Reserved, must be kept cleared.

Bit 16 **STRT:** Start
This bit triggers an erase operation when set. It is set only by software and cleared when the BSY bit is cleared.

Bits 15:10 Reserved, must be kept cleared.

Bits 9:8 **PSIZE[1:0]:** Program size
These bits select the program parallelism.
00 program x8
01 program x16
10 program x32
11 program x64

Bit 7 Reserved, must be kept cleared.

Bits 6:3 **SNB[3:0]:** Sector number
These bits select the sector to erase.
0000 sector 0
0001 sector 1
...
1011 sector 11
Others not allowed

Bit 2 **MER:** Mass Erase
Erase activated for all user sectors.

Bit 1 **SER:** Sector Erase
Sector Erase activated.

Bit 0 **PG:** Programming
Flash programming activated.

3.9.8 Flash control register (FLASH_CR) for STM32F42xxx and STM32F43xxx

The Flash control register is used to configure and start flash memory operations.

Address offset: 0x10

Reset value: 0x8000 0000

Access: no wait state when no flash memory operation is ongoing, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCK	Reserved					ERRIE	EOPIE	Reserved							STRT
rs						rw	rw								rs

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MER1	Reserved					PSIZE[1:0]		SNB[4:0]					MER	SER	PG
rw						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **LOCK:** Lock

Write to 1 only. When it is set, this bit indicates that the FLASH_CR register is locked. It is cleared by hardware after detecting the unlock sequence.
In the event of an unsuccessful unlock operation, this bit remains set until the next reset.

Bits 30:26 Reserved, must be kept cleared.

Bit 25 **ERRIE:** Error interrupt enable

This bit enables the interrupt generation when the OPERR bit in the FLASH_SR register is set to 1.

0: Error interrupt generation disabled
1: Error interrupt generation enabled

Bit 24 **EOPIE:** End of operation interrupt enable

This bit enables the interrupt generation when the EOP bit in the FLASH_SR register goes to 1.

0: Interrupt generation disabled
1: Interrupt generation enabled

Bits 23:17 Reserved, must be kept cleared.

Bit 16 **STRT:** Start

This bit triggers an erase operation when set. It is set only by software and cleared when the BSY bit is cleared.

Bit 15 **MER1:** Mass Erase of bank 2 sectors

Erase activated for bank 2 user sectors 12 to 23.

Bits 14:10 Reserved, must be kept cleared.

Bits 9:8 **PSIZE[1:0]:** Program size

These bits select the program parallelism.

00 program x8
01 program x16
10 program x32
11 program x64

Bits 7:3 **SNB[3:0]:** Sector number

These bits select the sector to erase.

0000: sector 0
0001: sector 1
...
01011: sector 11
01100: not allowed
01101: not allowed
01110: not allowed
01111: not allowed
10000: section 12
10001: section 13
...
11011 sector 23
11100: not allowed
11101: not allowed
11110: not allowed
11111: not allowed

Bit 2 **MER:** Mass Erase of bank 1 sectors

Erase activated of bank 1 sectors.

Bit 1 **SER:** Sector Erase

Sector Erase activated.

Bit 0 **PG:** Programming

Flash programming activated.

3.9.9 Flash option control register (FLASH_OPTCR) for STM32F405xx/07xx and STM32F415xx/17xx

The FLASH_OPTCR register is used to modify the user option bytes.

Address offset: 0x14

Reset value: 0x0FFF AAED. The option bits are loaded with values from flash memory at reset release.

Access: no wait state when no flash memory operation is ongoing, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				nWRP[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDP[7:0]								nRST_ STDBY	nRST_ STOP	WDG_ SW	Reserved	BOR_LEV		OPTSTRT	OPTLOCK
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rs	rs

Bits 31:28 Reserved, must be kept cleared.

Bits 27:16 **nWRP[11:0]:** Not write protect

These bits contain the value of the write-protection option bytes after reset. They can be written to program a new write protect value into flash memory.

0: Write protection active on selected sector
1: Write protection inactive on selected sector

Bits 15:8 **RDP[7:0]:** Read protect

These bits contain the value of the read-protection option level after reset. They can be written to program a new read protection value into flash memory.

0xAA: Level 0, read protection not active
0xCC: Level 2, chip read protection active
Others: Level 1, read protection of memories active

Bits 7:5 **USER[2:0]:** User option bytes

These bits contain the value of the user option byte after reset. They can be written to program a new user option byte value into flash memory.

Bit 7: nRST_STDBY
Bit 6: nRST_STOP
Bit 5: WDG_SW

Note: When changing the WDG mode from hardware to software or from software to hardware, a system reset is required to make the change effective.

Bit 4 Reserved, must be kept cleared. Always read as “0”.

Bits 3:2 **BOR_LEV[1:0]:** BOR reset Level

These bits contain the supply level threshold that activates/releases the reset. They can be written to program a new BOR level. By default, BOR is off. When the supply voltage (V_{DD}) drops below the selected BOR level, a device reset is generated.

00: BOR Level 3 (VBOR3), brownout threshold level 3
01: BOR Level 2 (VBOR2), brownout threshold level 2
10: BOR Level 1 (VBOR1), brownout threshold level 1
11: BOR off, POR/PDR reset threshold level is applied

Note: For full details about BOR characteristics, refer to the “Electrical characteristics” section in the device datasheet.

Bit 1 **OPTSTRT:** Option start

This bit triggers a user option operation when set. It is set only by software and cleared when the BSY bit is cleared.

Bit 0 **OPTLOCK:** Option lock

Write to 1 only. When this bit is set, it indicates that the FLASH_OPTCR register is locked. This bit is cleared by hardware after detecting the unlock sequence.

In the event of an unsuccessful unlock operation, this bit remains set until the next reset.

3.9.10 Flash option control register (FLASH_OPTCR) for STM32F42xxx and STM32F43xxx

The FLASH_OPTCR register is used to modify the user option bytes.

Address offset: 0x14

Reset value: 0x0FFF AAED. The option bits are loaded with values from flash memory at reset release.

Access: no wait state when no flash memory operation is ongoing, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SPRMOD	DB1M	Reserved		nWRP[11:0]											
rw	rw			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDP[7:0]								nRST_STDBY	nRST_STOP	WDG_SW	BFB2	BOR_LEV		OPTSTRT	OPTLOCK
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rs	rs

Bit 31 **SPRMOD**: Selection of protection mode for nWPRi bits

0: PCROP disabled. nWPRi bits used for Write protection on sector i.

1: PCROP enabled. nWPRi bits used for PCROP protection on sector i

Bit 30 **DB1M**: Dual-bank on 1 Mbyte flash memory devices

0: 1 Mbyte single bank flash memory (contiguous addresses in bank1)

1: 1 Mbyte dual bank flash memory. The flash memory is organized as two banks of 512 Kbytes each (see *Table 7: 1 Mbyte flash memory single bank vs dual bank organization (STM32F42xxx and STM32F43xxx)* and *Table 9: 1 Mbyte dual bank flash memory organization (STM32F42xxx and STM32F43xxx)*). To perform an erase operation, the right sector must be programmed (see *Table 7* for information on the sector numbering scheme).

Note: If DB1M is set and an erase operation is performed on Bank 2 while the default sector number is selected (as an example, sector 8 is configured instead of sector 12), the erase operation on Bank 2 sector is not performed.

Bits 29:28 Reserved, must be kept cleared.

Bits 27:16 **nWRP[11:0]:** Not write protect

These bits contain the value of the write-protection and read-protection (PCROP) option bytes for sectors 0 to 11 after reset. They can be written to program a new write-protect or PCROP value into flash memory.

If SPRMOD is reset:

0: Write protection active on sector i

1: Write protection not active on sector i

If SPRMOD is set:

0: PCROP protection not active on sector i

1: PCROP protection active on sector i

Bits 15:8 **RDP[7:0]:** Read protect

These bits contain the value of the read-protection option level after reset. They can be written to program a new read protection value into flash memory.

0xAA: Level 0, read protection not active

0xCC: Level 2, chip read protection active

Others: Level 1, read protection of memories active

Bits 7:5 **USER:** User option bytes

These bits contain the value of the user option byte after reset. They can be written to program a new user option byte value into flash memory.

Bit 7: nRST_STDBY

Bit 6: nRST_STOP

Bit 5: WDG_SW

Note: When changing the WDG mode from hardware to software or from software to hardware, a system reset is required to make the change effective.

Bit 4 **BFB2**: Dual-bank Boot option byte

0: Dual-bank boot disabled. Boot can be performed either from flash memory bank 1 or from system memory depending on boot pin state (default)

1: Dual-bank boot enabled. Boot is always performed from system memory.

Note: For STM32F42xx and STM32F43xx 1MB part numbers, this option bit must be kept cleared when DB1M=0.

Bits 3:2 **BOR_LEV:** BOR reset Level

These bits contain the supply level threshold that activates/releases the reset. They can be written to program a new BOR level. By default, BOR is off. When the supply voltage (V_{DD}) drops below the selected BOR level, a device reset is generated.

00: BOR Level 3 (VBOR3), brownout threshold level 3

01: BOR Level 2 (VBOR2), brownout threshold level 2

10: BOR Level 1 (VBOR1), brownout threshold level 1

11: BOR off, POR/PDR reset threshold level is applied

Note: For full details on BOR characteristics, refer to the “Electrical characteristics” section of the product datasheet.

Bit 1 **OPTSTRT:** Option start

This bit triggers a user option operation when set. It is set only by software and cleared when the BSY bit is cleared.

Bit 0 **OPTLOCK:** Option lock

Write to 1 only. When this bit is set, it indicates that the FLASH_OPTCR register is locked. This bit is cleared by hardware after detecting the unlock sequence.

In the event of an unsuccessful unlock operation, this bit remains set until the next reset.

3.9.11 Flash option control register (FLASH_OPTCR1) for STM32F42xxx and STM32F43xxx

This register is available only on STM32F42xxx and STM32F43xxx.

The FLASH_OPTCR1 register is used to modify the user option bytes for bank 2.

Address offset: 0x18

Reset value: 0x0FFF 0000. The option bits are loaded with values from flash memory at reset release.

Access: no wait state when no flash memory operation is ongoing, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				nWRP[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bits 31:28 Reserved, must be kept cleared.

Bits 27:16 **nWRP[11:0]:** Not write protect

These bits contain the value of the write-protection and read-protection (PCROP) option bytes for sectors 0 to 11 after reset. They can be written to program a new write-protect or PCROP value into flash memory.

If SPRMOD is reset (default value):

0: Write protection active on sector i.

1: Write protection not active on sector i.

If SPRMOD is set:

0: PCROP protection not active on sector i.

1: PCROP protection active on sector i.

Bits 15:0 Reserved, must be kept cleared.

3.9.12 Flash interface register map

Table 20. Flash register map and reset values (STM32F405xx/07xx and STM32F415xx/17xx)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	FLASH_ACR	Reserved																			DCRST	ICRST	DCEN	ICEN	PRFTEN	Reserved					LATENCY [2:0]		
	Reset value																				0	0	0	0	0						0	0	0
0x04	FLASH_ KEYR	KEY[31:16]																KEY[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	FLASH_OPT KEYR	OPTKEYR[31:16]																OPTKEYR[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	FLASH_SR	Reserved															BSY	Reserved								PGSERR	PGPERR	PGAERR	WRPERR	Reserved		OPERR	EOP
	Reset value																0									0	0	0	0			0	0
0x10	FLASH_CR	LOCK	Reserved						EOPIE	Reserved							STRT	Reserved						PSIZE[1:0]		Reserved	SNB[3:0]				MER	SER	PG
	Reset value	1							0								0							0	0		0	0	0	0		0	0
0x14	FLASH_ OPTCR	Reserved				nWRP[11:0]												RDP[7:0]								nRST_STDBY	nRST_STOP	WDG_SW	Reserved	BOR_LEV[1:0]		OPTSTRT	OPTLOCK
	Reset value					1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	0	1	0	1	1	1		1	1	0	1

Table 21. Flash register map and reset values (STM32F42xxx and STM32F43xxx)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	FLASH_ACR	Reserved																			DCRST	ICRST	DCEN	ICEN	PRFTEN	Reserved				LATENCY[3:0]			
	Reset value																				0	0	0	0	0					0	0	0	0
0x04	FLASH_KEYR	KEY[31:16]																KEY[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	FLASH_ OPTKEYR	OPTKEYR[31:16]																OPTKEYR[15:0]															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	FLASH_SR	Reserved															BSY	Reserved							RDERR	PGSERR	PGPERR	PGAERR	WRPERR	Reserved		OPERR	EOP
	Reset value																0								0	0	0	0	0			0	0

Table 21. Flash register map and reset values (STM32F42xxx and STM32F43xxx) (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x10	**FLASH_CR**	LOCK	Reserved						EOPIE	Reserved							STRT	MER1	Reserved					PSIZE[1:0]		SNB[4:0]					MER	SER	PG
	Reset value	1							0								0	0						0	0	0	0	0	0	0	0	0	0
0x14	**FLASH_OPTCR**	SPRMOD	DB1M	Reserved		nWRP[11:0]												RDP[7:0]								nRST_STDBY	nRST_STOP	WDG_SW	BFB2	BOR_LEV[1:0]		OPTSTRT	OPTLOCK
	Reset value	0	0			1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	0	1	0	1	1	1	0	1	1	0	1
0x18	**FLASH_OPTCR1**	Reserved				nWRP[11:0]												Reserved															
	Reset value					1	1	1	1	1	1	1	1	1	1	1	1																

4 CRC calculation unit

This section applies to the whole STM32F4xx family, unless otherwise specified.

4.1 CRC introduction

The CRC (cyclic redundancy check) calculation unit is used to get a CRC code from a 32-bit data word and a fixed generator polynomial.

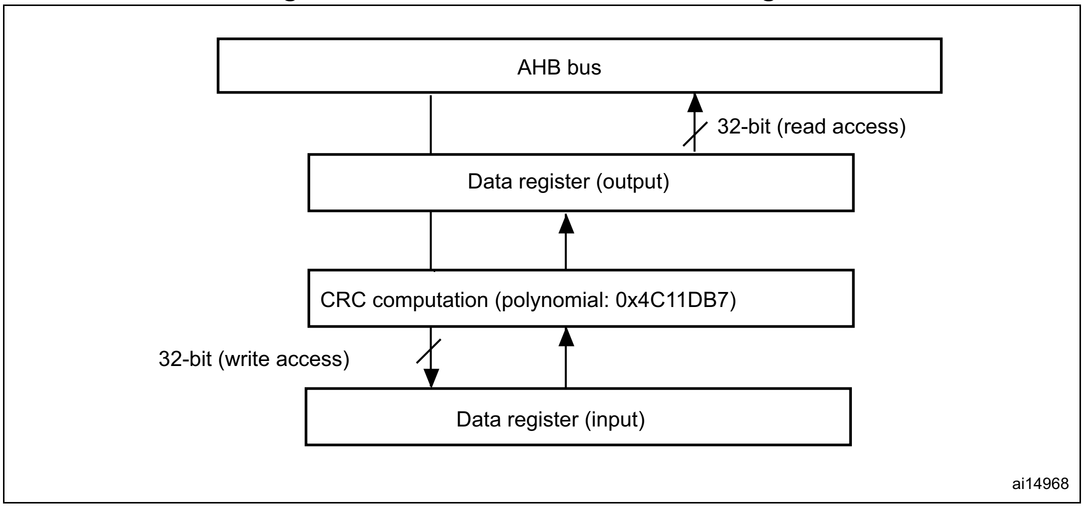
Among other applications, CRC-based techniques are used to verify data transmission or storage integrity. In the scope of the EN/IEC 60335-1 standard, they offer a means of verifying the flash memory integrity. The CRC calculation unit helps compute a signature of the software during runtime, to be compared with a reference signature generated at link-time and stored at a given memory location.

4.2 CRC main features

- Uses CRC-32 (Ethernet) polynomial: 0x4C11DB7
  - $X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$
- Single input/output 32-bit data register
- CRC computation done in 4 AHB clock cycles (HCLK)
- General-purpose 8-bit register (can be used for temporary storage)

The block diagram is shown in *Figure 8*.

Figure 8. CRC calculation unit block diagram



4.3 CRC functional description

The CRC calculation unit mainly consists of a single 32-bit data register, which:

- is used as an input register to enter new data in the CRC calculator (when writing into the register)
- holds the result of the previous CRC calculation (when reading the register)

Each write operation into the data register creates a combination of the previous CRC value and the new one (CRC computation is done on the whole 32-bit data word, and not byte per byte).

The write operation is stalled until the end of the CRC computation, thus allowing back-to-back write accesses or consecutive write and read accesses.

The CRC calculator can be reset to 0xFFFF FFFF with the RESET control bit in the CRC_CR register. This operation does not affect the contents of the CRC_IDR register.

4.4 CRC registers

The CRC calculation unit contains two data registers and a control register.The peripheral The CRC registers have to be accessed by words (32 bits).

4.4.1 Data register (CRC_DR)

Address offset: 0x00

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DR [31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
DR [15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **Data register bits**

Used as an input register when writing new data into the CRC calculator.
Holds the previous CRC calculation result when it is read.

4.4.2 Independent data register (CRC_IDR)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved								IDR[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **General-purpose 8-bit data register bits**

Can be used as a temporary storage location for one byte.

This register is not affected by CRC resets generated by the RESET bit in the CRC_CR register.

4.4.3 Control register (CRC_CR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved															RESET
															w

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **RESET bit**

Resets the CRC calculation unit and sets the data register to 0xFFFF FFFF.

This bit can only be set, it is automatically cleared by hardware.

4.4.4 CRC register map

The following table provides the CRC register map and reset values.

Table 22. CRC calculation unit register map and reset values

Offset	Register	31-24	23-16	15-8	7	6	5	4	3	2	1	0
0x00	**CRC_DR**	Data register										
	Reset value	0xFFFF FFFF										
0x04	**CRC_IDR**	Reserved			Independent data register							
	Reset value				0x00							
0x08	**CRC_CR**	Reserved										RESET
	Reset value											0

5 Power controller (PWR)

This section applies to the whole STM32F4xx family, unless otherwise specified.

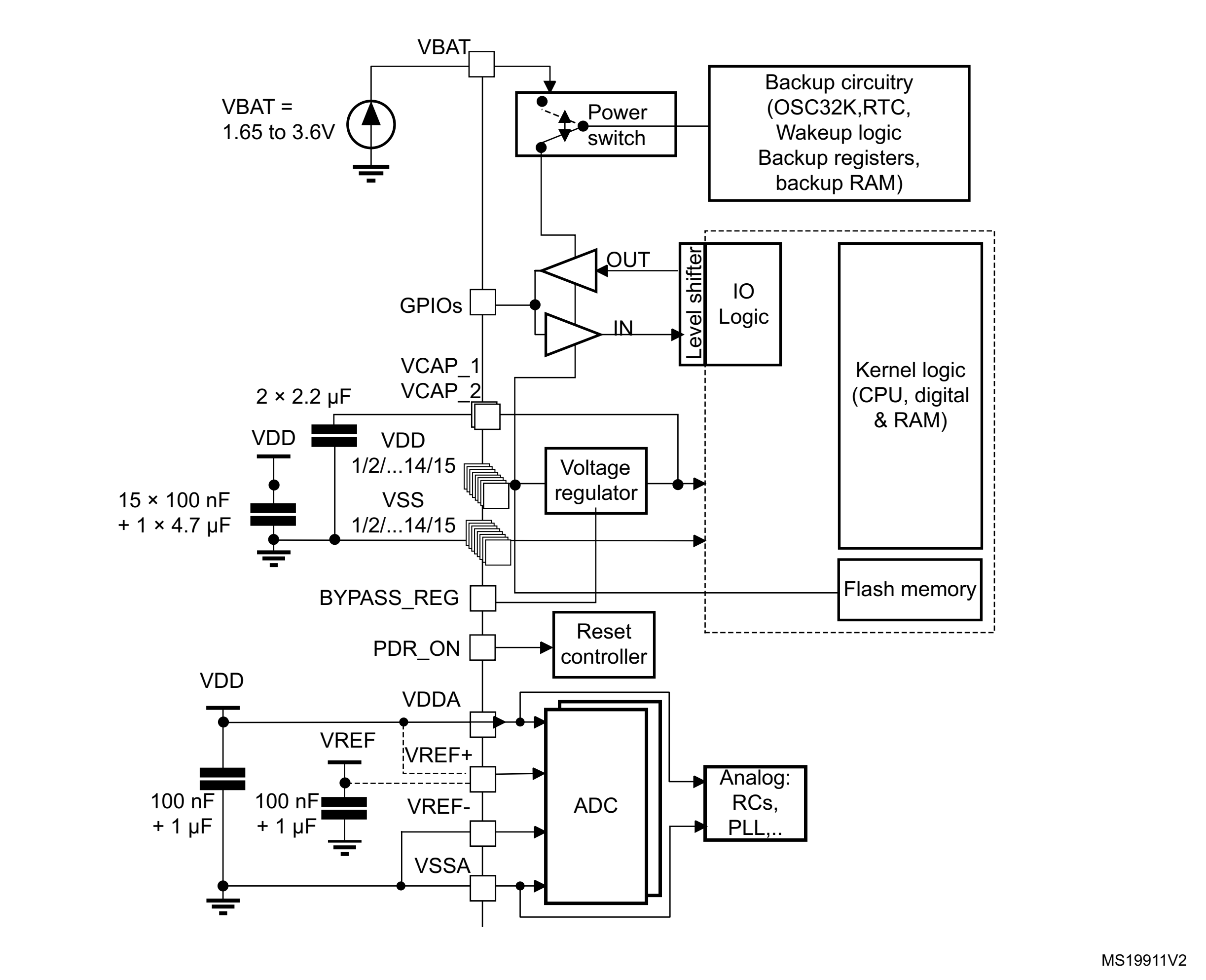
5.1 Power supplies

The device requires a 1.8 to 3.6 V operating voltage supply (V_{DD}). An embedded linear voltage regulator is used to supply the internal 1.2 V digital power.

The real-time clock (RTC), the RTC backup registers, and the backup SRAM (BKP SRAM) can be powered from the V_{BAT} voltage when the main V_{DD} supply is powered off.

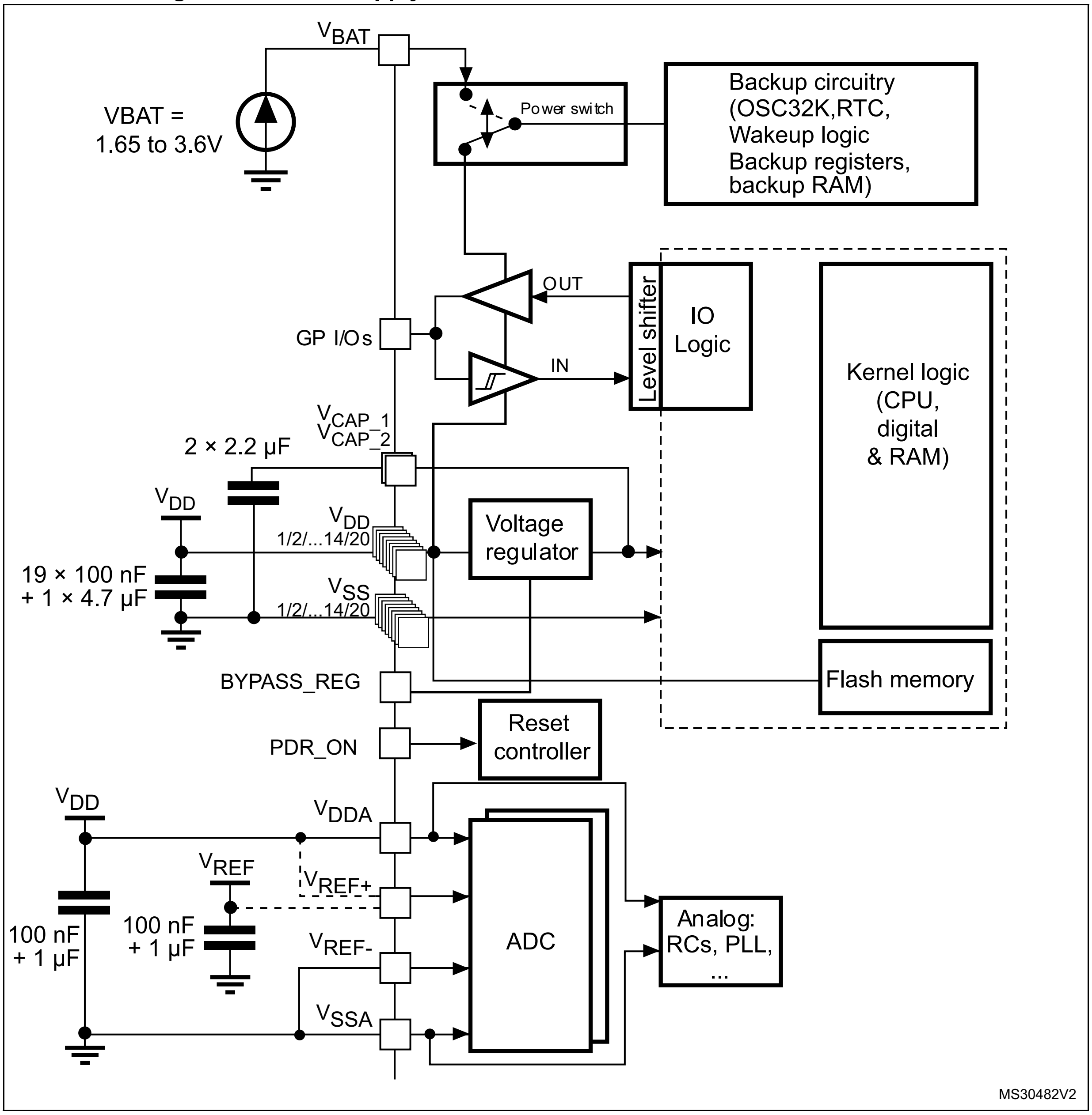
Note: Depending on the operating power supply range, some peripheral may be used with limited functionality and performance. For more details refer to section "General operating conditions" in STM32F4xx datasheets.

Figure 9. Power supply overview for STM32F405xx/07xx and STM32F415xx/17xx



1. V_{DDA} and V_{SSA} must be connected to V_{DD} and V_{SS}, respectively.

Figure 10. Power supply overview for STM32F42xxx and STM32F43xxx



1. V_{DDA} and V_{SSA} must be connected to V_{DD} and V_{SS}, respectively.

5.1.1 Independent A/D converter supply and reference voltage

To improve conversion accuracy, the ADC has an independent power supply which can be separately filtered and shielded from noise on the PCB.

- The ADC voltage supply input is available on a separate VDDA pin.
- An isolated supply ground connection is provided on VSSA pin.

To ensure a better accuracy of low voltage inputs, the user can connect a separate external reference voltage ADC input on V_{REF}. The voltage on V_{REF} ranges from 1.8 V to V_{DDA}.

5.1.2 Battery backup domain

Backup domain description

To retain the content of the RTC backup registers, backup SRAM, and supply the RTC when V_{DD} is turned off, VBAT pin can be connected to an optional standby voltage supplied by a battery or by another source.

To allow the RTC to operate even when the main digital supply (V_{DD}) is turned off, the VBAT pin powers the following blocks:

- The RTC
- The LSE oscillator
- The backup SRAM when the low-power backup regulator is enabled
- PC13 to PC15 I/Os, plus PI8 I/O (when available)

The switch to the V_{BAT} supply is controlled by the power-down reset embedded in the Reset block.

Warning: During $t_{RSTTEMPO}$ (temporization at V_{DD} startup) or after a PDR is detected, the power switch between V_{BAT} and V_{DD} remains connected to V_{BAT}.
During the startup phase, if V_{DD} is established in less than $t_{RSTTEMPO}$ (Refer to the datasheet for the value of $t_{RSTTEMPO}$) and $V_{DD} > V_{BAT} + 0.6$ V, a current may be injected into V_{BAT} through an internal diode connected between V_{DD} and the power switch (V_{BAT}).
If the power supply/battery connected to the VBAT pin cannot support this current injection, it is strongly recommended to connect an external low-drop diode between this power supply and the VBAT pin.

If no external battery is used in the application, it is recommended to connect the VBAT pin to V_{DD} supply, and add a 100 nF external decoupling ceramic capacitor on VBAT pin.

When the backup domain is supplied by V_{DD} (analog switch connected to V_{DD}), the following functions are available:

- PC14 and PC15 can be used as either GPIO or LSE pins
- PC13 can be used as a GPIOas the RTC_AF1 pin (refer to *Table 38: RTC_AF1 pin* for more details about this pin configuration)

Note: Due to the fact that the switch only sinks a limited amount of current (3 mA), the use of PI8 and PC13 to PC15 GPIOs in output mode is restricted: the speed has to be limited to 2 MHz with a maximum load of 30 pF and these I/Os must not be used as a current source (e.g. to drive an LED).

When the backup domain is supplied by V_{BAT} (analog switch connected to V_{BAT} because V_{DD} is not present), the following functions are available:

- PC14 and PC15 can be used as LSE pins only
- PC13 can be used as the RTC_AF1 pin (refer to *Table 38: RTC_AF1 pin* for more details about this pin configuration)
- PI8 can be used as RTC_AF2

Backup domain access

After reset, the backup domain (RTC registers, RTC backup register and backup SRAM) is protected against possible unwanted write accesses. To enable access to the backup domain, proceed as follows:

- Access to the RTC and RTC backup registers

1. Enable the power interface clock by setting the PWREN bits in the RCC_APB1ENR register (see *Section 7.3.13* and *Section 6.3.13*)
2. Set the DBP bit in the *Section 5.4.1* and *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx* to enable access to the backup domain
3. Select the RTC clock source: see *Section 7.2.8: RTC/AWU clock*
4. Enable the RTC clock by programming the RTCEN [15] bit in the *Section 7.3.20: RCC Backup domain control register (RCC_BDCR)*

- Access to the backup SRAM

1. Enable the power interface clock by setting the PWREN bits in the RCC_APB1ENR register (see *Section 7.3.13* and *Section 6.3.13* for STM32F405xx/07xx and STM32F415xx/17xx and STM32F42xxx and STM32F43xxx, respectively)
2. Set the DBP bit in the *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx* and *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx* to enable access to the backup domain
3. Enable the backup SRAM clock by setting BKPSRAMEN bit in the *RCC AHB1 peripheral clock enable register (RCC_AHB1ENR)*.

RTC and RTC backup registers

The real-time clock (RTC) is an independent BCD timer/counter. The RTC provides a time-of-day clock/calendar, two programmable alarm interrupts, and a periodic programmable wake-up flag with interrupt capability. The RTC contains 20 backup data registers (80 bytes) which are reset when a tamper detection event occurs. For more details refer to *Section 26: Real-time clock (RTC)*.

Backup SRAM

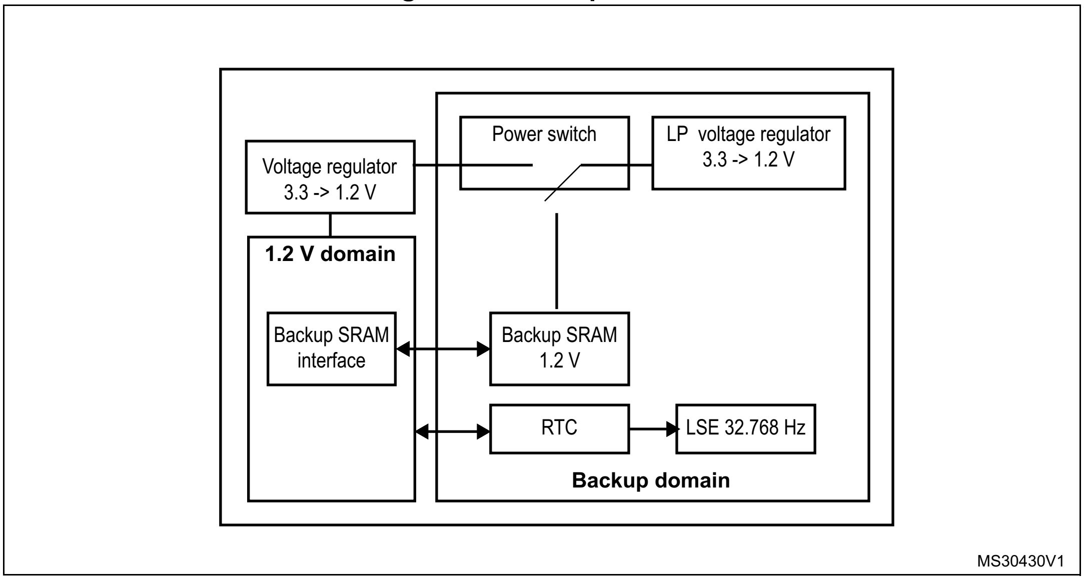
The backup domain includes 4 Kbytes of backup SRAM addressed in 32-bit, 16-bit or 8-bit mode. Its content is retained even in Standby or V_{BAT} mode when the low-power backup regulator is enabled. It can be considered as an internal EEPROM when V_{BAT} is always present.

When the backup domain is supplied by V_{DD} (analog switch connected to V_{DD}), the backup SRAM is powered from V_{DD} which replaces the V_{BAT} power supply to save battery life.

When the backup domain is supplied by V_{BAT} (analog switch connected to V_{BAT} because V_{DD} is not present), the backup SRAM is powered by a dedicated low-power regulator. This regulator can be ON or OFF depending whether the application needs the backup SRAM function in Standby and V_{BAT} modes or not. The power-down of this regulator is controlled by a dedicated bit, the BRE control bit of the PWR_CSR register (see *Section 5.4.2: PWR power control/status register (PWR_CSR) for STM32F405xx/07xx and STM32F415xx/17xx*).

The backup SRAM is not mass erased by an tamper event. When the Flash memory is read protected, the backup SRAM is also read protected to prevent confidential data, such as cryptographic private key, from being accessed. When the protection level change from level 1 to level 0 is requested, the backup SRAM content is erased.

Figure 11. Backup domain



5.1.3 Voltage regulator for STM32F405xx/07xx and STM32F415xx/17xx

An embedded linear voltage regulator supplies all the digital circuitries except for the backup domain and the Standby circuitry. The regulator output voltage is around 1.2 V.

This voltage regulator requires one or two external capacitors to be connected to one or two dedicated pins, V_{CAP_1} and V_{CAP_2} available in all packages. Specific pins must be connected either to V_{SS} or V_{DD} to activate or deactivate the voltage regulator. These pins depend on the package.

When activated by software, the voltage regulator is always enabled after Reset. It works in three different modes depending on the application modes.

- In **Run mode**, the regulator supplies full power to the 1.2 V domain (core, memories and digital peripherals). In this mode, the regulator output voltage (around 1.2 V) can be scaled by software to different voltage values:

    Scale 1 or scale 2 can be configured on the fly through VOS (bit 15 of the PWR_CR register).

  The voltage scaling allows optimizing the power consumption when the device is clocked below the maximum system frequency.
- In **Stop mode,** the main regulator or the low-power regulator supplies to the 1.2 V domain, thus preserving the content of registers and internal SRAM. The voltage

regulator can be put either in main regulator mode (MR) or in low-power mode (LPR). The programmed voltage scale remains the same during Stop mode:

The programmed voltage scale remains the same during Stop mode (see *Section 5.4.1: PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx*).

- In **Standby mode**, the regulator is powered down. The content of the registers and SRAM are lost except for the Standby circuitry and the backup domain.

Note: For more details, refer to the voltage regulator section in the STM32F405xx/07xx and STM32F415xx/17xx datasheets.

5.1.4 Voltage regulator for STM32F42xxx and STM32F43xxx

An embedded linear voltage regulator supplies all the digital circuitries except for the backup domain and the Standby circuitry. The regulator output voltage is around 1.2 V.

This voltage regulator requires two external capacitors to be connected to two dedicated pins, VCAP_1 and VCAP_2 available in all packages. Specific pins must be connected either to V_{SS} or V_{DD} to activate or deactivate the voltage regulator. These pins depend on the package.

When activated by software, the voltage regulator is always enabled after Reset. It works in three different modes depending on the application modes (Run, Stop, or Standby mode).

- In **Run mode**, the main regulator supplies full power to the 1.2 V domain (core, memories and digital peripherals). In this mode, the regulator output voltage (around 1.2 V) can be scaled by software to different voltage values (scale 1, scale 2, and scale 3 can be configured through VOS[1:0] bits of the PWR_CR register). The scale can be modified only when the PLL is OFF and the HSI or HSE clock source is selected as system clock source. The new value programmed is active only when the PLL is ON. When the PLL is OFF, the voltage scale 3 is automatically selected.

  The voltage scaling allows optimizing the power consumption when the device is clocked below the maximum system frequency. After exit from Stop mode, the voltage scale 3 is automatically selected.(see *Section 5.4.1: PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx*.

2 operating modes are available:

– **Normal mode**: The CPU and core logic operate at maximum frequency at a given voltage scaling (scale 1, scale 2 or scale 3)

– **Over-drive mode**: This mode allows the CPU and the core logic to operate at a higher frequency than the normal mode for the voltage scaling scale 1 and scale 2.

- In **Stop mode**: the main regulator or low-power regulator supplies a low-power voltage to the 1.2V domain, thus preserving the content of registers and internal SRAM.

  The voltage regulator can be put either in main regulator mode (MR) or in low-power mode (LPR). Both modes can be configured by software as follows:

  – **Normal mode**: the 1.2 V domain is preserved in nominal leakage mode. It is the default mode when the main regulator (MR) or the low-power regulator (LPR) is enabled.

  – **Under-drive mode**: the 1.2 V domain is preserved in reduced leakage mode. This mode is only available with the main regulator or in low-power regulator mode (see *Table 23*).

- In **Standby mode**: the regulator is powered down. The content of the registers and SRAM are lost except for the Standby circuitry and the backup domain.

Note: Over-drive and under-drive mode are not available when the regulator is bypassed.

For more details, refer to the voltage regulator section in the STM32F42xxx and STM32F43xxx datasheets.

Table 23. Voltage regulator configuration mode versus device operating mode(1)

Voltage regulator configuration	Run mode	Sleep mode	Stop mode	Standby mode
Normal mode	MR	MR	MR or LPR	-
Over-drive mode(2)	MR	MR	-	-
Under-drive mode	-	-	MR or LPR	-
Power-down mode	-	-	-	Yes

1. '-' means that the corresponding configuration is not available.
2. The over-drive mode is not available when V_{DD} = 1.8 to 2.1 V.

Entering Over-drive mode

It is recommended to enter Over-drive mode when the application is not running critical tasks and when the system clock source is either HSI or HSE. To optimize the configuration time, enable the Over-drive mode during the PLL lock phase.

To enter Over-drive mode, follow the sequence below:

1. Select HSI or HSE as system clock.
2. Configure RCC_PLLCFGR register and set PLLON bit of RCC_CR register.
3. Set ODEN bit of PWR_CR register to enable the Over-drive mode and wait for the ODRDY flag to be set in the PWR_CSR register.
4. Set the ODSW bit in the PWR_CR register to switch the voltage regulator from Normal mode to Over-drive mode. The System is stalled during the switch but the PLL clock system is still running during locking phase.
5. Wait for the ODSWRDY flag in the PWR_CSR to be set.
6. Select the required Flash latency as well as AHB and APB prescalers.
7. Wait for PLL lock.
8. Switch the system clock to the PLL.
9. Enable the peripherals that are not generated by the System PLL (I2S clock, LCD-TFT clock, SAI1 clock, USB_48MHz clock....).

Note: The PLLI2S and PLLSAI can be configured at the same time as the system PLL.

During the Over-drive switch activation, no peripheral clocks should be enabled. The peripheral clocks must be enabled once the Over-drive mode is activated.

Entering Stop mode disables the Over-drive mode, as well as the PLL. The application software has to configure again the Over-drive mode and the PLL after exiting from Stop mode.

Exiting from Over-drive mode

It is recommended to exit from Over-drive mode when the application is not running critical tasks and when the system clock source is either HSI or HSE.There are two sequences that allow exiting from over-drive mode:

- By resetting simultaneously the ODEN and ODSW bits bit in the PWR_CR register (sequence 1)
- By resetting first the ODSW bit to switch the voltage regulator to Normal mode and then resetting the ODEN bit to disable the Over-drive mode (sequence 2).

Example of sequence 1:

1. Select HSI or HSE as system clock source.
2. Disable the peripheral clocks that are not generated by the System PLL (I2S clock, LCD-TFT clock, SAI1 clock, USB_48MHz clock,....)
3. Reset simultaneously the ODEN and the ODSW bits in the PWR_CR register to switch back the voltage regulator to Normal mode and disable the Over-drive mode.
4. Wait for the ODWRDY flag of PWR_CSR to be reset.

Example of sequence 2:

1. Select HSI or HSE as system clock source.
2. Disable the peripheral clocks that are not generated by the System PLL (I2S clock, LCD-TFT clock, SAI1 clock, USB_48MHz clock,....).
3. Reset the ODSW bit in the PWR_CR register to switch back the voltage regulator to Normal mode. The system clock is stalled during voltage switching.
4. Wait for the ODWRDY flag of PWR_CSR to be reset.
5. Reset the ODEN bit in the PWR_CR register to disable the Over-drive mode.

Note: During step 3, the ODEN bit remains set and the Over-drive mode is still enabled but not active (ODSW bit is reset). If the ODEN bit is reset instead, the Over-drive mode is disabled and the voltage regulator is switched back to the initial voltage.

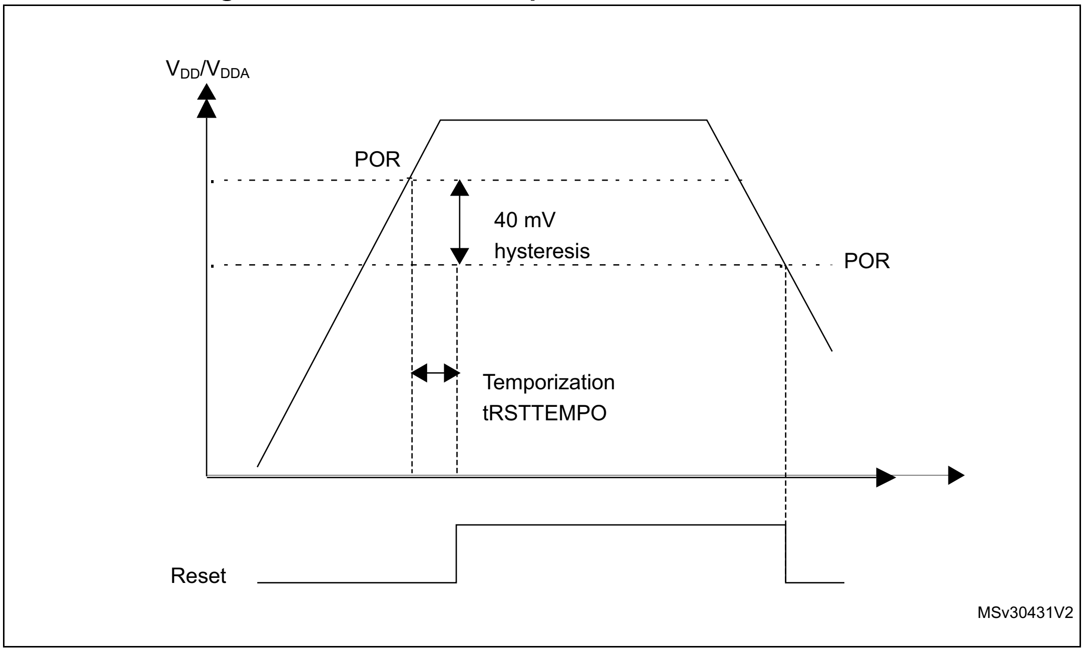
5.2 Power supply supervisor

5.2.1 Power-on reset (POR)/power-down reset (PDR)

The device has an integrated POR/PDR circuitry that allows proper operation starting from 1.8 V.

The device remains in Reset mode when V_{DD}/V_{DDA} is below a specified threshold, $V_{POR/PDR}$, without the need for an external reset circuit. For more details concerning the power on/power-down reset threshold, refer to the electrical characteristics of the datasheet.

Figure 12. Power-on reset/power-down reset waveform



5.2.2 Brownout reset (BOR)

During power on, the Brownout reset (BOR) keeps the device under reset until the supply voltage reaches the specified V_{BOR} threshold.

V_{BOR} is configured through device option bytes. By default, BOR is off. 3 programmable V_{BOR} threshold levels can be selected:

- BOR Level 3 (VBOR3). Brownout threshold level 3.
- BOR Level 2 (VBOR2). Brownout threshold level 2.
- BOR Level 1 (VBOR1). Brownout threshold level 1.

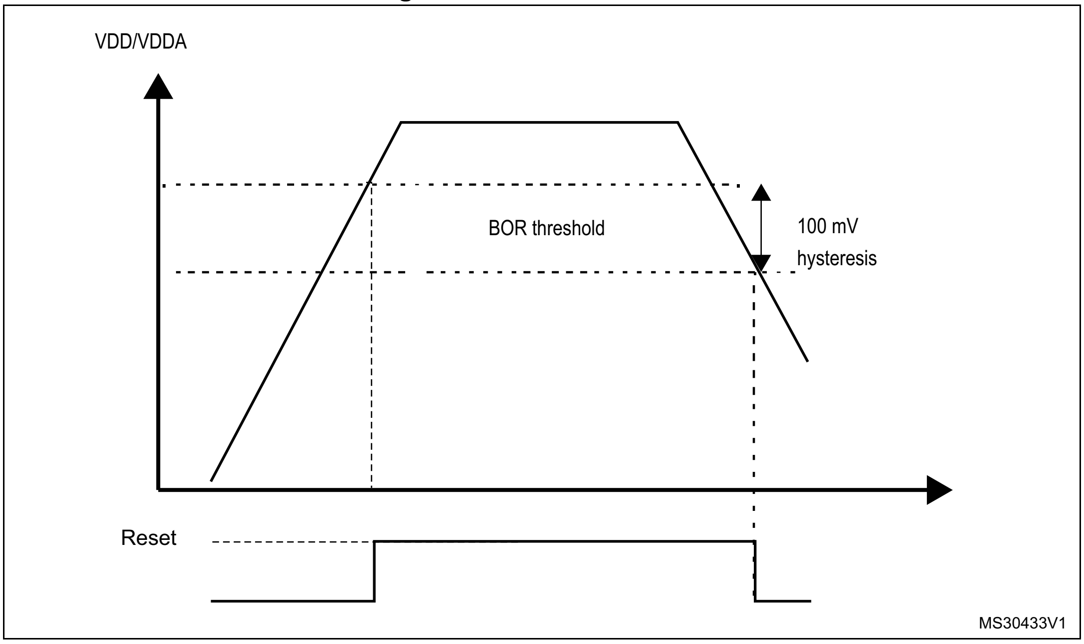
Note: For full details about BOR characteristics, refer to the "Electrical characteristics" section in the device datasheet.

When the supply voltage (V_{DD}) drops below the selected V_{BOR} threshold, a device reset is generated.

The BOR can be disabled by programming the device option bytes. In this case, the power-on and power-down is then monitored by the POR/ PDR (see *Section 5.2.1: Power-on reset (POR)/power-down reset (PDR)*).

The BOR threshold hysteresis is ~100 mV (between the rising and the falling edge of the supply voltage).

Figure 13. BOR thresholds



5.2.3 Programmable voltage detector (PVD)

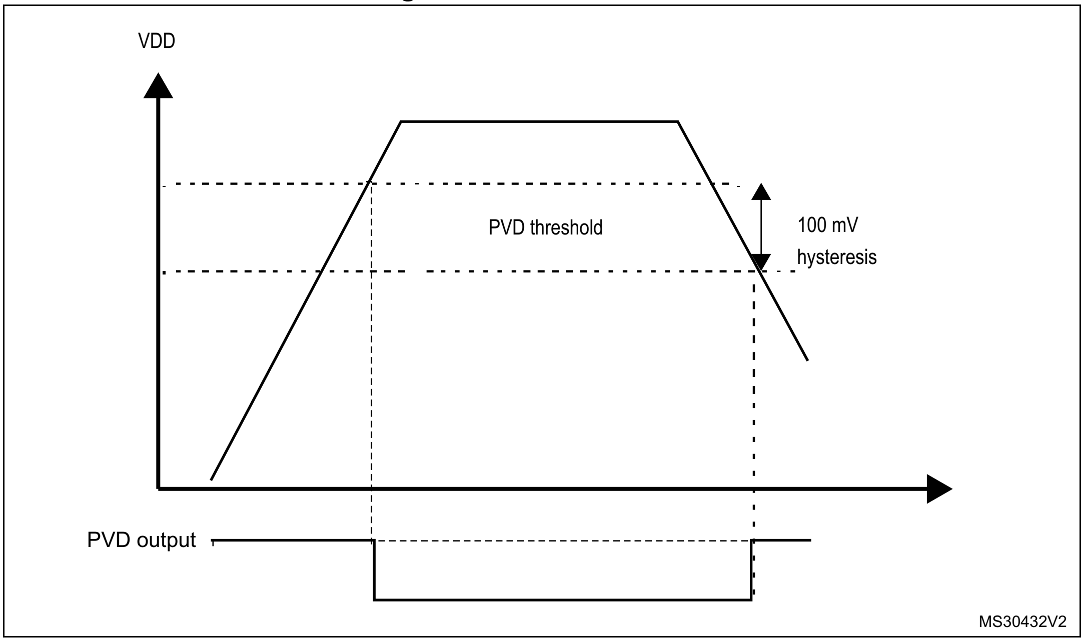
You can use the PVD to monitor the V_{DD} power supply by comparing it to a threshold selected by the PLS[2:0] bits in the *PWR power control register (PWR_CR) for*

STM32F405xx/07xx and STM32F415xx/17xx and *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx*.

The PVD is enabled by setting the PVDE bit.

A PVDO flag is available, in the *PWR power control/status register (PWR_CSR) for STM32F405xx/07xx and STM32F415xx/17xx*, to indicate if V_{DD} is higher or lower than the PVD threshold. This event is internally connected to the EXTI line16 and can generate an interrupt if enabled through the EXTI registers. The PVD output interrupt can be generated when V_{DD} drops below the PVD threshold and/or when V_{DD} rises above the PVD threshold depending on EXTI line16 rising/falling edge configuration. As an example the service routine could perform emergency shutdown tasks.

Figure 14. PVD thresholds



5.3 Low-power modes

By default, the microcontroller is in Run mode after a system or a power-on reset. In Run mode the CPU is clocked by HCLK and the program code is executed. Several low-power modes are available to save power when the CPU does not need to be kept running, for example when waiting for an external event. It is up to the user to select the mode that gives the best compromise between low-power consumption, short startup time and available wake-up sources.

The devices feature three low-power modes:

- Sleep mode (Cortex®-M4 with FPU core stopped, peripherals kept running)
- Stop mode (all clocks are stopped)
- Standby mode (1.2 V domain powered off)

In addition, the power consumption in Run mode can be reduce by one of the following means:

- Slowing down the system clocks
- Gating the clocks to the APBx and AHBx peripherals when they are unused.

Entering low-power mode

Low-power modes are entered by the MCU by executing the WFI (Wait For Interrupt), or WFE (Wait for Event) instructions, or when the SLEEPONEXIT bit in the Cortex®-M4 with FPU System Control register is set on Return from ISR.

Entering Low-power mode through WFI or WFE is executed only if no interrupt is pending or no event is pending.

Exiting low-power mode

The MCU exits from Sleep and Stop modes low-power mode depending on the way the low-power mode was entered:

- If the WFI instruction or Return from ISR was used to enter the low-power mode, any peripheral interrupt acknowledged by the NVIC can wake up the device.
- If the WFE instruction is used to enter the low-power mode, the MCU exits the low-power mode as soon as an event occurs. The wake-up event can be generated either by:
  – NVIC IRQ interrupt:

    When SEVONPEND = 0 in the Cortex®-M4 with FPU System Control register: by enabling an interrupt in the peripheral control register and in the NVIC. When the MCU resumes from WFE, the peripheral interrupt pending bit and the NVIC peripheral IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared. Only NVIC interrupts with sufficient priority wakes up and interrupts the MCU.

    When SEVONPEND = 1 in the Cortex®-M4 with FPU System Control register: by enabling an interrupt in the peripheral control register and optionally in the NVIC. When the MCU resumes from WFE, the peripheral interrupt pending bit and when enabled the NVIC peripheral IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared. All NVIC interrupts wakes up the MCU, even the disabled ones. Only enabled NVIC interrupts with sufficient priority wakes up and interrupts the MCU.

  – Event

    This is done by configuring a EXTI line in event mode. When the CPU resumes from WFE, it is not necessary to clear the EXTI peripheral interrupt pending bit or the NVIC IRQ channel pending bit as the pending bits corresponding to the event line is not set. It may be necessary to clear the interrupt flag in the peripheral.

The MCU exits from Standby low-power mode through an external reset (NRST pin), an IWDG reset, a rising edge on one of the enabled WKUPx pins or a RTC event occurs (see *Figure 237: RTC block diagram*).

After waking up from Standby mode, program execution restarts in the same way as after a Reset (boot pin sampling, option bytes loading, reset vector is fetched, etc.).

Only enabled NVIC interrupts with sufficient priority wakes up and interrupts the MCU.

Table 24. Low-power mode summary

Mode name	Entry	Wake-up	Effect on 1.2 V domain clocks	Effect on V_{DD} domain clocks	Voltage regulator
Sleep (Sleep now or Sleep-on-exit)	WFI or Return from ISR	Any interrupt	CPU CLK OFF no effect on other clocks or analog clock sources	None	ON
	WFE	Wake-up event			
Stop	PDDS and LPDS bits + SLEEPDEEP bit + WFI, Return from ISR or WFE	Any EXTI line (configured in the EXTI registers, internal and external lines)	All 1.2 V domain clocks OFF	HSI and HSE oscillators OFF	ON or in low- power mode (depends on *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx* and *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xxPWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx*
Standby	PDDS bit + SLEEPDEEP bit + WFI, Return from ISR or WFE	WKUP pin rising edge, RTC alarm (Alarm A or Alarm B), RTC Wake-up event, RTC tamper events, RTC time stamp event, external reset in NRST pin, IWDG reset			OFF

5.3.1 Slowing down system clocks

In Run mode the speed of the system clocks (SYSCLK, HCLK, PCLK1, PCLK2) can be reduced by programming the prescaler registers. These prescalers can also be used to slow down peripherals before entering Sleep mode.

For more details refer to *Section 7.3.3: RCC clock configuration register (RCC_CFGR)*.

5.3.2 Peripheral clock gating

In Run mode, the HCLKx and PCLKx for individual peripherals and memories can be stopped at any time to reduce power consumption.

To further reduce power consumption in Sleep mode the peripheral clocks can be disabled prior to executing the WFI or WFE instructions.

Peripheral clock gating is controlled by the AHB1 peripheral clock enable register (RCC_AHB1ENR), AHB2 peripheral clock enable register (RCC_AHB2ENR), AHB3 peripheral clock enable register (RCC_AHB3ENR) (see *Section 7.3.10: RCC AHB1 peripheral clock enable register (RCC_AHB1ENR)*, *Section 7.3.11: RCC AHB2 peripheral clock enable register (RCC_AHB2ENR)*, *Section 7.3.12: RCC AHB3 peripheral clock enable register (RCC_AHB3ENR)* for STM32F405xx/07xx and STM32F415xx/17xx, and *Section 6.3.10: RCC AHB1 peripheral clock register (RCC_AHB1ENR)*, *Section 6.3.11: RCC AHB2 peripheral clock enable register (RCC_AHB2ENR)*, and *Section 6.3.12: RCC AHB3 peripheral clock enable register (RCC_AHB3ENR)* for STM32F42xxx and STM32F43xxx).

Disabling the peripherals clocks in Sleep mode can be performed automatically by resetting the corresponding bit in RCC_AHBxLPENR and RCC_APBxLPENR registers.

5.3.3 Sleep mode

Entering Sleep mode

The Sleep mode is entered according to *Section : Entering low-power mode*, when the SLEEPDEEP bit in the Cortex®-M4 with FPU System Control register is cleared.

Refer to *Table 25* and *Table 26* for details on how to enter Sleep mode.

Note: All interrupt pending bits must be cleared before the sleep mode entry.

Exiting Sleep mode

The Sleep mode is exited according to *Section : Exiting low-power mode*.

Refer to *Table 25* and *Table 26* for more details on how to exit Sleep mode.

Table 25. Sleep-now entry and exit

Sleep-now mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP = 0, and – No interrupt (for WFI) or event (for WFE) is pending. Refer to the Cortex®-M4 with FPU System Control register.
	On Return from ISR while: – SLEEPDEEP = 0 and – SLEEPONEXIT = 1, – No interrupt is pending. Refer to the Cortex®-M4 with FPU System Control register.

Table 25. Sleep-now entry and exit (continued)

Sleep-now mode	Description
Mode exit	If WFI or Return from ISR was used for entry: Interrupt: Refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx* and *Table 63: Vector table for STM32F42xxx and STM32F43xxx* If WFE was used for entry and SEVONPEND = 0 Wake-up event: Refer to *Section 12.2.3: Wake-up event management* f WFE was used for entry and SEVONPEND = 1 Interrupt even when disabled in NVIC: refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx* and *Table 63: Vector table for STM32F42xxx and STM32F43xxx* or Wake-up event (see *Section 12.2.3: Wake-up event management*).
Wake-up latency	None

Table 26. Sleep-on-exit entry and exit

Sleep-on-exit	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP = 0, and – No interrupt (for WFI) or event (for WFE) is pending. Refer to the Cortex®-M4 with FPU System Control register.
	On Return from ISR while: – SLEEPDEEP = 0, and – SLEEPONEXIT = 1, and – No interrupt is pending. Refer to the Cortex®-M4 with FPU System Control register.
Mode exit	Interrupt: refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx* and *Table 63: Vector table for STM32F42xxx and STM32F43xxx*
Wake-up latency	None

5.3.4 Stop mode (STM32F405xx/07xx and STM32F415xx/17xx)

The Stop mode is based on the Cortex®-M4 with FPU deepsleep mode combined with peripheral clock gating. The voltage regulator can be configured either in normal or low-power mode. In Stop mode, all clocks in the 1.2 V domain are stopped, the PLLs, the HSI and the HSE RC oscillators are disabled. Internal SRAM and register contents are preserved.

By setting the FPDS bit in the PWR_CR register, the Flash memory also enters power-down mode when the device enters Stop mode. When the Flash memory is in power-down mode, an additional startup delay is incurred when waking up from Stop mode (see *Table 27: Stop operating modes (STM32F405xx/07xx and STM32F415xx/17xx)* and *Section 5.4.1: PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx*).

Table 27. Stop operating modes (STM32F405xx/07xx and STM32F415xx/17xx)

Stop mode	LPDS bit	FPDS bit	Wake-up latency
STOP MR (Main regulator)	0	0	HSI RC startup time
STOP MR-FPD	0	1	HSI RC startup time + Flash wake-up time from Power Down mode
STOP LP	1	0	HSI RC startup time + regulator wake-up time from LP mode
STOP LP-FPD	1	1	HSI RC startup time + Flash wake-up time from Power Down mode + regulator wake-up time from LP mode

Entering Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx)

The Stop mode is entered according to *Section : Entering low-power mode*, when the SLEEPDEEP bit in the Cortex®-M4 with FPU System Control register is set.

Refer to *Table 28* for details on how to enter the Stop mode.

To further reduce power consumption in Stop mode, the internal voltage regulator can be put in low-power mode. This is configured by the LPDS bit of the *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx* and *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx*.

If Flash memory programming is ongoing, the Stop mode entry is delayed until the memory access is finished.

If an access to the APB domain is ongoing, The Stop mode entry is delayed until the APB access is finished.

In Stop mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started it cannot be stopped except by a Reset. See *Section 21.3* in *Section 21: Independent watchdog (IWDG)*.
- Real-time clock (RTC): this is configured by the RTCEN bit in the *Section 7.3.20: RCC Backup domain control register (RCC_BDCR)*
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the *Section 7.3.21: RCC clock control & status register (RCC_CSR)*.
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the *Section 7.3.20: RCC Backup domain control register (RCC_BDCR)*.

The ADC or DAC can also consume power during the Stop mode, unless they are disabled before entering it. To disable them, the ADON bit in the ADC_CR2 register and the ENx bit in the DAC_CR register must both be written to 0.

Note: If the application needs to disable the external clock before entering Stop mode, the HSEON bit must first be disabled and the system clock switched to HSI.

Otherwise, if the HSEON bit is kept enabled while the external clock (external oscillator) can be removed before entering stop mode, the clock security system (CSS) feature must be enabled to detect any external oscillator failure and avoid a malfunction behavior when entering stop mode.

Exiting Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx)

The Stop mode is exited according to *Section : Exiting low-power mode*.

Refer to *Table 28* for more details on how to exit Stop mode.

When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as system clock.

When the voltage regulator operates in low-power mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

Table 28. Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx)

Stop mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – No interrupt (for WFI) or event (for WFE) is pending, – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, – PDDS bit is cleared in Power Control register (PWR_CR), – Select the voltage regulator mode by configuring LPDS bit in PWR_CR.
	On Return from ISR: – No interrupt is pending, – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, – SLEEPONEXIT = 1, – PDDS bit is cleared in Power Control register (PWR_CR).
	Note: To enter Stop mode, all EXTI Line pending bits (in Pending register (EXTI_PR)), all peripheral interrupts pending bits, the RTC Alarm (Alarm A and Alarm B), RTC wake-up, RTC tamper, and RTC time stamp flags, must be reset. Otherwise, the Stop mode entry procedure is ignored and program execution continues.

Table 28. Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx)

Stop mode	Description
Mode exit	If WFI or Return from ISR was used for entry: Any EXTI lines configured in Interrupt mode (the corresponding EXTI Interrupt vector must be enabled in the NVIC). The interrupt source can be external interrupts or peripherals with wake-up capability. Refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 375* and *Table 63: Vector table for STM32F42xxx and STM32F43xxx*. If WFE was used for entry and SEVONPEND = 0 Any EXTI lines configured in event mode. Refer to *Section 12.2.3: Wake-up event management on page 383*. If WFE was used for entry and SEVONPEND = 1: – Any EXTI lines configured in Interrupt mode (even if the corresponding EXTI Interrupt vector is disabled in the NVIC). The interrupt source can be an external interrupt or a peripheral with wake-up capability. Refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 375* and *Table 63: Vector table for STM32F42xxx and STM32F43xxx*. – Wake-up event: refer to *Section 12.2.3: Wake-up event management on page 383*.
Wake-up latency	*Table 27: Stop operating modes (STM32F405xx/07xx and STM32F415xx/17xx)*

5.3.5 Stop mode (STM32F42xxx and STM32F43xxx)

The Stop mode is based on the Cortex®-M4 with FPU deepsleep mode combined with peripheral clock gating. The voltage regulator can be configured either in normal or low-power mode. In Stop mode, all clocks in the 1.2 V domain are stopped, the PLLs, the HSI and the HSE RC oscillators are disabled. Internal SRAM and register contents are preserved.

In Stop mode, the power consumption can be further reduced by using additional settings in the PWR_CR register. However this induces an additional startup delay when waking up from Stop mode (see *Table 29*).

Table 29. Stop operating modes (STM32F42xxx and STM32F43xxx)

Voltage Regulator Mode		UDEN[1:0] bits	MRUDS bit	LPUDS bit	LPDS bit	FPDS bit	Wake-up latency
Normal mode	STOP MR (Main Regulator)	-	0	-	0	0	HSI RC startup time
	STOP MR- FPD	-	0	-	0	1	HSI RC startup time + Flash wake-up time from power-down mode
	STOP LP	-	0	0	1	0	HSI RC startup time + regulator wake-up time from LP mode
	STOP LP-FPD	-	-	0	1	1	HSI RC startup time + Flash wake-up time from power-down mode + regulator wake-up time from LP mode
Under-drive Mode	STOP UMR-FPD	3	1	-	0	-	HSI RC startup time + Flash wake-up time from power-down mode + Main regulator wake-up time from under-drive mode + Core logic to nominal mode
	STOP ULP-FPD	3	-	1	1	-	HSI RC startup time + Flash wake-up time from power-down mode + regulator wake-up time from LP under-drive mode + Core logic to nominal mode

Entering Stop mode (STM32F42xxx and STM32F43xxx)

The Stop mode is entered according to *Section : Entering low-power mode*, when the SLEEPDEEP bit in the Cortex®-M4 with FPU System Control register is set.

Refer to *Table 30* for details on how to enter the Stop mode.

When the microcontroller enters in Stop mode, the voltage scale 3 is automatically selected. To further reduce power consumption in Stop mode, the internal voltage regulator can be put in low voltage mode. This is configured by the LPDS, MRUDS, LPUDS and UDEN bits of the *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx*.

If Flash memory programming is ongoing, the Stop mode entry is delayed until the memory access is finished.

If an access to the APB domain is ongoing, The Stop mode entry is delayed until the APB access is finished.

If the Over-drive mode was enabled before entering Stop mode, it is automatically disabled during when the Stop mode is activated.

In Stop mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started it cannot be stopped except by a Reset. See *Section 21.3* in *Section 21: Independent watchdog (IWDG)*.
- Real-time clock (RTC): this is configured by the RTCEN bit in the *Section 7.3.20: RCC Backup domain control register (RCC_BDCR)*
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the *Section 7.3.21: RCC clock control & status register (RCC_CSR)*.
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the *RCC Backup domain control register (RCC_BDCR)*.

The ADC or DAC can also consume power during the Stop mode, unless they are disabled before entering it. To disable them, the ADON bit in the ADC_CR2 register and the ENx bit in the DAC_CR register must both be written to 0.

Note: Before entering Stop mode, it is recommended to enable the clock security system (CSS) feature to prevent external oscillator (HSE) failure from impacting the internal MCU behavior.

Exiting Stop mode (STM32F42xxx and STM32F43xxx)

The Stop mode is exited according to *Section : Exiting low-power mode*.

Refer to *Table 30* for more details on how to exit Stop mode.

When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as system clock.

If the Under-drive mode was enabled, it is automatically disabled after exiting Stop mode.

When the voltage regulator operates in low voltage mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

When the voltage regulator operates in Under-drive mode, an additional startup delay is induced when waking up from Stop mode.

Table 30. Stop mode entry and exit (STM32F42xxx and STM32F43xxx)

Stop mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – No interrupt or event is pending, – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, – PDDS bit is cleared in Power Control register (PWR_CR), – Select the voltage regulator mode by configuring LPDS, MRUDS, LPUDS and UDEN bits in PWR_CR (see *Table 29: Stop operating modes (STM32F42xxx and STM32F43xxx)*).
	On Return from ISR while: – No interrupt is pending, – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, and – SLEEPONEXIT = 1, and – PDDS is cleared in PWR_CR1.
	Note: To enter Stop mode, all EXTI Line pending bits (in Pending register (EXTI_PR)), all peripheral interrupts pending bits, the RTC Alarm (Alarm A and Alarm B), RTC wake-up, RTC tamper, and RTC time stamp flags, must be reset. Otherwise, the Stop mode entry procedure is ignored and program execution continues.
Mode exit	If WFI or Return from ISR was used for entry: All EXTI lines configured in Interrupt mode (the corresponding EXTI Interrupt vector must be enabled in the NVIC). The interrupt source can be external interrupts or peripherals with wake-up capability. Refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 375*. If WFE was used for entry and SEVONPEND = 0: All EXTI Lines configured in event mode. Refer to *Section 12.2.3: Wake-up event management on page 383* If WFE was used for entry and SEVONPEND = 1: – Any EXTI lines configured in Interrupt mode (even if the corresponding EXTI Interrupt vector is disabled in the NVIC). The interrupt source can be external interrupts or peripherals with wake-up capability. Refer to *Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 375* and *Table 63: Vector table for STM32F42xxx and STM32F43xxx*. – Wake-up event: refer to *Section 12.2.3: Wake-up event management on page 383*.
Wake-up latency	Refer to *Table 29: Stop operating modes (STM32F42xxx and STM32F43xxx)*

5.3.6 Standby mode

The Standby mode allows to achieve the lowest power consumption. It is based on the Cortex®-M4 with FPU deepsleep mode, with the voltage regulator disabled. The 1.2 V domain is consequently powered off. The PLLs, the HSI oscillator and the HSE oscillator are also switched off. SRAM and register contents are lost except for registers in the backup domain (RTC registers, RTC backup register and backup SRAM), and Standby circuitry (see *Figure 9*).

Entering Standby mode

The Standby mode is entered according to *Section : Entering low-power mode*, when the SLEEPDEEP bit in the Cortex®-M4 with FPU System Control register is set.

Refer to *Table 31* for more details on how to enter Standby mode.

In Standby mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started it cannot be stopped except by a reset. See *Section 21.3* in *Section 21: Independent watchdog (IWDG)*.
- Real-time clock (RTC): this is configured by the RTCEN bit in the backup domain control register (RCC_BDCR)
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the Control/status register (RCC_CSR).
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the backup domain control register (RCC_BDCR)

Exiting Standby mode

The Standby mode is exited according to *Section : Exiting low-power mode*. The SBF status flag in PWR_CR (see *Section 5.4.2: PWR power control/status register (PWR_CSR) for STM32F405xx/07xx and STM32F415xx/17xx*) indicates that the MCU was in Standby mode. All registers are reset after wake-up from Standby except for PWR_CR.

Refer to *Table 31* for more details on how to exit Standby mode.

Table 31. Standby mode entry and exit

Standby mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP is set in Cortex®-M4 with FPU System Control register, – PDDS bit is set in Power Control register (PWR_CR), – No interrupt (for WFI) or event (for WFE) is pending, – WUF bit is cleared in Power Control register (PWR_CR), – the RTC flag corresponding to the chosen wake-up source (RTC Alarm A, RTC Alarm B, RTC wake-up, Tamper or Timestamp flags) is cleared
	On return from ISR while: – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, and – SLEEPONEXIT = 1, and – PDDS bit is set in Power Control register (PWR_CR), and – No interrupt is pending, – WUF bit is cleared in Power Control/Status register (PWR_SR), – The RTC flag corresponding to the chosen wake-up source (RTC Alarm A, RTC Alarm B, RTC wake-up, Tamper or Timestamp flags) is cleared.
Mode exit	WKUP pin rising edge, RTC alarm (Alarm A and Alarm B), RTC wake-up, tamper event, time stamp event, external reset in NRST pin, IWDG reset.
Wake-up latency	Reset phase.

I/O states in Standby mode

In Standby mode, all I/O pins are high impedance except for:

- Reset pad (still available)
- RTC_AF1 pin (PC13) if configured for tamper, time stamp, RTC Alarm out, or RTC clock calibration out
- WKUP pin (PA0), if enabled

Debug mode

By default, the debug connection is lost if the application puts the MCU in Stop or Standby mode while the debug features are used. This is due to the fact that the Cortex®-M4 with FPU core is no longer clocked.

However, by setting some configuration bits in the DBGMCU_CR register, the software can be debugged even when using the low-power modes extensively. For more details, refer to *Section 38.16.1: Debug support for low-power modes*.

5.3.7 Programming the RTC alternate functions to wake up the device from the Stop and Standby modes

The MCU can be woken up from a low-power mode by an RTC alternate function.

The RTC alternate functions are the RTC alarms (Alarm A and Alarm B), RTC wake-up, RTC tamper event detection and RTC time stamp event detection.

These RTC alternate functions can wake up the system from the Stop and Standby low-power modes.

The system can also wake up from low-power modes without depending on an external interrupt (Auto-wake-up mode), by using the RTC alarm or the RTC wake-up events.

The RTC provides a programmable time base for waking up from the Stop or Standby mode at regular intervals.

For this purpose, two of the three alternate RTC clock sources can be selected by programming the RTCSEL[1:0] bits in the *Section 7.3.20: RCC Backup domain control register (RCC_BDCR)*:

- Low-power 32.768 kHz external crystal oscillator (LSE OSC)
  This clock source provides a precise time base with a very low-power consumption (additional consumption of less than 1 µA under typical conditions)
- Low-power internal RC oscillator (LSI RC)
  This clock source has the advantage of saving the cost of the 32.768 kHz crystal. This internal RC oscillator is designed to use minimum power.

RTC alternate functions to wake up the device from the Stop mode

- To wake up the device from the Stop mode with an RTC alarm event, it is necessary to:
  a) Configure the EXTI Line 17 to be sensitive to rising edges (Interrupt or Event modes)
  b) Enable the RTC Alarm Interrupt in the RTC_CR register
  c) Configure the RTC to generate the RTC alarm
- To wake up the device from the Stop mode with an RTC tamper or time stamp event, it is necessary to:
  a) Configure the EXTI Line 21 to be sensitive to rising edges (Interrupt or Event modes)
  b) Enable the RTC time stamp Interrupt in the RTC_CR register or the RTC tamper interrupt in the RTC_TAFCR register
  c) Configure the RTC to detect the tamper or time stamp event
- To wake up the device from the Stop mode with an RTC wake-up event, it is necessary to:
  a) Configure the EXTI Line 22 to be sensitive to rising edges (Interrupt or Event modes)
  b) Enable the RTC wake-up interrupt in the RTC_CR register
  c) Configure the RTC to generate the RTC Wake-up event

RTC alternate functions to wake up the device from the Standby mode

- To wake up the device from the Standby mode with an RTC alarm event, it is necessary to:
  a) Enable the RTC alarm interrupt in the RTC_CR register
  b) Configure the RTC to generate the RTC alarm
- To wake up the device from the Standby mode with an RTC tamper or time stamp event, it is necessary to:
  a) Enable the RTC time stamp interrupt in the RTC_CR register or the RTC tamper interrupt in the RTC_TAFCR register
  b) Configure the RTC to detect the tamper or time stamp event
- To wake up the device from the Standby mode with an RTC wake-up event, it is necessary to:
  a) Enable the RTC wake-up interrupt in the RTC_CR register
  b) Configure the RTC to generate the RTC wake-up event

Safe RTC alternate function wake-up flag clearing sequence

If the selected RTC alternate function is set before the PWR wake-up flag (WUTF) is cleared, it is not detected on the next event as detection is made once on the rising edge.

To avoid bouncing on the pins onto which the RTC alternate functions are mapped, and exit correctly from the Stop and Standby modes, it is recommended to follow the sequence below before entering the Standby mode:

- When using RTC alarm to wake up the device from the low-power modes:
  a) Disable the RTC alarm interrupt (ALRAIE or ALRBIE bits in the RTC_CR register)
  b) Clear the RTC alarm (ALRAF/ALRBF) flag

c) Clear the PWR Wake-up (WUF) flag
d) Enable the RTC alarm interrupt
e) Re-enter the low-power mode

- When using RTC wake-up to wake up the device from the low-power modes:
  a) Disable the RTC Wake-up interrupt (WUTIE bit in the RTC_CR register)
  b) Clear the RTC Wake-up (WUTF) flag
  c) Clear the PWR Wake-up (WUF) flag
  d) Enable the RTC Wake-up interrupt
  e) Re-enter the low-power mode
- When using RTC tamper to wake up the device from the low-power modes:
  a) Disable the RTC tamper interrupt (TAMPIE bit in the RTC_TAFCR register)
  b) Clear the Tamper (TAMP1F/TSF) flag
  c) Clear the PWR Wake-up (WUF) flag
  d) Enable the RTC tamper interrupt
  e) Re-enter the low-power mode
- When using RTC time stamp to wake up the device from the low-power modes:
  a) Disable the RTC time stamp interrupt (TSIE bit in RTC_CR)
  b) Clear the RTC time stamp (TSF) flag
  c) Clear the PWR Wake-up (WUF) flag
  d) Enable the RTC TimeStamp interrupt
  e) Re-enter the low-power mode

5.4 Power control registers (STM32F405xx/07xx and STM32F415xx/17xx)

5.4.1 PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx

Address offset: 0x00

Reset value: 0x0000 4000 (reset by wake-up from Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	VOS	Reserved				FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
	rw					rw	rw	rw	rw	rw	rw	w	w	rw	rw

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **VOS**: Regulator voltage scaling output selection

This bit controls the main internal voltage regulator output voltage to achieve a trade-off between performance and power consumption when the device does not operate at the maximum frequency.

0: Scale 2 mode

1: Scale 1 mode (default value at reset)

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 **FPDS**: Flash power-down in Stop mode

When set, the Flash memory enters power-down mode when the device enters Stop mode. This allows to achieve a lower consumption in stop mode but a longer restart time.

0: Flash memory not in power-down when the device is in Stop mode

1: Flash memory in power-down when the device is in Stop mode

Bit 8 **DBP**: Disable backup domain write protection

In reset state, the RCC_BDCR register, the RTC registers (including the backup registers), and the BRE bit of the PWR_CSR register, are protected against parasitic write access. This bit must be set to enable write access to these registers.

0: Access to RTC and RTC Backup registers and backup SRAM disabled

1: Access to RTC and RTC Backup registers and backup SRAM enabled

Note: Depending on the APB1 prescaler, there is a delay between writing to DBP and the effective disabling/enabling of the backup domain protection. Therefore, a dummy read operation to the PWR_CR register is required just after writing to the DBP bit.

Bits 7:5 **PLS[2:0]:** PVD level selection

These bits are written by software to select the voltage threshold detected by the programmable voltage detector

000: 2.0 V
001: 2.1 V
010: 2.3 V
011: 2.5 V
100: 2.6 V
101: 2.7 V
110: 2.8 V
111: 2.9 V

Note: Refer to the electrical characteristics of the datasheet for more details.

Bit 4 **PVDE:** Programmable voltage detector enable

This bit is set and cleared by software.
0: PVD disabled
1: PVD enabled

Bit 3 **CSBF**: Clear standby flag

This bit is always read as 0.
0: No effect
1: Clear the SBF Standby Flag (write).

Bit 2 **CWUF:** Clear wake-up flag

This bit is always read as 0.
0: No effect
1: Clear the WUF Wake-up Flag **after 2 System clock cycles**

Bit 1 **PDDS**: Power-down deepsleep

This bit is set and cleared by software. It works together with the LPDS bit.
0: Enter Stop mode when the CPU enters deepsleep. The regulator status depends on the LPDS bit.
1: Enter Standby mode when the CPU enters deepsleep.

Bit 0 **LPDS:** Low-power deepsleep

This bit is set and cleared by software. It works together with the PDDS bit.
0: Voltage regulator on during Stop mode
1: Voltage regulator in low-power mode during Stop mode

5.4.2 PWR power control/status register (PWR_CSR) for STM32F405xx/07xx and STM32F415xx/17xx

Address offset: 0x04

Reset value: 0x0000 0000 (not reset by wake-up from Standby mode)

Additional APB cycles are needed to read this register versus a standard APB read.

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td></tr>
<tr><td colspan="16">Reserved</td></tr>
<tr><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td rowspan="2">Res</td><td>VOS RDY</td><td colspan="4" rowspan="2">Reserved</td><td>BRE</td><td>EWUP</td><td colspan="4" rowspan="2">Reserved</td><td>BRR</td><td>PVDO</td><td>SBF</td><td>WUF</td></tr>
<tr><td>r</td><td>rw</td><td>rw</td><td>r</td><td>r</td><td>r</td><td>r</td></tr>
</table>

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **VOSRDY**: Regulator voltage scaling output selection ready bit

0: Not ready

1: Ready

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 **BRE**: Backup regulator enable

When set, the Backup regulator (used to maintain backup SRAM content in Standby and V_{BAT} modes) is enabled. If BRE is reset, the backup regulator is switched off. The backup SRAM can still be used but its content is lost in the Standby and V_{BAT} modes. Once set, the application must wait that the Backup Regulator Ready flag (BRR) is set to indicate that the data written into the RAM is maintained in the Standby and V_{BAT} modes.

0: Backup regulator disabled

1: Backup regulator enabled

Note: This bit is not reset when the device wakes up from Standby mode, by a system reset, or by a power reset.

The DBP bit of the PWR_CR register must be set before BRE can be written.

Bit 8 **EWUP**: Enable WKUP pin

This bit is set and cleared by software.

0: WKUP pin is used for general purpose I/O. An event on the WKUP pin does not wake up the device from Standby mode.

1: WKUP pin is used for wake-up from Standby mode and forced in input pull down configuration (rising edge on WKUP pin wakes-up the system from Standby mode).

Note: This bit is reset by a system reset.

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **BRR**: Backup regulator ready

Set by hardware to indicate that the Backup Regulator is ready.

0: Backup Regulator not ready

1: Backup Regulator ready

Note: This bit is not reset when the device wakes up from Standby mode or by a system reset or power reset.

Bit 2 **PVDO:** PVD output

This bit is set and cleared by hardware. It is valid only if PVD is enabled by the PVDE bit.

0: V_{DD} is higher than the PVD threshold selected with the PLS[2:0] bits.

1: V_{DD} is lower than the PVD threshold selected with the PLS[2:0] bits.

Note: The PVD is stopped by Standby mode. For this reason, this bit is equal to 0 after Standby or reset until the PVDE bit is set.

Bit 1 **SBF:** Standby flag

This bit is set by hardware and cleared only by a POR/PDR (power-on reset/power-down reset) or by setting the CSBF bit in the PWR_CR register.

0: Device has not been in Standby mode

1: Device has been in Standby mode

Bit 0 **WUF:** Wake-up flag

This bit is set by hardware and cleared either by a system reset or by setting the CWUF bit in the PWR_CR register.

0: No wake-up event occurred

1: A wake-up event was received from the WKUP pin or from the RTC alarm (Alarm A or Alarm B), RTC Tamper event, RTC TimeStamp event or RTC Wake-up).

Note: An additional wake-up event is detected if the WKUP pin is enabled (by setting the EWUP bit) when the WKUP pin level is already high.

5.5 Power control registers (STM32F42xxx and STM32F43xxx)

5.5.1 PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx

Address offset: 0x00

Reset value: 0x0000 C000 (reset by wake-up from Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												UDEN[1:0]		ODSWEN	ODEN
												rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VOS[1:0]		ADCDC1	Res.	MRUDS	LPUDS	FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
rw	rw	rw		rw	rw	rw	rw	rw	rw	rw	rw	rc_w1	rc_w1	rw	rw

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:18 **UDEN[1:0]**: Under-drive enable in stop mode

These bits are set by software. They allow to achieve a lower power consumption in Stop mode but with a longer wake-up time.

When set, the digital area has less leakage consumption when the device enters Stop mode.

00: Under-drive disable

01: Reserved

10: Reserved

11:Under-drive enable

Bit 17 **ODSWEN**: Over-drive switching enabled.

This bit is set by software. It is cleared automatically by hardware after exiting from Stop mode or when the ODEN bit is reset. When set, It is used to switch to Over-drive mode.

To set or reset the ODSWEN bit, the HSI or HSE must be selected as system clock.

The ODSWEN bit must only be set when the ODRDY flag is set to switch to Over-drive mode.

0: Over-drive switching disabled

1: Over-drive switching enabled

Note: On any over-drive switch (enabled or disabled), the system clock is stalled during the internal voltage set up.

Bit 16 **ODEN**: Over-drive enable

This bit is set by software. It is cleared automatically by hardware after exiting from Stop mode. It is used to enabled the Over-drive mode in order to reach a higher frequency.

To set or reset the ODEN bit, the HSI or HSE must be selected as system clock. When the ODEN bit is set, the application must first wait for the Over-drive ready flag (ODRDY) to be set before setting the ODSWEN bit.

0: Over-drive disabled

1: Over-drive enabled

Bits 15:14 **VOS[1:0]**: Regulator voltage scaling output selection

These bits control the main internal voltage regulator output voltage to achieve a trade-off between performance and power consumption when the device does not operate at the maximum frequency (refer to the STM32F42xx and STM32F43xx datasheets for more details).

These bits can be modified only when the PLL is OFF. The new value programmed is active only when the PLL is ON. When the PLL is OFF, the voltage scale 3 is automatically selected.

00: Reserved (Scale 3 mode selected)

01: Scale 3 mode

10: Scale 2 mode

11: Scale 1 mode (reset value)

Bit 13 **ADCDC1**:

0: No effect.

1: Refer to AN4073 for details on how to use this bit.

Note: This bit can only be set when operating at supply voltage range 2.7 to 3.6V and when the Prefetch is OFF.

Bit 12 Reserved, must be kept at reset value.

Bit 11 **MRUDS**: Main regulator in deepsleep under-drive mode

This bit is set and cleared by software.

0: Main regulator ON when the device is in Stop mode

1: Main Regulator in under-drive mode and Flash memory in power-down when the device is in Stop under-drive mode.

Bit 10 **LPUDS**: Low-power regulator in deepsleep under-drive mode

This bit is set and cleared by software.

0: Low-power regulator ON if LPDS bit is set when the device is in Stop mode

1: Low-power regulator in under-drive mode if LPDS bit is set and Flash memory in power-down when the device is in Stop under-drive mode.

Bit 9 **FPDS**: Flash power-down in Stop mode

When set, the Flash memory enters power-down mode when the device enters Stop mode. This allows to achieve a lower consumption in stop mode but a longer restart time.

0: Flash memory not in power-down when the device is in Stop mode

1: Flash memory in power-down when the device is in Stop mode

Bit 8 **DBP**: Disable backup domain write protection

In reset state, the RCC_BDCR register, the RTC registers (including the backup registers), and the BRE bit of the PWR_CSR register, are protected against parasitic write access. This bit must be set to enable write access to these registers.

0: Access to RTC and RTC Backup registers and backup SRAM disabled

1: Access to RTC and RTC Backup registers and backup SRAM enabled

Bits 7:5 **PLS[2:0]:** PVD level selection

These bits are written by software to select the voltage threshold detected by the programmable voltage detector

000: 2.0 V
001: 2.1 V
010: 2.3 V
011: 2.5 V
100: 2.6 V
101: 2.7 V
110: 2.8 V
111: 2.9 V

Note: Refer to the electrical characteristics of the datasheet for more details.

Bit 4 **PVDE:** Programmable voltage detector enable

This bit is set and cleared by software.

0: PVD disabled
1: PVD enabled

Bit 3 **CSBF**: Clear standby flag

This bit is always read as 0.

0: No effect
1: Clear the SBF Standby Flag (write).

Bit 2 **CWUF:** Clear wake-up flag

This bit is always read as 0.

0: No effect
1: Clear the WUF Wake-up Flag **after 2 System clock cycles**

Bit 1 **PDDS**: Power-down deepsleep

This bit is set and cleared by software. It works together with the LPDS bit.

0: Enter Stop mode when the CPU enters deepsleep. The regulator status depends on the LPDS bit.
1: Enter Standby mode when the CPU enters deepsleep.

Bit 0 **LPDS:** Low-power deepsleep

This bit is set and cleared by software. It works together with the PDDS bit.

0:Main voltage regulator ON during Stop mode
1: Low-power voltage regulator ON during Stop mode

5.5.2 PWR power control/status register (PWR_CSR) for STM32F42xxx and STM32F43xxx

Address offset: 0x04

Reset value: 0x0000 0000 (not reset by wake-up from Standby mode)

Additional APB cycles are needed to read this register versus a standard APB read.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												UDRDY[1:0]		ODSWRDY	ODRDY
												rc_w1	rc_w1	r	r

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	VOS RDY	Reserved				BRE	EWUP	Reserved.				BRR	PVDO	SBF	WUF
	r					rw	rw					r	r	r	r

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:18 **UDRDY[1:0]**: Under-drive ready flag

These bits are set by hardware when MCU entered stop Under-drive mode and exited. When the under-drive mode is enabled, these bits are not set as long as the MCU has not entered stop mode yet. They are cleared by programming them to 1.

00: Under-drive is disabled

01: Reserved

10: Reserved

11:Under-drive mode is activated in Stop mode.

Bit 17 **ODSWRDY**: Over-drive mode switching ready

0: Over-drive mode is not active.

1: Over-drive mode is active on digital area on 1.2 V domain

Bit 16 **ODRDY**: Over-drive mode ready

0: Over-drive mode not ready.

1: Over-drive mode ready

Bit 14 **VOSRDY**: Regulator voltage scaling output selection ready bit

0: Not ready

1: Ready

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 **BRE**: Backup regulator enable

When set, the Backup regulator (used to maintain backup SRAM content in Standby and V_{BAT} modes) is enabled. If BRE is reset, the backup regulator is switched off. The backup SRAM can still be used but its content is lost in the Standby and V_{BAT} modes. Once set, the application must wait that the Backup Regulator Ready flag (BRR) is set to indicate that the data written into the RAM is maintained in the Standby and V_{BAT} modes.

0: Backup regulator disabled

1: Backup regulator enabled

Note: This bit is not reset when the device wakes up from Standby mode, by a system reset, or by a power reset.

The DBP bit of the PWR_CR register must be set before BRE can be written.

Bit 8 **EWUP**: Enable WKUP pin

This bit is set and cleared by software.

0: WKUP pin is used for general purpose I/O. An event on the WKUP pin does not wake up the device from Standby mode.

1: WKUP pin is used for wake-up from Standby mode and forced in input pull down configuration (rising edge on WKUP pin wakes-up the system from Standby mode).

Note: This bit is reset by a system reset.

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **BRR**: Backup regulator ready

Set by hardware to indicate that the Backup Regulator is ready.

0: Backup Regulator not ready

1: Backup Regulator ready

Note: This bit is not reset when the device wakes up from Standby mode or by a system reset or power reset.

Bit 2 **PVDO:** PVD output

This bit is set and cleared by hardware. It is valid only if PVD is enabled by the PVDE bit.

0: V_{DD} is higher than the PVD threshold selected with the PLS[2:0] bits.

1: V_{DD} is lower than the PVD threshold selected with the PLS[2:0] bits.

Note: The PVD is stopped by Standby mode. For this reason, this bit is equal to 0 after Standby or reset until the PVDE bit is set.

Bit 1 **SBF:** Standby flag

This bit is set by hardware and cleared only by a POR/PDR (power-on reset/power-down reset) or by setting the CSBF bit in the *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx*

0: Device has not been in Standby mode

1: Device has been in Standby mode

Bit 0 **WUF:** Wake-up flag

This bit is set by hardware and cleared either by a system reset or by setting the CWUF bit in the PWR_CR register.

0: No wake-up event occurred

1: A wake-up event was received from the WKUP pin or from the RTC alarm (Alarm A or Alarm B), RTC Tamper event, RTC TimeStamp event or RTC Wake-up).

Note: An additional wake-up event is detected if the WKUP pin is enabled (by setting the EWUP bit) when the WKUP pin level is already high.

5.6 PWR register map

The following table summarizes the PWR registers.

Table 32. PWR - register map and reset values for STM32F405xx/07xx and STM32F415xx/17xx

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x000	**PWR_CR**	Reserved																	VOS	Reserved				FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
	Reset value																		1					0	0	0	0	0	0	0	0	0	0
0x004	**PWR_CSR**	Reserved																	VOSRDY	Reserved				BRE	EWUP	Reserved				BRR	PVDO	SBF	WUF
	Reset value																		0					0	0					0	0	0	0

Table 33. PWR - register map and reset values for STM32F42xxx and STM32F43xxx

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x000	PWR_CR	Reserved												UDEN[1:0]		ODSWEN	ODEN	VOS[1:0]		ADCDC1	Reserved	MRUDS	LPUDS	FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
	Reset value													0	0	0	0	1	1	0		0	0	0	0	0	0	0	0	0	0	0	0
0x004	PWR_CSR	Reserved												UDRDY[1:0]		ODSWRDY	ODRDY	Reserved	VOSRDY	Reserved				BRE	EWUP	Reserved				BRR	PVDO	SBF	WUF
	Reset value													0	0	0	0		0					0	0					0	0	0	0

Refer to *Section 2.3: Memory map* for the register boundary addresses.

6 Reset and clock control for STM32F42xxx and STM32F43xxx (RCC)

6.1 Reset

There are three types of reset, defined as system Reset, power Reset and backup domain Reset.

6.1.1 System reset

A system reset sets all registers to their reset values unless specified otherwise in the register description (see *Figure 10*).

A system reset is generated when one of the following events occurs:

1. A low level on the NRST pin (external reset)
2. Window watchdog end of count condition (WWDG reset)
3. Independent watchdog end of count condition (IWDG reset)
4. A software reset (SW reset) (see *Software reset*)
5. Low-power management reset (see *Low-power management reset*)

Software reset

The reset source can be identified by checking the reset flags in the *RCC clock control & status register (RCC_CSR)*.

The SYSRESETREQ bit in Cortex®-M4 with FPU Application Interrupt and Reset Control Register must be set to force a software reset on the device. Refer to the Cortex®-M4 with FPU technical reference manual for more details.

Low-power management reset

There are two ways of generating a low-power management reset:

1. Reset generated when entering the Standby mode:

   This type of reset is enabled by resetting the nRST_STDBY bit in the user option bytes. In this case, whenever a Standby mode entry sequence is successfully executed, the device is reset instead of entering the Standby mode.
2. Reset when entering the Stop mode:

   This type of reset is enabled by resetting the nRST_STOP bit in the user option bytes. In this case, whenever a Stop mode entry sequence is successfully executed, the device is reset instead of entering the Stop mode.

6.1.2 Power reset

A power reset is generated when one of the following events occurs:

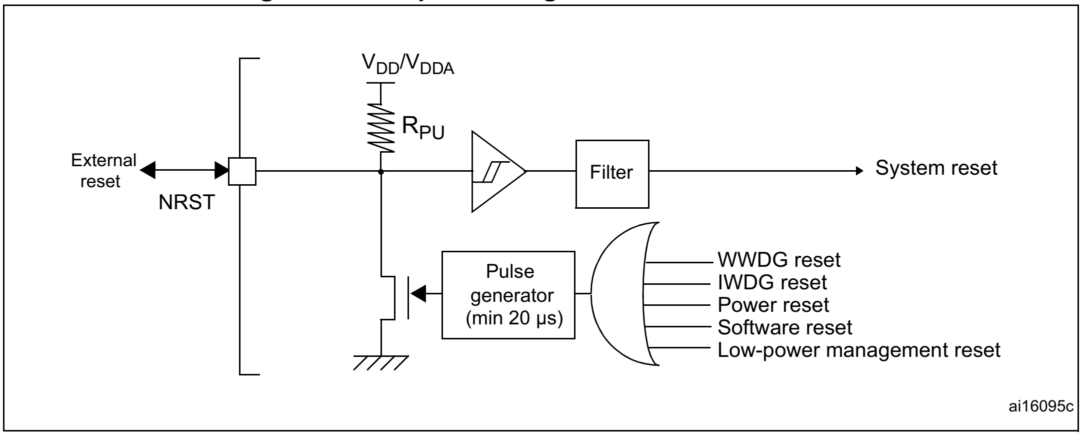
1. Power-on/power-down reset (POR/PDR reset) or brownout (BOR) reset
2. When exiting the Standby mode

A power reset sets all registers to their reset values except the Backup domain (see *Figure 10*)

These sources act on the NRST pin and it is always kept low during the delay phase. The RESET service routine vector is fixed at address 0x0000_0004 in the memory map.

The system reset signal provided to the device is output on the NRST pin. The pulse generator guarantees a minimum reset pulse duration of 20 µs for each internal reset source. In case of an external reset, the reset pulse is generated while the NRST pin is asserted low.

Figure 15. Simplified diagram of the reset circuit



The Backup domain has two specific resets that affect only the Backup domain (see *Figure 10*).

6.1.3 Backup domain reset

The backup domain reset sets all RTC registers, the RCC_BDCR register, and the BRE bit of the PWR_CSR register to their reset values. The BKPSRAM is not affected by this reset. The only way of resetting the BKPSRAM is through the Flash interface by requesting a protection level change from 1 to 0.

A backup domain reset is generated when one of the following events occurs:

1. Software reset, triggered by setting the BDRST bit in the *RCC Backup domain control register (RCC_BDCR)*.
2. V_{DD} or V_{BAT} power on, if both supplies have previously been powered off.

Note: The DBP bit of the PWR_CR register must be set to generate a backup domain reset.

6.2 Clocks

Three different clock sources can be used to drive the system clock (SYSCLK):

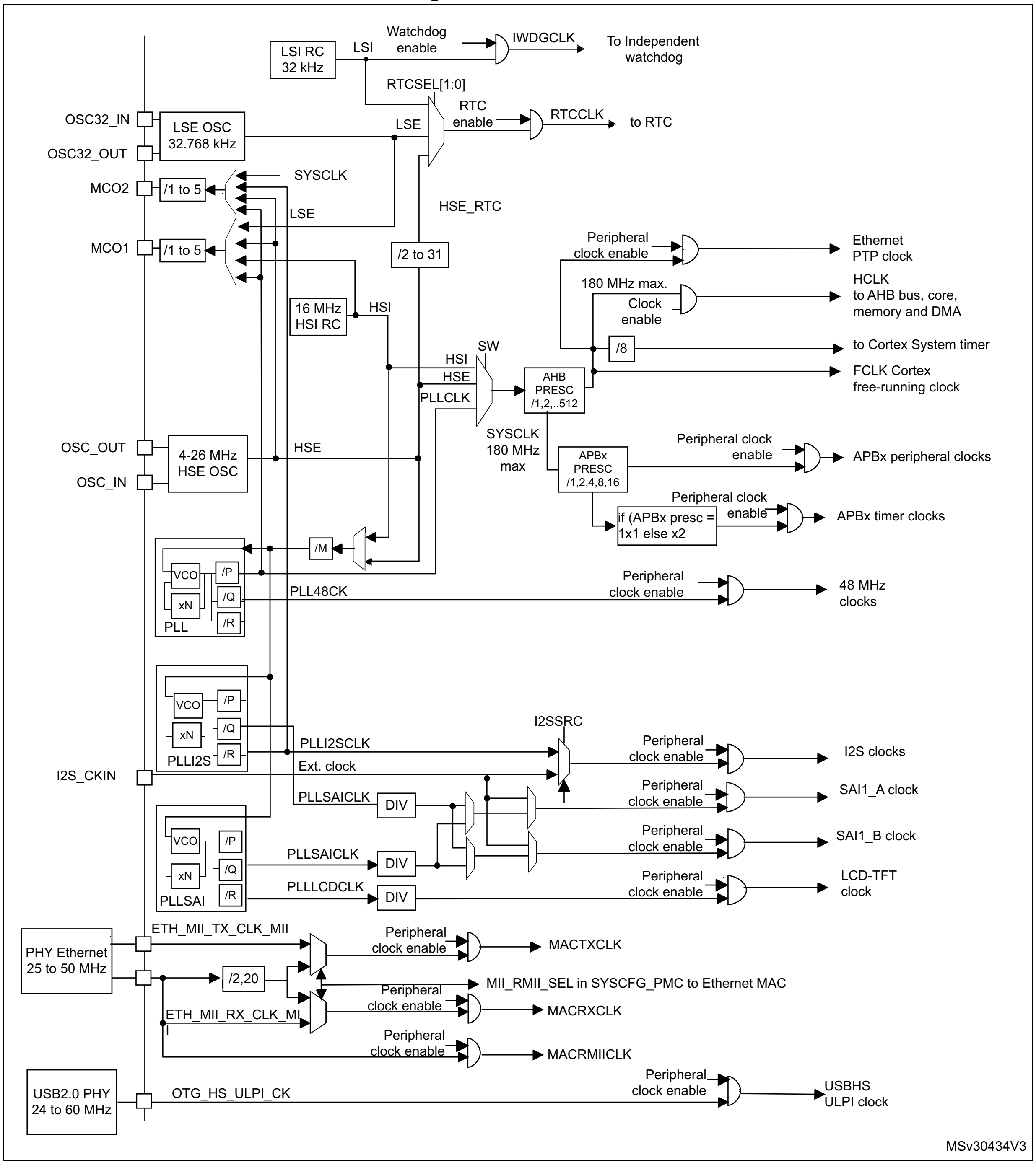
- HSI oscillator clock
- HSE oscillator clock
- Main PLL (PLL) clock

The devices have the two following secondary clock sources:

- 32 kHz low-speed internal RC (LSI RC) which drives the independent watchdog and, optionally, the RTC used for Auto-wakeup from the Stop/Standby mode.
- 32.768 kHz low-speed external crystal (LSE crystal) which optionally drives the RTC clock (RTCCLK)

Each clock source can be switched on or off independently when it is not used, to optimize power consumption.

Figure 16. Clock tree



1. For full details about the internal and external clock source characteristics, refer to the Electrical characteristics section in the device datasheet.
2. When TIMPRE bit of the RCC_DCKCFGR register is reset, if APBx prescaler is 1, then TIMxCLK = PCLKx, otherwise TIMxCLK = 2x PCLKx.
3. When TIMPRE bit in the RCC_DCKCFGR register is set, if APBx prescaler is 1,2 or 4, then TIMxCLK = HCLK, otherwise TIMxCLK = 4x PCLKx.

The clock controller provides a high degree of flexibility to the application in the choice of the external crystal or the oscillator to run the core and peripherals at the highest frequency and, guarantee the appropriate frequency for peripherals that need a specific clock like Ethernet, USB OTG FS and HS, I2S, SAI, LTDC, and SDIO.

Several prescalers are used to configure the AHB frequency, the high-speed APB (APB2) and the low-speed APB (APB1) domains. The maximum frequency of the AHB domain is 180 MHz. The maximum allowed frequency of the high-speed APB2 domain is 90 MHz. The maximum allowed frequency of the low-speed APB1 domain is 45 MHz

All peripheral clocks are derived from the system clock (SYSCLK) except for:

- The USB OTG FS clock (48 MHz), the random analog generator (RNG) clock (≤48 MHz) and the SDIO clock (≤48 MHz) which are coming from a specific output of PLL (PLL48CLK)
- I2S clock

  To achieve high-quality audio performance, the I2S clock can be derived either from a specific PLL (PLLI2S) or from an external clock mapped on the I2S_CKIN pin. For more information about I2S clock frequency and precision, refer to *Section 28.4.4: Clock generator*.
- SAI1 clock

  The SAI1 clock is generated from a specific PLL (PLLSAI or PLLI2S) or from an external clock mapped on the I2S_CKIN pin.

  The PLLSAI can be used as clock source for SAI1 peripheral in case the PLLI2S is programmed to achieve another audio sampling frequency (49.152 MHz or 11.2896 MHz), and the application requires both frequencies at the same time.
- LTDC clock

  The LTDC clock is generated from a specific PLL (PLLSAI).
- The USB OTG HS (60 MHz) clock which is provided from the external PHY
- The Ethernet MAC clocks (TX, RX and RMII) which are provided from the external PHY. For further information on the Ethernet configuration, please refer to *Section 33.4.4: MII/RMII selection* in the Ethernet peripheral description. When the Ethernet is used, the AHB clock frequency must be at least 25 MHz.

The RCC feeds the external clock of the Cortex System Timer (SysTick) with the AHB clock (HCLK) divided by 8. The SysTick can work either with this clock or with the Cortex clock (HCLK), configurable in the SysTick control and status register.

The timer clock frequencies for STM32F42xxx and STM32F43xxx are automatically set by hardware. There are two cases depending on the value of TIMPRE bit in RCC_CFGR register:

- If TIMPRE bit in RCC_DKCFGR register is reset:

  If the APB prescaler is configured to a division factor of 1, the timer clock frequencies (TIMxCLK) are set to PCLKx. Otherwise, the timer clock frequencies are twice the frequency of the APB domain to which the timers are connected: TIMxCLK = 2xPCLKx.
- If TIMPRE bit in RCC_DKCFGR register is set:

  If the APB prescaler is configured to a division factor of 1, 2 or 4, the timer clock frequencies (TIMxCLK) are set to HCLK. Otherwise, the timer clock frequencies is four times the frequency of the APB domain to which the timers are connected: TIMxCLK = 4xPCLKx.

FCLK acts as Cortex®-M4 with FPU free-running clock. For more details, refer to the Cortex®-M4 with FPU technical reference manual.

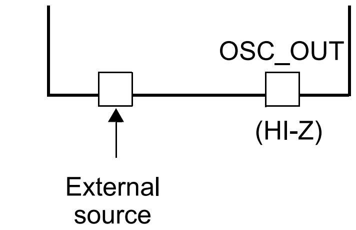
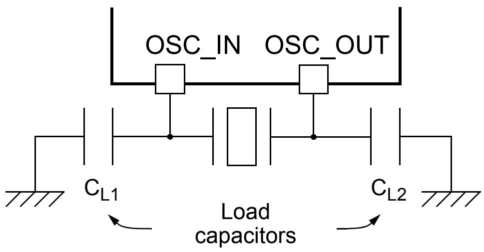
6.2.1 HSE clock

The high speed external clock signal (HSE) can be generated from two possible clock sources:

- HSE external crystal/ceramic resonator
- HSE external user clock

The resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. The loading capacitance values must be adjusted according to the selected oscillator.

Figure 17. HSE/ LSE clock sources

	Hardware configuration
External clock	
Crystal/ceramic resonators	

External source (HSE bypass)

In this mode, an external clock source must be provided. You select this mode by setting the HSEBYP and HSEON bits in the *RCC clock control register (RCC_CR)*. The external clock signal (square, sinus or triangle) with ~50% duty cycle has to drive the OSC_IN pin while the OSC_OUT pin should be left HI-Z. See *Figure 17*.

External crystal/ceramic resonator (HSE crystal)

The HSE has the advantage of producing a very accurate rate on the main clock.

The associated hardware configuration is shown in *Figure 17*. Refer to the electrical characteristics section of the *datasheet* for more details.

The HSERDY flag in the *RCC clock control register (RCC_CR)* indicates if the high-speed external oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the *RCC clock interrupt register (RCC_CIR)*.

The HSE Crystal can be switched on and off using the HSEON bit in the *RCC clock control register (RCC_CR)*.

6.2.2 HSI clock

The HSI clock signal is generated from an internal 16 MHz RC oscillator and can be used directly as a system clock, or used as PLL input.

The HSI RC oscillator has the advantage of providing a clock source at low cost (no external components). It also has a faster startup time than the HSE crystal oscillator however, even with calibration the frequency is less accurate than an external crystal oscillator or ceramic resonator.

Calibration

RC oscillator frequencies can vary from one chip to another due to manufacturing process variations, this is why each device is factory calibrated by ST for 1% accuracy at T_A= 25 °C.

After reset, the factory calibration value is loaded in the HSICAL[7:0] bits in the *RCC clock control register (RCC_CR)*.

If the application is subject to voltage or temperature variations this may affect the RC oscillator speed. You can trim the HSI frequency in the application using the HSITRIM[4:0] bits in the *RCC clock control register (RCC_CR)*.

The HSIRDY flag in the *RCC clock control register (RCC_CR)* indicates if the HSI RC is stable or not. At startup, the HSI RC output clock is not released until this bit is set by hardware.

The HSI RC can be switched on and off using the HSION bit in the *RCC clock control register (RCC_CR)*.

The HSI signal can also be used as a backup source (Auxiliary clock) if the HSE crystal oscillator fails. Refer to *Section 6.2.7: Clock security system (CSS) on page 159*.

6.2.3 PLL configuration

The STM32F4xx devices feature three PLLs:

- A main PLL (PLL) clocked by the HSE or HSI oscillator and featuring two different output clocks:
  - The first output is used to generate the high speed system clock (up to 180 MHz)
  - The second output is used to generate the clock for the USB OTG FS (48 MHz), the random analog generator (≤48 MHz) and the SDIO (≤48 MHz).
- Two dedicated PLLs (PLLI2S and PLLSAI) used to generate an accurate clock to achieve high-quality audio performance on the I2S and SAI1 interfaces. PLLSAI is also used for the LCD-TFT clock.

Since the main-PLL configuration parameters cannot be changed once PLL is enabled, it is recommended to configure PLL before enabling it (selection of the HSI or HSE oscillator as PLL clock source, and configuration of division factors M, N, P, and Q).

The PLLI2S and PLLSAI use the same input clock as PLL (PLLM[5:0] and PLLSRC bits are common to both PLLs). However, the PLLI2S and PLLSAI have dedicated enable/disable and division factors (N and R) configuration bits. Once the PLLI2S and PLLSAI are enabled, the configuration parameters cannot be changed.

The three PLLs are disabled by hardware when entering Stop and Standby modes, or when an HSE failure occurs when HSE or PLL (clocked by HSE) are used as system clock. *RCC PLL configuration register (RCC_PLLCFGR)*,*RCC clock configuration register (RCC_CFGR)*, and *RCC Dedicated Clock Configuration Register (RCC_DCKCFGR)* can be used to configure PLL, PLLI2S, and PLLSAI.

6.2.4 LSE clock

The LSE clock is generated from a 32.768 kHz low-speed external crystal or ceramic resonator. It has the advantage providing a low-power but highly accurate clock source to the real-time clock peripheral (RTC) for clock/calendar or other timing functions.

The LSE oscillator is switched on and off using the LSEON bit in *RCC Backup domain control register (RCC_BDCR)*.

The LSERDY flag in the *RCC Backup domain control register (RCC_BDCR)* indicates if the LSE crystal is stable or not. At startup, the LSE crystal output clock signal is not released until this bit is set by hardware. An interrupt can be generated if enabled in the *RCC clock interrupt register (RCC_CIR)*.

External source (LSE bypass)

In this mode, an external clock source must be provided. It must have a frequency up to 1 MHz. You select this mode by setting the LSEBYP and LSEON bits in the *RCC Backup domain control register (RCC_BDCR)*. The external clock signal (square, sinus or triangle) with ~50% duty cycle has to drive the OSC32_IN pin while the OSC32_OUT pin should be left HI-Z. See *Figure 17*.

6.2.5 LSI clock

The LSI RC acts as an low-power clock source that can be kept running in Stop and Standby mode for the independent watchdog (IWDG) and Auto-wakeup unit (AWU). The clock frequency is around 32 kHz. For more details, refer to the electrical characteristics section of the datasheets.

The LSI RC can be switched on and off using the LSION bit in the *RCC clock control & status register (RCC_CSR)*.

The LSIRDY flag in the *RCC clock control & status register (RCC_CSR)* indicates if the low-speed internal oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the *RCC clock interrupt register (RCC_CIR)*.

6.2.6 System clock (SYSCLK) selection

After a system reset, the HSI oscillator is selected as the system clock. When a clock source is used directly or through PLL as the system clock, it is not possible to stop it.

A switch from one clock source to another occurs only if the target clock source is ready (clock stable after startup delay or PLL locked). If a clock source that is not yet ready is selected, the switch occurs when the clock source is ready. Status bits in the *RCC clock control register (RCC_CR)* indicate which clock(s) is (are) ready and which clock is currently used as the system clock.

6.2.7 Clock security system (CSS)

The clock security system can be activated by software. In this case, the clock detector is enabled after the HSE oscillator startup delay, and disabled when this oscillator is stopped.

If a failure is detected on the HSE clock, this oscillator is automatically disabled, a clock failure event is sent to the break inputs of advanced-control timers TIM1 and TIM8, and an interrupt is generated to inform the software about the failure (clock security system interrupt CSSI), allowing the MCU to perform rescue operations. The CSSI is linked to the Cortex®-M4 with FPU NMI (non-maskable interrupt) exception vector.

Note: When the CSS is enabled, if the HSE clock happens to fail, the CSS generates an interrupt, which causes the automatic generation of an NMI. The NMI is executed indefinitely unless the CSS interrupt pending bit is cleared. As a consequence, the application has to clear the CSS interrupt in the NMI ISR by setting the CSSC bit in the Clock interrupt register (RCC_CIR).

If the HSE oscillator is used directly or indirectly as the system clock (indirectly meaning that it is directly used as PLL input clock, and that PLL clock is the system clock) and a failure is detected, then the system clock switches to the HSI oscillator and the HSE oscillator is disabled.

If the HSE oscillator clock was the clock source of PLL used as the system clock when the failure occurred, PLL is also disabled. In this case, if the PLLI2S was enabled, it is also disabled when the HSE fails.

6.2.8 RTC/AWU clock

Once the RTCCLK clock source has been selected, the only possible way of modifying the selection is to reset the power domain.

The RTCCLK clock source can be either the HSE 1 MHz (HSE divided by a programmable prescaler), the LSE or the LSI clock. This is selected by programming the RTCSEL[1:0] bits in the *RCC Backup domain control register (RCC_BDCR)* and the RTCPRE[4:0] bits in *RCC clock configuration register (RCC_CFGR)*. This selection cannot be modified without resetting the Backup domain.

If the LSE is selected as the RTC clock, the RTC operates normally if the backup or the system supply disappears. If the LSI is selected as the AWU clock, the AWU state is not guaranteed if the system supply disappears. If the HSE oscillator divided by a value between 2 and 31 is used as the RTC clock, the RTC state is not guaranteed if the backup or the system supply disappears.

The LSE clock is in the Backup domain, whereas the HSE and LSI clocks are not. As a consequence:

- If LSE is selected as the RTC clock:
  – The RTC continues to work even if the V_{DD} supply is switched off, provided the V_{BAT} supply is maintained.
- If LSI is selected as the Auto-wakeup unit (AWU) clock:
  – The AWU state is not guaranteed if the V_{DD} supply is powered off. Refer to *Section 6.2.5: LSI clock on page 158* for more details on LSI calibration.
- If the HSE clock is used as the RTC clock:
  – The RTC state is not guaranteed if the V_{DD} supply is powered off or if the internal voltage regulator is powered off (removing power from the 1.2 V domain).

Note: To read the RTC calendar register when the APB1 clock frequency is less than seven times the RTC clock frequency ($f_{APB1} < 7xf_{RTCLCK}$), the software must read the calendar time and date registers twice. The data are correct if the second read access to RTC_TR gives the same result than the first one. Otherwise a third read access must be performed.

6.2.9 Watchdog clock

If the independent watchdog (IWDG) is started by either hardware option or software access, the LSI oscillator is forced ON and cannot be disabled. After the LSI oscillator temporization, the clock is provided to the IWDG.

6.2.10 Clock-out capability

Two microcontroller clock output (MCO) pins are available:

- MCO1
  You can output four different clock sources onto the MCO1 pin (PA8) using the configurable prescaler (from 1 to 5):
  - HSI clock
  - LSE clock
  - HSE clock
  - PLL clock

  The desired clock source is selected using the MCO1PRE[2:0] and MCO1[1:0] bits in the *RCC clock configuration register (RCC_CFGR)*.
- MCO2
  You can output four different clock sources onto the MCO2 pin (PC9) using the configurable prescaler (from 1 to 5):
  - HSE clock
  - PLL clock
  - System clock (SYSCLK)
  - PLLI2S clock

  The desired clock source is selected using the MCO2PRE[2:0] and MCO2 bits in the *RCC clock configuration register (RCC_CFGR)*.

For the different MCO pins, the corresponding GPIO port has to be programmed in alternate function mode.

The selected clock to output onto MCO must not exceed 100 MHz (the maximum I/O speed).

6.2.11 Internal/external clock measurement using TIM5/TIM11

It is possible to indirectly measure the frequencies of all on-board clock source generators by means of the input capture of TIM5 channel4 and TIM11 channel1 as shown in *Figure 18* and *Figure 19*.

Internal/external clock measurement using TIM5 channel4

TIM5 has an input multiplexer which allows choosing whether the input capture is triggered by the I/O or by an internal clock. This selection is performed through the TI4_RMP [1:0] bits in the TIM5_OR register.

The primary purpose of having the LSE connected to the channel4 input capture is to be able to precisely measure the HSI (this requires to have the HSI used as the system clock source). The number of HSI clock counts between consecutive edges of the LSE signal provides a measurement of the internal clock period. Taking advantage of the high precision of LSE crystals (typically a few tens of ppm) we can determine the internal clock frequency with the same resolution, and trim the source to compensate for manufacturing-process and/or temperature- and voltage-related frequency deviations.

The HSI oscillator has dedicated, user-accessible calibration bits for this purpose.

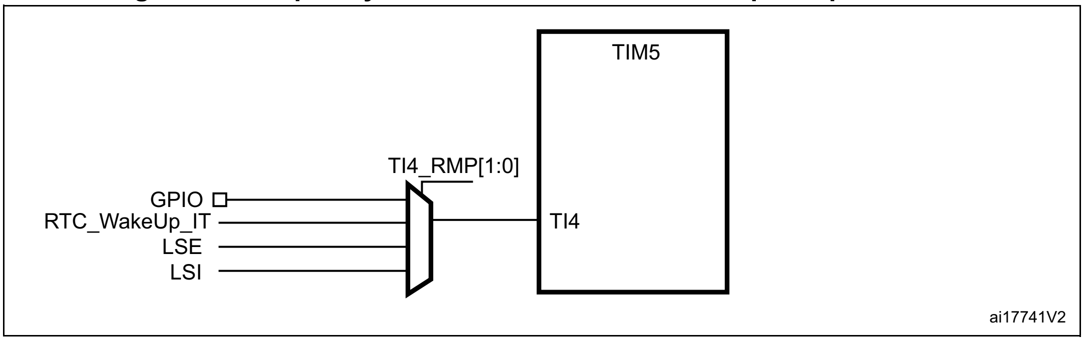
The basic concept consists in providing a relative measurement (e.g. HSI/LSE ratio): the precision is therefore tightly linked to the ratio between the two clock sources. The greater the ratio, the better the measurement.

It is also possible to measure the LSI frequency: this is useful for applications that do not have a crystal. The ultralow-power LSI oscillator has a large manufacturing process deviation: by measuring it versus the HSI clock source, it is possible to determine its frequency with the precision of the HSI. The measured value can be used to have more accurate RTC time base timeouts (when LSI is used as the RTC clock source) and/or an IWDG timeout with an acceptable accuracy.

Use the following procedure to measure the LSI frequency:

1. Enable the TIM5 timer and configure channel4 in Input capture mode.
2. This bit is set the TI4_RMP bits in the TIM5_OR register to 0x01 to connect the LSI clock internally to TIM5 channel4 input capture for calibration purposes.
3. Measure the LSI clock frequency using the TIM5 capture/compare 4 event or interrupt.
4. Use the measured LSI frequency to update the prescaler of the RTC depending on the desired time base and/or to compute the IWDG timeout.

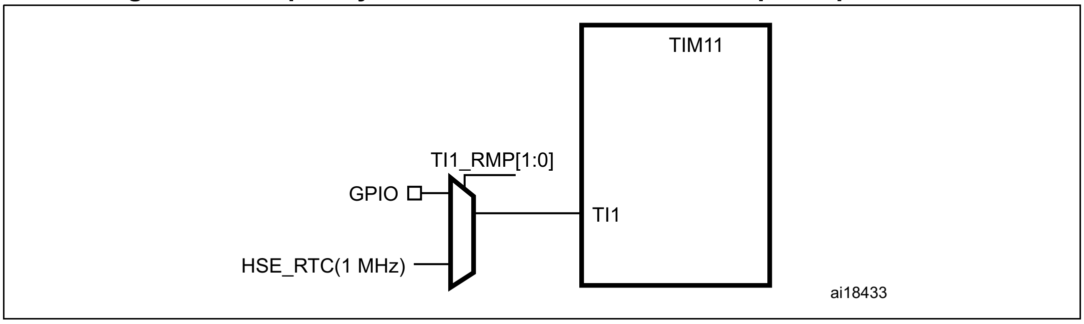
Figure 18. Frequency measurement with TIM5 in Input capture mode



Internal/external clock measurement using TIM11 channel1

TIM11 has an input multiplexer which allows choosing whether the input capture is triggered by the I/O or by an internal clock. This selection is performed through TI1_RMP [1:0] bits in the TIM11_OR register. The HSE_RTC clock (HSE divided by a programmable prescaler) is connected to channel 1 input capture to have a rough indication of the external crystal frequency. This requires that the HSI is the system clock source. This can be useful for instance to ensure compliance with the IEC 60730/IEC 61335 standards which require to be able to determine harmonic or subharmonic frequencies (–50/+100% deviations).

Figure 19. Frequency measurement with TIM11 in Input capture mode



6.3 RCC registers

Refer to *Section 1.1: List of abbreviations for registers* for a list of abbreviations used in register descriptions.

6.3.1 RCC clock control register (RCC_CR)

Address offset: 0x00

Reset value: 0x0000 XX83 where X is undefined.

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		PLLSAI RDY	PLLSAI ON	PLLI2S RDY	PLLI2S ON	PLLRDY	PLLON	Reserved				CSS ON	HSE BYP	HSE RDY	HSE ON
		r	rw	r	rw	r	rw					rw	rw	r	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Res.	HSI RDY	HSION
r	r	r	r	r	r	r	r	rw	rw	rw	rw	rw		r	rw

Bits 31:28 Reserved, must be kept at reset value.

Bit 29 **PLLSAIRDY**: PLLSAI clock ready flag
Set by hardware to indicate that the PLLSAI is locked.
0: PLLSAI unlocked
1: PLLSAI locked

Bit 28 **PLLSAION**: PLLSAI enable
Set and cleared by software to enable PLLSAI.
Cleared by hardware when entering Stop or Standby mode.
0: PLLSAI OFF
1: PLLSAI ON

Bit 27 **PLLI2SRDY**: PLLI2S clock ready flag
Set by hardware to indicate that the PLLI2S is locked.
0: PLLI2S unlocked
1: PLLI2S locked

Bit 26 **PLLI2SON**: PLLI2S enable
Set and cleared by software to enable PLLI2S.
Cleared by hardware when entering Stop or Standby mode.
0: PLLI2S OFF
1: PLLI2S ON

Bit 25 **PLLRDY**: Main PLL (PLL) clock ready flag
Set by hardware to indicate that PLL is locked.
0: PLL unlocked
1: PLL locked

Bit 24 **PLLON**: Main PLL (PLL) enable
Set and cleared by software to enable PLL.
Cleared by hardware when entering Stop or Standby mode. This bit cannot be reset if PLL clock is used as the system clock.
0: PLL OFF
1: PLL ON

Bits 23:20 Reserved, must be kept at reset value.

Bit 19 **CSSON**: Clock security system enable

Set and cleared by software to enable the clock security system. When CSSON is set, the clock detector is enabled by hardware when the HSE oscillator is ready, and disabled by hardware if an oscillator failure is detected.

0: Clock security system OFF (Clock detector OFF)

1: Clock security system ON (Clock detector ON if HSE oscillator is stable, OFF if not)

Bit 18 **HSEBYP**: HSE clock bypass

Set and cleared by software to bypass the oscillator with an external clock. The external clock must be enabled with the HSEON bit, to be used by the device.

The HSEBYP bit can be written only if the HSE oscillator is disabled.

0: HSE oscillator not bypassed

1: HSE oscillator bypassed with an external clock

Bit 17 **HSERDY**: HSE clock ready flag

Set by hardware to indicate that the HSE oscillator is stable. After the HSEON bit is cleared, HSERDY goes low after 6 HSE oscillator clock cycles.

0: HSE oscillator not ready

1: HSE oscillator ready

Bit 16 **HSEON**: HSE clock enable

Set and cleared by software.

Cleared by hardware to stop the HSE oscillator when entering Stop or Standby mode. This bit cannot be reset if the HSE oscillator is used directly or indirectly as the system clock.

0: HSE oscillator OFF

1: HSE oscillator ON

Bits 15:8 **HSICAL[7:0]**: Internal high-speed clock calibration

These bits are initialized automatically at startup.

Bits 7:3 **HSITRIM[4:0]**: Internal high-speed clock trimming

These bits provide an additional user-programmable trimming value that is added to the HSICAL[7:0] bits. It can be programmed to adjust to variations in voltage and temperature that influence the frequency of the internal HSI RC.

Bit 2 Reserved, must be kept at reset value.

Bit 1 **HSIRDY**: Internal high-speed clock ready flag

Set by hardware to indicate that the HSI oscillator is stable. After the HSION bit is cleared, HSIRDY goes low after 6 HSI clock cycles.

0: HSI oscillator not ready

1: HSI oscillator ready

Bit 0 **HSION**: Internal high-speed clock enable

Set and cleared by software.

Set by hardware to force the HSI oscillator ON when leaving the Stop or Standby mode or in case of a failure of the HSE oscillator used directly or indirectly as the system clock. This bit cannot be cleared if the HSI is used directly or indirectly as the system clock.

0: HSI oscillator OFF

1: HSI oscillator ON

6.3.2 RCC PLL configuration register (RCC_PLLCFGR)

Address offset: 0x04

Reset value: 0x2400 3010

Access: no wait state, word, half-word and byte access.

This register is used to configure the PLL clock outputs according to the formulas:

- $f_{(VCO\ clock)} = f_{(PLL\ clock\ input)} \times (PLLN / PLLM)$
- $f_{(PLL\ general\ clock\ output)} = f_{(VCO\ clock)} / PLLP$
- $f_{(USB\ OTG\ FS,\ SDIO,\ RNG\ clock\ output)} = f_{(VCO\ clock)} / PLLQ$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLQ3	PLLQ2	PLLQ1	PLLQ0	Reserved	PLLSRC	Reserved				PLLP1	PLLP0
				rw	rw	rw	rw		rw					rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved	PLLN									PLLM5	PLLM4	PLLM3	PLLM2	PLLM1	PLLM0
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **PLLQ:** Main PLL (PLL) division factor for USB OTG FS, SDIO and random number generator clocks

Set and cleared by software to control the frequency of USB OTG FS clock, the random number generator clock and the SDIO clock. These bits should be written only if PLL is disabled.

Caution: The USB OTG FS requires a 48 MHz clock to work correctly. The SDIO and the random number generator need a frequency lower than or equal to 48 MHz to work correctly.

USB OTG FS clock frequency = VCO frequency / PLLQ with 2 ≤PLLQ ≤15

0000: PLLQ = 0, wrong configuration

0001: PLLQ = 1, wrong configuration

0010: PLLQ = 2

0011: PLLQ = 3

0100: PLLQ = 4

...

1111: PLLQ = 15

Bit 23 Reserved, must be kept at reset value.

Bit 22 **PLLSRC:** Main PLL(PLL) and audio PLL (PLLI2S) entry clock source

Set and cleared by software to select PLL and PLLI2S clock source. This bit can be written only when PLL and PLLI2S are disabled.

0: HSI clock selected as PLL and PLLI2S clock entry

1: HSE oscillator clock selected as PLL and PLLI2S clock entry

Bits 21:18 Reserved, must be kept at reset value.

Bits 17:16 **PLLP:** Main PLL (PLL) division factor for main system clock

Set and cleared by software to control the frequency of the general PLL output clock. These bits can be written only if PLL is disabled.

Caution: The software has to set these bits correctly not to exceed 180 MHz on this domain.

PLL output clock frequency = VCO frequency / PLLP with PLLP = 2, 4, 6, or 8

00: PLLP = 2
01: PLLP = 4
10: PLLP = 6
11: PLLP = 8

Bits 14:6 **PLLN:** Main PLL (PLL) multiplication factor for VCO

Set and cleared by software to control the multiplication factor of the VCO. These bits can be written only when PLL is disabled. Only half-word and word accesses are allowed to write these bits.

Caution: The software has to set these bits correctly to ensure that the VCO output frequency is between 100 and 432 MHz.

VCO output frequency = VCO input frequency × PLLN with $50 \le PLLN \le 432$

000000000: PLLN = 0, wrong configuration
000000001: PLLN = 1, wrong configuration
...
000110010: PLLN = 50
...
001100011: PLLN = 99
001100100: PLLN = 100
...
110110000: PLLN = 432
110110001: PLLN = 433, wrong configuration
...
111111111: PLLN = 511, wrong configuration

Note: Multiplication factors ranging from 50 and 99 are possible for VCO input frequency higher than 1 MHz. However care must be taken that the minimum VCO output frequency respects the value specified above.

Bits 5:0 **PLLM:** Division factor for the main PLL (PLL) and audio PLL (PLLI2S) input clock

Set and cleared by software to divide the PLL and PLLI2S input clock before the VCO. These bits can be written only when the PLL and PLLI2S are disabled.

Caution: The software has to set these bits correctly to ensure that the VCO input frequency ranges from 1 to 2 MHz. It is recommended to select a frequency of 2 MHz to limit PLL jitter.

VCO input frequency = PLL input clock frequency / PLLM with $2 \le PLLM \le 63$

000000: PLLM = 0, wrong configuration
000001: PLLM = 1, wrong configuration
000010: PLLM = 2
000011: PLLM = 3
000100: PLLM = 4
...
111110: PLLM = 62
111111: PLLM = 63

6.3.3 RCC clock configuration register (RCC_CFGR)

Address offset: 0x08

Reset value: 0x0000 0000

Access: 0 ≤ wait state ≤ 2, word, half-word and byte access

1 or 2 wait states inserted only if the access occurs during a clock source switch.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MCO2		MCO2 PRE[2:0]			MCO1 PRE[2:0]			I2SSCR	MCO1		RTCPRE[4:0]				
rw		rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PPRE2[2:0]			PPRE1[2:0]			Reserved		HPRE[3:0]				SWS1	SWS0	SW1	SW0
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	r	r	rw	rw

Bits 31:30 **MCO2[1:0]:** Microcontroller clock output 2

Set and cleared by software. Clock source selection may generate glitches on MCO2. It is highly recommended to configure these bits only after reset before enabling the external oscillators and the PLLs.

00: System clock (SYSCLK) selected
01: PLLI2S clock selected
10: HSE oscillator clock selected
11: PLL clock selected

Bits 27:29 **MCO2PRE:** MCO2 prescaler

Set and cleared by software to configure the prescaler of the MCO2. Modification of this prescaler may generate glitches on MCO2. It is highly recommended to change this prescaler only after reset before enabling the external oscillators and the PLLs.

0xx: no division
100: division by 2
101: division by 3
110: division by 4
111: division by 5

Bits 24:26 **MCO1PRE:** MCO1 prescaler

Set and cleared by software to configure the prescaler of the MCO1. Modification of this prescaler may generate glitches on MCO1. It is highly recommended to change this prescaler only after reset before enabling the external oscillators and the PLL.

0xx: no division
100: division by 2
101: division by 3
110: division by 4
111: division by 5

Bit 23 **I2SSRC**: I2S clock selection

Set and cleared by software. This bit allows to select the I2S clock source between the PLLI2S clock and the external clock. It is highly recommended to change this bit only after reset and before enabling the I2S module.

0: PLLI2S clock used as I2S clock source
1: External clock mapped on the I2S_CKIN pin used as I2S clock source

Bits 22:21 **MCO1:** Microcontroller clock output 1

Set and cleared by software. Clock source selection may generate glitches on MCO1. It is highly recommended to configure these bits only after reset before enabling the external oscillators and PLL.

00: HSI clock selected
01: LSE oscillator selected
10: HSE oscillator clock selected
11: PLL clock selected

Bits 20:16 **RTCPRE:** HSE division factor for RTC clock

Set and cleared by software to divide the HSE clock input clock to generate a 1 MHz clock for RTC.

Caution: The software has to set these bits correctly to ensure that the clock supplied to the RTC is 1 MHz. These bits must be configured if needed before selecting the RTC clock source.

00000: no clock
00001: no clock
00010: HSE/2
00011: HSE/3
00100: HSE/4
...
11110: HSE/30
11111: HSE/31

Bits 15:13 **PPRE2:** APB high-speed prescaler (APB2)

Set and cleared by software to control APB high-speed clock division factor.

Caution: The software has to set these bits correctly not to exceed 90 MHz on this domain. The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after PPRE2 write.

0xx: AHB clock not divided
100: AHB clock divided by 2
101: AHB clock divided by 4
110: AHB clock divided by 8
111: AHB clock divided by 16

Bits 12:10 **PPRE1:** APB Low speed prescaler (APB1)

Set and cleared by software to control APB low-speed clock division factor.

Caution: The software has to set these bits correctly not to exceed 45 MHz on this domain. The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after PPRE1 write.

0xx: AHB clock not divided
100: AHB clock divided by 2
101: AHB clock divided by 4
110: AHB clock divided by 8
111: AHB clock divided by 16

Bits 9:8 Reserved, must be kept at reset value.

Bits 7:4 **HPRE:** AHB prescaler

Set and cleared by software to control AHB clock division factor.

Caution: The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after HPRE write.

Caution: The AHB clock frequency must be at least 25 MHz when the Ethernet is used.

0xxx: system clock not divided
1000: system clock divided by 2
1001: system clock divided by 4
1010: system clock divided by 8
1011: system clock divided by 16
1100: system clock divided by 64
1101: system clock divided by 128
1110: system clock divided by 256
1111: system clock divided by 512

Bits 3:2 **SWS:** System clock switch status

Set and cleared by hardware to indicate which clock source is used as the system clock.
00: HSI oscillator used as the system clock
01: HSE oscillator used as the system clock
10: PLL used as the system clock
11: not applicable

Bits 1:0 **SW:** System clock switch

Set and cleared by software to select the system clock source.
Set by hardware to force the HSI selection when leaving the Stop or Standby mode or in case of failure of the HSE oscillator used directly or indirectly as the system clock.
00: HSI oscillator selected as system clock
01: HSE oscillator selected as system clock
10: PLL selected as system clock
11: not allowed

6.3.4 RCC clock interrupt register (RCC_CIR)

Address offset: 0x0C

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CSSC	PLLSAI RDYC	PLLI2S RDYC	PLL RDYC	HSE RDYC	HSI RDYC	LSE RDYC	LSI RDYC
								w	w	w	w	w	w	w	w

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLSAI RDYIE	PLLI2S RDYIE	PLL RDYIE	HSE RDYIE	HSI RDYIE	LSE RDYIE	LSI RDYIE	CSSF	PLLSAI RDYF	PLLI2S RDYF	PLL RDYF	HSE RDYF	HSI RDYF	LSE RDYF	LSI RDYF
	rw	rw	rw	rw	rw	rw	rw	r	r	r	r	r	r	r	r

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **CSSC:** Clock security system interrupt clear
This bit is set by software to clear the CSSF flag.
0: No effect
1: Clear CSSF flag

Bit 22 **PLLSAIRDYC**: PLLSAI Ready Interrupt Clear
This bit is set by software to clear PLLSAIRDYF flag. It is reset by hardware when the PLLSAIRDYF is cleared.
0: PLLSAIRDYF not cleared
1: PLLSAIRDYF cleared

Bit 21 **PLLI2SRDYC:** PLLI2S ready interrupt clear
This bit is set by software to clear the PLLI2SRDYF flag.
0: No effect
1: PLLI2SRDYF cleared

Bit 20 **PLLRDYC:** Main PLL(PLL) ready interrupt clear
This bit is set by software to clear the PLLRDYF flag.
0: No effect
1: PLLRDYF cleared

Bit 19 **HSERDYC:** HSE ready interrupt clear
This bit is set by software to clear the HSERDYF flag.
0: No effect
1: HSERDYF cleared

Bit 18 **HSIRDYC:** HSI ready interrupt clear
This bit is set software to clear the HSIRDYF flag.
0: No effect
1: HSIRDYF cleared

Bit 17 **LSERDYC:** LSE ready interrupt clear
This bit is set by software to clear the LSERDYF flag.
0: No effect
1: LSERDYF cleared

Bit 16 **LSIRDYC:** LSI ready interrupt clear
This bit is set by software to clear the LSIRDYF flag.
0: No effect
1: LSIRDYF cleared

Bit 15 Reserved, must be kept at reset value.

Bit 14 **PLLSAIRDYIE**: PLLSAI Ready Interrupt Enable
This bit is set and reset by software to enable/disable interrupt caused by PLLSAI lock.
0: PLLSAI lock interrupt disabled
1: PLLSAI lock interrupt enabled

Bit 13 **PLLI2SRDYIE:** PLLI2S ready interrupt enable
This bit is set and cleared by software to enable/disable interrupt caused by PLLI2S lock.
0: PLLI2S lock interrupt disabled
1: PLLI2S lock interrupt enabled

Bit 12 **PLLRDYIE:** Main PLL (PLL) ready interrupt enable

This bit is set and cleared by software to enable/disable interrupt caused by PLL lock.
0: PLL lock interrupt disabled
1: PLL lock interrupt enabled

Bit 11 **HSERDYIE:** HSE ready interrupt enable

This bit is set and cleared by software to enable/disable interrupt caused by the HSE oscillator stabilization.
0: HSE ready interrupt disabled
1: HSE ready interrupt enabled

Bit 10 **HSIRDYIE:** HSI ready interrupt enable

This bit is set and cleared by software to enable/disable interrupt caused by the HSI oscillator stabilization.
0: HSI ready interrupt disabled
1: HSI ready interrupt enabled

Bit 9 **LSERDYIE:** LSE ready interrupt enable

This bit is set and cleared by software to enable/disable interrupt caused by the LSE oscillator stabilization.
0: LSE ready interrupt disabled
1: LSE ready interrupt enabled

Bit 8 **LSIRDYIE:** LSI ready interrupt enable

This bit is set and cleared by software to enable/disable interrupt caused by LSI oscillator stabilization.
0: LSI ready interrupt disabled
1: LSI ready interrupt enabled

Bit 7 **CSSF:** Clock security system interrupt flag

This bit is set by hardware when a failure is detected in the HSE oscillator.
It is cleared by software by setting the CSSC bit.
0: No clock security interrupt caused by HSE clock failure
1: Clock security interrupt caused by HSE clock failure

Bit 6 **PLLSAIRDYF**: PLLSAI Ready Interrupt flag

This bit is set by hardware when the PLLSAI is locked and PLLSAIRDYIE is set.
It is cleared by software by setting the PLLSAIRDYC bit.
0: No clock ready interrupt caused by PLLSAI lock
1: Clock ready interrupt caused by PLLSAI lock

Bit 5 **PLLI2SRDYF:** PLLI2S ready interrupt flag

This bit is set by hardware when the PLLI2S is locked and PLLI2SRDYIE is set.
It is cleared by software by setting the PLLRI2SDYC bit.
0: No clock ready interrupt caused by PLLI2S lock
1: Clock ready interrupt caused by PLLI2S lock

Bit 4 **PLLRDYF:** Main PLL (PLL) ready interrupt flag

This bit is set by hardware when PLL is locked and PLLRDYIE is set.
It is cleared by software setting the PLLRDYC bit.
0: No clock ready interrupt caused by PLL lock
1: Clock ready interrupt caused by PLL lock

Bit 3 **HSERDYF:** HSE ready interrupt flag
This bit is set by hardware when External High Speed clock becomes stable and HSERDYIE is set.
It is cleared by software by setting the HSERDYC bit.
0: No clock ready interrupt caused by the HSE oscillator
1: Clock ready interrupt caused by the HSE oscillator

Bit 2 **HSIRDYF:** HSI ready interrupt flag
This bit is set by hardware when the Internal High Speed clock becomes stable and HSIRDYIE is set.
It is cleared by software by setting the HSIRDYC bit.
0: No clock ready interrupt caused by the HSI oscillator
1: Clock ready interrupt caused by the HSI oscillator

Bit 1 **LSERDYF:** LSE ready interrupt flag
This bit is set by hardware when the External Low Speed clock becomes stable and LSERDYIE is set.
It is cleared by software by setting the LSERDYC bit.
0: No clock ready interrupt caused by the LSE oscillator
1: Clock ready interrupt caused by the LSE oscillator

Bit 0 **LSIRDYF:** LSI ready interrupt flag
This bit is set by hardware when the internal low speed clock becomes stable and LSIRDYIE is set.
It is cleared by software by setting the LSIRDYC bit.
0: No clock ready interrupt caused by the LSI oscillator
1: Clock ready interrupt caused by the LSI oscillator

6.3.5 RCC AHB1 peripheral reset register (RCC_AHB1RSTR)

Address offset: 0x10

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		OTGHS RST	Reserved			ETHMAC RST	Res.	DMA2D RST	DMA2 RST	DMA1 RST	Reserved				
		rw				rw		rw	rw	rw					
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved			CRCRST	Res.	GPIOK RST	GPIOJ RST	GPIOI RST	GPIOH RST	GPIOGG RST	GPIOF RST	GPIOE RST	GPIOD RST	GPIOC RST	GPIOB RST	GPIOA RST
			rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **OTGHSRST:** USB OTG HS module reset
This bit is set and cleared by software.
0: does not reset the USB OTG HS module
1: resets the USB OTG HS module

Bits 28:26 Reserved, must be kept at reset value.

Bit 25 **ETHMACRST:** Ethernet MAC reset
This bit is set and cleared by software.
0: does not reset Ethernet MAC
1: resets Ethernet MAC

Bit 24 Reserved, must be kept at reset value.

Bit 23 **DMA2DRST**: DMA2D reset
This bit is set and reset by software.
0: does not reset DMA2D
1: resets DMA2D

Bit 22 **DMA2RST:** DMA2 reset
This bit is set and cleared by software.
0: does not reset DMA2
1: resets DMA2

Bit 21 **DMA1RST:** DMA2 reset
This bit is set and cleared by software.
0: does not reset DMA2
1: resets DMA2

Bits 20:13 Reserved, must be kept at reset value.

Bit 12 **CRCRST:** CRC reset
This bit is set and cleared by software.
0: does not reset CRC
1: resets CRC

Bit 11 Reserved, must be kept at reset value.

Bit 10 **GPIOKRST**: IO port K reset
This bit is set and cleared by software.
0: does not reset IO port K
1: resets IO port K

Bit 9 **GPIOJRST**: IO port J reset
This bit is set and cleared by software.
0: does not reset IO port J
1: resets IO port J

Bit 8 **GPIOIRST:** IO port I reset
This bit is set and cleared by software.
0: does not reset IO port I
1: resets IO port I

Bit 7 **GPIOHRST:** IO port H reset
This bit is set and cleared by software.
0: does not reset IO port H
1: resets IO port H

Bit 6 **GPIOGRST:** IO port G reset
This bit is set and cleared by software.
0: does not reset IO port G
1: resets IO port G

Bit 5 **GPIOFRST:** IO port F reset
This bit is set and cleared by software.
0: does not reset IO port F
1: resets IO port F

Bit 4 **GPIOERST:** IO port E reset
This bit is set and cleared by software.
0: does not reset IO port E
1: resets IO port E

Bit 3 **GPIODRST:** IO port D reset
This bit is set and cleared by software.
0: does not reset IO port D
1: resets IO port D

Bit 2 **GPIOCRST:** IO port C reset
This bit is set and cleared by software.
0: does not reset IO port C
1: resets IO port C

Bit 1 **GPIOBRST:** IO port B reset
This bit is set and cleared by software.
0: does not reset IO port B
1:resets IO port B

Bit 0 **GPIOARST:** IO port A reset
This bit is set and cleared by software.
0: does not reset IO port A
1: resets IO port A

6.3.6 RCC AHB2 peripheral reset register (RCC_AHB2RSTR)

Address offset: 0x14

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								OTGFS RST	RNG RST	HASH RST	CRYP RST	Reserved			DCMI RST
								rw	rw	rw	rw				rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **OTGFSRST:** USB OTG FS module reset
Set and cleared by software.
0: does not reset the USB OTG FS module
1: resets the USB OTG FS module

Bit 6 **RNGRST:** Random number generator module reset
Set and cleared by software.
0: does not reset the random number generator module
1: resets the random number generator module

Bit 5 **HASHRST:** Hash module reset
Set and cleared by software.
0: does not reset the HASH module
1: resets the HASH module

Bit 4 **CRYPRST:** Cryptographic module reset
Set and cleared by software.
0: does not reset the cryptographic module
1: resets the cryptographic module

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMIRST:** Camera interface reset
Set and cleared by software.
0: does not reset the Camera interface
1: resets the Camera interface

6.3.7 RCC AHB3 peripheral reset register (RCC_AHB3RSTR)

Address offset: 0x18

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															FMCRST
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **FMCRST:** Flexible memory controller module reset
Set and cleared by software.
0: does not reset the FMC module
1: resets the FMC module

6.3.8 RCC APB1 peripheral reset register (RCC_APB1RSTR)

Address offset: 0x20

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UART8RST	UART7RST	DACRST	PWR RST	Reser-ved	CAN2 RST	CAN1 RST	Reser-ved	I2C3 RST	I2C2 RST	I2C1 RST	UART5 RST	UART4 RST	UART3 RST	UART2 RST	Reser-ved
rw	rw	rw	rw			rw		rw	rw	rw	rw	rw	rw	rw	
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
SPI3 RST	SPI2 RST	Reserved		WWDG RST	Reserved		TIM14 RST	TIM13 RST	TIM12 RST	TIM7 RST	TIM6 RST	TIM5 RST	TIM4 RST	TIM3 RST	TIM2 RST
rw	rw			rw			rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UART8RST**: UART8 reset
Set and cleared by software.
0: does not reset UART8
1: resets UART8

Bit 30 **UART7RST**: UART7 reset
Set and cleared by software.
0: does not reset UART7
1: resets UART7

Bit 29 **DACRST:** DAC reset
Set and cleared by software.
0: does not reset the DAC interface
1: resets the DAC interface

Bit 28 **PWRRST:** Power interface reset
Set and cleared by software.
0: does not reset the power interface
1: resets the power interface

Bit 27 Reserved, must be kept at reset value.

Bit 26 **CAN2RST:** CAN2 reset
Set and cleared by software.
0: does not reset CAN2
1: resets CAN2

Bit 25 **CAN1RST:** CAN1 reset
Set and cleared by software.
0: does not reset CAN1
1: resets CAN1

Bit 24 Reserved, must be kept at reset value.

Bit 23 **I2C3RST:** I2C3 reset
Set and cleared by software.
0: does not reset I2C3
1: resets I2C3

Bit 22 **I2C2RST:** I2C2 reset
Set and cleared by software.
0: does not reset I2C2
1: resets I2C2

Bit 21 **I2C1RST:** I2C1 reset
Set and cleared by software.
0: does not reset I2C1
1: resets I2C1

Bit 20 **UART5RST:** UART5 reset
Set and cleared by software.
0: does not reset UART5
1: resets UART5

Bit 19 **UART4RST:** USART4 reset
Set and cleared by software.
0: does not reset UART4
1: resets UART4

Bit 18 **USART3RST:** USART3 reset
Set and cleared by software.
0: does not reset USART3
1: resets USART3

Bit 17 **USART2RST:** USART2 reset
Set and cleared by software.
0: does not reset USART2
1: resets USART2

Bit 16 Reserved, must be kept at reset value.

Bit 15 **SPI3RST:** SPI3 reset
Set and cleared by software.
0: does not reset SPI3
1: resets SPI3

Bit 14 **SPI2RST:** SPI2 reset
Set and cleared by software.
0: does not reset SPI2
1: resets SPI2

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGRST:** Window watchdog reset
Set and cleared by software.
0: does not reset the window watchdog
1: resets the window watchdog

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **TIM14RST:** TIM14 reset
Set and cleared by software.
0: does not reset TIM14
1: resets TIM14

Bit 7 **TIM13RST:** TIM13 reset
Set and cleared by software.
0: does not reset TIM13
1: resets TIM13

Bit 6 **TIM12RST:** TIM12 reset
Set and cleared by software.
0: does not reset TIM12
1: resets TIM12

Bit 5 **TIM7RST:** TIM7 reset
Set and cleared by software.
0: does not reset TIM7
1: resets TIM7

Bit 4 **TIM6RST:** TIM6 reset
Set and cleared by software.
0: does not reset TIM6
1: resets TIM6

Bit 3 **TIM5RST:** TIM5 reset
Set and cleared by software.
0: does not reset TIM5
1: resets TIM5

Bit 2 **TIM4RST:** TIM4 reset
Set and cleared by software.
0: does not reset TIM4
1: resets TIM4

Bit 1 **TIM3RST:** TIM3 reset
Set and cleared by software.
0: does not reset TIM3
1: resets TIM3

Bit 0 **TIM2RST:** TIM2 reset
Set and cleared by software.
0: does not reset TIM2
1: resets TIM2

6.3.9 RCC APB2 peripheral reset register (RCC_APB2RSTR)

Address offset: 0x24

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					LTDCRST	Reserved			SAI1RST	SPI6RST	SPI5RST	Res.	TIM11RST	TIM10RST	TIM9RST
					rw				rw	rw	rw		rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser-ved	SYSCFG RST	SPI4 RST	SPI1 RST	SDIO RST	Reserved		ADC RST	Reserved		USART6 RST	USART1 RST	Reserved		TIM8 RST	TIM1 RST
	rw	rw	rw	rw			rw			rw	rw			rw	rw

Bits 31:27 Reserved, must be kept at reset value.

Bit 26 **LTDCRST**: LTDC reset
This bit is set and reset by software.
0: does not reset LCD-TFT
1: resets LCD-TFT

Bits 27:23 Reserved, must be kept at reset value.

Bit 22 **SAI1RST**: SAI1 reset
This bit is set and reset by software.
0: does not reset SAI1
1: resets SAI1

Bit 21 **SPI6RST**: SPI6 reset
This bit is set and cleared by software.
0: does not reset SPI6
1: resets SPI6

Bit 20 **SPI5RST**: SPI5 reset
This bit is set and cleared by software.
0: does not reset SPI5
1: resets SPI5

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TIM11RST:** TIM11 reset
This bit is set and cleared by software.
0: does not reset TIM11
1: resets TIM14

Bit 17 **TIM10RST:** TIM10 reset
This bit is set and cleared by software.
0: does not reset TIM10
1: resets TIM10

Bit 16 **TIM9RST:** TIM9 reset
This bit is set and cleared by software.
0: does not reset TIM9
1: resets TIM9

Bit 15 Reserved, must be kept at reset value.

Bit 14 **SYSCFGRST:** System configuration controller reset
This bit is set and cleared by software.
0: does not reset the System configuration controller
1: resets the System configuration controller

Bit 13 **SPI4RST**: SPI4 reset
This bit is set and cleared by software.
0: does not reset SPI4
1: resets SPI4

Bit 12 **SPI1RST:** SPI1 reset
This bit is set and cleared by software.
0: does not reset SPI1
1: resets SPI1

Bit 11 **SDIORST:** SDIO reset
This bit is set and cleared by software.
0: does not reset the SDIO module
1: resets the SDIO module

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **ADCRST:** ADC interface reset (common to all ADCs)
This bit is set and cleared by software.
0: does not reset the ADC interface
1: resets the ADC interface

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **USART6RST:** USART6 reset
This bit is set and cleared by software.
0: does not reset USART6
1: resets USART6

Bit 4 **USART1RST:** USART1 reset
This bit is set and cleared by software.
0: does not reset USART1
1: resets USART1

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8RST:** TIM8 reset
This bit is set and cleared by software.
0: does not reset TIM8
1: resets TIM8

Bit 0 **TIM1RST:** TIM1 reset
This bit is set and cleared by software.
0: does not reset TIM1
1: resets TIM1

6.3.10 RCC AHB1 peripheral clock register (RCC_AHB1ENR)

Address offset: 0x30

Reset value: 0x0010 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reser-ved	OTGHS ULPIEN	OTGHSEN	ETHMACPTPEN	ETHMACRXEN	ETHMACTXEN	ETHMACEN	Res.	**DMA2DEN**	DMA2EN	DMA1EN	CCMDATARAMEN	Res.	BKPSRAMEN	Reserved	
	rw	rw	rw	rw	rw	rw		rw	rw	rw			rw		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CRCEN	Res.	GPIOKEN	GPIOJEN	GPIOIEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
			rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 Reserved, must be kept at reset value.

Bit 30 **OTGHSULPIEN:** USB OTG HSULPI clock enable

This bit is set and cleared by software. It must be cleared when the OTG_HS is used in FS mode.
0: USB OTG HS ULPI clock disabled
1: USB OTG HS ULPI clock enabled

Bit 29 **OTGHSEN:** USB OTG HS clock enable

This bit is set and cleared by software.
0: USB OTG HS clock disabled
1: USB OTG HS clock enabled

Bit 28 **ETHMACPTPEN:** Ethernet PTP clock enable

This bit is set and cleared by software.
0: Ethernet PTP clock disabled
1: Ethernet PTP clock enabled

Bit 27 **ETHMACRXEN:** Ethernet Reception clock enable

This bit is set and cleared by software.
0: Ethernet Reception clock disabled
1: Ethernet Reception clock enabled

Bit 26 **ETHMACTXEN:** Ethernet Transmission clock enable

This bit is set and cleared by software.
0: Ethernet Transmission clock disabled
1: Ethernet Transmission clock enabled

Bit 25 **ETHMACEN:** Ethernet MAC clock enable

This bit is set and cleared by software.
0: Ethernet MAC clock disabled
1: Ethernet MAC clock enabled

Bit 24 Reserved, must be kept at reset value.

Bit 23 **DMA2DEN:** DMA2D clock enable

This bit is set and cleared by software.
0: DMA2D clock disabled
1: DMA2D clock enabled

Bit 22 **DMA2EN:** DMA2 clock enable
This bit is set and cleared by software.
0: DMA2 clock disabled
1: DMA2 clock enabled

Bit 21 **DMA1EN:** DMA1 clock enable
This bit is set and cleared by software.
0: DMA1 clock disabled
1: DMA1 clock enabled

Bit 20 **CCMDATARAMEN**: CCM data RAM clock enable
This bit is set and cleared by software.
0: CCM data RAM clock disabled
1: CCM data RAM clock enabled

Bit 19 Reserved, must be kept at reset value.

Bit 18 **BKPSRAMEN:** Backup SRAM interface clock enable
This bit is set and cleared by software.
0: Backup SRAM interface clock disabled
1: Backup SRAM interface clock enabled

Bits 17:13 Reserved, must be kept at reset value.

Bit 12 **CRCEN:** CRC clock enable
This bit is set and cleared by software.
0: CRC clock disabled
1: CRC clock enabled

Bit 11 Reserved, must be kept at reset value.

Bit 10 **GPIOKEN:** IO port K clock enable
This bit is set and cleared by software.
0: IO port K clock disabled
1: IO port K clock enabled

Bit 9 **GPIOJEN:** IO port J clock enable
This bit is set and cleared by software.
0: IO port J clock disabled
1: IO port J clock enabled

Bit 8 **GPIOIEN:** IO port I clock enable
This bit is set and cleared by software.
0: IO port I clock disabled
1: IO port I clock enabled

Bit 7 **GPIOHEN:** IO port H clock enable
This bit is set and cleared by software.
0: IO port H clock disabled
1: IO port H clock enabled

Bit 6 **GPIOGEN:** IO port G clock enable
This bit is set and cleared by software.
0: IO port G clock disabled
1: IO port G clock enabled

Bit 5 **GPIOFEN:** IO port F clock enable
This bit is set and cleared by software.
0: IO port F clock disabled
1: IO port F clock enabled

Bit 4 **GPIOEEN:** IO port E clock enable
This bit is set and cleared by software.
0: IO port E clock disabled
1: IO port E clock enabled

Bit 3 **GPIODEN:** IO port D clock enable
This bit is set and cleared by software.
0: IO port D clock disabled
1: IO port D clock enabled

Bit 2 **GPIOCEN:** IO port C clock enable
This bit is set and cleared by software.
0: IO port C clock disabled
1: IO port C clock enabled

Bit 1 **GPIOBEN:** IO port B clock enable
This bit is set and cleared by software.
0: IO port B clock disabled
1: IO port B clock enabled

Bit 0 **GPIOAEN:** IO port A clock enable
This bit is set and cleared by software.
0: IO port A clock disabled
1: IO port A clock enabled

6.3.11 RCC AHB2 peripheral clock enable register (RCC_AHB2ENR)

Address offset: 0x34

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved								OTGFS EN	RNG EN	HASH EN	CRYP EN	Reserved			DCMI EN
								rw	rw	rw	rw				rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **OTGFSEN:** USB OTG FS clock enable
This bit is set and cleared by software.
0: USB OTG FS clock disabled
1: USB OTG FS clock enabled

Bit 6 **RNGEN:** Random number generator clock enable
This bit is set and cleared by software.
0: Random number generator clock disabled
1: Random number generator clock enabled

Bit 5 **HASHEN:** Hash modules clock enable
This bit is set and cleared by software.
0: Hash modules clock disabled
1: Hash modules clock enabled

Bit 4 **CRYPEN:** Cryptographic modules clock enable
This bit is set and cleared by software.
0: cryptographic module clock disabled
1: cryptographic module clock enabled

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMIEN:** Camera interface enable
This bit is set and cleared by software.
0: Camera interface clock disabled
1: Camera interface clock enabled

6.3.12 RCC AHB3 peripheral clock enable register (RCC_AHB3ENR)

Address offset: 0x38

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved															FMCEN
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **FMCEN:** Flexible memory controller module clock enable
This bit is set and cleared by software.
0: FMC module clock disabled
1: FMC module clock enabled

6.3.13 RCC APB1 peripheral clock enable register (RCC_APB1ENR)

Address offset: 0x40

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UART8 EN	UART7 EN	DAC EN	PWR EN	Reser-ved	CAN2 EN	CAN1 EN	Reser-ved	I2C3 EN	I2C2 EN	I2C1 EN	UART5 EN	UART4 EN	USART 3 EN	USART 2 EN	Reser-ved
rw	rw	rw	rw		rw	rw		rw	rw	rw	rw	rw	rw	rw	
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
SPI3 EN	SPI2 EN	Reserved		WWDG EN	Reserved		TIM14 EN	TIM13 EN	TIM12 EN	TIM7 EN	TIM6 EN	TIM5 EN	TIM4 EN	TIM3 EN	TIM2 EN
rw	rw			rw			rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UART8EN**: UART8 clock enable
This bit is set and cleared by software.
0: UART8 clock disabled
1: UART8 clock enabled

Bit 30 **UART7EN**: UART7 clock enable
This bit is set and cleared by software.
0: UART7 clock disabled
1: UART7 clock enabled

Bit 29 **DACEN:** DAC interface clock enable
This bit is set and cleared by software.
0: DAC interface clock disabled
1: DAC interface clock enable

Bit 28 **PWREN:** Power interface clock enable
This bit is set and cleared by software.
0: Power interface clock disabled
1: Power interface clock enable

Bit 27 Reserved, must be kept at reset value.

Bit 26 **CAN2EN:** CAN 2 clock enable
This bit is set and cleared by software.
0: CAN 2 clock disabled
1: CAN 2 clock enabled

Bit 25 **CAN1EN:** CAN 1 clock enable
This bit is set and cleared by software.
0: CAN 1 clock disabled
1: CAN 1 clock enabled

Bit 24 Reserved, must be kept at reset value.

Bit 23 **I2C3EN:** I2C3 clock enable
This bit is set and cleared by software.
0: I2C3 clock disabled
1: I2C3 clock enabled

Bit 22 **I2C2EN:** I2C2 clock enable
This bit is set and cleared by software.
0: I2C2 clock disabled
1: I2C2 clock enabled

Bit 21 **I2C1EN:** I2C1 clock enable
This bit is set and cleared by software.
0: I2C1 clock disabled
1: I2C1 clock enabled

Bit 20 **UART5EN:** UART5 clock enable
This bit is set and cleared by software.
0: UART5 clock disabled
1: UART5 clock enabled

Bit 19 **UART4EN:** UART4 clock enable
This bit is set and cleared by software.
0: UART4 clock disabled
1: UART4 clock enabled

Bit 18 **USART3EN:** USART3 clock enable
This bit is set and cleared by software.
0: USART3 clock disabled
1: USART3 clock enabled

Bit 17 **USART2EN:** USART2 clock enable
This bit is set and cleared by software.
0: USART2 clock disabled
1: USART2 clock enabled

Bit 16 Reserved, must be kept at reset value.

Bit 15 **SPI3EN:** SPI3 clock enable
This bit is set and cleared by software.
0: SPI3 clock disabled
1: SPI3 clock enabled

Bit 14 **SPI2EN:** SPI2 clock enable
This bit is set and cleared by software.
0: SPI2 clock disabled
1: SPI2 clock enabled

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGEN:** Window watchdog clock enable
This bit is set and cleared by software.
0: Window watchdog clock disabled
1: Window watchdog clock enabled

Bit 10:9 Reserved, must be kept at reset value.

Bit 8 **TIM14EN:** TIM14 clock enable
This bit is set and cleared by software.
0: TIM14 clock disabled
1: TIM14 clock enabled

Bit 7 **TIM13EN:** TIM13 clock enable
This bit is set and cleared by software.
0: TIM13 clock disabled
1: TIM13 clock enabled

Bit 6 **TIM12EN:** TIM12 clock enable
This bit is set and cleared by software.
0: TIM12 clock disabled
1: TIM12 clock enabled

Bit 5 **TIM7EN:** TIM7 clock enable
This bit is set and cleared by software.
0: TIM7 clock disabled
1: TIM7 clock enabled

Bit 4 **TIM6EN:** TIM6 clock enable
This bit is set and cleared by software.
0: TIM6 clock disabled
1: TIM6 clock enabled

Bit 3 **TIM5EN:** TIM5 clock enable
This bit is set and cleared by software.
0: TIM5 clock disabled
1: TIM5 clock enabled

Bit 2 **TIM4EN:** TIM4 clock enable
This bit is set and cleared by software.
0: TIM4 clock disabled
1: TIM4 clock enabled

Bit 1 **TIM3EN:** TIM3 clock enable
This bit is set and cleared by software.
0: TIM3 clock disabled
1: TIM3 clock enabled

Bit 0 **TIM2EN:** TIM2 clock enable
This bit is set and cleared by software.
0: TIM2 clock disabled
1: TIM2 clock enabled

6.3.14 RCC APB2 peripheral clock enable register (RCC_APB2ENR)

Address offset: 0x44

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					LTDC EN	Reserved			SAI1EN	SPI6EN	SPI5EN	Res.	TIM11 EN	TIM10 EN	TIM9 EN
					rw				rw	rw	rw		rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser-ved	SYSCF G EN	SPI4E N	SPI1 EN	SDIO EN	ADC3 EN	ADC2 EN	ADC1 EN	Reserved		USART 6 EN	USART 1 EN	Reserved		TIM8 EN	TIM1 EN
	rw	rw	rw	rw	rw	rw	rw			rw	rw			rw	rw

Bits 31:27 Reserved, must be kept at reset value.

Bit 26 **LTDCEN:** LTDC clock enable
This bit is set and cleared by software.
0: LTDC clock disabled
1: LTDC clock enabled

Bits 27: 23 Reserved, must be kept at reset value.

Bit 22 **SAI1EN:** SAI1 clock enable
This bit is set and cleared by software.
0: SAI1 clock disabled
1: SAI1 clock enabled

Bit 21 **SPI6EN:** SPI6 clock enable
This bit is set and cleared by software.
0: SPI6 clock disabled
1: SPI6 clock enabled

Bit 20 **SPI5EN:** SPI5 clock enable
This bit is set and cleared by software.
0: SPI5 clock disabled
1: SPI5 clock enabled

Bit 18 **TIM11EN:** TIM11 clock enable
This bit is set and cleared by software.
0: TIM11 clock disabled
1: TIM11 clock enabled

Bit 17 **TIM10EN:** TIM10 clock enable
This bit is set and cleared by software.
0: TIM10 clock disabled
1: TIM10 clock enabled

Bit 16 **TIM9EN:** TIM9 clock enable
This bit is set and cleared by software.
0: TIM9 clock disabled
1: TIM9 clock enabled

Bit 15 Reserved, must be kept at reset value.

Bit 14 **SYSCFGEN:** System configuration controller clock enable
This bit is set and cleared by software.
0: System configuration controller clock disabled
1: System configuration controller clock enabled

Bit 13 **SPI4EN:** SPI4 clock enable
This bit is set and cleared by software.
0: SPI4 clock disabled
1: SPI4 clock enabled

Bit 12 **SPI1EN:** SPI1 clock enable
This bit is set and cleared by software.
0: SPI1 clock disabled
1: SPI1 clock enabled

Bit 11 **SDIOEN:** SDIO clock enable
This bit is set and cleared by software.
0: SDIO module clock disabled
1: SDIO module clock enabled

Bit 10 **ADC3EN:** ADC3 clock enable
This bit is set and cleared by software.
0: ADC3 clock disabled
1: ADC3 clock enabled

Bit 9 **ADC2EN:** ADC2 clock enable
This bit is set and cleared by software.
0: ADC2 clock disabled
1: ADC2 clock enabled

Bit 8 **ADC1EN:** ADC1 clock enable
This bit is set and cleared by software.
0: ADC1 clock disabled
1: ADC1 clock enabled

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **USART6EN:** USART6 clock enable
This bit is set and cleared by software.
0: USART6 clock disabled
1: USART6 clock enabled

Bit 4 **USART1EN:** USART1 clock enable
This bit is set and cleared by software.
0: USART1 clock disabled
1: USART1 clock enabled

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8EN:** TIM8 clock enable
This bit is set and cleared by software.
0: TIM8 clock disabled
1: TIM8 clock enabled

Bit 0 **TIM1EN:** TIM1 clock enable
This bit is set and cleared by software.
0: TIM1 clock disabled
1: TIM1 clock enabled

6.3.15 RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR)

Address offset: 0x50

Reset value: 0x7EEF 97FF

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	OTGHSULPILPEN	OTGHSLPEN	ETHPTPLPEN	ETHRXLPEN	ETHTXLPEN	ETHMACLPEN	Res.	DMA2DLPEN	DMA2LPEN	DMA1LPEN	Res.	SRAM3LPEN	BKPSRAMLPEN	SRAM2LPEN	SRAM1LPEN
	rw	rw	rw	rw	rw	rw		rw	rw	rw		rw	rw	rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
FLITFLPEN	Reserved		CRCLPEN	Res.	GPIOKLPEN	GPIOIJLPEN	GPIOILPEN	GPIOHLPEN	GPIOGLPEN	GPIOFLPEN	GPIOELPEN	GPIODLPEN	GPIOCLPEN	GPIOBLPEN	GPIOALPEN
rw			rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 Reserved, must be kept at reset value.

Bit 30 **OTGHSULPILPEN:** USB OTG HS ULPI clock enable during Sleep mode
This bit is set and cleared by software. It must be cleared when the OTG_HS is used in FS mode.
0: USB OTG HS ULPI clock disabled during Sleep mode
1: USB OTG HS ULPI clock enabled during Sleep mode

Bit 29 **OTGHSLPEN:** USB OTG HS clock enable during Sleep mode
This bit is set and cleared by software.
0: USB OTG HS clock disabled during Sleep mode
1: USB OTG HS clock enabled during Sleep mode

Bit 28 **ETHMACPTPLPEN:** Ethernet PTP clock enable during Sleep mode
This bit is set and cleared by software.
0: Ethernet PTP clock disabled during Sleep mode
1: Ethernet PTP clock enabled during Sleep mode

Bit 27 **ETHMACRXLPEN:** Ethernet reception clock enable during Sleep mode
This bit is set and cleared by software.
0: Ethernet reception clock disabled during Sleep mode
1: Ethernet reception clock enabled during Sleep mode

Bit 26 **ETHMACTXLPEN:** Ethernet transmission clock enable during Sleep mode
This bit is set and cleared by software.
0: Ethernet transmission clock disabled during sleep mode
1: Ethernet transmission clock enabled during sleep mode

Bit 25 **ETHMACLPEN:** Ethernet MAC clock enable during Sleep mode
This bit is set and cleared by software.
0: Ethernet MAC clock disabled during Sleep mode
1: Ethernet MAC clock enabled during Sleep mode

Bit 24 Reserved, must be kept at reset value.

Bit 23 **DMA2DLPEN:** DMA2D clock enable during Sleep mode
This bit is set and cleared by software.
0: DMA2D clock disabled during Sleep mode
1: DMA2D clock enabled during Sleep mode

Bit 22 **DMA2LPEN:** DMA2 clock enable during Sleep mode
This bit is set and cleared by software.
0: DMA2 clock disabled during Sleep mode
1: DMA2 clock enabled during Sleep mode

Bit 21 **DMA1LPEN:** DMA1 clock enable during Sleep mode
This bit is set and cleared by software.
0: DMA1 clock disabled during Sleep mode
1: DMA1 clock enabled during Sleep mode

Bit 20 Reserved, must be kept at reset value.

Bit 19 **SRAM3LPEN:** SRAM3 interface clock enable during Sleep mode
This bit is set and cleared by software.
0: SRAM3 interface clock disabled during Sleep mode
1: SRAM3 interface clock enabled during Sleep mode

Bit 18 **BKPSRAMLPEN:** Backup SRAM interface clock enable during Sleep mode
This bit is set and cleared by software.
0: Backup SRAM interface clock disabled during Sleep mode
1: Backup SRAM interface clock enabled during Sleep mode

Bit 17 **SRAM2LPEN:** SRAM2 interface clock enable during Sleep mode
This bit is set and cleared by software.
0: SRAM2 interface clock disabled during Sleep mode
1: SRAM2 interface clock enabled during Sleep mode

Bit 16 **SRAM1LPEN:** SRAM1 interface clock enable during Sleep mode
This bit is set and cleared by software.
0: SRAM1 interface clock disabled during Sleep mode
1: SRAM1 interface clock enabled during Sleep mode

Bit 15 **FLITFLPEN:** Flash interface clock enable during Sleep mode
This bit is set and cleared by software.
0: Flash interface clock disabled during Sleep mode
1: Flash interface clock enabled during Sleep mode

Bits 14:13 Reserved, must be kept at reset value.

Bit 12 **CRCLPEN:** CRC clock enable during Sleep mode
This bit is set and cleared by software.
0: CRC clock disabled during Sleep mode
1: CRC clock enabled during Sleep mode

Bit 11 Reserved, must be kept at reset value.

Bit 10 **GPIOKLPEN:** IO port K clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port K clock disabled during Sleep mode
1: IO port K clock enabled during Sleep mode

Bit 9 **GPIOJLPEN:** IO port J clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port J clock disabled during Sleep mode
1: IO port J clock enabled during Sleep mode

Bit 8 **GPIOILPEN:** IO port I clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port I clock disabled during Sleep mode
1: IO port I clock enabled during Sleep mode

Bit 7 **GPIOHLPEN:** IO port H clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port H clock disabled during Sleep mode
1: IO port H clock enabled during Sleep mode

Bits 6 **GPIOGLPEN:** IO port G clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port G clock disabled during Sleep mode
1: IO port G clock enabled during Sleep mode

Bit 5 **GPIOFLPEN:** IO port F clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port F clock disabled during Sleep mode
1: IO port F clock enabled during Sleep mode

Bit 4 **GPIOELPEN:** IO port E clock enable during Sleep mode
Set and cleared by software.
0: IO port E clock disabled during Sleep mode
1: IO port E clock enabled during Sleep mode

Bit 3 **GPIODLPEN:** IO port D clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port D clock disabled during Sleep mode
1: IO port D clock enabled during Sleep mode

Bit 2 **GPIOCLPEN:** IO port C clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port C clock disabled during Sleep mode
1: IO port C clock enabled during Sleep mode

Bit 1 **GPIOBLPEN:** IO port B clock enable during Sleep mode
This bit is set and cleared by software.
0: IO port B clock disabled during Sleep mode
1: IO port B clock enabled during Sleep mode

Bit 0 **GPIOALPEN:** IO port A clock enable during sleep mode
This bit is set and cleared by software.
0: IO port A clock disabled during Sleep mode
1: IO port A clock enabled during Sleep mode

6.3.16 RCC AHB2 peripheral clock enable in low power mode register (RCC_AHB2LPENR)

Address offset: 0x54

Reset value: 0x0000 00F1

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								OTGFS LPEN	RNG LPEN	HASH LPEN	CRYP LPEN	Reserved			DCMI LPEN
								rw	rw	rw	rw				rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **OTGFSLPEN:** USB OTG FS clock enable during Sleep mode
This bit is set and cleared by software.
0: USB OTG FS clock disabled during Sleep mode
1: USB OTG FS clock enabled during Sleep mode

Bit 6 **RNGLPEN:** Random number generator clock enable during Sleep mode
This bit is set and cleared by software.
0: Random number generator clock disabled during Sleep mode
1: Random number generator clock enabled during Sleep mode

Bit 5 **HASHLPEN:** Hash modules clock enable during Sleep mode
This bit is set and cleared by software.
0: Hash modules clock disabled during Sleep mode
1: Hash modules clock enabled during Sleep mode

Bit 4 **CRYPLPEN:** Cryptography modules clock enable during Sleep mode
This bit is set and cleared by software.
0: cryptography modules clock disabled during Sleep mode
1: cryptography modules clock enabled during Sleep mode

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMILPEN:** Camera interface enable during Sleep mode
This bit is set and cleared by software.
0: Camera interface clock disabled during Sleep mode
1: Camera interface clock enabled during Sleep mode

6.3.17 RCC AHB3 peripheral clock enable in low power mode register (RCC_AHB3LPENR)

Address offset: 0x58

Reset value: 0x0000 0001

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															FMC LPEN
															rw

Bits 31:1Reserved, must be kept at reset value.

Bit 0 **FMCLPEN:** Flexible memory controller module clock enable during Sleep mode
This bit is set and cleared by software.
0: FMC module clock disabled during Sleep mode
1: FMC module clock enabled during Sleep mode

6.3.18 RCC APB1 peripheral clock enable in low power mode register (RCC_APB1LPENR)

Address offset: 0x60

Reset value: 0xF6FE C9FF

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UART8 LPEN	UART7 LPEN	DAC LPEN	PWR LPEN	RESER VED	CAN2 LPEN	CAN1 LPEN	Reser-ved	I2C3 LPEN	I2C2 LPEN	I2C1 LPEN	UART5 LPEN	UART4 LPEN	USART 3 LPEN	USART 2 LPEN	Reser-ved
rw	rw	rw	rw		rw	rw		rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 LPEN	SPI2 LPEN	Reserved		WWDG LPEN	Reserved		TIM14 LPEN	TIM13 LPEN	TIM12 LPEN	TIM7 LPEN	TIM6 LPEN	TIM5 LPEN	TIM4 LPEN	TIM3 LPEN	TIM2 LPEN
rw	rw			rw			rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **UART8LPEN:** UART8 clock enable during Sleep mode
This bit is set and cleared by software.
0: UART8 clock disabled during Sleep mode
1: UART8 clock enabled during Sleep mode

Bit 30 **UART7LPEN:** UART7 clock enable during Sleep mode
This bit is set and cleared by software.
0: UART7 clock disabled during Sleep mode
1: UART7 clock enabled during Sleep mode

Bit 29 **DACLPEN:** DAC interface clock enable during Sleep mode
This bit is set and cleared by software.
0: DAC interface clock disabled during Sleep mode
1: DAC interface clock enabled during Sleep mode

Bit 28 **PWRLPEN:** Power interface clock enable during Sleep mode
This bit is set and cleared by software.
0: Power interface clock disabled during Sleep mode
1: Power interface clock enabled during Sleep mode

Bit 27 Reserved, must be kept at reset value.

Bit 26 **CAN2LPEN:** CAN 2 clock enable during Sleep mode
This bit is set and cleared by software.
0: CAN 2 clock disabled during sleep mode
1: CAN 2 clock enabled during sleep mode

Bit 25 **CAN1LPEN:** CAN 1 clock enable during Sleep mode
This bit is set and cleared by software.
0: CAN 1 clock disabled during Sleep mode
1: CAN 1 clock enabled during Sleep mode

Bit 24 Reserved, must be kept at reset value.

Bit 23 **I2C3LPEN:** I2C3 clock enable during Sleep mode
This bit is set and cleared by software.
0: I2C3 clock disabled during Sleep mode
1: I2C3 clock enabled during Sleep mode

Bit 22 **I2C2LPEN:** I2C2 clock enable during Sleep mode
This bit is set and cleared by software.
0: I2C2 clock disabled during Sleep mode
1: I2C2 clock enabled during Sleep mode

Bit 21 **I2C1LPEN:** I2C1 clock enable during Sleep mode
This bit is set and cleared by software.
0: I2C1 clock disabled during Sleep mode
1: I2C1 clock enabled during Sleep mode

Bit 20 **UART5LPEN:** UART5 clock enable during Sleep mode
This bit is set and cleared by software.
0: UART5 clock disabled during Sleep mode
1: UART5 clock enabled during Sleep mode

Bit 19 **UART4LPEN:** UART4 clock enable during Sleep mode
This bit is set and cleared by software.
0: UART4 clock disabled during Sleep mode
1: UART4 clock enabled during Sleep mode

Bit 18 **USART3LPEN:** USART3 clock enable during Sleep mode
This bit is set and cleared by software.
0: USART3 clock disabled during Sleep mode
1: USART3 clock enabled during Sleep mode

Bit 17 **USART2LPEN:** USART2 clock enable during Sleep mode
This bit is set and cleared by software.
0: USART2 clock disabled during Sleep mode
1: USART2 clock enabled during Sleep mode

Bit 16 Reserved, must be kept at reset value.

Bit 15 **SPI3LPEN:** SPI3 clock enable during Sleep mode
This bit is set and cleared by software.
0: SPI3 clock disabled during Sleep mode
1: SPI3 clock enabled during Sleep mode

Bit 14 **SPI2LPEN:** SPI2 clock enable during Sleep mode
This bit is set and cleared by software.
0: SPI2 clock disabled during Sleep mode
1: SPI2 clock enabled during Sleep mode

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGLPEN:** Window watchdog clock enable during Sleep mode
This bit is set and cleared by software.
0: Window watchdog clock disabled during sleep mode
1: Window watchdog clock enabled during sleep mode

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **TIM14LPEN:** TIM14 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM14 clock disabled during Sleep mode
1: TIM14 clock enabled during Sleep mode

Bit 7 **TIM13LPEN:** TIM13 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM13 clock disabled during Sleep mode
1: TIM13 clock enabled during Sleep mode

Bit 6 **TIM12LPEN:** TIM12 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM12 clock disabled during Sleep mode
1: TIM12 clock enabled during Sleep mode

Bit 5 **TIM7LPEN:** TIM7 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM7 clock disabled during Sleep mode
1: TIM7 clock enabled during Sleep mode

Bit 4 **TIM6LPEN:** TIM6 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM6 clock disabled during Sleep mode
1: TIM6 clock enabled during Sleep mode

Bit 3 **TIM5LPEN:** TIM5 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM5 clock disabled during Sleep mode
1: TIM5 clock enabled during Sleep mode

Bit 2 **TIM4LPEN:** TIM4 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM4 clock disabled during Sleep mode
1: TIM4 clock enabled during Sleep mode

Bit 1 **TIM3LPEN:** TIM3 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM3 clock disabled during Sleep mode
1: TIM3 clock enabled during Sleep mode

Bit 0 **TIM2LPEN:** TIM2 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM2 clock disabled during Sleep mode
1: TIM2 clock enabled during Sleep mode

6.3.19 RCC APB2 peripheral clock enabled in low power mode register (RCC_APB2LPENR)

Address offset: 0x64

Reset value: 0x0x0477 7F33

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					LTDC LPEN	Reserved			SAI1 LPEN	SPI6 LPEN	SPI5 LPEN	Reser-ved	TIM11 LPEN	TIM10 LPEN	TIM9 LPEN
					rw				rw	rw	rw		rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser-ved	SYSC FG LPEN	SPI4 LPEN	SPI1 LPEN	SDIO LPEN	ADC3 LPEN	ADC2 LPEN	ADC1 LPEN	Reserved		USART 6 LPEN	USART 1 LPEN	Reserved		TIM8 LPEN	TIM1 LPEN
	rw	rw	rw	rw	rw	rw	rw			rw	rw			rw	rw

Bits 31:27 Reserved, must be kept at reset value.

Bit 26 **LTDCLPEN:** LTDC clock enable during Sleep mode
This bit is set and cleared by software.
0: LTDC clock disabled during Sleep mode
1: LTDC clock enabled during Sleep mode

Bits 25:23 Reserved, must be kept at reset value.

Bit 22 **SAI1LPEN:** SAI1 clock enable during Sleep mode
This bit is set and cleared by software.
0: SAI1 clock disabled during Sleep mode
1: SAI1 clock enabled during Sleep mode

Bit 21 **SPI6LPEN:** SPI6 clock enable during Sleep mode
This bit is set and cleared by software.
0: SPI6 clock disabled during Sleep mode
1: SPI6 clock enabled during Sleep mode

Bit 20 **SPI5LPEN:** SPI5 clock enable during Sleep mode
This bit is set and cleared by software.
0: SPI5 clock disabled during Sleep mode
1: SPI5 clock enabled during Sleep mode

Bit 19 Reserved, must be kept at reset value.

Bit 18 **TIM11LPEN:** TIM11 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM11 clock disabled during Sleep mode
1: TIM11 clock enabled during Sleep mode

Bit 17 **TIM10LPEN:** TIM10 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM10 clock disabled during Sleep mode
1: TIM10 clock enabled during Sleep mode

Bit 16 **TIM9LPEN:** TIM9 clock enable during sleep mode
This bit is set and cleared by software.
0: TIM9 clock disabled during Sleep mode
1: TIM9 clock enabled during Sleep mode

Bit 15 Reserved, must be kept at reset value.

Bit 14 **SYSCFGLPEN:** System configuration controller clock enable during Sleep mode
This bit is set and cleared by software.
0: System configuration controller clock disabled during Sleep mode
1: System configuration controller clock enabled during Sleep mode

Bit 13 **SPI4LPEN:** SPI4 clock enable during Sleep mode
This bit is set and cleared by software.
0: SPI4 clock disabled during Sleep mode
1: SPI4 clock enabled during Sleep mode

Bit 12 **SPI1LPEN:** SPI1 clock enable during Sleep mode
This bit is set and cleared by software.
0: SPI1 clock disabled during Sleep mode
1: SPI1 clock enabled during Sleep mode

Bit 11 **SDIOLPEN:** SDIO clock enable during Sleep mode
This bit is set and cleared by software.
0: SDIO module clock disabled during Sleep mode
1: SDIO module clock enabled during Sleep mode

Bit 10 **ADC3LPEN:** ADC 3 clock enable during Sleep mode
This bit is set and cleared by software.
0: ADC 3 clock disabled during Sleep mode
1: ADC 3 clock enabled during Sleep mode

Bit 9 **ADC2LPEN:** ADC2 clock enable during Sleep mode
This bit is set and cleared by software.
0: ADC2 clock disabled during Sleep mode
1: ADC2 clock enabled during Sleep mode

Bit 8 **ADC1LPEN:** ADC1 clock enable during Sleep mode
This bit is set and cleared by software.
0: ADC1 clock disabled during Sleep mode
1: ADC1 clock enabled during Sleep mode

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **USART6LPEN:** USART6 clock enable during Sleep mode
This bit is set and cleared by software.
0: USART6 clock disabled during Sleep mode
1: USART6 clock enabled during Sleep mode

Bit 4 **USART1LPEN:** USART1 clock enable during Sleep mode
This bit is set and cleared by software.
0: USART1 clock disabled during Sleep mode
1: USART1 clock enabled during Sleep mode

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8LPEN:** TIM8 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM8 clock disabled during Sleep mode
1: TIM8 clock enabled during Sleep mode

Bit 0 **TIM1LPEN:** TIM1 clock enable during Sleep mode
This bit is set and cleared by software.
0: TIM1 clock disabled during Sleep mode
1: TIM1 clock enabled during Sleep mode

6.3.20 RCC Backup domain control register (RCC_BDCR)

Address offset: 0x70

Reset value: 0x0000 0000, reset by Backup domain reset.
Access: $0 \leq$ wait state ≤ 3, word, half-word and byte access
Wait states are inserted in case of successive accesses to this register.

The LSEON, LSEBYP, RTCSEL and RTCEN bits in the *RCC Backup domain control register (RCC_BDCR)* are in the Backup domain. As a result, after Reset, these bits are write-protected and the DBP bit in the *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx* has to be set before these can be modified. Refer to *Section 6.1.1: System reset on page 152* for further information. These bits are only reset after a Backup domain Reset (see *Section 6.1.3: Backup domain reset*). Any internal or external Reset has no effect on these bits.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															BDRST
															rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTCEN	Reserved					RTCSEL[1:0]		Reserved					LSEBYP	LSERDY	LSEON
rw						rw	rw						rw	r	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **BDRST:** Backup domain software reset
This bit is set and cleared by software.
0: Reset not activated
1: Resets the entire Backup domain

Note: The BKPSRAM is not affected by this reset, the only way of resetting the BKPSRAM is through the Flash interface when a protection level change from level 1 to level 0 is requested.

The backup domain software reset does not take effect until the DBP bit of the PWR_CR register is set.

Bit 15 **RTCEN:** RTC clock enable
This bit is set and cleared by software.
0: RTC clock disabled
1: RTC clock enabled

Bits 14:10 Reserved, must be kept at reset value.

Bits 9:8 **RTCSEL[1:0]:** RTC clock source selection

These bits are set by software to select the clock source for the RTC. Once the RTC clock source has been selected, it cannot be changed anymore unless the Backup domain is reset. The BDRST bit can be used to reset them.

00: No clock
01: LSE oscillator clock used as the RTC clock
10: LSI oscillator clock used as the RTC clock
11: HSE oscillator clock divided by a programmable prescaler (selection through the RTCPRE[4:0] bits in the RCC clock configuration register (RCC_CFGR)) used as the RTC clock

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 **LSEBYP:** External low-speed oscillator bypass

This bit is set and cleared by software to bypass the oscillator. This bit can be written only when the LSE clock is disabled.

0: LSE oscillator not bypassed
1: LSE oscillator bypassed

Bit 1 **LSERDY:** External low-speed oscillator ready

This bit is set and cleared by hardware to indicate when the external 32 kHz oscillator is stable. After the LSEON bit is cleared, LSERDY goes low after 6 external low-speed oscillator clock cycles.

0: LSE clock not ready
1: LSE clock ready

Bit 0 **LSEON:** External low-speed oscillator enable

This bit is set and cleared by software.

0: LSE clock OFF
1: LSE clock ON

6.3.21 RCC clock control & status register (RCC_CSR)

Address offset: 0x74

Reset value: 0x0E00 0000, reset by system reset, except reset flags by power reset only.

Access: $0 \leq$ wait state ≤ 3, word, half-word and byte access

Wait states are inserted in case of successive accesses to this register.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LPWR RSTF	WWDG RSTF	IWDG RSTF	SFT RSTF	POR RSTF	PIN RSTF	BORRS TF	RMVF	Reserved							
r	r	r	r	r	r	r	rt_w								

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														LSIRDY	LSION
														r	rw

Bit 31 **LPWRRSTF:** Low-power reset flag

This bit is set by hardware when a Low-power management reset occurs.
Cleared by writing to the RMVF bit.

0: No Low-power management reset occurred
1: Low-power management reset occurred

For further information on Low-power management reset, refer to *Low-power management reset*.

Bit 30 **WWDGRSTF:** Window watchdog reset flag

This bit is set by hardware when a window watchdog reset occurs.
Cleared by writing to the RMVF bit.

0: No window watchdog reset occurred
1: Window watchdog reset occurred

Bit 29 **IWDGRSTF**: Independent watchdog reset flag

This bit is set by hardware when an independent watchdog reset from V_{DD} domain occurs.
Cleared by writing to the RMVF bit.

0: No watchdog reset occurred
1: Watchdog reset occurred

Bit 28 **SFTRSTF:** Software reset flag

This bit is set by hardware when a software reset occurs.
Cleared by writing to the RMVF bit.

0: No software reset occurred
1: Software reset occurred

Bit 27 **PORRSTF:** POR/PDR reset flag

This bit is set by hardware when a POR/PDR reset occurs.
Cleared by writing to the RMVF bit.

0: No POR/PDR reset occurred
1: POR/PDR reset occurred

Bit 26 **PINRSTF:** PIN reset flag

This bit is set by hardware when a reset from the NRST pin occurs.
Cleared by writing to the RMVF bit.

0: No reset from NRST pin occurred
1: Reset from NRST pin occurred

Bit 25 **BORRSTF:** BOR reset flag

Cleared by software by writing the RMVF bit.
This bit is set by hardware when a POR/PDR or BOR reset occurs.

0: No POR/PDR or BOR reset occurred
1: POR/PDR or BOR reset occurred

Bit 24 **RMVF:** Remove reset flag

This bit is set by software to clear the reset flags.

0: No effect
1: Clear the reset flags

Bits 23:2 Reserved, must be kept at reset value.

Bit 1 **LSIRDY:** Internal low-speed oscillator ready

This bit is set and cleared by hardware to indicate when the internal RC 40 kHz oscillator is stable. After the LSION bit is cleared, LSIRDY goes low after 3 LSI clock cycles.
0: LSI RC oscillator not ready
1: LSI RC oscillator ready

Bit 0 **LSION:** Internal low-speed oscillator enable

This bit is set and cleared by software.
0: LSI RC oscillator OFF
1: LSI RC oscillator ON

6.3.22 RCC spread spectrum clock generation register (RCC_SSCGR)

Address offset: 0x80

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

The spread spectrum clock generation is available only for the main PLL.

The RCC_SSCGR register must be written either before the main PLL is enabled or after the main PLL disabled.

Note: For full details about PLL spread spectrum clock generation (SSCG) characteristics, refer to the "Electrical characteristics" section in your device datasheet.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SSCGEN	SPREADSEL	Reserved		INCSTEP											
rw	rw			rw	rw	rw		rw	rw	rw	rw	rw	rw	rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
INCSTEP			MODPER												
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **SSCGEN:** Spread spectrum modulation enable

This bit is set and cleared by software.
0: Spread spectrum modulation DISABLE. (To write after clearing CR[24]=PLLON bit)
1: Spread spectrum modulation ENABLE. (To write before setting CR[24]=PLLON bit)

Bit 30 **SPREADSEL:** Spread Select

This bit is set and cleared by software.
To write before to set CR[24]=PLLON bit.
0: Center spread
1: Down spread

Bits 29:28 Reserved, must be kept at reset value.

Bits 27:13 **INCSTEP:** Incrementation step

These bits are set and cleared by software. To write before setting CR[24]=PLLON bit.
Configuration input for modulation profile amplitude.

Bits 12:0 **MODPER:** Modulation period

These bits are set and cleared by software. To write before setting CR[24]=PLLON bit.
Configuration input for modulation profile period.

6.3.23 RCC PLLI2S configuration register (RCC_PLLI2SCFGR)

Address offset: 0x84

Reset value: 0x2400 3000

Access: no wait state, word, half-word and byte access.

This register is used to configure the PLLI2S clock outputs according to the formulas:

$$f_{(\text{VCO clock})} = f_{(\text{PLLI2S clock input})} \times (\text{PLLI2SN} / \text{PLLM})$$

$$f_{(\text{PLL I2S clock output})} = f_{(\text{VCO clock})} / \text{PLLI2SR}$$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	PLLI2SR2	PLLI2SR1	PLLI2SR0	PLLI2SQ				Reserved							
	rw	rw	rw	rw	rw	rw	rw								

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLI2SN8	PLLI2SN7	PLLI2SN6	PLLI2SN5	PLLI2SN4	PLLI2SN3	PLLI2SN2	PLLI2SN1	PLLI2SN0	Reserved					
	rw	rw	rw	rw	rw	rw	rw	rw	rw						

Bit 31 Reserved, must be kept at reset value.

Bits 30:28 **PLLI2SR:** PLLI2S division factor for I2S clocks

These bits are set and cleared by software to control the I2S clock frequency. These bits should be written only if the PLLI2S is disabled. The factor must be chosen in accordance with the prescaler values inside the I2S peripherals, to reach 0.3% error when using standard crystals and 0% error with audio crystals. For more information about I2S clock frequency and precision, refer to *Section 28.4.4: Clock generator* in the I2S chapter.

Caution: The I2Ss requires a frequency lower than or equal to 192 MHz to work correctly.

I2S clock frequency = VCO frequency / PLLR with $2 \leq PLLR \leq 7$

000: PLLR = 0, wrong configuration
001: PLLR = 1, wrong configuration
010: PLLR = 2
...
111: PLLR = 7

Bits 27:24 **PLLI2SQ**: PLLI2S division factor for SAI1 clock

These bits are set and cleared by software to control the SAI1 clock frequency.
They should be written when the PLLI2S is disabled.
SAI1 clock frequency = VCO frequency / PLLI2SQ with 2 <= PLLI2SIQ <= 15
0000: PLLI2SQ = 0, wrong configuration
0001: PLLI2SQ = 1, wrong configuration
0010: PLLI2SQ = 2
0011: PLLI2SQ = 3
0100: PLLI2SQ = 4
0101: PLLI2SQ = 5
...
1111: PLLI2SQ = 15

Bits 23:15 Reserved, must be kept at reset value.

Bits 14:6 **PLLI2SN:** PLLI2S multiplication factor for VCO

These bits are set and cleared by software to control the multiplication factor of the VCO. These bits can be written only when PLLI2S is disabled. Only half-word and word accesses are allowed to write these bits.

Caution: The software has to set these bits correctly to ensure that the VCO output frequency is between 100 and 432 MHz.

VCO output frequency = VCO input frequency × PLLI2SN with 50 ≤ PLLI2SN ≤ 432

000000000: PLLI2SN = 0, wrong configuration

000000001: PLLI2SN = 1, wrong configuration

...

000110010: PLLI2SN = 50

...

001100011: PLLI2SN = 99

001100100: PLLI2SN = 100

001100101: PLLI2SN = 101

001100110: PLLI2SN = 102

...

110110000: PLLI2SN = 432

110110001: PLLI2SN = 433, wrong configuration

...

111111111: PLLI2SN = 511, wrong configuration

Note: Multiplication factors ranging from 50 and 99 are possible for VCO input frequency higher than 1 MHz. However care must be taken that the minimum VCO output frequency respects the value specified above.

Bits 5:0 Reserved, must be kept at reset value.

6.3.24 RCC PLL configuration register (RCC_PLLSAICFGR)

Address offset: 0x88

Reset value: 0x2400 3000

Access: no wait state, word, half-word and byte access.

This register is used to configure the PLLSAI clock outputs according to the formulas:

- $f_{(VCO\ clock)} = f_{(PLLSAI\ clock\ input)} \times (PLLSAIN / PLLM)$
- $f_{(PLLSAI1\ clock\ output)} = f_{(VCO\ clock)} / PLLSAIQ$
- $f_{(PLL\ LCD\ clock\ output)} = f_{(VCO\ clock)} / PLLSAIR$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	PLLSAIR			PLLSAIQ				Reserved							
	rw	rw	rw	rw	rw	rw	rw								

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLSAIN									Reserved					
	rw	rw	rw	rw	rw	rw	rw	rw	rw						

Bit 31 Reserved, must be kept at reset value.

Bits 30:28 **PLLSAIR:** PLLSAI division factor for LCD clock

Set and reset by software to control the LCD clock frequency.
These bits should be written when the PLLSAI is disabled.
LCD clock frequency = VCO frequency / PLLSAIR with $2 \leq PLLSAIR \leq 7$
000: PLLSAIR = 0, wrong configuration
001: PLLSAIR = 1, wrong configuration
010: PLLSAIR = 2
...
111: PLLSAIR = 7

Bits 27:24 **PLLSAIQ:** PLLSAI division factor for SAI1 clock

Set and reset by software to control the frequency of SAI1 clock.
These bits should be written when the PLLSAI is disabled.
SAI1 clock frequency = VCO frequency / PLLSAIQ with $2 \leq PLLSAIQ \leq 15$
0000: PLLSAIQ = 0, wrong configuration
0001: PLLSAIQ = 1, wrong configuration
...
0010: PLLSAIQ = 2
0011: PLLSAIQ = 3
0100: PLLSAIQ = 4
0101: PLLSAIQ = 5
...
1111: PLLSAIQ = 15

Bits 23:15 Reserved, must be kept at reset value.

Bits 14:6 **PLLSAIN:** PLLSAI division factor for VCO

These bits are set and cleared by software to control the multiplication factor of the VCO. These bits can be written only when PLLSAI is disabled. Only half-word and word accesses are allowed to write these bits.

Caution: The software has to set these bits correctly to ensure that the VCO output frequency is between 100 and 432 MHz.

VCO output frequency = VCO input frequency × PLLISAIN with 50 ≤PLLISAIN ≤432

000000000: PLLISAIN = 0, wrong configuration
000000001: PLLISAIN = 1, wrong configuration
...
000110010: PLLISAIN = 50
...
001100011: PLLISAIN = 99
001100100: PLLISAIN = 100
001100101: PLLISAIN = 101
001100110: PLLISAIN = 102
...
110110000: PLLISAIN = 432
110110001: PLLISAIN = 433, wrong configuration
...
111111111: PLLISAIN = 511, wrong configuration

Note: Multiplication factors ranging from 50 and 99 are possible for VCO input frequency higher than 1 MHz. However care must be taken that the minimum VCO output frequency respects the value specified above.

Bits 5:0 Reserved, must be kept at reset value

6.3.25 RCC Dedicated Clock Configuration Register (RCC_DCKCFGR)

Address offset: 0x8C

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

The RCC_DCKCFGR register allows to configure the timer clock prescalers and the PLLSAI and PLLI2S output clock dividers for SAI1 and LTDC peripherals according to the following formula:

$$f_{(PLLSAIDIVQ\ clock\ output)} = f_{(PLLSAI\_Q)} / PLLSAIDIVQ$$
$$f_{(PLLSAIDIVR\ clock\ output)} = f_{(PLLSAI\_R)} / PLLSAIDIVR$$
$$f_{(PLLI2SDIVQ\ clock\ output)} = f_{(PLLI2S\_Q)} / PLLI2SDIVQ$$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							TIMPRE	SAI1BSRC		SAI1ASRC		Reserved		PLLSAIDIVR	
							rw	rw	rw	rw	rw			rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			PLLSAIDIVQ					Reserved			PLLS2DIVQ				
			rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bit 24 **TIMPRE**: Timers clocks prescalers selection

This bit is set and reset by software to control the clock frequency of all the timers connected to APB1 and APB2 domain.

0: If the APB prescaler (PPRE1, PPRE2 in the RCC_CFGR register) is configured to a division factor of 1, TIMxCLK = PCLKx. Otherwise, the timer clock frequencies are set to twice to the frequency of the APB domain to which the timers are connected:

TIMxCLK = 2xPCLKx.

1:If the APB prescaler (PPRE1, PPRE2 in the RCC_CFGR register) is configured to a division factor of 1, 2 or 4, TIMxCLK = HCLK. Otherwise, the timer clock frequencies are set to four times to the frequency of the APB domain to which the timers are connected: TIMxCLK = 4xPCLKx.

Bits 23:22 **SAI1BSRC**: SAI1-B clock source selection

These bits are set and cleared by software to control the SAI1-B clock frequency.
They should be written when the PLLSAI and PLLI2S are disabled.
00: SAI1-B clock frequency = f(PLLSAI_Q) / PLLSAIDIVQ
01: SAI1-B clock frequency = f(PLLI2S_Q) / PLLI2SDIVQ
10: SAI1-B clock frequency = Alternate function input frequency
11: wrong configuration

Bits 21:20 **SAI1ASRC**: SAI1-A clock source selection

These bits are set and cleared by software to control the SAI1-A clock frequency.
They should be written when the PLLSAI and PLLI2S are disabled.
00: SAI1-A clock frequency = f(PLLSAI_Q) / PLLSAIDIVQ
01: SAI1-A clock frequency = f(PLLI2S_Q) / PLLI2SDIVQ
10: SAI1-A clock frequency = Alternate function input frequency
11: wrong configuration

Bits 19: 18 Reserved, must be kept at reset value.

Bits 17:16 **PLLSAIDIVR**: division factor for LCD_CLK

These bits are set and cleared by software to control the frequency of LCD_CLK.
They should be written only if PLLSAI is disabled.
LCD_CLK frequency = f(PLLSAI_R) / PLLSAIDIVR with 2 ≤ PLLSAIDIVR ≤ 16
00: PLLSAIDIVR = /2
01: PLLSAIDIVR = /4
10: PLLSAIDIVR = /8
11: PLLSAIDIVR = /16

Bits 15: 13 Reserved, must be kept at reset value.

Bits 12:8 **PLLSAIDIVQ**: PLLSAI division factor for SAI1 clock

These bits are set and reset by software to control the SAI1 clock frequency.
They should be written only if PLLSAI is disabled.
SAI1 clock frequency = f(PLLSAI_Q) / PLLSAIDIVQ with 1 ≤PLLSAIDIVQ ≤31
00000: PLLSAIDIVQ = /1
00001: PLLSAIDIVQ = /2
00010: PLLSAIDIVQ = /3
00011: PLLSAIDIVQ = /4
00100: PLLSAIDIVQ = /5
...
11111: PLLSAIDIVQ = /32

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **PLLI2SDIVQ**: PLLI2S division factor for SAI1 clock

These bits are set and reset by software to control the SAI1 clock frequency.
They should be written only if PLLI2S is disabled.
SAI1 clock frequency = f(PLLI2S_Q) / PLLI2SDIVQ with 1 <= PLLI2SDIVQ <= 31
00000: PLLI2SDIVQ = /1
00001: PLLI2SDIVQ = /2
00010: PLLI2SDIVQ = /3
00011: PLLI2SDIVQ = /4
00100: PLLI2SDIVQ = /5
...
11111: PLLI2SDIVQ = /32

6.3.26 RCC register map

Table 34 gives the register map and reset values.

Table 34. RCC register map and reset values for STM32F42xxx and STM32F43xxx

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	RCC_CR	Reserved		PLL SAIRDY	PLL SAION	PLL I2SRDY	PLL I2SON	PLL RDY	PLL ON	Reserved				CSSON	HSEBYP	HSERDY	HSEON	HSICAL 7	HSICAL 6	HSICAL 5	HSICAL 4	HSICAL 3	HSICAL 2	HSICAL 1	HSICAL 0	HSITRIM 4	HSITRIM 3	HSITRIM 2	HSITRIM 1	HSITRIM 0	Reserved	HSIRDY	HSION
	Reset value			0	0	0	0	0	0					0	0	0	0	X	X	X	X	X	X	X	X	1	0	0	0	0	0	1	1
0x04	RCC_ PLLCFGR	Reserved				PLLQ 3	PLLQ 2	PLLQ 1	PLLQ 0	Reserved	PLLSRC	Reserved				PLLP 1	PLLP 0	Reserved	PLLN 8	PLLN 7	PLLN 6	PLLN 5	PLLN 4	PLLN 3	PLLN 2	PLLN 1	PLLN 0	PLLM 5	PLLM 4	PLLM 3	PLLM 2	PLLM 1	PLLM 0
	Reset value					0	1	0	0		0					0	0		0	1	1	0	0	0	0	0	0	0	1	0	0	0	0
0x08	RCC_CFGR	MCO2 1	MCO2 0	MCO2PRE2	MCO2PRE1	MCO2PRE0	MCO1PRE2	MCO1PRE1	MCO1PRE0	I2SSRC	MCO1 1	MCO1 0	RTCPRE 4	RTCPRE 3	RTCPRE 2	RTCPRE 1	RTCPRE 0	PPRE2 2	PPRE2 1	PPRE2 0	PPRE1 2	PPRE1 1	PPRE1 0	Reserved		HPRE 3	HPRE 2	HPRE 1	HPRE 0	SWS 1	SWS 0	SW 1	SW 0
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0
0x0C	RCC_CIR	Reserved								CSSC	PLLSAIRDYC	PLLI2SRDYC	PLLRDYC	HSERDYC	HSIRDYC	LSERDYC	LSIRDYC	Reserved	PLLSAIRDYIE	PLLI2SRDYIE	PLLRDYIE	HSERDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE	CSSF	PLLSAIRDYF	PLLI2SRDYF	PLLRDYF	HSERDYF	HSIRDYF	LSERDYF	LSIRDYF
	Reset value									0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	RCC_ AHB1RSTR	Reserved		OTGHSRST	Reserved			ETHMACRST	Reserved	DMA2DRST	DMA2RST	DMA1RST	Reserved								CRCRST	Reserved	GPIOKRST	GPIOJRST	GPIOIRST	GPIOHRST	GPIOGRST	GPIOFRST	GPIOERST	GPIODRST	GPIOCRST	GPIOBRST	GPIOARST
	Reset value			0				0		0	0	0									0		0	0	0	0	0	0	0	0	0	0	0
0x14	RCC_ AHB2RSTR	Reserved																								OTGFSRS	RNGRST	HSAHRST	CRYPRST	Reserved			DCMIRST
	Reset value																									0	0	0	0				0
0x18	RCC_ AHB3RSTR	Reserved																															FMCRS
	Reset value																																0
0x1C	Reserved	Reserved																															
0x20	RCC_ APB1RSTR	UART8RST	UART7RST	DACRST	PWRRST	Reserved	CAN2RST	CAN1RST	Reserved	I2C3RST	I2C2RST	I2C1RST	UART5RST	UART4RST	UART3RST	UART2RST	Reserved	SPI3RST	SPI2RST	Reserved		WWDGRST	Reserved		TIM14RST	TIM13RST	TIM12RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST
	Reset value	0	0	0	0		0	0		0	0	0	0	0	0	0		0	0			0			0	0	0	0	0	0	0	0	0
0x24	RCC_ APB2RSTR	Reserved					LTDCRST	Reserved			SAI1RST	SPI6RST	SPI5RST	Reserved	TIM11RST	TIM10RST	TIM9RST	Reserved	SYSCFGRST	SP45RST	SPI1RST	SDIORST	Reserved		ADCRST	Reserved		USART6RST	USART1RST	Reserved		TIM8RST	TIM1RST
	Reset value						0				0	0	0		0	0	0		0	0	0	0			0			0	0			0	0
0x28	Reserved	Reserved																															
0x2C	Reserved	Reserved																															
0x30	RCC_ AHB1ENR	Reserved	OTGHSULPIEN	OTGHSEN	ETHMACPTPEN	ETHMACRXEN	ETHMACTXEN	ETHMACEN	Reserved	DMA2DEN	DMA2EN	DMA1EN	CCMDATARAMEN	Reserved	BKPSRAMEN	Reserved					CRCEN	Reserved	GPIOKEN	GPIOJEN	GPIOIEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
	Reset value		0	0	0	0	0	0		0	0	0	1		0						0		0	0	0	0	0	0	0	0	0	0	0

Table 34. RCC register map and reset values for STM32F42xxx and STM32F43xxx (continued)

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x34	RCC_ AHB2ENR	Reserved																								OTGFSEN	RNGEN	HASHEN	CRYPEN	Reserved			DCMIEN
	Reset value																									0	0	0	0				0
0x38	RCC_ AHB3ENR	Reserved																															FMCEN
	Reset value																																0
0x3C	Reserved	Reserved																															
0x40	RCC_ APB1ENR	UART8EN	UART7EN	DACEN	PWREN	Reserved	CAN2EN	CAN1EN	Reserved	I2C3EN	I2C2EN	I2C1EN	UART5EN	UART4EN	USART3EN	USART2EN	Reserved	SPI3EN	SPI2EN	Reserved		WWDGEN	Reserved		TIM14EN	TIM13EN	TIM12EN	TIM7EN	TIM6EN	TIM5EN	TIM4EN	TIM3EN	TIM2EN
	Reset value	0	0	0	0		0	0		0	0	0	0	0	0	0		0	0			0			0	0	0	0	0	0	0	0	0
0x44	RCC_ APB2ENR	Reserved					LTDCEN	Reserved			SAI1EN	SPI6EN	SPI5EN	Reserved	TIM11EN	TIM10EN	TIM9EN	Reserved	SYSCFGEN	SPI4EN	SPI1EN	SDIOEN	ADC3EN	ADC2EN	ADC1EN	Reserved		USART6EN	USART1EN	Reserved		TIM8EN	TIM1EN
	Reset value						0				0	0	0		0	0	0		0	0	0	0	0	0	0			0	0			0	0
0x48	Reserved	Reserved																															
0x4C	Reserved	Reserved																															
0x50	RCC_ AHB1LPENR	Reserved	OTGHSULPILPEN	OTGHSLPEN	ETHMACPTPLPEN	ETHMACRXLPEN	ETHMACTXLPEN	ETHMACLPEN	Reserved	DMA2DLPEN	DMA2LPEN	DMA1LPEN	Reserved	SRAM3LPEN	BKPSRAMLPEN	SRAM2LPEN	SRAM1LPEN	FLITFLPEN	Reserved		CRCLPEN	Reserved	GPIOKLPEN	GPIOJLPEN	GPIOILPEN	GPIOHLPEN	GPIOGLPEN	GPIOFLPEN	GPIOELPEN	GPIODLPEN	GPIOCLPEN	GPIOBLPEN	GPIOALPEN
	Reset value		1	1	1	1	1	1		1	1	1		1	1	1	1	1			1		1	1	1	1	1	1	1	1	1	1	1
0x54	RCC_ AHB2LPENR	Reserved																								OTGFSLPEN	RNGLPEN	HASHLPEN	CRYPLPEN	Reserved			DCMILPEN
	Reset value																									1	1	1	1				1
0x58	RCC_ AHB3LPENR	Reserved																															FMCLPEN
	Reset value																																1
0x5C	Reserved	Reserved																															
0x60	RCC_ APB1LPENR	UART8LPEN	UART7LPEN	DACLPEN	PWRLPEN	Reserved	CAN2LPEN	CAN1LPEN	Reserved	I2C3LPEN	I2C2LPEN	I2C1LPEN	UART5LPEN	UART4LPEN	USART3LPEN	USART2LPEN	Reserved	SPI3LPEN	SPI2LPEN	Reserved		WWDGLPEN	Reserved		TIM14LPEN	TIM13LPEN	TIM12LPEN	TIM7LPEN	TIM6LPEN	TIM5LPEN	TIM4LPEN	TIM3LPEN	TIM2LPEN
	Reset value	1	1	1	1		1	1		1	1	1	1	1	1	1		1	1			1			1	1	1	1	1	1	1	1	1
0x64	RCC_ APB2LPENR	Reserved					LTDCLPEN	Reserved			SAI1LPEN	SPI6LPEN	SPI5LPEN	Reserved	TIM11LPEN	TIM10LPEN	TIM9LPEN	Reserved	SYSCFGLPEN	SPI4LPEN	SPI1LPEN	SDIOLPEN	ADC3LPEN	ADC2LPEN	ADC1LPEN	Reserved		USART6LPEN	USART1LPEN	Reserved		TIM8LPEN	TIM1LPEN
	Reset value						1				1	0	1		1	0	1		1	0	1	1	1	1	1			1	1			1	1
0x68	Reserved	Reserved																															
0x6C	Reserved	Reserved																															
0x70	RCC_BDCR	Reserved															BDRST	RTCEN	Reserved					RTCSEL 1	RTCSEL 0	Reserved					LSEBYP	LSERDY	LSEON
	Reset value																0	0						0	0						0	0	0

Table 34. RCC register map and reset values for STM32F42xxx and STM32F43xxx (continued)

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x74	RCC_CSR	LPWRRSTF	WWDGRSTF	WDGRSTF	SFTRSTF	PORRSTF	PADRSTF	BORRSTF	RMVF	Reserved																						LSIRDY	LSION
	Reset value	0	0	0	0	1	1	1	1																							0	0
0x78	Reserved	Reserved																															
0x7C	Reserved	Reserved																															
0x80	RCC_SSCGR	SSCGEN	SPREADSEL	Reserved		INCSTEP															MODPER												
	Reset value	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x84	RCC_ PLLI2SCFGR	Reserved	PLLI2SRx			PLLI2SQ				Reserved									PLLI2SNx									Reserved					
	Reset value		0	1	0	0	1	0	0										0	1	1	0	0	0	0	0	0						
0x88	RCC_ PLLSAICFGR	Reserved	PLLSAIR			PLLSAIQ				Reserved									PLLSAIN									Reserved					
	Reset value		0	1	0	0	1	0	0										0	1	1	0	0	0	0	0	0						
0x8C	RCC_ DCKCFGR	Reserved							TIMPRE	SAI1BSCR		SAI1ASCR		Reserved		PLLSAIDIVR		Reserved			0	0	0	0	0	Reserved			PLLI2SDIVQ				
	Reset value								0	0	0	0	0			0	0				0	0	0	0	0				0	0	0	0	0

Refer to *Section 2.3: Memory map* for the register boundary addresses.

7 Reset and clock control for STM32F405xx/07xx and STM32F415xx/17xx(RCC)

7.1 Reset

There are three types of reset, defined as system Reset, power Reset and backup domain Reset.

7.1.1 System reset

A system reset sets all registers to their reset values unless specified otherwise in the register description (see *Figure 9*).

A system reset is generated when one of the following events occurs:

1. A low level on the NRST pin (external reset)
2. Window watchdog end of count condition (WWDG reset)
3. Independent watchdog end of count condition (IWDG reset)
4. A software reset (SW reset) (see *Software reset*)
5. Low-power management reset (see *Low-power management reset*)

Software reset

The reset source can be identified by checking the reset flags in the *RCC clock control & status register (RCC_CSR)*.

The SYSRESETREQ bit in Cortex®-M4 with FPU Application Interrupt and Reset Control Register must be set to force a software reset on the device. Refer to the Cortex®-M4 with FPU technical reference manual for more details.

Low-power management reset

There are two ways of generating a low-power management reset:

1. Reset generated when entering the Standby mode:

   This type of reset is enabled by resetting the nRST_STDBY bit in the user option bytes. In this case, whenever a Standby mode entry sequence is successfully executed, the device is reset instead of entering the Standby mode.

2. Reset when entering the Stop mode:

   This type of reset is enabled by resetting the nRST_STOP bit in the user option bytes. In this case, whenever a Stop mode entry sequence is successfully executed, the device is reset instead of entering the Stop mode.

7.1.2 Power reset

A power reset is generated when one of the following events occurs:

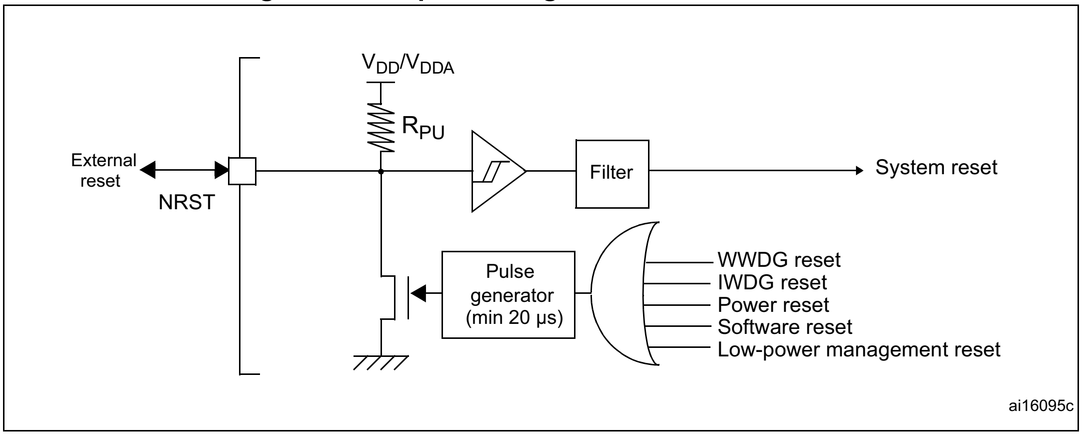
1. Power-on/power-down reset (POR/PDR reset) or brownout (BOR) reset
2. When exiting the Standby mode

A power reset sets all registers to their reset values except the Backup domain (see *Figure 9*)

These sources act on the NRST pin and it is always kept low during the delay phase. The RESET service routine vector is fixed at address 0x0000_0004 in the memory map.

The system reset signal provided to the device is output on the NRST pin. The pulse generator guarantees a minimum reset pulse duration of 20 µs for each internal reset source. In case of an external reset, the reset pulse is generated while the NRST pin is asserted low.

Figure 20. Simplified diagram of the reset circuit



The Backup domain has two specific resets that affect only the Backup domain (see *Figure 9*).

7.1.3 Backup domain reset

The backup domain reset sets all RTC registers, the RCC_BDCR register, and the BRE bit of the PWR_CSR register to their reset values. The BKPSRAM is not affected by this reset. The only way of resetting the BKPSRAM is through the Flash interface by requesting a protection level change from 1 to 0.

A backup domain reset is generated when one of the following events occurs:

1. Software reset, triggered by setting the BDRST bit in the *RCC Backup domain control register (RCC_BDCR)*.
2. V_{DD} or V_{BAT} power on, if both supplies have previously been powered off.

Note: The DBP bit of the PWR_CR register must be set to generate a backup domain reset.

7.2 Clocks

Three different clock sources can be used to drive the system clock (SYSCLK):

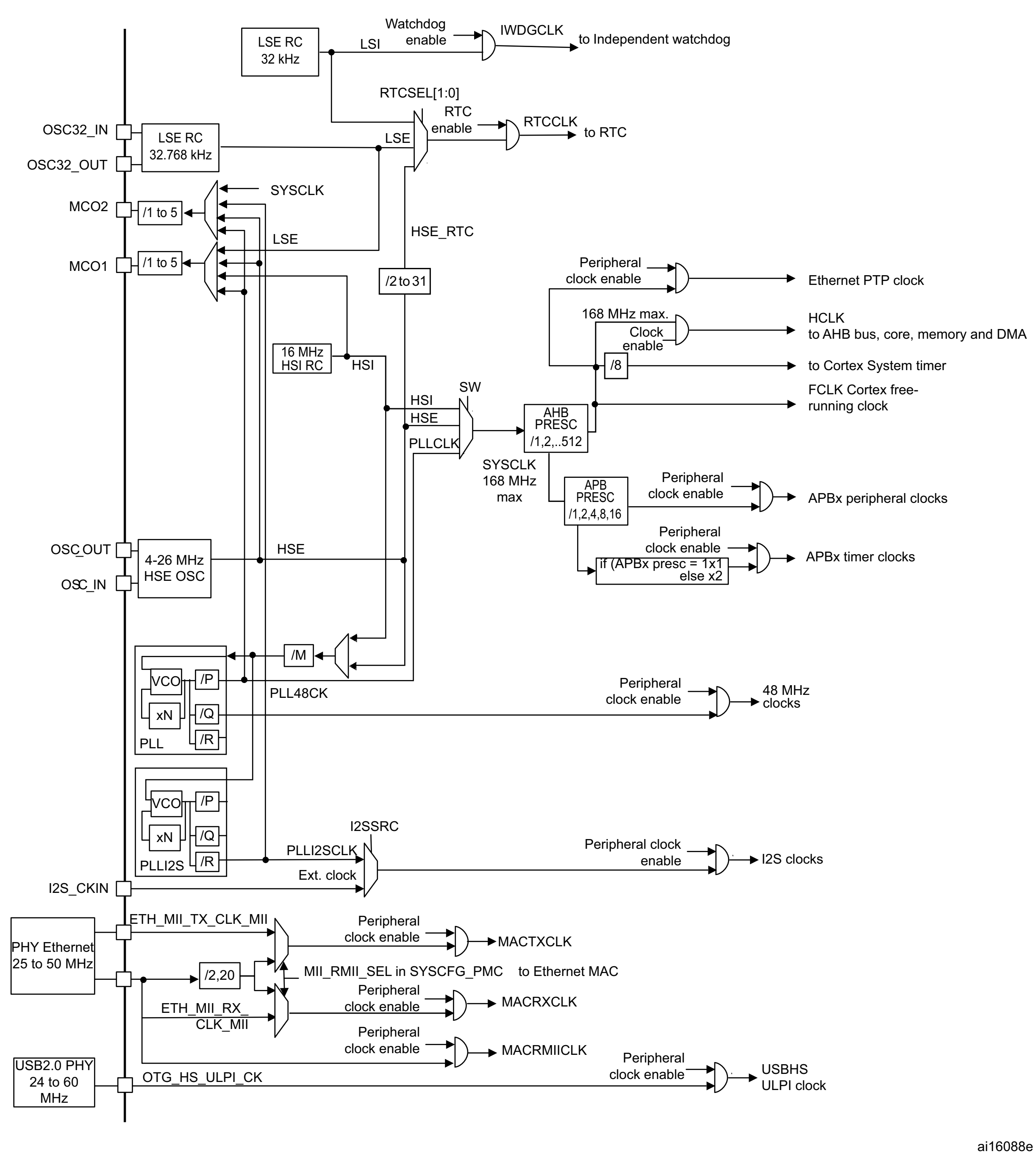
- HSI oscillator clock
- HSE oscillator clock
- Main PLL (PLL) clock

The devices have the two following secondary clock sources:

- 32 kHz low-speed internal RC (LSI RC) which drives the independent watchdog and, optionally, the RTC used for Auto-wakeup from the Stop/Standby mode.
- 32.768 kHz low-speed external crystal (LSE crystal) which optionally drives the RTC clock (RTCCLK)

Each clock source can be switched on or off independently when it is not used, to optimize power consumption.

Figure 21. Clock tree



1. For full details about the internal and external clock source characteristics, refer to the Electrical characteristics section in the device datasheet.

The clock controller provides a high degree of flexibility to the application in the choice of the external crystal or the oscillator to run the core and peripherals at the highest frequency and, guarantee the appropriate frequency for peripherals that need a specific clock like Ethernet, USB OTG FS and HS, I2S and SDIO.

Several prescalers are used to configure the AHB frequency, the high-speed APB (APB2) and the low-speed APB (APB1) domains. The maximum frequency of the AHB domain is 168 MHz. The maximum allowed frequency of the high-speed APB2 domain is 84 MHz. The maximum allowed frequency of the low-speed APB1 domain is 42 MHz

All peripheral clocks are derived from the system clock (SYSCLK) except for:

- The USB OTG FS clock (48 MHz), the random analog generator (RNG) clock (≤48 MHz) and the SDIO clock (≤48 MHz) which are coming from a specific output of PLL (PLL48CLK)
- The I2S clock

  To achieve high-quality audio performance, the I2S clock can be derived either from a specific PLL (PLLI2S) or from an external clock mapped on the I2S_CKIN pin. For more information about I2S clock frequency and precision, refer to *Section 28.4.4: Clock generator*.
- The USB OTG HS (60 MHz) clock which is provided from the external PHY
- The Ethernet MAC clocks (TX, RX and RMII) which are provided from the external PHY. For further information on the Ethernet configuration, please refer to *Section 33.4.4: MII/RMII selection* in the Ethernet peripheral description. When the Ethernet is used, the AHB clock frequency must be at least 25 MHz.

The RCC feeds the external clock of the Cortex System Timer (SysTick) with the AHB clock (HCLK) divided by 8. The SysTick can work either with this clock or with the Cortex clock (HCLK), configurable in the SysTick control and status register.

The timer clock frequencies are automatically set by hardware. There are two cases:

1. If the APB prescaler is 1, the timer clock frequencies are set to the same frequency as that of the APB domain to which the timers are connected.
2. Otherwise, they are set to twice (×2) the frequency of the APB domain to which the timers are connected.

FCLK acts as Cortex®-M4 with FPU free-running clock. For more details, refer to the Cortex®-M4 with FPU technical reference manual.

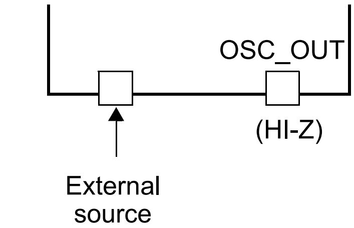
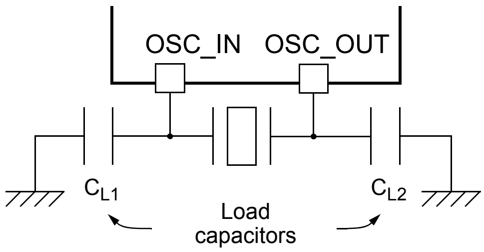
7.2.1 HSE clock

The high speed external clock signal (HSE) can be generated from two possible clock sources:

- HSE external crystal/ceramic resonator
- HSE external user clock

The resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. The loading capacitance values must be adjusted according to the selected oscillator.

Figure 22. HSE/ LSE clock sources

	Hardware configuration
External clock	
Crystal/ceramic resonators	

External source (HSE bypass)

In this mode, an external clock source must be provided. You select this mode by setting the HSEBYP and HSEON bits in the *RCC clock control register (RCC_CR)*. The external clock signal (square, sinus or triangle) with ~50% duty cycle has to drive the OSC_IN pin while the OSC_OUT pin should be left HI-Z. See *Figure 22*.

External crystal/ceramic resonator (HSE crystal)

The HSE has the advantage of producing a very accurate rate on the main clock.

The associated hardware configuration is shown in *Figure 22*. Refer to the electrical characteristics section of the *datasheet* for more details.

The HSERDY flag in the *RCC clock control register (RCC_CR)* indicates if the high-speed external oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the *RCC clock interrupt register (RCC_CIR)*.

The HSE Crystal can be switched on and off using the HSEON bit in the *RCC clock control register (RCC_CR)*.

7.2.2 HSI clock

The HSI clock signal is generated from an internal 16 MHz RC oscillator and can be used directly as a system clock, or used as PLL input.

The HSI RC oscillator has the advantage of providing a clock source at low cost (no external components). It also has a faster startup time than the HSE crystal oscillator however, even with calibration the frequency is less accurate than an external crystal oscillator or ceramic resonator.

Calibration

RC oscillator frequencies can vary from one chip to another due to manufacturing process variations, this is why each device is factory calibrated by ST for 1% accuracy at T_A= 25 °C.

After reset, the factory calibration value is loaded in the HSICAL[7:0] bits in the *RCC clock control register (RCC_CR)*.

If the application is subject to voltage or temperature variations this may affect the RC oscillator speed. You can trim the HSI frequency in the application using the HSITRIM[4:0] bits in the *RCC clock control register (RCC_CR)*.

The HSIRDY flag in the *RCC clock control register (RCC_CR)* indicates if the HSI RC is stable or not. At startup, the HSI RC output clock is not released until this bit is set by hardware.

The HSI RC can be switched on and off using the HSION bit in the *RCC clock control register (RCC_CR)*.

The HSI signal can also be used as a backup source (Auxiliary clock) if the HSE crystal oscillator fails. Refer to *Section 7.2.7: Clock security system (CSS) on page 222*.

7.2.3 PLL configuration

The STM32F4xx devices feature two PLLs:

- A main PLL (PLL) clocked by the HSE or HSI oscillator and featuring two different output clocks:
  - The first output is used to generate the high speed system clock (up to 168 MHz)
  - The second output is used to generate the clock for the USB OTG FS (48 MHz), the random analog generator (≤48 MHz) and the SDIO (≤48 MHz).
- A dedicated PLL (PLLI2S) used to generate an accurate clock to achieve high-quality audio performance on the I2S interface.

Since the main-PLL configuration parameters cannot be changed once PLL is enabled, it is recommended to configure PLL before enabling it (selection of the HSI or HSE oscillator as PLL clock source, and configuration of division factors M, N, P, and Q).

The PLLI2S uses the same input clock as PLL (PLLM[5:0] and PLLSRC bits are common to both PLLs). However, the PLLI2S has dedicated enable/disable and division factors (N and R) configuration bits. Once the PLLI2S is enabled, the configuration parameters cannot be changed.

The two PLLs are disabled by hardware when entering Stop and Standby modes, or when an HSE failure occurs when HSE or PLL (clocked by HSE) are used as system clock. *RCC PLL configuration register (RCC_PLLCFGR)* and *RCC clock configuration register (RCC_CFGR)* can be used to configure PLL and PLLI2S, respectively.

7.2.4 LSE clock

The LSE clock is generated from a 32.768 kHz low-speed external crystal or ceramic resonator. It has the advantage providing a low-power but highly accurate clock source to the real-time clock peripheral (RTC) for clock/calendar or other timing functions.

The LSE oscillator is switched on and off using the LSEON bit in *RCC Backup domain control register (RCC_BDCR)*.

The LSERDY flag in the *RCC Backup domain control register (RCC_BDCR)* indicates if the LSE crystal is stable or not. At startup, the LSE crystal output clock signal is not released until this bit is set by hardware. An interrupt can be generated if enabled in the *RCC clock interrupt register (RCC_CIR)*.

External source (LSE bypass)

In this mode, an external clock source must be provided. It must have a frequency up to 1 MHz. You select this mode by setting the LSEBYP and LSEON bits in the *RCC Backup domain control register (RCC_BDCR)*. The external clock signal (square, sinus or triangle) with ~50% duty cycle has to drive the OSC32_IN pin while the OSC32_OUT pin should be left HI-Z. See *Figure 22*.

7.2.5 LSI clock

The LSI RC acts as an low-power clock source that can be kept running in Stop and Standby mode for the independent watchdog (IWDG) and Auto-wakeup unit (AWU). The clock frequency is around 32 kHz. For more details, refer to the electrical characteristics section of the datasheets.

The LSI RC can be switched on and off using the LSION bit in the *RCC clock control & status register (RCC_CSR)*.

The LSIRDY flag in the *RCC clock control & status register (RCC_CSR)* indicates if the low-speed internal oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the *RCC clock interrupt register (RCC_CIR)*.

7.2.6 System clock (SYSCLK) selection

After a system reset, the HSI oscillator is selected as the system clock. When a clock source is used directly or through PLL as the system clock, it is not possible to stop it.

A switch from one clock source to another occurs only if the target clock source is ready (clock stable after startup delay or PLL locked). If a clock source that is not yet ready is selected, the switch occurs when the clock source is ready. Status bits in the *RCC clock control register (RCC_CR)* indicate which clock(s) is (are) ready and which clock is currently used as the system clock.

7.2.7 Clock security system (CSS)

The clock security system can be activated by software. In this case, the clock detector is enabled after the HSE oscillator startup delay, and disabled when this oscillator is stopped.

If a failure is detected on the HSE clock, this oscillator is automatically disabled, a clock failure event is sent to the break inputs of advanced-control timers TIM1 and TIM8, and an interrupt is generated to inform the software about the failure (clock security system interrupt CSSI), allowing the MCU to perform rescue operations. The CSSI is linked to the Cortex®-M4 with FPU NMI (non-maskable interrupt) exception vector.

Note: When the CSS is enabled, if the HSE clock happens to fail, the CSS generates an interrupt, which causes the automatic generation of an NMI. The NMI is executed indefinitely unless the CSS interrupt pending bit is cleared. As a consequence, the application has to clear the CSS interrupt in the NMI ISR by setting the CSSC bit in the Clock interrupt register (RCC_CIR).

If the HSE oscillator is used directly or indirectly as the system clock (indirectly meaning that it is directly used as PLL input clock, and that PLL clock is the system clock) and a failure is detected, then the system clock switches to the HSI oscillator and the HSE oscillator is disabled.

If the HSE oscillator clock was the clock source of PLL used as the system clock when the failure occurred, PLL is also disabled. In this case, if the PLLI2S was enabled, it is also disabled when the HSE fails.

7.2.8 RTC/AWU clock

Once the RTCCLK clock source has been selected, the only possible way of modifying the selection is to reset the power domain.

The RTCCLK clock source can be either the HSE 1 MHz (HSE divided by a programmable prescaler), the LSE or the LSI clock. This is selected by programming the RTCSEL[1:0] bits in the *RCC Backup domain control register (RCC_BDCR)* and the RTCPRE[4:0] bits in *RCC clock configuration register (RCC_CFGR)*. This selection cannot be modified without resetting the Backup domain.

If the LSE is selected as the RTC clock, the RTC operates normally if the backup or the system supply disappears. If the LSI is selected as the AWU clock, the AWU state is not guaranteed if the system supply disappears. If the HSE oscillator divided by a value between 2 and 31 is used as the RTC clock, the RTC state is not guaranteed if the backup or the system supply disappears.

The LSE clock is in the Backup domain, whereas the HSE and LSI clocks are not. As a consequence:

- If LSE is selected as the RTC clock:
  - The RTC continues to work even if the V_{DD} supply is switched off, provided the V_{BAT} supply is maintained.
- If LSI is selected as the Auto-wakeup unit (AWU) clock:
  - The AWU state is not guaranteed if the V_{DD} supply is powered off. Refer to *Section 7.2.5: LSI clock on page 222* for more details on LSI calibration.
- If the HSE clock is used as the RTC clock:
  - The RTC state is not guaranteed if the V_{DD} supply is powered off or if the internal voltage regulator is powered off (removing power from the 1.2 V domain).

Note: To read the RTC calendar register when the APB1 clock frequency is less than seven times the RTC clock frequency ($f_{APB1} < 7xf_{RTCLCK}$), the software must read the calendar time and date registers twice. The data are correct if the second read access to RTC_TR gives the same result than the first one. Otherwise a third read access must be performed.

7.2.9 Watchdog clock

If the independent watchdog (IWDG) is started by either hardware option or software access, the LSI oscillator is forced ON and cannot be disabled. After the LSI oscillator temporization, the clock is provided to the IWDG.

7.2.10 Clock-out capability

Two microcontroller clock output (MCO) pins are available:

- MCO1

  You can output four different clock sources onto the MCO1 pin (PA8) using the configurable prescaler (from 1 to 5):
  - HSI clock
  - LSE clock
  - HSE clock
  - PLL clock

  The desired clock source is selected using the MCO1PRE[2:0] and MCO1[1:0] bits in the *RCC clock configuration register (RCC_CFGR)*.
- MCO2

  You can output four different clock sources onto the MCO2 pin (PC9) using the configurable prescaler (from 1 to 5):
  - HSE clock
  - PLL clock
  - System clock (SYSCLK)
  - PLLI2S clock

  The desired clock source is selected using the MCO2PRE[2:0] and MCO2 bits in the *RCC clock configuration register (RCC_CFGR)*.

For the different MCO pins, the corresponding GPIO port has to be programmed in alternate function mode.

The selected clock to output onto MCO must not exceed 100 MHz (the maximum I/O speed).

7.2.11 Internal/external clock measurement using TIM5/TIM11

It is possible to indirectly measure the frequencies of all on-board clock source generators by means of the input capture of TIM5 channel4 and TIM11 channel1 as shown in *Figure 23* and *Figure 24*.

Internal/external clock measurement using TIM5 channel4

TIM5 has an input multiplexer which allows choosing whether the input capture is triggered by the I/O or by an internal clock. This selection is performed through the TI4_RMP [1:0] bits in the TIM5_OR register.

The primary purpose of having the LSE connected to the channel4 input capture is to be able to precisely measure the HSI (this requires to have the HSI used as the system clock source). The number of HSI clock counts between consecutive edges of the LSE signal provides a measurement of the internal clock period. Taking advantage of the high precision of LSE crystals (typically a few tens of ppm) we can determine the internal clock frequency with the same resolution, and trim the source to compensate for manufacturing-process and/or temperature- and voltage-related frequency deviations.

The HSI oscillator has dedicated, user-accessible calibration bits for this purpose.

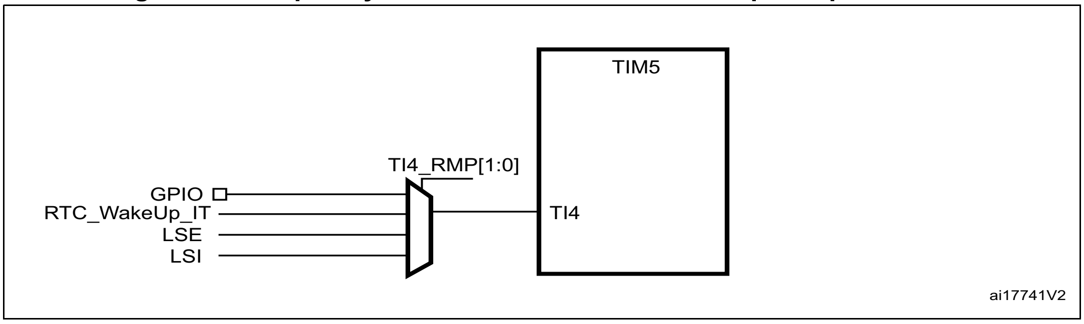
The basic concept consists in providing a relative measurement (e.g. HSI/LSE ratio): the precision is therefore tightly linked to the ratio between the two clock sources. The greater the ratio, the better the measurement.

It is also possible to measure the LSI frequency: this is useful for applications that do not have a crystal. The ultralow-power LSI oscillator has a large manufacturing process deviation: by measuring it versus the HSI clock source, it is possible to determine its frequency with the precision of the HSI. The measured value can be used to have more accurate RTC time base timeouts (when LSI is used as the RTC clock source) and/or an IWDG timeout with an acceptable accuracy.

Use the following procedure to measure the LSI frequency:

1. Enable the TIM5 timer and configure channel4 in Input capture mode.
2. Set the TI4_RMP bits in the TIM5_OR register to 0x01 to connect the LSI clock internally to TIM5 channel4 input capture for calibration purposes.
3. Measure the LSI clock frequency using the TIM5 capture/compare 4 event or interrupt.
4. Use the measured LSI frequency to update the prescaler of the RTC depending on the desired time base and/or to compute the IWDG timeout.

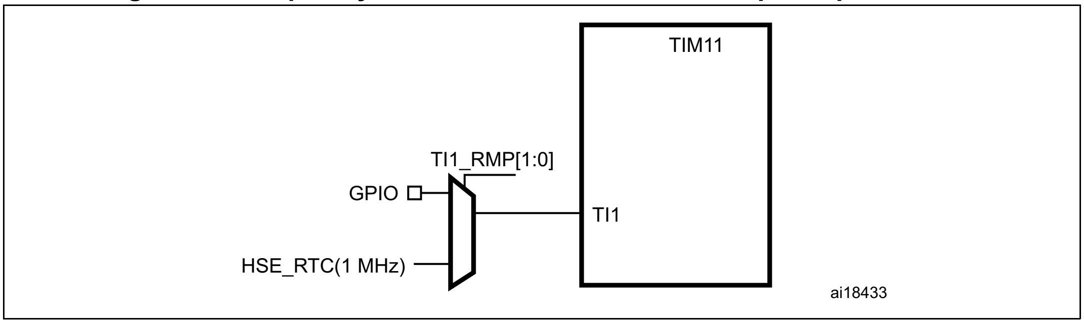
Figure 23. Frequency measurement with TIM5 in Input capture mode



Internal/external clock measurement using TIM11 channel1

TIM11 has an input multiplexer which allows choosing whether the input capture is triggered by the I/O or by an internal clock. This selection is performed through TI1_RMP [1:0] bits in the TIM11_OR register. The HSE_RTC clock (HSE divided by a programmable prescaler) is connected to channel 1 input capture to have a rough indication of the external crystal frequency. This requires that the HSI is the system clock source. This can be useful for instance to ensure compliance with the IEC 60730/IEC 61335 standards which require to be able to determine harmonic or subharmonic frequencies (–50/+100% deviations).

Figure 24. Frequency measurement with TIM11 in Input capture mode



7.3 RCC registers

Refer to *Section 1.1: List of abbreviations for registers* for a list of abbreviations used in register descriptions.

7.3.1 RCC clock control register (RCC_CR)

Address offset: 0x00

Reset value: 0x0000 XX83 where X is undefined.

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLI2S RDY	PLLI2S ON	PLLRD Y	PLLON	Reserved				CSS ON	HSE BYP	HSE RDY	HSE ON
				r	rw	r	rw					rw	rw	r	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Res.	HSI RDY	HSION
r	r	r	r	r	r	r	r	rw	rw	rw	rw	rw		r	rw

Bits 31:28 Reserved, must be kept at reset value.

Bit 27 **PLLI2SRDY**: PLLI2S clock ready flag
Set by hardware to indicate that the PLLI2S is locked.
0: PLLI2S unlocked
1: PLLI2S locked

Bit 26 **PLLI2SON**: PLLI2S enable
Set and cleared by software to enable PLLI2S.
Cleared by hardware when entering Stop or Standby mode.
0: PLLI2S OFF
1: PLLI2S ON

Bit 25 **PLLRDY**: Main PLL (PLL) clock ready flag
Set by hardware to indicate that PLL is locked.
0: PLL unlocked
1: PLL locked

Bit 24 **PLLON**: Main PLL (PLL) enable
Set and cleared by software to enable PLL.
Cleared by hardware when entering Stop or Standby mode. This bit cannot be reset if PLL clock is used as the system clock.
0: PLL OFF
1: PLL ON

Bits 23:20 Reserved, must be kept at reset value.

Bit 19 **CSSON**: Clock security system enable
Set and cleared by software to enable the clock security system. When CSSON is set, the clock detector is enabled by hardware when the HSE oscillator is ready, and disabled by hardware if an oscillator failure is detected.
0: Clock security system OFF (Clock detector OFF)
1: Clock security system ON (Clock detector ON if HSE oscillator is stable, OFF if not)

Bit 18 **HSEBYP**: HSE clock bypass

Set and cleared by software to bypass the oscillator with an external clock. The external clock must be enabled with the HSEON bit, to be used by the device.

The HSEBYP bit can be written only if the HSE oscillator is disabled.

0: HSE oscillator not bypassed

1: HSE oscillator bypassed with an external clock

Bit 17 **HSERDY**: HSE clock ready flag

Set by hardware to indicate that the HSE oscillator is stable. After the HSEON bit is cleared, HSERDY goes low after 6 HSE oscillator clock cycles.

0: HSE oscillator not ready

1: HSE oscillator ready

Bit 16 **HSEON**: HSE clock enable

Set and cleared by software.

Cleared by hardware to stop the HSE oscillator when entering Stop or Standby mode. This bit cannot be reset if the HSE oscillator is used directly or indirectly as the system clock.

0: HSE oscillator OFF

1: HSE oscillator ON

Bits 15:8 **HSICAL[7:0]**: Internal high-speed clock calibration

These bits are initialized automatically at startup.

Bits 7:3 **HSITRIM[4:0]**: Internal high-speed clock trimming

These bits provide an additional user-programmable trimming value that is added to the HSICAL[7:0] bits. It can be programmed to adjust to variations in voltage and temperature that influence the frequency of the internal HSI RC.

Bit 2 Reserved, must be kept at reset value.

Bit 1 **HSIRDY**: Internal high-speed clock ready flag

Set by hardware to indicate that the HSI oscillator is stable. After the HSION bit is cleared, HSIRDY goes low after 6 HSI clock cycles.

0: HSI oscillator not ready

1: HSI oscillator ready

Bit 0 **HSION**: Internal high-speed clock enable

Set and cleared by software.

Set by hardware to force the HSI oscillator ON when leaving the Stop or Standby mode or in case of a failure of the HSE oscillator used directly or indirectly as the system clock. This bit cannot be cleared if the HSI is used directly or indirectly as the system clock.

0: HSI oscillator OFF

1: HSI oscillator ON

7.3.2 RCC PLL configuration register (RCC_PLLCFGR)

Address offset: 0x04

Reset value: 0x2400 3010

Access: no wait state, word, half-word and byte access.

This register is used to configure the PLL clock outputs according to the formulas:

- $f_{(VCO\ clock)} = f_{(PLL\ clock\ input)} \times (PLLN / PLLM)$
- $f_{(PLL\ general\ clock\ output)} = f_{(VCO\ clock)} / PLLP$
- $f_{(USB\ OTG\ FS,\ SDIO,\ RNG\ clock\ output)} = f_{(VCO\ clock)} / PLLQ$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLQ3	PLLQ2	PLLQ1	PLLQ0	Reserved	PLLSRC	Reserved				PLLP1	PLLP0
				rw	rw	rw	rw		rw					rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLN									PLLM5	PLLM4	PLLM3	PLLM2	PLLM1	PLLM0
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:24 **PLLQ:** Main PLL (PLL) division factor for USB OTG FS, SDIO and random number generator clocks

Set and cleared by software to control the frequency of USB OTG FS clock, the random number generator clock and the SDIO clock. These bits should be written only if PLL is disabled.

Caution: The USB OTG FS requires a 48 MHz clock to work correctly. The SDIO and the random number generator need a frequency lower than or equal to 48 MHz to work correctly.

USB OTG FS clock frequency = VCO frequency / PLLQ with 2 ≤PLLQ ≤15

0000: PLLQ = 0, wrong configuration
0001: PLLQ = 1, wrong configuration
0010: PLLQ = 2
0011: PLLQ = 3
0100: PLLQ = 4
...
1111: PLLQ = 15

Bit 23 Reserved, must be kept at reset value.

Bit 22 **PLLSRC:** Main PLL(PLL) and audio PLL (PLLI2S) entry clock source

Set and cleared by software to select PLL and PLLI2S clock source. This bit can be written only when PLL and PLLI2S are disabled.
0: HSI clock selected as PLL and PLLI2S clock entry
1: HSE oscillator clock selected as PLL and PLLI2S clock entry

Bits 21:18 Reserved, must be kept at reset value.

Bits 17:16 **PLLP:** Main PLL (PLL) division factor for main system clock

Set and cleared by software to control the frequency of the general PLL output clock. These bits can be written only if PLL is disabled.

Caution: The software has to set these bits correctly not to exceed 168 MHz on this domain.

PLL output clock frequency = VCO frequency / PLLP with PLLP = 2, 4, 6, or 8

00: PLLP = 2
01: PLLP = 4
10: PLLP = 6
11: PLLP = 8

Bits 14:6 **PLLN:** Main PLL (PLL) multiplication factor for VCO

Set and cleared by software to control the multiplication factor of the VCO. These bits can be written only when PLL is disabled. Only half-word and word accesses are allowed to write these bits.

Caution: The software has to set these bits correctly to ensure that the VCO output frequency is between 100 and 432 MHz.

VCO output frequency = VCO input frequency × PLLN with $50 \leq PLLN \leq 432$

000000000: PLLN = 0, wrong configuration
000000001: PLLN = 1, wrong configuration
...
000110010: PLLN = 50
...
001100011: PLLN = 99
001100100: PLLN = 100
...
110110000: PLLN = 432
110110001: PLLN = 433, wrong configuration
...
111111111: PLLN = 511, wrong configuration

Note: Multiplication factors ranging from 50 and 99 are possible for VCO input frequency higher than 1 MHz. However care must be taken that the minimum VCO output frequency respects the value specified above.

Bits 5:0 **PLLM:** Division factor for the main PLL (PLL) and audio PLL (PLLI2S) input clock

Set and cleared by software to divide the PLL and PLLI2S input clock before the VCO. These bits can be written only when the PLL and PLLI2S are disabled.

Caution: The software has to set these bits correctly to ensure that the VCO input frequency ranges from 1 to 2 MHz. It is recommended to select a frequency of 2 MHz to limit PLL jitter.

VCO input frequency = PLL input clock frequency / PLLM with $2 \leq PLLM \leq 63$

000000: PLLM = 0, wrong configuration
000001: PLLM = 1, wrong configuration
000010: PLLM = 2
000011: PLLM = 3
000100: PLLM = 4
...
111110: PLLM = 62
111111: PLLM = 63

7.3.3 RCC clock configuration register (RCC_CFGR)

Address offset: 0x08

Reset value: 0x0000 0000

Access: 0 ≤ wait state ≤ 2, word, half-word and byte access

1 or 2 wait states inserted only if the access occurs during a clock source switch.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MCO2		MCO2 PRE[2:0]			MCO1 PRE[2:0]			I2SSCR	MCO1		RTCPRE[4:0]				
rw		rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PPRE2[2:0]			PPRE1[2:0]			Reserved		HPRE[3:0]				SWS1	SWS0	SW1	SW0
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	r	r	rw	rw

Bits 31:30 **MCO2[1:0]:** Microcontroller clock output 2

Set and cleared by software. Clock source selection may generate glitches on MCO2. It is highly recommended to configure these bits only after reset before enabling the external oscillators and the PLLs.

00: System clock (SYSCLK) selected
01: PLLI2S clock selected
10: HSE oscillator clock selected
11: PLL clock selected

Bits 27:29 **MCO2PRE:** MCO2 prescaler

Set and cleared by software to configure the prescaler of the MCO2. Modification of this prescaler may generate glitches on MCO2. It is highly recommended to change this prescaler only after reset before enabling the external oscillators and the PLLs.

0xx: no division
100: division by 2
101: division by 3
110: division by 4
111: division by 5

Bits 24:26 **MCO1PRE:** MCO1 prescaler

Set and cleared by software to configure the prescaler of the MCO1. Modification of this prescaler may generate glitches on MCO1. It is highly recommended to change this prescaler only after reset before enabling the external oscillators and the PLL.

0xx: no division
100: division by 2
101: division by 3
110: division by 4
111: division by 5

Bit 23 **I2SSRC**: I2S clock selection

Set and cleared by software. This bit allows to select the I2S clock source between the PLLI2S clock and the external clock. It is highly recommended to change this bit only after reset and before enabling the I2S module.

0: PLLI2S clock used as I2S clock source
1: External clock mapped on the I2S_CKIN pin used as I2S clock source

Bits 22:21 **MCO1:** Microcontroller clock output 1

Set and cleared by software. Clock source selection may generate glitches on MCO1. It is highly recommended to configure these bits only after reset before enabling the external oscillators and PLL.

00: HSI clock selected
01: LSE oscillator selected
10: HSE oscillator clock selected
11: PLL clock selected

Bits 20:16 **RTCPRE:** HSE division factor for RTC clock

Set and cleared by software to divide the HSE clock input clock to generate a 1 MHz clock for RTC.

Caution: The software has to set these bits correctly to ensure that the clock supplied to the RTC is 1 MHz. These bits must be configured if needed before selecting the RTC clock source.

00000: no clock
00001: no clock
00010: HSE/2
00011: HSE/3
00100: HSE/4
...
11110: HSE/30
11111: HSE/31

Bits 15:13 **PPRE2:** APB high-speed prescaler (APB2)

Set and cleared by software to control APB high-speed clock division factor.

Caution: The software has to set these bits correctly not to exceed 84 MHz on this domain. The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after PPRE2 write.

0xx: AHB clock not divided
100: AHB clock divided by 2
101: AHB clock divided by 4
110: AHB clock divided by 8
111: AHB clock divided by 16

Bits 12:10 **PPRE1:** APB Low speed prescaler (APB1)

Set and cleared by software to control APB low-speed clock division factor.

Caution: The software has to set these bits correctly not to exceed 42 MHz on this domain. The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after PPRE1 write.

0xx: AHB clock not divided
100: AHB clock divided by 2
101: AHB clock divided by 4
110: AHB clock divided by 8
111: AHB clock divided by 16

Bits 9:8 Reserved, must be kept at reset value.

Bits 7:4 **HPRE:** AHB prescaler

Set and cleared by software to control AHB clock division factor.

Caution: The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after HPRE write.

Caution: The AHB clock frequency must be at least 25 MHz when the Ethernet is used.

0xxx: system clock not divided
1000: system clock divided by 2
1001: system clock divided by 4
1010: system clock divided by 8
1011: system clock divided by 16
1100: system clock divided by 64
1101: system clock divided by 128
1110: system clock divided by 256
1111: system clock divided by 512

Bits 3:2 **SWS:** System clock switch status

Set and cleared by hardware to indicate which clock source is used as the system clock.
00: HSI oscillator used as the system clock
01: HSE oscillator used as the system clock
10: PLL used as the system clock
11: not applicable

Bits 1:0 **SW:** System clock switch

Set and cleared by software to select the system clock source.
Set by hardware to force the HSI selection when leaving the Stop or Standby mode or in case of failure of the HSE oscillator used directly or indirectly as the system clock.
00: HSI oscillator selected as system clock
01: HSE oscillator selected as system clock
10: PLL selected as system clock
11: not allowed

7.3.4 RCC clock interrupt register (RCC_CIR)

Address offset: 0x0C

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CSSC	Reserved	PLLI2S RDYC	PLL RDYC	HSE RDYC	HSI RDYC	LSE RDYC	LSI RDYC
								w		w	w	w	w	w	w

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		PLLI2S RDYIE	PLL RDYIE	HSE RDYIE	HSI RDYIE	LSE RDYIE	LSI RDYIE	CSSF	Reserved	PLLI2S RDYF	PLL RDYF	HSE RDYF	HSI RDYF	LSE RDYF	LSI RDYF
		rw	rw	rw	rw	rw	rw	r		r	r	r	r	r	r

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **CSSC:** Clock security system interrupt clear
This bit is set by software to clear the CSSF flag.
0: No effect
1: Clear CSSF flag

Bits 22 Reserved, must be kept at reset value.

Bit 21 **PLLI2SRDYC:** PLLI2S ready interrupt clear
This bit is set by software to clear the PLLI2SRDYF flag.
0: No effect
1: PLLI2SRDYF cleared

Bit 20 **PLLRDYC:** Main PLL(PLL) ready interrupt clear
This bit is set by software to clear the PLLRDYF flag.
0: No effect
1: PLLRDYF cleared

Bit 19 **HSERDYC:** HSE ready interrupt clear
This bit is set by software to clear the HSERDYF flag.
0: No effect
1: HSERDYF cleared

Bit 18 **HSIRDYC:** HSI ready interrupt clear
This bit is set software to clear the HSIRDYF flag.
0: No effect
1: HSIRDYF cleared

Bit 17 **LSERDYC:** LSE ready interrupt clear
This bit is set by software to clear the LSERDYF flag.
0: No effect
1: LSERDYF cleared

Bit 16 **LSIRDYC:** LSI ready interrupt clear
This bit is set by software to clear the LSIRDYF flag.
0: No effect
1: LSIRDYF cleared

Bits 15:12 Reserved, must be kept at reset value.

Bit 13 **PLLI2SRDYIE:** PLLI2S ready interrupt enable
Set and cleared by software to enable/disable interrupt caused by PLLI2S lock.
0: PLLI2S lock interrupt disabled
1: PLLI2S lock interrupt enabled

Bit 12 **PLLRDYIE:** Main PLL (PLL) ready interrupt enable
Set and cleared by software to enable/disable interrupt caused by PLL lock.
0: PLL lock interrupt disabled
1: PLL lock interrupt enabled

Bit 11 **HSERDYIE:** HSE ready interrupt enable
Set and cleared by software to enable/disable interrupt caused by the HSE oscillator stabilization.
0: HSE ready interrupt disabled
1: HSE ready interrupt enabled

Bit 10 **HSIRDYIE:** HSI ready interrupt enable

Set and cleared by software to enable/disable interrupt caused by the HSI oscillator stabilization.

0: HSI ready interrupt disabled

1: HSI ready interrupt enabled

Bit 9 **LSERDYIE:** LSE ready interrupt enable

Set and cleared by software to enable/disable interrupt caused by the LSE oscillator stabilization.

0: LSE ready interrupt disabled

1: LSE ready interrupt enabled

Bit 8 **LSIRDYIE:** LSI ready interrupt enable

Set and cleared by software to enable/disable interrupt caused by LSI oscillator stabilization.

0: LSI ready interrupt disabled

1: LSI ready interrupt enabled

Bit 7 **CSSF:** Clock security system interrupt flag

Set by hardware when a failure is detected in the HSE oscillator.

Cleared by software setting the CSSC bit.

0: No clock security interrupt caused by HSE clock failure

1: Clock security interrupt caused by HSE clock failure

Bits 6 Reserved, must be kept at reset value.

Bit 5 **PLLI2SRDYF:** PLLI2S ready interrupt flag

Set by hardware when the PLLI2S locks and PLLI2SRDYIE is set.

Cleared by software setting the PLLRI2SDYC bit.

0: No clock ready interrupt caused by PLLI2S lock

1: Clock ready interrupt caused by PLLI2S lock

Bit 4 **PLLRDYF:** Main PLL (PLL) ready interrupt flag

Set by hardware when PLL locks and PLLRDYIE is set.

Cleared by software setting the PLLRDYC bit.

0: No clock ready interrupt caused by PLL lock

1: Clock ready interrupt caused by PLL lock

Bit 3 **HSERDYF:** HSE ready interrupt flag

Set by hardware when External High Speed clock becomes stable and HSERDYIE is set.

Cleared by software setting the HSERDYC bit.

0: No clock ready interrupt caused by the HSE oscillator

1: Clock ready interrupt caused by the HSE oscillator

Bit 2 **HSIRDYF:** HSI ready interrupt flag

Set by hardware when the Internal High Speed clock becomes stable and HSIRDYIE is set.
Cleared by software setting the HSIRDYC bit.
0: No clock ready interrupt caused by the HSI oscillator
1: Clock ready interrupt caused by the HSI oscillator

Bit 1 **LSERDYF:** LSE ready interrupt flag

Set by hardware when the External Low Speed clock becomes stable and LSERDYIE is set.
Cleared by software setting the LSERDYC bit.
0: No clock ready interrupt caused by the LSE oscillator
1: Clock ready interrupt caused by the LSE oscillator

Bit 0 **LSIRDYF:** LSI ready interrupt flag

Set by hardware when the internal low speed clock becomes stable and LSIRDYIE is set.
Cleared by software setting the LSIRDYC bit.
0: No clock ready interrupt caused by the LSI oscillator
1: Clock ready interrupt caused by the LSI oscillator

7.3.5 RCC AHB1 peripheral reset register (RCC_AHB1RSTR)

Address offset: 0x10

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		OTGHS RST	Reserved			ETHMAC RST	Reserved		DMA2 RST	DMA1 RST	Reserved				
		rw				rw			rw	rw					

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CRCRST	Reserved			GPIOI RST	GPIOH RST	GPIOGG RST	GPIOF RST	GPIOE RST	GPIOD RST	GPIOC RST	GPIOB RST	GPIOA RST
			rw				rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **OTGHSRST:** USB OTG HS module reset

Set and cleared by software.
0: does not reset the USB OTG HS module
1: resets the USB OTG HS module

Bits 28:26 Reserved, must be kept at reset value.

Bit 25 **ETHMACRST:** Ethernet MAC reset

Set and cleared by software.
0: does not reset Ethernet MAC
1: resets Ethernet MAC

Bits 24:23 Reserved, must be kept at reset value.

Bit 22 **DMA2RST:** DMA2 reset
Set and cleared by software.
0: does not reset DMA2
1: resets DMA2

Bit 21 **DMA1RST:** DMA1 reset
Set and cleared by software.
0: does not reset DMA1
1: resets DMA1

Bits 20:13 Reserved, must be kept at reset value.

Bit 12 **CRCRST:** CRC reset
Set and cleared by software.
0: does not reset CRC
1: resets CRC

Bits 11:9 Reserved, must be kept at reset value.

Bit 8 **GPIOIRST:** IO port I reset
Set and cleared by software.
0: does not reset IO port I
1: resets IO port I

Bit 7 **GPIOHRST:** IO port H reset
Set and cleared by software.
0: does not reset IO port H
1: resets IO port H

Bits 6 **GPIOGRST:** IO port G reset
Set and cleared by software.
0: does not reset IO port G
1: resets IO port G

Bit 5 **GPIOFRST:** IO port F reset
Set and cleared by software.
0: does not reset IO port F
1: resets IO port F

Bit 4 **GPIOERST:** IO port E reset
Set and cleared by software.
0: does not reset IO port E
1: resets IO port E

Bit 3 **GPIODRST:** IO port D reset
Set and cleared by software.
0: does not reset IO port D
1: resets IO port D

Bit 2 **GPIOCRST:** IO port C reset
Set and cleared by software.
0: does not reset IO port C
1: resets IO port C

Bit 1 **GPIOBRST:** IO port B reset
Set and cleared by software.
0: does not reset IO port B
1:resets IO port B

Bit 0 **GPIOARST:** IO port A reset
Set and cleared by software.
0: does not reset IO port A
1: resets IO port A

7.3.6 RCC AHB2 peripheral reset register (RCC_AHB2RSTR)

Address offset: 0x14

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								OTGFS RST	RNG RST	HASH RST	CRYP RST	Reserved			DCMI RST
								rw	rw	rw	rw				rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **OTGFSRST:** USB OTG FS module reset
Set and cleared by software.
0: does not reset the USB OTG FS module
1: resets the USB OTG FS module

Bit 6 **RNGRST:** Random number generator module reset
Set and cleared by software.
0: does not reset the random number generator module
1: resets the random number generator module

Bit 5 **HASHRST:** Hash module reset
Set and cleared by software.
0: does not reset the HASH module
1: resets the HASH module

Bit 4 **CRYPRST:** Cryptographic module reset
Set and cleared by software.
0: does not reset the cryptographic module
1: resets the cryptographic module

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMIRST:** Camera interface reset
Set and cleared by software.
0: does not reset the Camera interface
1: resets the Camera interface

7.3.7 RCC AHB3 peripheral reset register (RCC_AHB3RSTR)

Address offset: 0x18

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved															FSMCRST
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **FSMCRST:** Flexible static memory controller module reset
Set and cleared by software.
0: does not reset the FSMC module
1: resets the FSMC module

7.3.8 RCC APB1 peripheral reset register (RCC_APB1RSTR)

Address offset: 0x20

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		DACRST	PWR RST	Reser-ved	CAN2 RST	CAN1 RST	Reser-ved	I2C3 RST	I2C2 RST	I2C1 RST	UART5 RST	UART4 RST	UART3 RST	UART2 RST	Reser-ved
		rw	rw			rw		rw	rw	rw	rw	rw	rw	rw	
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
SPI3 RST	SPI2 RST	Reserved		WWDG RST	Reserved		TIM14 RST	TIM13 RST	TIM12 RST	TIM7 RST	TIM6 RST	TIM5 RST	TIM4 RST	TIM3 RST	TIM2 RST
rw	rw			rw			rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **DACRST:** DAC reset
Set and cleared by software.
0: does not reset the DAC interface
1: resets the DAC interface

Bit 28 **PWRRST:** Power interface reset
Set and cleared by software.
0: does not reset the power interface
1: resets the power interface

Bit 27 Reserved, must be kept at reset value.

Bit 26 **CAN2RST:** CAN2 reset
Set and cleared by software.
0: does not reset CAN2
1: resets CAN2

Bit 25 **CAN1RST:** CAN1 reset
Set and cleared by software.
0: does not reset CAN1
1: resets CAN1

Bit 24 Reserved, must be kept at reset value.

Bit 23 **I2C3RST:** I2C3 reset
Set and cleared by software.
0: does not reset I2C3
1: resets I2C3

Bit 22 **I2C2RST:** I2C2 reset
Set and cleared by software.
0: does not reset I2C2
1: resets I2C2

Bit 21 **I2C1RST:** I2C1 reset
Set and cleared by software.
0: does not reset I2C1
1: resets I2C1

Bit 20 **UART5RST:** UART5 reset
Set and cleared by software.
0: does not reset UART5
1: resets UART5

Bit 19 **UART4RST:** USART4 reset
Set and cleared by software.
0: does not reset UART4
1: resets UART4

Bit 18 **USART3RST:** USART3 reset
Set and cleared by software.
0: does not reset USART3
1: resets USART3

Bit 17 **USART2RST:** USART2 reset
Set and cleared by software.
0: does not reset USART2
1: resets USART2

Bit 16 Reserved, must be kept at reset value.

Bit 15 **SPI3RST:** SPI3 reset
Set and cleared by software.
0: does not reset SPI3
1: resets SPI3

Bit 14 **SPI2RST:** SPI2 reset
Set and cleared by software.
0: does not reset SPI2
1: resets SPI2

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGRST:** Window watchdog reset
Set and cleared by software.
0: does not reset the window watchdog
1: resets the window watchdog

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **TIM14RST:** TIM14 reset
Set and cleared by software.
0: does not reset TIM14
1: resets TIM14

Bit 7 **TIM13RST:** TIM13 reset
Set and cleared by software.
0: does not reset TIM13
1: resets TIM13

Bit 6 **TIM12RST:** TIM12 reset
Set and cleared by software.
0: does not reset TIM12
1: resets TIM12

Bit 5 **TIM7RST:** TIM7 reset
Set and cleared by software.
0: does not reset TIM7
1: resets TIM7

Bit 4 **TIM6RST:** TIM6 reset
Set and cleared by software.
0: does not reset TIM6
1: resets TIM6

Bit 3 **TIM5RST:** TIM5 reset
Set and cleared by software.
0: does not reset TIM5
1: resets TIM5

Bit 2 **TIM4RST:** TIM4 reset
Set and cleared by software.
0: does not reset TIM4
1: resets TIM4

Bit 1 **TIM3RST:** TIM3 reset
Set and cleared by software.
0: does not reset TIM3
1: resets TIM3

Bit 0 **TIM2RST:** TIM2 reset
Set and cleared by software.
0: does not reset TIM2
1: resets TIM2

7.3.9 RCC APB2 peripheral reset register (RCC_APB2RSTR)

Address offset: 0x24

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													TIM11 RST	TIM10 RST	TIM9 RST
													rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser-ved	SYSCFG RST	Reser-ved	SPI1 RST	SDIO RST	Reserved		ADC RST	Reserved		USART 6 RST	USART 1 RST	Reserved		TIM8 RST	TIM1 RST
	rw		rw	rw			rw			rw	rw			rw	rw

Bits 31:19 Reserved, must be kept at reset value.

Bit 18 **TIM11RST:** TIM11 reset
Set and cleared by software.
0: does not reset TIM11
1: resets TIM14

Bit 17 **TIM10RST:** TIM10 reset
Set and cleared by software.
0: does not reset TIM10
1: resets TIM10

Bit 16 **TIM9RST:** TIM9 reset
Set and cleared by software.
0: does not reset TIM9
1: resets TIM9

Bit 15 Reserved, must be kept at reset value.

Bit 14 **SYSCFGRST:** System configuration controller reset
Set and cleared by software.
0: does not reset the System configuration controller
1: resets the System configuration controller

Bit 13 Reserved, must be kept at reset value.

Bit 12 **SPI1RST:** SPI1 reset
Set and cleared by software.
0: does not reset SPI1
1: resets SPI1

Bit 11 **SDIORST:** SDIO reset
Set and cleared by software.
0: does not reset the SDIO module
1: resets the SDIO module

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **ADCRST:** ADC interface reset (common to all ADCs)
Set and cleared by software.
0: does not reset the ADC interface
1: resets the ADC interface

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **USART6RST:** USART6 reset
Set and cleared by software.
0: does not reset USART6
1: resets USART6

Bit 4 **USART1RST:** USART1 reset
Set and cleared by software.
0: does not reset USART1
1: resets USART1

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8RST:** TIM8 reset
Set and cleared by software.
0: does not reset TIM8
1: resets TIM8

Bit 0 **TIM1RST:** TIM1 reset
Set and cleared by software.
0: does not reset TIM1
1: resets TIM1

7.3.10 RCC AHB1 peripheral clock enable register (RCC_AHB1ENR)

Address offset: 0x30

Reset value: 0x0010 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	OTGHSULPIEN	OTGHSEN	ETHMACPTPEN	ETHMACRXEN	ETHMACTXEN	ETHMACEN	Reserved		DMA2EN	DMA1EN	CCMDATARAMEN	Res.	BKPSRAMEN	Reserved	
	rw	rw	rw	rw	rw	rw			rw	rw			rw		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CRCEN	Reserved			GPIOIEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
			rw				rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 Reserved, must be kept at reset value.

Bit 30 **OTGHSULPIEN:** USB OTG HSULPI clock enable
Set and cleared by software. This bit must be cleared when the OTG_HS is used in FS mode.
0: USB OTG HS ULPI clock disabled
1: USB OTG HS ULPI clock enabled

Bit 29 **OTGHSEN:** USB OTG HS clock enable
Set and cleared by software.
0: USB OTG HS clock disabled
1: USB OTG HS clock enabled

Bit 28 **ETHMACPTPEN:** Ethernet PTP clock enable
Set and cleared by software.
0: Ethernet PTP clock disabled
1: Ethernet PTP clock enabled

Bit 27 **ETHMACRXEN:** Ethernet Reception clock enable
Set and cleared by software.
0: Ethernet Reception clock disabled
1: Ethernet Reception clock enabled

Bit 26 **ETHMACTXEN:** Ethernet Transmission clock enable
Set and cleared by software.
0: Ethernet Transmission clock disabled
1: Ethernet Transmission clock enabled

Bit 25 **ETHMACEN:** Ethernet MAC clock enable
Set and cleared by software.
0: Ethernet MAC clock disabled
1: Ethernet MAC clock enabled

Bits 24:23 Reserved, must be kept at reset value.

Bit 22 **DMA2EN:** DMA2 clock enable
Set and cleared by software.
0: DMA2 clock disabled
1: DMA2 clock enabled

Bit 21 **DMA1EN:** DMA1 clock enable
Set and cleared by software.
0: DMA1 clock disabled
1: DMA1 clock enabled

Bit 20 **CCMDATARAMEN**: CCM data RAM clock enable
Set and cleared by software.
0: CCM data RAM clock disabled
1: CCM data RAM clock enabled

Bit 19 Reserved, must be kept at reset value.

Bit 18 **BKPSRAMEN:** Backup SRAM interface clock enable
Set and cleared by software.
0: Backup SRAM interface clock disabled
1: Backup SRAM interface clock enabled

Bits 17:13 Reserved, must be kept at reset value.

Bit 12 **CRCEN:** CRC clock enable
Set and cleared by software.
0: CRC clock disabled
1: CRC clock enabled

Bits 11:9 Reserved, must be kept at reset value.

Bit 8 **GPIOIEN:** IO port I clock enable
Set and cleared by software.
0: IO port I clock disabled
1: IO port I clock enabled

Bit 7 **GPIOHEN:** IO port H clock enable
Set and cleared by software.
0: IO port H clock disabled
1: IO port H clock enabled

Bit 6 **GPIOGEN:** IO port G clock enable
Set and cleared by software.
0: IO port G clock disabled
1: IO port G clock enabled

Bit 5 **GPIOFEN:** IO port F clock enable
Set and cleared by software.
0: IO port F clock disabled
1: IO port F clock enabled

Bit 4 **GPIOEEN:** IO port E clock enable
Set and cleared by software.
0: IO port E clock disabled
1: IO port E clock enabled

Bit 3 **GPIODEN:** IO port D clock enable
Set and cleared by software.
0: IO port D clock disabled
1: IO port D clock enabled

Bit 2 **GPIOCEN:** IO port C clock enable
Set and cleared by software.
0: IO port C clock disabled
1: IO port C clock enabled

Bit 1 **GPIOBEN:** IO port B clock enable
Set and cleared by software.
0: IO port B clock disabled
1: IO port B clock enabled

Bit 0 **GPIOAEN:** IO port A clock enable
Set and cleared by software.
0: IO port A clock disabled
1: IO port A clock enabled

7.3.11 RCC AHB2 peripheral clock enable register (RCC_AHB2ENR)

Address offset: 0x34

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved								OTGFS EN	RNG EN	HASH EN	CRYP EN	Reserved			DCMI EN
								rw	rw	rw	rw				rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **OTGFSEN:** USB OTG FS clock enable
Set and cleared by software.
0: USB OTG FS clock disabled
1: USB OTG FS clock enabled

Bit 6 **RNGEN:** Random number generator clock enable
Set and cleared by software.
0: Random number generator clock disabled
1: Random number generator clock enabled

Bit 5 **HASHEN:** Hash modules clock enable
Set and cleared by software.
0: Hash modules clock disabled
1: Hash modules clock enabled

Bit 4 **CRYPEN:** Cryptographic modules clock enable
Set and cleared by software.
0: cryptographic module clock disabled
1: cryptographic module clock enabled

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMIEN:** Camera interface enable
Set and cleared by software.
0: Camera interface clock disabled
1: Camera interface clock enabled

7.3.12 RCC AHB3 peripheral clock enable register (RCC_AHB3ENR)

Address offset: 0x38

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															FSMCEN
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **FSMCEN:** Flexible static memory controller module clock enable
Set and cleared by software.
0: FSMC module clock disabled
1: FSMC module clock enabled

7.3.13 RCC APB1 peripheral clock enable register (RCC_APB1ENR)

Address offset: 0x40

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		DAC EN	PWR EN	Reser- ved	CAN2 EN	CAN1 EN	Reser- ved	I2C3 EN	I2C2 EN	I2C1 EN	UART5 EN	UART4 EN	USART 3 EN	USART 2 EN	Reser- ved
		rw	rw		rw	rw		rw	rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 EN	SPI2 EN	Reserved		WWDG EN	Reserved		TIM14 EN	TIM13 EN	TIM12 EN	TIM7 EN	TIM6 EN	TIM5 EN	TIM4 EN	TIM3 EN	TIM2 EN
rw	rw			rw			rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **DACEN:** DAC interface clock enable
Set and cleared by software.
0: DAC interface clock disabled
1: DAC interface clock enable

Bit 28 **PWREN:** Power interface clock enable
Set and cleared by software.
0: Power interface clock disabled
1: Power interface clock enable

Bit 27 Reserved, must be kept at reset value.

Bit 26 **CAN2EN:** CAN 2 clock enable
Set and cleared by software.
0: CAN 2 clock disabled
1: CAN 2 clock enabled

Bit 25 **CAN1EN:** CAN 1 clock enable
Set and cleared by software.
0: CAN 1 clock disabled
1: CAN 1 clock enabled

Bit 24 Reserved, must be kept at reset value.

Bit 23 **I2C3EN:** I2C3 clock enable
Set and cleared by software.
0: I2C3 clock disabled
1: I2C3 clock enabled

Bit 22 **I2C2EN:** I2C2 clock enable
Set and cleared by software.
0: I2C2 clock disabled
1: I2C2 clock enabled

Bit 21 **I2C1EN:** I2C1 clock enable
Set and cleared by software.
0: I2C1 clock disabled
1: I2C1 clock enabled

Bit 20 **UART5EN:** UART5 clock enable
Set and cleared by software.
0: UART5 clock disabled
1: UART5 clock enabled

Bit 19 **UART4EN:** UART4 clock enable
Set and cleared by software.
0: UART4 clock disabled
1: UART4 clock enabled

Bit 18 **USART3EN:** USART3 clock enable
Set and cleared by software.
0: USART3 clock disabled
1: USART3 clock enabled

Bit 17 **USART2EN:** USART2 clock enable
Set and cleared by software.
0: USART2 clock disabled
1: USART2 clock enabled

Bit 16 Reserved, must be kept at reset value.

Bit 15 **SPI3EN:** SPI3 clock enable
Set and cleared by software.
0: SPI3 clock disabled
1: SPI3 clock enabled

Bit 14 **SPI2EN:** SPI2 clock enable
Set and cleared by software.
0: SPI2 clock disabled
1: SPI2 clock enabled

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGEN:** Window watchdog clock enable
Set and cleared by software.
0: Window watchdog clock disabled
1: Window watchdog clock enabled

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **TIM14EN:** TIM14 clock enable
Set and cleared by software.
0: TIM14 clock disabled
1: TIM14 clock enabled

Bit 7 **TIM13EN:** TIM13 clock enable
Set and cleared by software.
0: TIM13 clock disabled
1: TIM13 clock enabled

Bit 6 **TIM12EN:** TIM12 clock enable
Set and cleared by software.
0: TIM12 clock disabled
1: TIM12 clock enabled

Bit 5 **TIM7EN:** TIM7 clock enable
Set and cleared by software.
0: TIM7 clock disabled
1: TIM7 clock enabled

Bit 4 **TIM6EN:** TIM6 clock enable
Set and cleared by software.
0: TIM6 clock disabled
1: TIM6 clock enabled

Bit 3 **TIM5EN:** TIM5 clock enable
Set and cleared by software.
0: TIM5 clock disabled
1: TIM5 clock enabled

Bit 2 **TIM4EN:** TIM4 clock enable
Set and cleared by software.
0: TIM4 clock disabled
1: TIM4 clock enabled

Bit 1 **TIM3EN:** TIM3 clock enable
Set and cleared by software.
0: TIM3 clock disabled
1: TIM3 clock enabled

Bit 0 **TIM2EN:** TIM2 clock enable
Set and cleared by software.
0: TIM2 clock disabled
1: TIM2 clock enabled

7.3.14 RCC APB2 peripheral clock enable register (RCC_APB2ENR)

Address offset: 0x44

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													TIM11 EN	TIM10 EN	TIM9 EN
													rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser-ved	SYSCFG EN	Reser-ved	SPI1 EN	SDIO EN	ADC3 EN	ADC2 EN	ADC1 EN	Reserved		USART 6 EN	USART 1 EN	Reserved		TIM8 EN	TIM1 EN
	rw		rw	rw	rw	rw	rw			rw	rw			rw	rw

Bits 31:19 Reserved, must be kept at reset value.

Bit 18 **TIM11EN:** TIM11 clock enable
Set and cleared by software.
0: TIM11 clock disabled
1: TIM11 clock enabled

Bit 17 **TIM10EN:** TIM10 clock enable
Set and cleared by software.
0: TIM10 clock disabled
1: TIM10 clock enabled

Bit 16 **TIM9EN:** TIM9 clock enable
Set and cleared by software.
0: TIM9 clock disabled
1: TIM9 clock enabled

Bit 15 Reserved, must be kept at reset value.

Bit 14 **SYSCFGEN:** System configuration controller clock enable
Set and cleared by software.
0: System configuration controller clock disabled
1: System configuration controller clock enabled

Bit 13 Reserved, must be kept at reset value.

Bit 12 **SPI1EN:** SPI1 clock enable
Set and cleared by software.
0: SPI1 clock disabled
1: SPI1 clock enabled

Bit 11 **SDIOEN:** SDIO clock enable
Set and cleared by software.
0: SDIO module clock disabled
1: SDIO module clock enabled

Bit 10 **ADC3EN:** ADC3 clock enable
Set and cleared by software.
0: ADC3 clock disabled
1: ADC3 clock disabled

Bit 9 **ADC2EN:** ADC2 clock enable
Set and cleared by software.
0: ADC2 clock disabled
1: ADC2 clock disabled

Bit 8 **ADC1EN:** ADC1 clock enable
Set and cleared by software.
0: ADC1 clock disabled
1: ADC1 clock disabled

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **USART6EN:** USART6 clock enable
Set and cleared by software.
0: USART6 clock disabled
1: USART6 clock enabled

Bit 4 **USART1EN:** USART1 clock enable
Set and cleared by software.
0: USART1 clock disabled
1: USART1 clock enabled

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8EN:** TIM8 clock enable
Set and cleared by software.
0: TIM8 clock disabled
1: TIM8 clock enabled

Bit 0 **TIM1EN:** TIM1 clock enable
Set and cleared by software.
0: TIM1 clock disabled
1: TIM1 clock enabled

7.3.15 RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR)

Address offset: 0x50

Reset value: 0x7E67 91FF

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reser-ved	OTGHS ULPILPEN	OTGHS LPEN	ETHPTP LPEN	ETHRX LPEN	ETHTX LPEN	ETHMAC LPEN	Reserved		DMA2 LPEN	DMA1 LPEN	Reserved		BKPSRAM LPEN	SRAM2 LPEN	SRAM1 LPEN
	rw	rw	rw	rw	rw	rw			rw	rw			rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLITF LPEN	Reserved		CRC LPEN	Reserved			GPIOI LPEN	GPIOH LPEN	GPIOG LPEN	GPIOF LPEN	GPIOE LPEN	GPIOD LPEN	GPIOC LPEN	GPIOB LPEN	GPIOA LPEN
rw			rw				rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 Reserved, must be kept at reset value.

Bit 30 **OTGHSULPILPEN:** USB OTG HS ULPI clock enable during Sleep mode
Set and cleared by software. This bit must be cleared when the OTG_HS is used in FS mode.
0: USB OTG HS ULPI clock disabled during Sleep mode
1: USB OTG HS ULPI clock enabled during Sleep mode

Bit 29 **OTGHSLPEN:** USB OTG HS clock enable during Sleep mode
Set and cleared by software.
0: USB OTG HS clock disabled during Sleep mode
1: USB OTG HS clock enabled during Sleep mode

Bit 28 **ETHMACPTPLPEN:** Ethernet PTP clock enable during Sleep mode
Set and cleared by software.
0: Ethernet PTP clock disabled during Sleep mode
1: Ethernet PTP clock enabled during Sleep mode

Bit 27 **ETHMACRXLPEN:** Ethernet reception clock enable during Sleep mode
Set and cleared by software.
0: Ethernet reception clock disabled during Sleep mode
1: Ethernet reception clock enabled during Sleep mode

Bit 26 **ETHMACTXLPEN:** Ethernet transmission clock enable during Sleep mode
Set and cleared by software.
0: Ethernet transmission clock disabled during sleep mode
1: Ethernet transmission clock enabled during sleep mode

Bit 25 **ETHMACLPEN:** Ethernet MAC clock enable during Sleep mode
Set and cleared by software.
0: Ethernet MAC clock disabled during Sleep mode
1: Ethernet MAC clock enabled during Sleep mode

Bits 24:23 Reserved, must be kept at reset value.

Bit 22 **DMA2LPEN:** DMA2 clock enable during Sleep mode
Set and cleared by software.
0: DMA2 clock disabled during Sleep mode
1: DMA2 clock enabled during Sleep mode

Bit 21 **DMA1LPEN:** DMA1 clock enable during Sleep mode
Set and cleared by software.
0: DMA1 clock disabled during Sleep mode
1: DMA1 clock enabled during Sleep mode

Bits 20:19 Reserved, must be kept at reset value.

Bit 18 **BKPSRAMLPEN:** Backup SRAM interface clock enable during Sleep mode
Set and cleared by software.
0: Backup SRAM interface clock disabled during Sleep mode
1: Backup SRAM interface clock enabled during Sleep mode

Bit 17 **SRAM2LPEN:** SRAM 2 interface clock enable during Sleep mode
Set and cleared by software.
0: SRAM 2 interface clock disabled during Sleep mode
1: SRAM 2 interface clock enabled during Sleep mode

Bit 16 **SRAM1LPEN:** SRAM 1interface clock enable during Sleep mode
Set and cleared by software.
0: SRAM 1 interface clock disabled during Sleep mode
1: SRAM 1 interface clock enabled during Sleep mode

Bit 15 **FLITFLPEN:** Flash interface clock enable during Sleep mode
Set and cleared by software.
0: Flash interface clock disabled during Sleep mode
1: Flash interface clock enabled during Sleep mode

Bits 14:13 Reserved, must be kept at reset value.

Bit 12 **CRCLPEN:** CRC clock enable during Sleep mode
Set and cleared by software.
0: CRC clock disabled during Sleep mode
1: CRC clock enabled during Sleep mode

Bits 11:9 Reserved, must be kept at reset value.

Bit 8 **GPIOILPEN:** IO port I clock enable during Sleep mode
Set and cleared by software.
0: IO port I clock disabled during Sleep mode
1: IO port I clock enabled during Sleep mode

Bit 7 **GPIOHLPEN:** IO port H clock enable during Sleep mode
Set and cleared by software.
0: IO port H clock disabled during Sleep mode
1: IO port H clock enabled during Sleep mode

Bits 6 **GPIOGLPEN:** IO port G clock enable during Sleep mode
Set and cleared by software.
0: IO port G clock disabled during Sleep mode
1: IO port G clock enabled during Sleep mode

Bit 5 **GPIOFLPEN:** IO port F clock enable during Sleep mode
Set and cleared by software.
0: IO port F clock disabled during Sleep mode
1: IO port F clock enabled during Sleep mode

Bit 4 **GPIOELPEN:** IO port E clock enable during Sleep mode
Set and cleared by software.
0: IO port E clock disabled during Sleep mode
1: IO port E clock enabled during Sleep mode

Bit 3 **GPIODLPEN:** IO port D clock enable during Sleep mode
Set and cleared by software.
0: IO port D clock disabled during Sleep mode
1: IO port D clock enabled during Sleep mode

Bit 2 **GPIOCLPEN:** IO port C clock enable during Sleep mode
Set and cleared by software.
0: IO port C clock disabled during Sleep mode
1: IO port C clock enabled during Sleep mode

Bit 1 **GPIOBLPEN:** IO port B clock enable during Sleep mode
Set and cleared by software.
0: IO port B clock disabled during Sleep mode
1: IO port B clock enabled during Sleep mode

Bit 0 **GPIOALPEN:** IO port A clock enable during sleep mode
Set and cleared by software.
0: IO port A clock disabled during Sleep mode
1: IO port A clock enabled during Sleep mode

7.3.16 RCC AHB2 peripheral clock enable in low power mode register (RCC_AHB2LPENR)

Address offset: 0x54

Reset value: 0x0000 00F1

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved								OTGFS LPEN	RNG LPEN	HASH LPEN	CRYP LPEN	Reserved			DCMI LPEN
								rw	rw	rw	rw				rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **OTGFSLPEN:** USB OTG FS clock enable during Sleep mode
Set and cleared by software.
0: USB OTG FS clock disabled during Sleep mode
1: USB OTG FS clock enabled during Sleep mode

Bit 6 **RNGLPEN:** Random number generator clock enable during Sleep mode
Set and cleared by software.
0: Random number generator clock disabled during Sleep mode
1: Random number generator clock enabled during Sleep mode

Bit 5 **HASHLPEN:** Hash modules clock enable during Sleep mode
Set and cleared by software.
0: Hash modules clock disabled during Sleep mode
1: Hash modules clock enabled during Sleep mode

Bit 4 **CRYPLPEN:** Cryptography modules clock enable during Sleep mode
Set and cleared by software.
0: cryptography modules clock disabled during Sleep mode
1: cryptography modules clock enabled during Sleep mode

Bits 3:1 Reserved, must be kept at reset value.

Bit 0 **DCMILPEN:** Camera interface enable during Sleep mode
Set and cleared by software.
0: Camera interface clock disabled during Sleep mode
1: Camera interface clock enabled during Sleep mode

7.3.17 RCC AHB3 peripheral clock enable in low power mode register (RCC_AHB3LPENR)

Address offset: 0x58

Reset value: 0x0000 0001

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved															FSMC LPEN
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **FSMCLPEN:** Flexible static memory controller module clock enable during Sleep mode
Set and cleared by software.
0: FSMC module clock disabled during Sleep mode
1: FSMC module clock enabled during Sleep mode

7.3.18 RCC APB1 peripheral clock enable in low power mode register (RCC_APB1LPENR)

Address offset: 0x60

Reset value: 0x36FE C9FF

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		DAC LPEN	PWR LPEN	RESER VED	CAN2 LPEN	CAN1 LPEN	Reser-ved	I2C3 LPEN	I2C2 LPEN	I2C1 LPEN	UART5 LPEN	UART4 LPEN	USART 3 LPEN	USART 2 LPEN	Reser-ved
		rw	rw		rw	rw		rw	rw	rw	rw	rw	rw	rw	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3 LPEN	SPI2 LPEN	Reserved		WWDG LPEN	Reserved		TIM14 LPEN	TIM13 LPEN	TIM12 LPEN	TIM7 LPEN	TIM6 LPEN	TIM5 LPEN	TIM4 LPEN	TIM3 LPEN	TIM2 LPEN
rw	rw			rw			rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **DACLPEN:** DAC interface clock enable during Sleep mode
Set and cleared by software.
0: DAC interface clock disabled during Sleep mode
1: DAC interface clock enabled during Sleep mode

Bit 28 **PWRLPEN:** Power interface clock enable during Sleep mode
Set and cleared by software.
0: Power interface clock disabled during Sleep mode
1: Power interface clock enabled during Sleep mode

Bit 27 Reserved, must be kept at reset value.

Bit 26 **CAN2LPEN:** CAN 2 clock enable during Sleep mode
Set and cleared by software.
0: CAN 2 clock disabled during sleep mode
1: CAN 2 clock enabled during sleep mode

Bit 25 **CAN1LPEN:** CAN 1 clock enable during Sleep mode
Set and cleared by software.
0: CAN 1 clock disabled during Sleep mode
1: CAN 1 clock enabled during Sleep mode

Bit 24 Reserved, must be kept at reset value.

Bit 23 **I2C3LPEN:** I2C3 clock enable during Sleep mode
Set and cleared by software.
0: I2C3 clock disabled during Sleep mode
1: I2C3 clock enabled during Sleep mode

Bit 22 **I2C2LPEN:** I2C2 clock enable during Sleep mode
Set and cleared by software.
0: I2C2 clock disabled during Sleep mode
1: I2C2 clock enabled during Sleep mode

Bit 21 **I2C1LPEN:** I2C1 clock enable during Sleep mode
Set and cleared by software.
0: I2C1 clock disabled during Sleep mode
1: I2C1 clock enabled during Sleep mode

Bit 20 **UART5LPEN:** UART5 clock enable during Sleep mode
Set and cleared by software.
0: UART5 clock disabled during Sleep mode
1: UART5 clock enabled during Sleep mode

Bit 19 **UART4LPEN:** UART4 clock enable during Sleep mode
Set and cleared by software.
0: UART4 clock disabled during Sleep mode
1: UART4 clock enabled during Sleep mode

Bit 18 **USART3LPEN:** USART3 clock enable during Sleep mode
Set and cleared by software.
0: USART3 clock disabled during Sleep mode
1: USART3 clock enabled during Sleep mode

Bit 17 **USART2LPEN:** USART2 clock enable during Sleep mode
Set and cleared by software.
0: USART2 clock disabled during Sleep mode
1: USART2 clock enabled during Sleep mode

Bit 16 Reserved, must be kept at reset value.

Bit 15 **SPI3LPEN:** SPI3 clock enable during Sleep mode
Set and cleared by software.
0: SPI3 clock disabled during Sleep mode
1: SPI3 clock enabled during Sleep mode

Bit 14 **SPI2LPEN:** SPI2 clock enable during Sleep mode
Set and cleared by software.
0: SPI2 clock disabled during Sleep mode
1: SPI2 clock enabled during Sleep mode

Bits 13:12 Reserved, must be kept at reset value.

Bit 11 **WWDGLPEN:** Window watchdog clock enable during Sleep mode
Set and cleared by software.
0: Window watchdog clock disabled during sleep mode
1: Window watchdog clock enabled during sleep mode

Bits 10:9 Reserved, must be kept at reset value.

Bit 8 **TIM14LPEN:** TIM14 clock enable during Sleep mode
Set and cleared by software.
0: TIM14 clock disabled during Sleep mode
1: TIM14 clock enabled during Sleep mode

Bit 7 **TIM13LPEN:** TIM13 clock enable during Sleep mode
Set and cleared by software.
0: TIM13 clock disabled during Sleep mode
1: TIM13 clock enabled during Sleep mode

Bit 6 **TIM12LPEN:** TIM12 clock enable during Sleep mode
Set and cleared by software.
0: TIM12 clock disabled during Sleep mode
1: TIM12 clock enabled during Sleep mode

Bit 5 **TIM7LPEN:** TIM7 clock enable during Sleep mode
Set and cleared by software.
0: TIM7 clock disabled during Sleep mode
1: TIM7 clock enabled during Sleep mode

Bit 4 **TIM6LPEN:** TIM6 clock enable during Sleep mode
Set and cleared by software.
0: TIM6 clock disabled during Sleep mode
1: TIM6 clock enabled during Sleep mode

Bit 3 **TIM5LPEN:** TIM5 clock enable during Sleep mode
Set and cleared by software.
0: TIM5 clock disabled during Sleep mode
1: TIM5 clock enabled during Sleep mode

Bit 2 **TIM4LPEN:** TIM4 clock enable during Sleep mode
Set and cleared by software.
0: TIM4 clock disabled during Sleep mode
1: TIM4 clock enabled during Sleep mode

Bit 1 **TIM3LPEN:** TIM3 clock enable during Sleep mode
Set and cleared by software.
0: TIM3 clock disabled during Sleep mode
1: TIM3 clock enabled during Sleep mode

Bit 0 **TIM2LPEN:** TIM2 clock enable during Sleep mode
Set and cleared by software.
0: TIM2 clock disabled during Sleep mode
1: TIM2 clock enabled during Sleep mode

7.3.19 RCC APB2 peripheral clock enabled in low power mode register (RCC_APB2LPENR)

Address offset: 0x64

Reset value: 0x0007 5F33

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													TIM11 LPEN	TIM10 LPEN	TIM9 LPEN
													rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser-ved	SYSC FG LPEN	Reser-ved	SPI1 LPEN	SDIO LPEN	ADC3 LPEN	ADC2 LPEN	ADC1 LPEN	Reserved		USART 6 LPEN	USART 1 LPEN	Reserved		TIM8 LPEN	TIM1 LPEN
	rw		rw	rw	rw	rw	rw			rw	rw			rw	rw

Bits 31:19 Reserved, must be kept at reset value.

Bit 18 **TIM11LPEN:** TIM11 clock enable during Sleep mode
Set and cleared by software.
0: TIM11 clock disabled during Sleep mode
1: TIM11 clock enabled during Sleep mode

Bit 17 **TIM10LPEN:** TIM10 clock enable during Sleep mode
Set and cleared by software.
0: TIM10 clock disabled during Sleep mode
1: TIM10 clock enabled during Sleep mode

Bit 16 **TIM9LPEN:** TIM9 clock enable during sleep mode
Set and cleared by software.
0: TIM9 clock disabled during Sleep mode
1: TIM9 clock enabled during Sleep mode

Bit 15 Reserved, must be kept at reset value.

Bit 14 **SYSCFGLPEN:** System configuration controller clock enable during Sleep mode
Set and cleared by software.
0: System configuration controller clock disabled during Sleep mode
1: System configuration controller clock enabled during Sleep mode

Bit 13 Reserved, must be kept at reset value.

Bit 12 **SPI1LPEN:** SPI1 clock enable during Sleep mode
Set and cleared by software.
0: SPI1 clock disabled during Sleep mode
1: SPI1 clock enabled during Sleep mode

Bit 11 **SDIOLPEN:** SDIO clock enable during Sleep mode
Set and cleared by software.
0: SDIO module clock disabled during Sleep mode
1: SDIO module clock enabled during Sleep mode

Bit 10 **ADC3LPEN:** ADC 3 clock enable during Sleep mode
Set and cleared by software.
0: ADC 3 clock disabled during Sleep mode
1: ADC 3 clock disabled during Sleep mode

Bit 9 **ADC2LPEN:** ADC2 clock enable during Sleep mode
Set and cleared by software.
0: ADC2 clock disabled during Sleep mode
1: ADC2 clock disabled during Sleep mode

Bit 8 **ADC1LPEN:** ADC1 clock enable during Sleep mode
Set and cleared by software.
0: ADC1 clock disabled during Sleep mode
1: ADC1 clock disabled during Sleep mode

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **USART6LPEN:** USART6 clock enable during Sleep mode
Set and cleared by software.
0: USART6 clock disabled during Sleep mode
1: USART6 clock enabled during Sleep mode

Bit 4 **USART1LPEN:** USART1 clock enable during Sleep mode
Set and cleared by software.
0: USART1 clock disabled during Sleep mode
1: USART1 clock enabled during Sleep mode

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **TIM8LPEN:** TIM8 clock enable during Sleep mode
Set and cleared by software.
0: TIM8 clock disabled during Sleep mode
1: TIM8 clock enabled during Sleep mode

Bit 0 **TIM1LPEN:** TIM1 clock enable during Sleep mode
Set and cleared by software.
0: TIM1 clock disabled during Sleep mode
1: TIM1 clock enabled during Sleep mode

7.3.20 RCC Backup domain control register (RCC_BDCR)

Address offset: 0x70

Reset value: 0x0000 0000, reset by Backup domain reset.
Access: 0 ≤ wait state ≤ 3, word, half-word and byte access
Wait states are inserted in case of successive accesses to this register.

The LSEON, LSEBYP, RTCSEL and RTCEN bits in the *RCC Backup domain control register (RCC_BDCR)* are in the Backup domain. As a result, after Reset, these bits are write-protected and the DBP bit in the *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx* has to be set before these can be modified. Refer to *Section 7.1.1: System reset on page 215* for further information. These bits are only reset after a Backup domain Reset (see *Section 7.1.3: Backup domain reset*). Any internal or external Reset has no effect on these bits.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															BDRST
															rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RTCEN	Reserved					RTCSEL[1:0]		Reserved					LSEBYP	LSERDY	LSEON
rw						rw	rw						rw	r	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **BDRST:** Backup domain software reset

Set and cleared by software.
0: Reset not activated
1: Resets the entire Backup domain

Note: The BKPSRAM is not affected by this reset, the only way of resetting the BKPSRAM is through the Flash interface when a protection level change from level 1 to level 0 is requested.

The backup domain software reset does not take effect until the DBP bit of the PWR_CR register is set.

Bit 15 **RTCEN:** RTC clock enable

Set and cleared by software.
0: RTC clock disabled
1: RTC clock enabled

Bits 14:10 Reserved, must be kept at reset value.

Bits 9:8 **RTCSEL[1:0]:** RTC clock source selection

Set by software to select the clock source for the RTC. Once the RTC clock source has been selected, it cannot be changed anymore unless the Backup domain is reset. The BDRST bit can be used to reset them.
00: No clock
01: LSE oscillator clock used as the RTC clock
10: LSI oscillator clock used as the RTC clock
11: HSE oscillator clock divided by a programmable prescaler (selection through the RTCPRE[4:0] bits in the RCC clock configuration register (RCC_CFGR)) used as the RTC clock

Bits 7:3 Reserved, must be kept at reset value.

Bit 2 **LSEBYP:** External low-speed oscillator bypass

Set and cleared by software to bypass the oscillator. This bit can be written only when the LSE clock is disabled.
0: LSE oscillator not bypassed
1: LSE oscillator bypassed

Bit 1 **LSERDY:** External low-speed oscillator ready

Set and cleared by hardware to indicate when the external 32 kHz oscillator is stable. After the LSEON bit is cleared, LSERDY goes low after 6 external low-speed oscillator clock cycles.
0: LSE clock not ready
1: LSE clock ready

Bit 0 **LSEON:** External low-speed oscillator enable

Set and cleared by software.
0: LSE clock OFF
1: LSE clock ON

7.3.21 RCC clock control & status register (RCC_CSR)

Address offset: 0x74

Reset value: 0x0E00 0000, reset by system reset, except reset flags by power reset only.

Access: $0 \leq$ wait state ≤ 3, word, half-word and byte access

Wait states are inserted in case of successive accesses to this register.

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td></tr>
<tr><td>LPWR RSTF</td><td>WWDG RSTF</td><td>IWDG RSTF</td><td>SFT RSTF</td><td>POR RSTF</td><td>PIN RSTF</td><td>BORRS TF</td><td>RMVF</td><td colspan="8" rowspan="2">Reserved</td></tr>
<tr><td>r</td><td>r</td><td>r</td><td>r</td><td>r</td><td>r</td><td>r</td><td>rt_w</td></tr>
<tr><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="14" rowspan="2">Reserved</td><td>LSIRDY</td><td>LSION</td></tr>
<tr><td>r</td><td>rw</td></tr>
</table>

Bit 31 **LPWRRSTF:** Low-power reset flag

Set by hardware when a Low-power management reset occurs.
Cleared by writing to the RMVF bit.
0: No Low-power management reset occurred
1: Low-power management reset occurred
For further information on Low-power management reset, refer to *Low-power management reset*.

Bit 30 **WWDGRSTF:** Window watchdog reset flag

Set by hardware when a window watchdog reset occurs.
Cleared by writing to the RMVF bit.
0: No window watchdog reset occurred
1: Window watchdog reset occurred

Bit 29 **IWDGRSTF**: Independent watchdog reset flag

Set by hardware when an independent watchdog reset from V_{DD} domain occurs.
Cleared by writing to the RMVF bit.
0: No watchdog reset occurred
1: Watchdog reset occurred

Bit 28 **SFTRSTF:** Software reset flag
Set by hardware when a software reset occurs.
Cleared by writing to the RMVF bit.
0: No software reset occurred
1: Software reset occurred

Bit 27 **PORRSTF:** POR/PDR reset flag
Set by hardware when a POR/PDR reset occurs.
Cleared by writing to the RMVF bit.
0: No POR/PDR reset occurred
1: POR/PDR reset occurred

Bit 26 **PINRSTF:** PIN reset flag
Set by hardware when a reset from the NRST pin occurs.
Cleared by writing to the RMVF bit.
0: No reset from NRST pin occurred
1: Reset from NRST pin occurred

Bit 25 **BORRSTF:** BOR reset flag
Cleared by software by writing the RMVF bit.
Set by hardware when a POR/PDR or BOR reset occurs.
0: No POR/PDR or BOR reset occurred
1: POR/PDR or BOR reset occurred

Bit 24 **RMVF:** Remove reset flag
Set by software to clear the reset flags.
0: No effect
1: Clear the reset flags

Bits 23:2 Reserved, must be kept at reset value.

Bit 1 **LSIRDY:** Internal low-speed oscillator ready
Set and cleared by hardware to indicate when the internal RC 40 kHz oscillator is stable.
After the LSION bit is cleared, LSIRDY goes low after 3 LSI clock cycles.
0: LSI RC oscillator not ready
1: LSI RC oscillator ready

Bit 0 **LSION:** Internal low-speed oscillator enable
Set and cleared by software.
0: LSI RC oscillator OFF
1: LSI RC oscillator ON

7.3.22 RCC spread spectrum clock generation register (RCC_SSCGR)

Address offset: 0x80

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

The spread spectrum clock generation is available only for the main PLL.

The RCC_SSCGR register must be written either before the main PLL is enabled or after the main PLL disabled.

Note: For full details about PLL spread spectrum clock generation (SSCG) characteristics, refer to the “Electrical characteristics” section in your device datasheet.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SSCGEN	SPREADSEL	Reserved		INCSTEP											
rw	rw			rw	rw	rw		rw	rw	rw	rw	rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INCSTEP			MODPER												
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **SSCGEN:** Spread spectrum modulation enable
Set and cleared by software.
0: Spread spectrum modulation DISABLE. (To write after clearing CR[24]=PLLON bit)
1: Spread spectrum modulation ENABLE. (To write before setting CR[24]=PLLON bit)

Bit 30 **SPREADSEL:** Spread Select
Set and cleared by software.
To write before to set CR[24]=PLLON bit.
0: Center spread
1: Down spread

Bits 29:28 Reserved, must be kept at reset value.

Bits 27:13 **INCSTEP:** Incrementation step
Set and cleared by software. To write before setting CR[24]=PLLON bit.
Configuration input for modulation profile amplitude.

Bits 12:0 **MODPER:** Modulation period
Set and cleared by software. To write before setting CR[24]=PLLON bit.
Configuration input for modulation profile period.

7.3.23 RCC PLLI2S configuration register (RCC_PLLI2SCFGR)

Address offset: 0x84

Reset value: 0x2000 3000

Access: no wait state, word, half-word and byte access.

This register is used to configure the PLLI2S clock outputs according to the formulas:

- $f_{(VCO\ clock)} = f_{(PLLI2S\ clock\ input)} \times (PLLI2SN / PLLM)$
- $f_{(PLL\ I2S\ clock\ output)} = f_{(VCO\ clock)} / PLLI2SR$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	PLLI2SR2	PLLI2SR1	PLLI2SR0	Reserved											
	rw	rw	rw												

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLI2SN8	PLLI2SN7	PLLI2SN6	PLLI2SN5	PLLI2SN4	PLLI2SN3	PLLI2SN2	PLLI2SN1	PLLI2SN0	Reserved					
	rw	rw	rw	rw	rw	rw	rw	rw	rw						

Bit 31 Reserved, must be kept at reset value.

Bits 30:28 **PLLI2SR:** PLLI2S division factor for I2S clocks

Set and cleared by software to control the I2S clock frequency. These bits should be written only if the PLLI2S is disabled. The factor must be chosen in accordance with the prescaler values inside the I2S peripherals, to reach 0.3% error when using standard crystals and 0% error with audio crystals. For more information about I2S clock frequency and precision, refer to *Section 28.4.4: Clock generator* in the I2S chapter.

Caution: The I2Ss requires a frequency lower than or equal to 192 MHz to work correctly.

I2S clock frequency = VCO frequency / PLLR with $2 \leq PLLR \leq 7$

000: PLLR = 0, wrong configuration

001: PLLR = 1, wrong configuration

010: PLLR = 2

...

111: PLLR = 7

Bits 27:15 Reserved, must be kept at reset value.

Bits 14:6 **PLLI2SN:** PLLI2S multiplication factor for VCO

These bits are set and cleared by software to control the multiplication factor of the VCO. These bits can be written only when PLLI2S is disabled. Only half-word and word accesses are allowed to write these bits.

Caution: The software has to set these bits correctly to ensure that the VCO output frequency is between 100 and 432 MHz.

VCO output frequency = VCO input frequency × PLLI2SN with 50 ≤PLLI2SN ≤432

000000000: PLLI2SN = 0, wrong configuration
000000001: PLLI2SN = 1, wrong configuration
...
000110010: PLLI2SN = 50
...
001100011: PLLI2SN = 99
001100100: PLLI2SN = 100
001100101: PLLI2SN = 101
001100110: PLLI2SN = 102
...
110110000: PLLI2SN = 432
110110001: PLLI2SN = 433, wrong configuration
...
111111111: PLLI2SN = 511, wrong configuration

Note: Multiplication factors ranging from 50 and 99 are possible for VCO input frequency higher than 1 MHz. However care must be taken that the minimum VCO output frequency respects the value specified above.

Bits 5:0 Reserved, must be kept at reset value.

7.3.24 RCC register map

Table 35 gives the register map and reset values.

Table 35. RCC register map and reset values

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	RCC_CR	Reserved				PLL I2SRDY	PLL I2SON	PLL RDY	PLL ON	Reserved				CSSON	HSEBYP	HSERDY	HSEON	HSICAL 7	HSICAL 6	HSICAL 5	HSICAL 4	HSICAL 3	HSICAL 2	HSICAL 1	HSICAL 0	HSITRIM 4	HSITRIM 3	HSITRIM 2	HSITRIM 1	HSITRIM 0	Reserved	HSIRDY	HSION
	Reset value					0	0	0	0					0	0	0	0	X	X	X	X	X	X	X	X	1	0	0	0	0	0	1	1
0x04	RCC_ PLLCFGR	Reserved				PLLQ 3	PLLQ 2	PLLQ 1	PLLQ 0	Reserved	PLLSRC	Reserved				PLLP 1	PLLP 0	Reserved	PLLN 8	PLLN 7	PLLN 6	PLLN 5	PLLN 4	PLLN 3	PLLN 2	PLLN 1	PLLN 0	PLLM 5	PLLM 4	PLLM 3	PLLM 2	PLLM 1	PLLM 0
	Reset value					0	1	0	0		0					0	0		0	1	1	0	0	0	0	0	0	0	1	0	0	0	0
0x08	RCC_CFGR	MCO2 1	MCO2 0	MCO2PRE2	MCO2PRE1	MCO2PRE0	MCO1PRE2	MCO1PRE1	MCO1PRE0	I2SSRC	MCO1 1	MCO1 0	RTCPRE 4	RTCPRE 3	RTCPRE 2	RTCPRE 1	RTCPRE 0	PPRE2 2	PPRE2 1	PPRE2 0	PPRE1 2	PPRE1 1	PPRE1 0	Reserved		HPRE 3	HPRE 2	HPRE 1	HPRE 0	SWS 1	SWS 0	SW 1	SW 0
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0
0x0C	RCC_CIR	Reserved								CSSC	Reserved	PLLI2SRDYC	PLLRDYC	HSERDYC	HSIRDYC	LSERDYC	LSIRDYC	Reserved		PLLI2SRDYIE	PLLRDYIE	HSERDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE	CSSF	Reserved	PLLI2SRDYF	PLLRDYF	HSERDYF	HSIRDYF	LSERDYF	LSIRDYF
	Reset value									0		0	0	0	0	0	0			0	0	0	0	0	0	0		0	0	0	0	0	0
0x10	RCC_ AHB1RSTR	Reserved		OTGHSRST	Reserved			ETHMACRST	Reserved		DMA2RST	DMA1RST	Reserved								CRCRST	Reserved			GPIOIRST	GPIOHRST	GPIOGRST	GPIOFRST	GPIOERST	GPIODRST	GPIOCRST	GPIOBRST	GPIOARST
	Reset value			0				0			0	0									0				0	0	0	0	0	0	0	0	0
0x14	RCC_ AHB2RSTR	Reserved																								OTGFSRS	RNGRST	HSAHRST	CRYPRST	Reserved			DCMIRST
	Reset value																																
0x18	RCC_ AHB3RSTR	Reserved																															FSMCRST
	Reset value																																0
0x1C	Reserved	Reserved																															
0x20	RCC_ APB1RSTR	Reserved		DACRST	PWRRST	Reserved	CAN2RST	CAN1RST	Reserved	I2C3RST	I2C2RST	I2C1RST	UART5RST	UART4RST	UART3RST	UART2RST	Reserved	SPI3RST	SPI2RST	Reserved		WWDGRST	Reserved		TIM14RST	TIM13RST	TIM12RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST
	Reset value			0	0		0	0		0	0	0	0	0	0	0		0	0			0			0	0	0	0	0	0	0	0	0
0x24	RCC_ APB2RSTR	Reserved													TIM11RST	TIM10RST	TIM9RST	Reserved	SYSCFGRST	Reserved	SPI1RST	SDIORST	Reserved		ADCRST	Reserved		USART6RST	USART1RST	Reserved		TIM8RST	TIM1RST
	Reset value														0	0	0		0		0	0			0			0	0			0	0
0x28	Reserved	Reserved																															
0x2C	Reserved	Reserved																															
0x30	RCC_ AHB1ENR	Reserved	OTGHSULPIEN	OTGHSEN	ETHMACPTPEN	ETHMACRXEN	ETHMACTXEN	ETHMACEN	Reserved		DMA2EN	DMA1EN	CCMDATARAMEN	Reserved	BKPSRAMEN	Reserved					CRCEN	Reserved			GPIOIEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
	Reset value		0	0	0	0	0	0			0	0	1		0						0				0	0	0	0	0	0	0	0	0

Table 35. RCC register map and reset values (continued)

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x34	RCC_ AHB2ENR	Reserved																								OTGFSEN	RNGEN	HASHEN	CRYPEN	Reserved			DCMIEN
	Reset value																									0	0	0	0				0
0x38	RCC_ AHB3ENR	Reserved																															FSMCEN
	Reset value																																0
0x3C	Reserved	Reserved																															
0x40	RCC_ APB1ENR	Reserved		DACEN	PWREN	Reserved	CAN2EN	CAN1EN	Reserved	I2C3EN	I2C2EN	I2C1EN	UART5EN	UART4EN	USART3EN	USART2EN	Reserved	SPI3EN	SPI2EN	Reserved		WWDGEN	Reserved		TIM14EN	TIM13EN	TIM12EN	TIM7EN	TIM6EN	TIM5EN	TIM4EN	TIM3EN	TIM2EN
	Reset value			0	0		0	0		0	0	0	0	0	0	0		0	0			0			0	0	0	0	0	0	0	0	0
0x44	RCC_ APB2ENR	Reserved													TIM11EN	TIM10EN	TIM9EN	Reserved	SYSCFGEN	Reserved	SPI1EN	SDIOEN	ADC3EN	ADC2EN	ADC1EN	Reserved		USART6EN	USART1EN	Reserved		TIM8EN	TIM1EN
	Reset value														0	0	0		0		0	0	0	0	0			0	0			0	0
0x48	Reserved	Reserved																															
0x4C	Reserved	Reserved																															
0x50	RCC_ AHB1LPENR	Reserved	OTGHSULPILPEN	OTGHSLPEN	ETHMACPTPLPEN	ETHMACRXLPEN	ETHMACTXLPEN	ETHMACLPEN	Reserved		DMA2LPEN	DMA1LPEN	Reserved		BKPSRAMLPEN	SRAM2LPEN	SRAM1LPEN	FLITFLPEN	Reserved		CRCLPEN	Reserved			GPIOILPEN	GPIOHLPEN	GPIOGLPEN	GPIOFLPEN	GPIOELPEN	GPIODLPEN	GPIOCLPEN	GPIOBLPEN	GPIOALPEN
	Reset value		1	1	1	1	1	1			1	1			1	1	1	1			1				1	1	1	1	1	1	1	1	1
0x54	RCC_ AHB2LPENR	Reserved																								OTGFSLPEN	RNGLPEN	HASHLPEN	CRYPLPEN	Reserved			DCMILPEN
	Reset value																									1	1	1	1				1
0x58	RCC_ AHB3LPENR	Reserved																															FSMCLPEN
	Reset value																																1
0x5C	Reserved	Reserved																															
0x60	RCC_ APB1LPENR	Reserved		DACLPEN	PWRLPEN	Reserved	CAN2LPEN	CAN1LPEN	Reserved	I2C3LPEN	I2C2LPEN	I2C1LPEN	UART5LPEN	UART4LPEN	USART3LPEN	USART2LPEN	Reserved	SPI3LPEN	SPI2LPEN	Reserved		WWDGLPEN	Reserved		TIM14LPEN	TIM13LPEN	TIM12LPEN	TIM7LPEN	TIM6LPEN	TIM5LPEN	TIM4LPEN	TIM3LPEN	TIM2LPEN
	Reset value			1	1		1	1		1	1	1	1	1	1	1		1	1			1			1	1	1	1	1	1	1	1	1
0x64	RCC_ APB2LPENR	Reserved													TIM11LPEN	TIM10LPEN	TIM9LPEN	Reserved	SYSCFGLPEN	Reserved	SPI1LPEN	SDIOLPEN	ADC3LPEN	ADC2LPEN	ADC1LPEN	Reserved		USART6LPEN	USART1LPEN	Reserved		TIM8LPEN	TIM1LPEN
	Reset value														1	0	1		1		1	1	1	1	1			1	1			1	1
0x68	Reserved	Reserved																															
0x6C	Reserved	Reserved																															
0x70	RCC_BDCR	Reserved															BDRST	RTCEN	Reserved					RTCSEL 1	RTCSEL 0	Reserved					LSEBYP	LSERDY	LSEON
	Reset value																0	0						0	0						0	0	0

Table 35. RCC register map and reset values (continued)

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x74	RCC_CSR	LPWRRSTF	WWDGRSTF	WDGRSTF	SFTRSTF	PORRSTF	PADRSTF	BORRSTF	RMVF	Reserved																						LSIRDY	LSION
	Reset value	0	0	0	0	1	1	1	1																							0	0
0x78	Reserved	Reserved																															
0x7C	Reserved	Reserved																															
0x80	RCC_SSCGR	SSCGEN	SPREADSEL	Reserved		INCSTEP															MODPER												
	Reset value	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x84	RCC_PLLI2SCFGR	Reserved	PLLI2SRx			Reserved													PLLI2SNx									Reserved					
	Reset value		0	1	0														0	1	1	0	0	0	0	0	0						

Refer to *Section 2.3: Memory map* for the register boundary addresses.

8 General-purpose I/Os (GPIO)

This section applies to the whole STM32F4xx family, unless otherwise specified.

8.1 GPIO introduction

Each general-purpose I/O port has four 32-bit configuration registers (GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR and GPIOx_PUPDR), two 32-bit data registers (GPIOx_IDR and GPIOx_ODR), a 32-bit set/reset register (GPIOx_BSRR), a 32-bit locking register (GPIOx_LCKR) and two 32-bit alternate function selection register (GPIOx_AFRH and GPIOx_AFRL).

8.2 GPIO main features

- Up to 16 I/Os under control
- Output states: push-pull or open drain + pull-up/down
- Output data from output data register (GPIOx_ODR) or peripheral (alternate function output)
- Speed selection for each I/O
- Input states: floating, pull-up/down, analog
- Input data to input data register (GPIOx_IDR) or peripheral (alternate function input)
- Bit set and reset register (GPIOx_BSRR) for bitwise write access to GPIOx_ODR
- Locking mechanism (GPIOx_LCKR) provided to freeze the I/O configuration
- Analog function
- Alternate function input/output selection registers (at most 16 AFs per I/O)
- Fast toggle capable of changing every two clock cycles
- Highly flexible pin multiplexing allows the use of I/O pins as GPIOs or as one of several peripheral functions

8.3 GPIO functional description

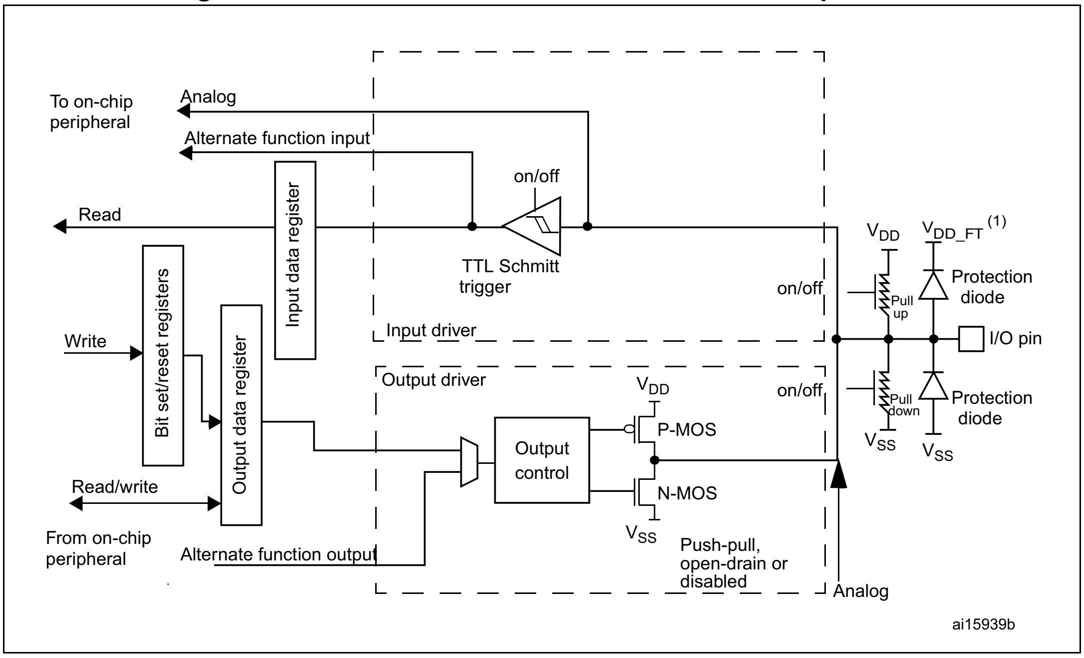
Subject to the specific hardware characteristics of each I/O port listed in the datasheet, each port bit of the general-purpose I/O (GPIO) ports can be individually configured by software in several modes:

- Input floating
- Input pull-up
- Input-pull-down
- Analog
- Output open-drain with pull-up or pull-down capability
- Output push-pull with pull-up or pull-down capability
- Alternate function push-pull with pull-up or pull-down capability
- Alternate function open-drain with pull-up or pull-down capability

Each I/O port bit is freely programmable, however the I/O port registers have to be accessed as 32-bit words, half-words or bytes. The purpose of the GPIOx_BSRR register is to allow atomic read/modify accesses to any of the GPIO registers. In this way, there is no risk of an IRQ occurring between the read and the modify access.

Figure 25 shows the basic structure of a 5 V tolerant I/O port bit. *Table 40* gives the possible port bit configurations.

Figure 25. Basic structure of a five-volt tolerant I/O port bit



1. V_{DD_FT} is a potential specific to five-volt tolerant I/Os and different from V_{DD}.

Table 36. Port bit configuration table[(1)]

MODER(i) [1:0]	OTYPER(i)	OSPEEDR(i) [B:A]	PUPDR(i) [1:0]		I/O configuration	
01	0	SPEED [B:A]	0	0	GP output	PP
	0		0	1	GP output	PP + PU
	0		1	0	GP output	PP + PD
	0		1	1	Reserved	
	1		0	0	GP output	OD
	1		0	1	GP output	OD + PU
	1		1	0	GP output	OD + PD
	1		1	1	Reserved (GP output OD)	

Table 36. Port bit configuration table(1) (continued)

MODER(i) [1:0]	OTYPER(i)	OSPEEDR(i) [B:A]		PUPDR(i) [1:0]		I/O configuration	
10	0	SPEED [B:A]		0	0	AF	PP
	0			0	1	AF	PP + PU
	0			1	0	AF	PP + PD
	0			1	1	Reserved	
	1			0	0	AF	OD
	1			0	1	AF	OD + PU
	1			1	0	AF	OD + PD
	1			1	1	Reserved	
00	x	x	x	0	0	Input	Floating
	x	x	x	0	1	Input	PU
	x	x	x	1	0	Input	PD
	x	x	x	1	1	Reserved (input floating)	
11	x	x	x	0	0	Input/output	Analog
	x	x	x	0	1	Reserved	
	x	x	x	1	0		
	x	x	x	1	1		

1. GP = general-purpose, PP = push-pull, PU = pull-up, PD = pull-down, OD = open-drain, AF = alternate function.

8.3.1 General-purpose I/O (GPIO)

During and just after reset, the alternate functions are not active and the I/O ports are configured in input floating mode.

The debug pins are in AF pull-up/pull-down after reset:

- PA15: JTDI in pull-up
- PA14: JTCK/SWCLK in pull-down
- PA13: JTMS/SWDAT in pull-up
- PB4: NJTRST in pull-up
- PB3: JTDO in floating state

When the pin is configured as output, the value written to the output data register (GPIOx_ODR) is output on the I/O pin. It is possible to use the output driver in push-pull mode or open-drain mode (only the N-MOS is activated when 0 is output).

The input data register (GPIOx_IDR) captures the data present on the I/O pin at every AHB1 clock cycle.

All GPIO pins have weak internal pull-up and pull-down resistors, which can be activated or not depending on the value in the GPIOx_PUPDR register.

8.3.2 I/O pin multiplexer and mapping

The microcontroller I/O pins are connected to onboard peripherals/modules through a multiplexer that allows only one peripheral's alternate function (AF) connected to an I/O pin at a time. In this way, there can be no conflict between peripherals sharing the same I/O pin.

Each I/O pin has a multiplexer with sixteen alternate function inputs (AF0 to AF15) that can be configured through the GPIOx_AFRL (for pin 0 to 7) and GPIOx_AFRH (for pin 8 to 15) registers:

- After reset all I/Os are connected to the system's alternate function 0 (AF0)
- The peripherals' alternate functions are mapped from AF1 to AF13
- Cortex®-M4 with FPU EVENTOUT is mapped on AF15

This structure is shown in *Figure 26* below.

In addition to this flexible I/O multiplexing architecture, each peripheral has alternate functions mapped onto different I/O pins to optimize the number of peripherals available in smaller packages.

To use an I/O in a given configuration, proceed as follows:

- **System function**

  Connect the I/O to AF0 and configure it depending on the function used:
  - JTAG/SWD, after each device reset these pins are assigned as dedicated pins immediately usable by the debugger host (not controlled by the GPIO controller)
  - RTC_REFIN: this pin should be configured in Input floating mode
  - MCO1 and MCO2: these pins have to be configured in alternate function mode.

Note: The user can disable some or all of the JTAG/SWD pins and so release the associated pins for GPIO usage (released pins highlighted in gray in the table).

For more details refer to *Section 7.2.10: Clock-out capability* and *Section 6.2.10: Clock-out capability*.

Table 37. Flexible SWJ-DP pin assignment

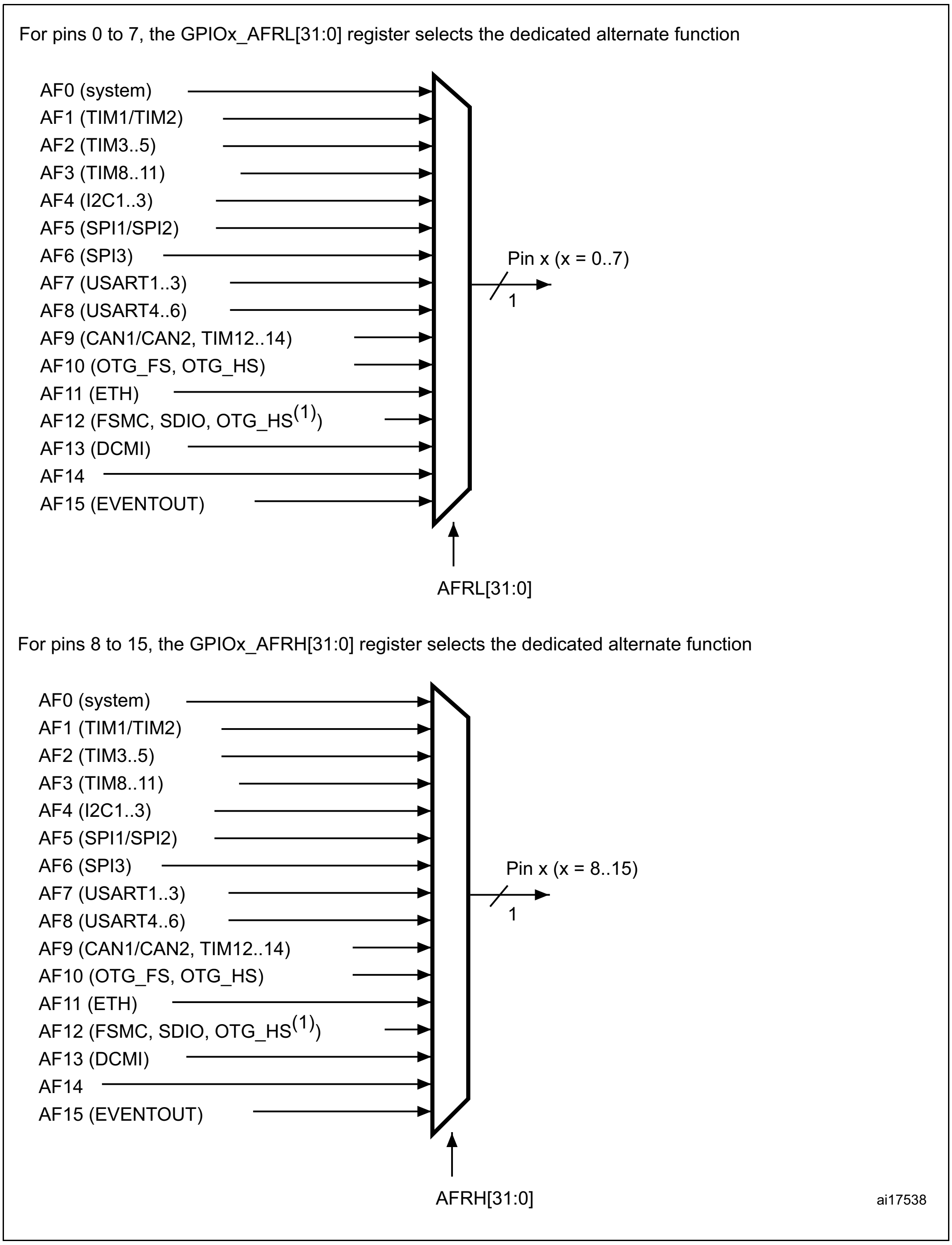
Available debug ports	SWJ I/O pin assigned				
	PA13 / JTMS/ SWDIO	**PA14 / JTCK/ SWCLK**	**PA15 / JTDI**	**PB3 / JTDO**	**PB4/ NJTRST**
Full SWJ (JTAG-DP + SW-DP) - Reset state	X	X	X	X	X
Full SWJ (JTAG-DP + SW-DP) but without NJTRST	X	X	X	X	
JTAG-DP Disabled and SW-DP Enabled	X	X			
JTAG-DP Disabled and SW-DP Disabled	Released				

- **GPIO**
  Configure the desired I/O as output or input in the GPIOx_MODER register.
- **Peripheral alternate function**
  For the ADC and DAC, configure the desired I/O as analog in the GPIOx_MODER register.
  For other peripherals:
  - Configure the desired I/O as an alternate function in the GPIOx_MODER register
  - Select the type, pull-up/pull-down and output speed via the GPIOx_OTYPER, GPIOx_PUPDR and GPIOx_OSPEEDR registers, respectively
  - Connect the I/O to the desired AFx in the GPIOx_AFRL or GPIOx_AFRH register
- **EVENTOUT**
  Configure the I/O pin used to output the Cortex®-M4 with FPU EVENTOUT signal by connecting it to AF15

Note: EVENTOUT is not mapped onto the following I/O pins: PC13, PC14, PC15, PH0, PH1 and PI8.

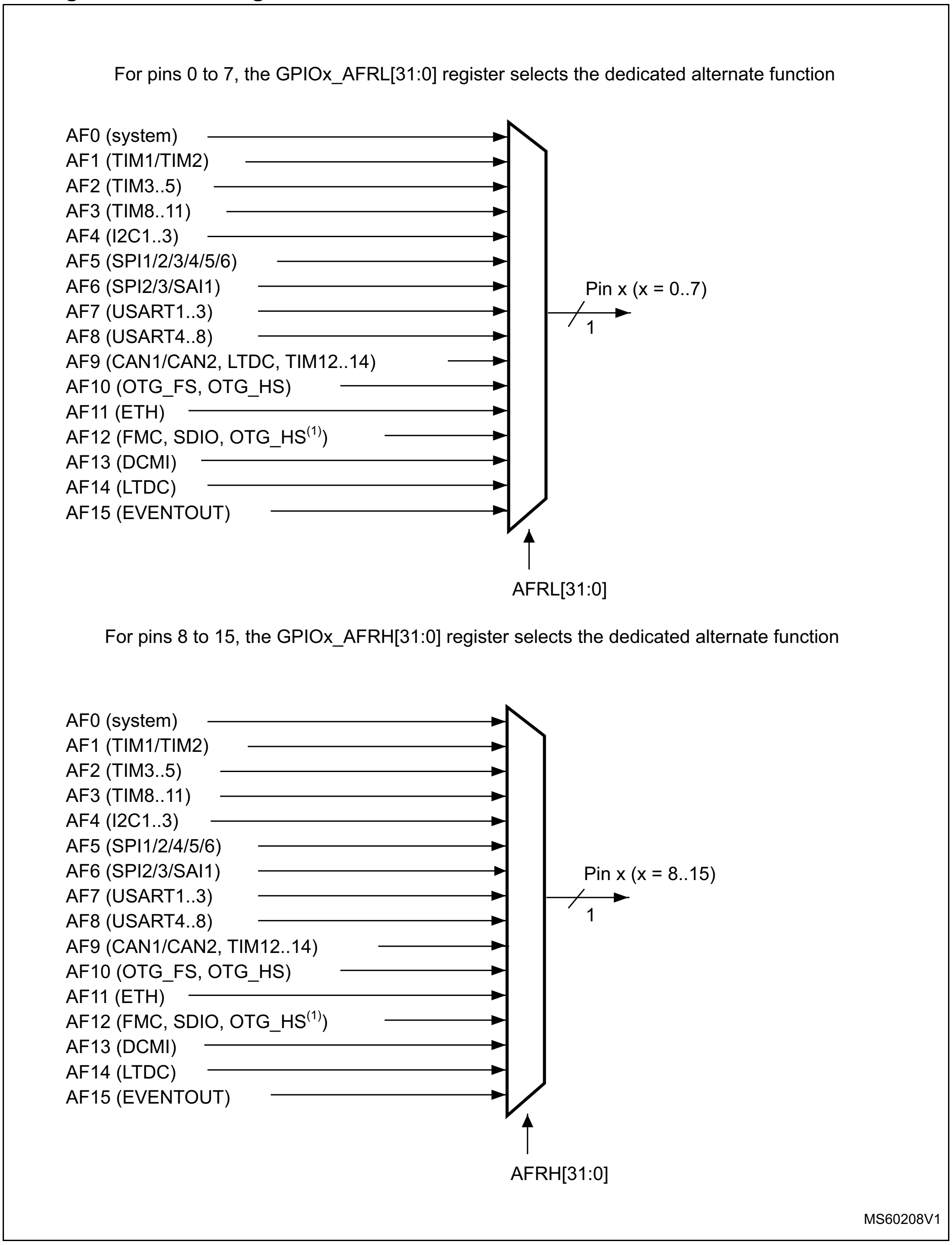
Refer to the “Alternate function mapping” table in the datasheets for the detailed mapping of the system and peripherals’ alternate function I/O pins.

Figure 26. Selecting an alternate function on STM32F405xx/07xx and STM32F415xx/17xx



1. Configured in FS.

Figure 27. Selecting an alternate function on STM32F42xxx and STM32F43xxx



1. Configured in FS.

8.3.3 I/O port control registers

Each of the GPIOs has four 32-bit memory-mapped control registers (GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR) to configure up to 16 I/Os.

The GPIOx_MODER register is used to select the I/O direction (input, output, AF, analog). The GPIOx_OTYPER and GPIOx_OSPEEDR registers are used to select the output type (push-pull or open-drain) and speed (the I/O speed pins are directly connected to the corresponding GPIOx_OSPEEDR register bits whatever the I/O direction). The GPIOx_PUPDR register is used to select the pull-up/pull-down whatever the I/O direction.

8.3.4 I/O port data registers

Each GPIO has two 16-bit memory-mapped data registers: input and output data registers (GPIOx_IDR and GPIOx_ODR). GPIOx_ODR stores the data to be output, it is read/write accessible. The data input through the I/O are stored into the input data register (GPIOx_IDR), a read-only register.

See *Section 8.4.5: GPIO port input data register (GPIOx_IDR) (x = A..I/J/K)* and *Section 8.4.6: GPIO port output data register (GPIOx_ODR) (x = A..I/J/K)* for the register descriptions.

8.3.5 I/O data bitwise handling

The bit set reset register (GPIOx_BSRR) is a 32-bit register which allows the application to set and reset each individual bit in the output data register (GPIOx_ODR). The bit set reset register has twice the size of GPIOx_ODR.

To each bit in GPIOx_ODR, correspond two control bits in GPIOx_BSRR: BSRR(i) and BSRR(i+SIZE). When written to 1, bit BSRR(i) sets the corresponding ODR(i) bit. When written to 1, bit BSRR(i+SIZE) resets the ODR(i) corresponding bit.

Writing any bit to 0 in GPIOx_BSRR does not have any effect on the corresponding bit in GPIOx_ODR. If there is an attempt to both set and reset a bit in GPIOx_BSRR, the set action takes priority.

Using the GPIOx_BSRR register to change the values of individual bits in GPIOx_ODR is a “one-shot” effect that does not lock the GPIOx_ODR bits. The GPIOx_ODR bits can always be accessed directly. The GPIOx_BSRR register provides a way of performing atomic bitwise handling.

There is no need for the software to disable interrupts when programming the GPIOx_ODR at bit level: it is possible to modify one or more bits in a single atomic AHB1 write access.

8.3.6 GPIO locking mechanism

It is possible to freeze the GPIO control registers by applying a specific write sequence to the GPIOx_LCKR register. The frozen registers are GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR, GPIOx_AFRL and GPIOx_AFRH.

To write the GPIOx_LCKR register, a specific write / read sequence has to be applied. When the right LOCK sequence is applied to bit 16 in this register, the value of LCKR[15:0] is used to lock the configuration of the I/Os (during the write sequence the LCKR[15:0] value must be the same). When the LOCK sequence has been applied to a port bit, the value of the port bit can no longer be modified until the next MCU or peripheral reset. Each GPIOx_LCKR bit freezes the corresponding bit in the control registers (GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR, GPIOx_AFRL and GPIOx_AFRH).

The LOCK sequence (refer to *Section 8.4.8: GPIO port configuration lock register (GPIOx_LCKR) (x = A..I/J/K)*) can only be performed using a word (32-bit long) access to the GPIOx_LCKR register due to the fact that GPIOx_LCKR bit 16 has to be set at the same time as the [15:0] bits.

For more details refer to LCKR register description in *Section 8.4.8: GPIO port configuration lock register (GPIOx_LCKR) (x = A..I/J/K)*.

8.3.7 I/O alternate function input/output

Two registers are provided to select one out of the sixteen alternate function inputs/outputs available for each I/O. With these registers, you can connect an alternate function to some other pin as required by your application.
This means that a number of possible peripheral functions are multiplexed on each GPIO using the GPIOx_AFRL and GPIOx_AFRH alternate function registers. The application can thus select any one of the possible functions for each I/O. The AF selection signal being common to the alternate function input and alternate function output, a single channel is selected for the alternate function input/output of one I/O.

To know which functions are multiplexed on each GPIO pin, refer to the datasheets.

Note: The application is allowed to select one of the possible peripheral functions for each I/O at a time.

8.3.8 External interrupt/wake-up lines

All ports have external interrupt capability. To use external interrupt lines, the port must be configured in input mode, refer to *Section 12.2: External interrupt/event controller (EXTI)* and *Section 12.2.3: Wake-up event management*.

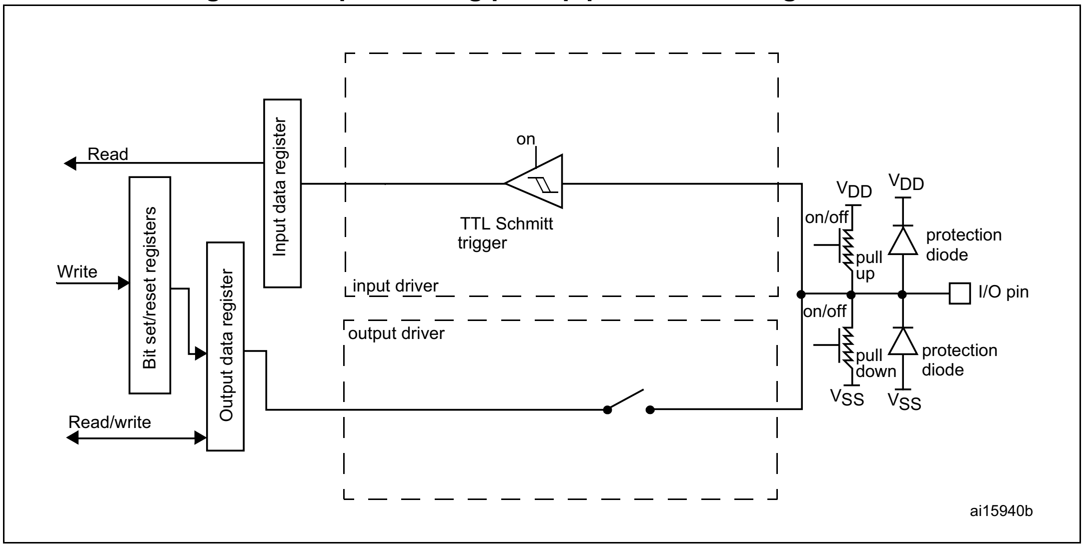
8.3.9 Input configuration

When the I/O port is programmed as Input:

- the output buffer is disabled
- the Schmitt trigger input is activated
- the pull-up and pull-down resistors are activated depending on the value in the GPIOx_PUPDR register
- The data present on the I/O pin are sampled into the input data register every AHB1 clock cycle
- A read access to the input data register provides the I/O State

Figure 28 shows the input configuration of the I/O port bit.

Figure 28. Input floating/pull up/pull down configurations



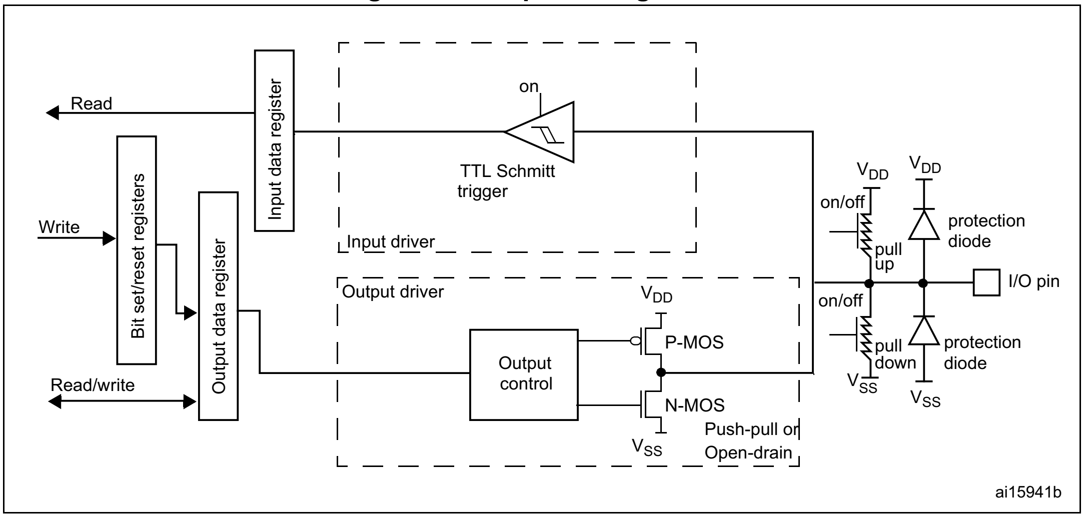
8.3.10 Output configuration

When the I/O port is programmed as output:

- The output buffer is enabled:
  - Open drain mode: A “0” in the Output register activates the N-MOS whereas a “1” in the Output register leaves the port in Hi-Z (the P-MOS is never activated)
  - Push-pull mode: A “0” in the Output register activates the N-MOS whereas a “1” in the Output register activates the P-MOS
- The Schmitt trigger input is activated
- The weak pull-up and pull-down resistors are activated or not depending on the value in the GPIOx_PUPDR register
- The data present on the I/O pin are sampled into the input data register every AHB1 clock cycle
- A read access to the input data register gets the I/O state
- A read access to the output data register gets the last written value

Figure 29 shows the output configuration of the I/O port bit.

Figure 29. Output configuration



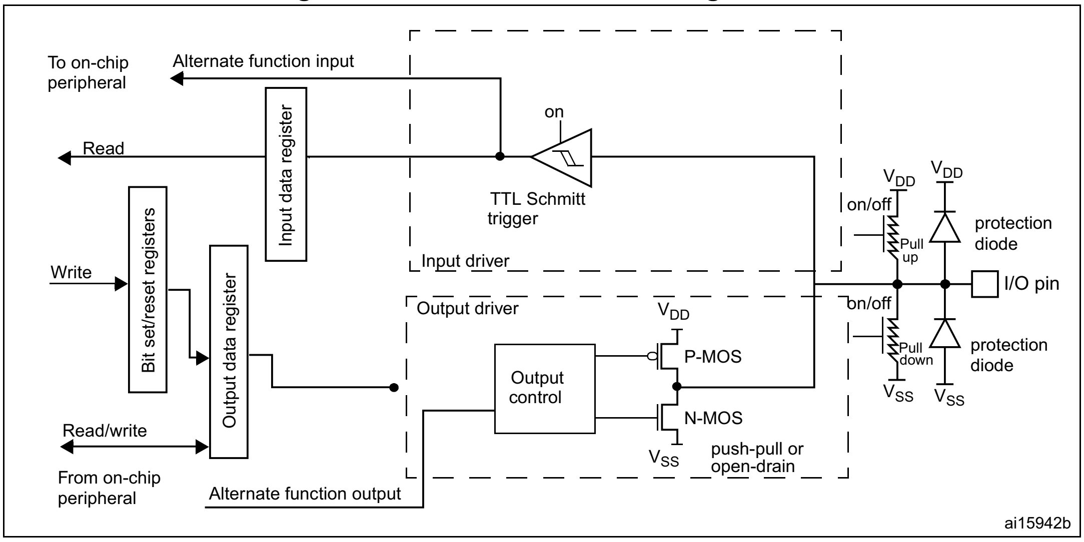
8.3.11 Alternate function configuration

When the I/O port is programmed as alternate function:

- The output buffer can be configured as open-drain or push-pull
- The output buffer is driven by the signal coming from the peripheral (transmitter enable and data)
- The Schmitt trigger input is activated
- The weak pull-up and pull-down resistors are activated or not depending on the value in the GPIOx_PUPDR register
- The data present on the I/O pin are sampled into the input data register every AHB1 clock cycle
- A read access to the input data register gets the I/O state

Figure 30 shows the Alternate function configuration of the I/O port bit.

Figure 30. Alternate function configuration



8.3.12 Analog configuration

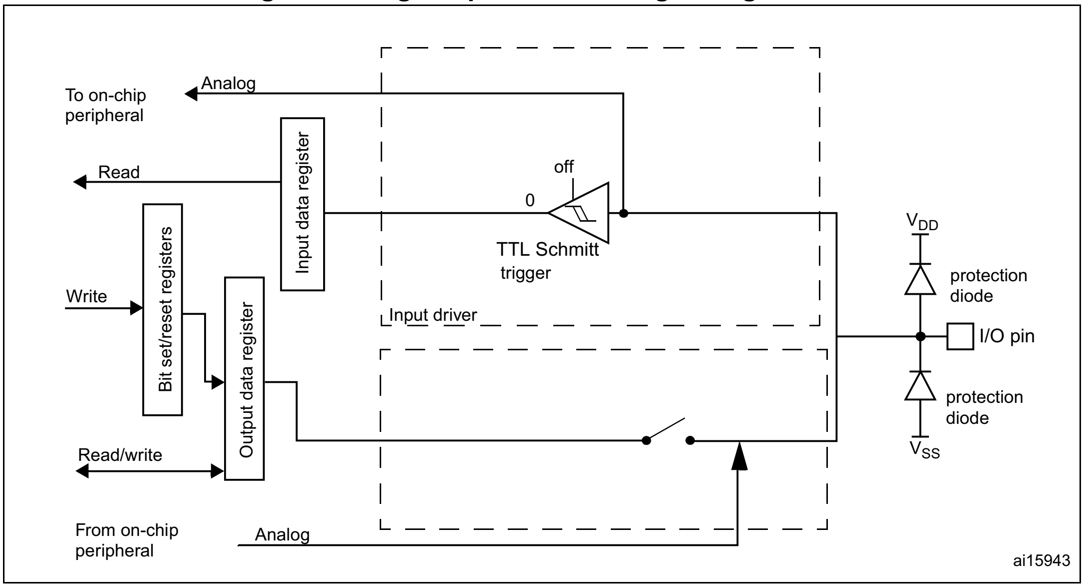
When the I/O port is programmed as analog configuration:

- The output buffer is disabled
- The Schmitt trigger input is deactivated, providing zero consumption for every analog value of the I/O pin. The output of the Schmitt trigger is forced to a constant value (0).
- The weak pull-up and pull-down resistors are disabled
- Read access to the input data register gets the value “0”

Note: In the analog configuration, the I/O pins cannot be 5 Volt tolerant.

Figure 31 shows the high-impedance, analog-input configuration of the I/O port bit.

Figure 31. High impedance-analog configuration



8.3.13 Using the OSC32_IN/OSC32_OUT pins as GPIO PC14/PC15 port pins

The LSE oscillator pins OSC32_IN and OSC32_OUT can be used as general-purpose PC14 and PC15 I/Os, respectively, when the LSE oscillator is off. The PC14 and PC15 I/Os are only configured as LSE oscillator pins OSC32_IN and OSC32_OUT when the LSE oscillator is ON. This is done by setting the LSEON bit in the RCC_BDCR register. The LSE has priority over the GPIO function.

Note: The PC14/PC15 GPIO functionality is lost when the 1.2 V domain is powered off (by the device entering the standby mode) or when the backup domain is supplied by V_{BAT} (V_{DD} no more supplied). In this case the I/Os are set in analog input mode.

8.3.14 Using the OSC_IN/OSC_OUT pins as GPIO PH0/PH1 port pins

The HSE oscillator pins OSC_IN/OSC_OUT can be used as general-purpose PH0/PH1 I/Os, respectively, when the HSE oscillator is OFF. (after reset, the HSE oscillator is off). The PH0/PH1 I/Os are only configured as OSC_IN/OSC_OUT HSE oscillator pins when the HSE oscillator is ON. This is done by setting the HSEON bit in the RCC_CR register. The HSE has priority over the GPIO function.

8.3.15 Selection of RTC_AF1 and RTC_AF2 alternate functions

The STM32F4xx feature two GPIO pins RTC_AF1 and RTC_AF2 that can be used for the detection of a tamper or time stamp event, or RTC_ALARM, or RTC_CALIB RTC outputs.

- The RTC_AF1 (PC13) can be used for the following purposes:

RTC_ALARM output: this output can be RTC Alarm A, RTC Alarm B or RTC Wake-up depending on the OSEL[1:0] bits in the RTC_CR register

- RTC_CALIB output: this feature is enabled by setting the COE[23] in the RTC_CR register
- RTC_TAMP1: tamper event detection
- RTC_TS: time stamp event detection

The RTC_AF2 (PI8) can be used for the following purposes:

- RTC_TAMP1: tamper event detection
- RTC_TAMP2: tamper event detection
- RTC_TS: time stamp event detection

The selection of the corresponding pin is performed through the RTC_TAFCR register as follows:

- TAMP1INSEL is used to select which pin is used as the RTC_TAMP1 tamper input
- TSINSEL is used to select which pin is used as the RTC_TS time stamp input
- ALARMOUTTYPE is used to select whether the RTC_ALARM is output in push-pull or open-drain mode

The output mechanism follows the priority order listed in *Table 38* and *Table 39*.

Table 38. RTC_AF1 pin(1)

Pin configuration and function	RTC_ALARM enabled	RTC_CALIB enabled	Tamper enabled	Time stamp enabled	TAMP1INSEL TAMPER1 pin selection	TSINSEL TIMESTAMP pin selection	ALARMOUTTYPE RTC_ALARM configuration
Alarm out output OD	1	Don't care	Don't care	Don't care	Don't care	Don't care	0
Alarm out output PP	1	Don't care	Don't care	Don't care	Don't care	Don't care	1
Calibration out output PP	0	1	Don't care	Don't care	Don't care	Don't care	Don't care
TAMPER1 input floating	0	0	1	0	0	Don't care	Don't care
TIMESTAMP and TAMPER1 input floating	0	0	1	1	0	0	Don't care
TIMESTAMP input floating	0	0	0	1	Don't care	0	Don't care
Standard GPIO	0	0	0	0	Don't care	Don't care	Don't care

1. OD: open drain; PP: push-pull.

Table 39. RTC_AF2 pin

Pin configuration and function	Tamper enabled	Time stamp enabled	TAMP1INSEL TAMPER1 pin selection	TSINSEL TIMESTAMP pin selection	ALARMOUTTYPE RTC_ALARM configuration
TAMPER1 input floating	1	0	1	Don’t care	Don’t care
TIMESTAMP and TAMPER1 input floating	1	1	1	1	Don’t care
TIMESTAMP input floating	0	1	Don’t care	1	Don’t care
Standard GPIO	0	0	Don’t care	Don’t care	Don’t care

8.4 GPIO registers

This section gives a detailed description of the GPIO registers.
For a summary of register bits, register address offsets and reset values, refer to *Table 40*.

The GPIO registers can be accessed by byte (8 bits), half-words (16 bits) or words (32 bits).

8.4.1 GPIO port mode register (GPIOx_MODER) (x = A..I/J/K)

Address offset: 0x00

Reset values:

- 0xA800 0000 for port A
- 0x0000 0280 for port B
- 0x0000 0000 for other ports

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **MODERy[1:0]:** Port x configuration bits (y = 0..15)

These bits are written by software to configure the I/O direction mode.
00: Input (reset state)
01: General purpose output mode
10: Alternate function mode
11: Analog mode

8.4.2 GPIO port output type register (GPIOx_OTYPER) (x = A..I/J/K)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **OTy**: Port x configuration bits (y = 0..15)

These bits are written by software to configure the output type of the I/O port.
0: Output push-pull (reset state)
1: Output open-drain

8.4.3 GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K)

Address offset: 0x08

Reset values:

- 0x0C00 0000 for port A
- 0x0000 00C0 for port B
- 0x0000 0000 for other ports

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OSPEEDR15 [1:0]		OSPEEDR14 [1:0]		OSPEEDR13 [1:0]		OSPEEDR12 [1:0]		OSPEEDR11 [1:0]		OSPEEDR10 [1:0]		OSPEEDR9 [1:0]		OSPEEDR8 [1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
OSPEEDR7[1:0]		OSPEEDR6[1:0]		OSPEEDR5[1:0]		OSPEEDR4[1:0]		OSPEEDR3[1:0]		OSPEEDR2[1:0]		OSPEEDR1 [1:0]		OSPEEDR0 1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **OSPEEDRy[1:0]:** Port x configuration bits (y = 0..15)

These bits are written by software to configure the I/O output speed.

00: Low speed
01: Medium speed
10: High speed
11: Very high speed

Note: Refer to the product datasheets for the values of OSPEEDRy bits versus V_{DD} range and external load.

8.4.4 GPIO port pull-up/pull-down register (GPIOx_PUPDR) (x = A..I/J/K)

Address offset: 0x0C

Reset values:

- 0x6400 0000 for port A
- 0x0000 0100 for port B
- 0x0000 0000 for other ports

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PUPDR15[1:0]		PUPDR14[1:0]		PUPDR13[1:0]		PUPDR12[1:0]		PUPDR11[1:0]		PUPDR10[1:0]		PUPDR9[1:0]		PUPDR8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PUPDR7[1:0]		PUPDR6[1:0]		PUPDR5[1:0]		PUPDR4[1:0]		PUPDR3[1:0]		PUPDR2[1:0]		PUPDR1[1:0]		PUPDR0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **PUPDRy[1:0]:** Port x configuration bits (y = 0..15)

These bits are written by software to configure the I/O pull-up or pull-down

00: No pull-up, pull-down

01: Pull-up

10: Pull-down

11: Reserved

8.4.5 GPIO port input data register (GPIOx_IDR) (x = A..I/J/K)

Address offset: 0x10

Reset value: 0x0000 XXXX (where X means undefined)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **IDRy**: Port input data (y = 0..15)

These bits are read-only and can be accessed in word mode only. They contain the input value of the corresponding I/O port.

8.4.6 GPIO port output data register (GPIOx_ODR) (x = A..I/J/K)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODRy**: Port output data (y = 0..15)

These bits can be read and written by software.

Note: For atomic bit set/reset, the ODR bits can be individually set and reset by writing to the GPIOx_BSRR register (x = A..I/J/K).

8.4.7 GPIO port bit set/reset register (GPIOx_BSRR) (x = A..I/J/K)

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 **BRy:** Port x reset bit y (y = 0..15)

These bits are write-only and can be accessed in word, half-word or byte mode. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODRx bit

1: Resets the corresponding ODRx bit

Note: If both BSx and BRx are set, BSx has priority.

Bits 15:0 **BSy:** Port x set bit y (y= 0..15)

These bits are write-only and can be accessed in word, half-word or byte mode. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODRx bit

1: Sets the corresponding ODRx bit

8.4.8 GPIO port configuration lock register (GPIOx_LCKR) (x = A..I/J/K)

This register is used to lock the configuration of the port bits when a correct write sequence is applied to bit 16 (LCKK). The value of bits [15:0] is used to lock the configuration of the GPIO. During the write sequence, the value of LCKR[15:0] must not change. When the LOCK sequence has been applied on a port bit, the value of this port bit can no longer be modified until the next MCU or peripheral reset.

Note: A specific write sequence is used to write to the GPIOx_LCKR register. Only word access (32-bit long) is allowed during this write sequence.

Each lock bit freezes a specific configuration register (control and alternate function registers).

Address offset: 0x1C

Reset value: 0x0000 0000

Access: 32-bit word only, read/write register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															LCKK
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCK15	LCK14	LCK13	LCK12	LCK11	LCK10	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **LCKK[16]:** Lock key

This bit can be read any time. It can only be modified using the lock key write sequence.
0: Port configuration lock key not active
1: Port configuration lock key active. The GPIOx_LCKR register is locked until an MCU reset or a peripheral reset occurs.

LOCK key write sequence:
WR LCKR[16] = ‘1’ + LCKR[15:0]
WR LCKR[16] = ‘0’ + LCKR[15:0]
WR LCKR[16] = ‘1’ + LCKR[15:0]
RD LCKR
RD LCKR[16] = ‘1’ (this read operation is optional but it confirms that the lock is active)

Note: During the LOCK key write sequence, the value of LCK[15:0] must not change.
Any error in the lock sequence aborts the lock.
After the first lock sequence on any bit of the port, any read access on the LCKK bit returns ‘1’ until the next CPU reset.

Bits 15:0 **LCKy:** Port x lock bit y (y= 0..15)

These bits are read/write but can only be written when the LCKK bit is ‘0.
0: Port configuration not locked
1: Port configuration locked

8.4.9 GPIO alternate function low register (GPIOx_AFRL) (x = A..I/J/K)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFRL7[3:0]				AFRL6[3:0]				AFRL5[3:0]				AFRL4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFRL3[3:0]				AFRL2[3:0]				AFRL1[3:0]				AFRL0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **AFRLy:** Alternate function selection for port x bit y (y = 0..7)

These bits are written by software to configure alternate function I/Os

AFRLy selection:

0000: AF0	1000: AF8
0001: AF1	1001: AF9
0010: AF2	1010: AF10
0011: AF3	1011: AF11
0100: AF4	1100: AF12
0101: AF5	1101: AF13
0110: AF6	1110: AF14
0111: AF7	1111: AF15

8.4.10 GPIO alternate function high register (GPIOx_AFRH) (x = A..I/J)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFRH15[3:0]				AFRH14[3:0]				AFRH13[3:0]				AFRH12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFRH11[3:0]				AFRH10[3:0]				AFRH9[3:0]				AFRH8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **AFRHy:** Alternate function selection for port x bit y (y = 8..15)

These bits are written by software to configure alternate function I/Os

AFRHy selection:

0000: AF0	1000: AF8
0001: AF1	1001: AF9
0010: AF2	1010: AF10
0011: AF3	1011: AF11
0100: AF4	1100: AF12
0101: AF5	1101: AF13
0110: AF6	1110: AF14
0111: AF7	1111: AF15

8.4.11 GPIO register map

The following table gives the GPIO register map and the reset values.

Table 40. GPIO register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	GPIOA_MODER	MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]		MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
	Reset value	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00	GPIOB_MODER	MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]		MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0
0x00	GPIOx_MODER (where x = C..I/J/K)	MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]		MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x04	GPIOx_OTYPER (where x = A..I/J/K)	Reserved																OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	GPIOx_OSPEEDR (where x = A..I/J/K except B)	OSPEEDR15[1:0]		OSPEEDR14[1:0]		OSPEEDR13[1:0]		OSPEEDR12[1:0]		OSPEEDR11[1:0]		OSPEEDR10[1:0]		OSPEEDR9[1:0]		OSPEEDR8[1:0]		OSPEEDR7[1:0]		OSPEEDR6[1:0]		OSPEEDR5[1:0]		OSPEEDR4[1:0]		OSPEEDR3[1:0]		OSPEEDR2[1:0]		OSPEEDR1[1:0]		OSPEEDR0[1:0]	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	GPIOB_OSPEEDR	OSPEEDR15[1:0]		OSPEEDR14[1:0]		OSPEEDR13[1:0]		OSPEEDR12[1:0]		OSPEEDR11[1:0]		OSPEEDR10[1:0]		OSPEEDR9[1:0]		OSPEEDR8[1:0]		OSPEEDR7[1:0]		OSPEEDR6[1:0]		OSPEEDR5[1:0]		OSPEEDR4[1:0]		OSPEEDR3[1:0]		OSPEEDR2[1:0]		OSPEEDR1[1:0]		OSPEEDR0[1:0]	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
0x0C	GPIOA_PUPDR	PUPDR15[1:0]		PUPDR14[1:0]		PUPDR13[1:0]		PUPDR12[1:0]		PUPDR11[1:0]		PUPDR10[1:0]		PUPDR9[1:0]		PUPDR8[1:0]		PUPDR7[1:0]		PUPDR6[1:0]		PUPDR5[1:0]		PUPDR4[1:0]		PUPDR3[1:0]		PUPDR2[1:0]		PUPDR1[1:0]		PUPDR0[1:0]	
	Reset value	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	GPIOB_PUPDR	PUPDR15[1:0]		PUPDR14[1:0]		PUPDR13[1:0]		PUPDR12[1:0]		PUPDR11[1:0]		PUPDR10[1:0]		PUPDR9[1:0]		PUPDR8[1:0]		PUPDR7[1:0]		PUPDR6[1:0]		PUPDR5[1:0]		PUPDR4[1:0]		PUPDR3[1:0]		PUPDR2[1:0]		PUPDR1[1:0]		PUPDR0[1:0]	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Table 40. GPIO register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0C	**GPIOx_PUPDR** (where x = C..I/J/K)	PUPDR15[1:0]		PUPDR14[1:0]		PUPDR13[1:0]		PUPDR12[1:0]		PUPDR11[1:0]		PUPDR10[1:0]		PUPDR9[1:0]		PUPDR8[1:0]		PUPDR7[1:0]		PUPDR6[1:0]		PUPDR5[1:0]		PUPDR4[1:0]		PUPDR3[1:0]		PUPDR2[1:0]		PUPDR1[1:0]		PUPDR0[1:0]	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	**GPIOx_IDR** (where x = A..I/J/K)	Reserved																IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0
	Reset value																	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
0x14	**GPIOx_ODR** (where x = A..I/J/K)	Reserved																ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x18	**GPIOx_BSRR** (where x = A..I/J/K)	BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0	BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1C	**GPIOx_LCKR** (where x = A..I/J/K)	Reserved															LCKK	LCK15	LCK14	LCK13	LCK12	LCK11	LCK10	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	**GPIOx_AFRL** (where x = A..I/J/K)	AFRL7[3:0]				AFRL6[3:0]				AFRL5[3:0]				AFRL4[3:0]				AFRL3[3:0]				AFRL2[3:0]				AFRL1[3:0]				AFRL0[3:0]			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x24	**GPIOx_AFRH** (where x = A..I/J)	AFRH15[3:0]				AFRH14[3:0]				AFRH13[3:0]				AFRH12[3:0]				AFRH11[3:0]				AFRH10[3:0]				AFRH9[3:0]				AFRH8[3:0]			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Refer to *Section 2.3: Memory map* for the register boundary addresses.

9 System configuration controller (SYSCFG)

The system configuration controller is mainly used to remap the memory accessible in the code area, select the Ethernet PHY interface and manage the external interrupt line connection to the GPIOs.

This section applies to the whole STM32F4xx family, unless otherwise specified.

9.1 I/O compensation cell

By default the I/O compensation cell is not used. However when the I/O output buffer speed is configured in 50 MHz or 100 MHz mode, it is recommended to use the compensation cell for slew rate control on I/O $t_{f(IO)out}$/$t_{r(IO)out}$ commutation to reduce the I/O noise on power supply.

When the compensation cell is enabled, a READY flag is set to indicate that the compensation cell is ready and can be used. The I/O compensation cell can be used only when the supply voltage ranges from 2.4 to 3.6 V.

9.2 SYSCFG registers for STM32F405xx/07xx and STM32F415xx/17xx

9.2.1 SYSCFG memory remap register (SYSCFG_MEMRMP)

This register is used for specific configurations on memory remap:

- Two bits are used to configure the type of memory accessible at address 0x0000 0000. These bits are used to select the physical remap by software and so, bypass the BOOT pins.
- After reset these bits take the value selected by the BOOT pins. When booting from main Flash memory with BOOT pins set to 10 [(BOOT1,BOOT0) = (1,0)] this register takes the value 0x00.

When the FSMC is remapped at address 0x0000 0000, only the first two regions of Bank 1 memory controller (Bank1 NOR/PSRAM 1 and NOR/PSRAM 2) can be remapped. In remap mode, the CPU can access the external memory via ICode bus instead of System bus which boosts up the performance.

Address offset: 0x00

Reset value: 0x0000 000X (X is the memory mode selected by the BOOT pins)

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td></tr>
<tr><td colspan="16">Reserved</td></tr>
<tr><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="14" rowspan="2">Reserved</td><td colspan="2">MEM_MODE</td></tr>
<tr><td>rw</td><td>rw</td></tr>
</table>

Bits 31:2 Reserved, must be kept at reset value.

Bits 1:0 **MEM_MODE:** Memory mapping selection

Set and cleared by software. This bit controls the memory internal mapping at address 0x0000 0000. After reset these bits take the value selected by the Boot pins (except for FSMC).

00: Main Flash memory mapped at 0x0000 0000
01: System Flash memory mapped at 0x0000 0000
10: FSMC Bank1 (NOR/PSRAM 1 and 2) mapped at 0x0000 0000
11: Embedded SRAM (SRAM1) mapped at 0x0000 0000

Note: Refer to Section 2.3: Memory map for details about the memory mapping at address 0x0000 0000.

9.2.2 SYSCFG peripheral mode configuration register (SYSCFG_PMC)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								MII_RMII_SEL	Reserved						
								rw							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **MII_RMII_SEL:** Ethernet PHY interface selection

Set and Cleared by software.These bits control the PHY interface for the Ethernet MAC.

0: MII interface is selected
1: RMII PHY interface is selected

Note: This configuration must be done while the MAC is under reset and before enabling the MAC clocks.

Bits 22:0 Reserved, must be kept at reset value.

9.2.3 SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 0 to 3)

These bits are written by software to select the source input for the EXTIx external interrupt.

0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin

9.2.4 SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 4 to 7)

These bits are written by software to select the source input for the EXTIx external interrupt.
0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin

9.2.5 SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 8 to 11)

These bits are written by software to select the source input for the EXTIx external interrupt.
0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin

9.2.6 SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 12 to 15)

These bits are written by software to select the source input for the EXTIx external interrupt.

0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin

Note: PI[15:12] are not used.

9.2.7 Compensation cell control register (SYSCFG_CMPCR)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							READY	Reserved							CMP_PD
							r								rw

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 **READY:** Compensation cell ready flag

0: I/O compensation cell not ready
1: O compensation cell ready

Bits 7:2 Reserved, must be kept at reset value.

Bit 0 **CMP_PD:** Compensation cell power-down

0: I/O compensation cell power-down mode
1: I/O compensation cell enabled

9.2.8 SYSCFG register maps for STM32F405xx/07xx and STM32F415xx/17xx

The following table gives the SYSCFG register map and the reset values.

Table 41. SYSCFG register map and reset values (STM32F405xx/07xx and STM32F415xx/17xx)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	SYSCFG_MEMRMP	Reserved																														MEM_MODE	
	Reset value																															x	x
0x04	SYSCFG_PMC	Reserved								MII_RMII_SEL	Reserved							Reserved															
	Reset value									0																							
0x08	SYSCFG_EXTICR1	Reserved																EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	SYSCFG_EXTICR2	Reserved																EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	SYSCFG_EXTICR3	Reserved																EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x14	SYSCFG_EXTICR4	Reserved																EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	SYSCFG_CMPCR	Reserved																								READY	Reserved						CMP_PD
	Reset value																									0							0

Refer to *Section 2.3: Memory map* for the register boundary addresses.

9.3 SYSCFG registers for STM32F42xxx and STM32F43xxx

9.3.1 SYSCFG memory remap register (SYSCFG_MEMRMP)

This register is used for specific configurations on memory remap:

- Three bits are used to configure the type of memory accessible at address 0x0000 0000. These bits are used to select the physical remap by software and so, bypass the BOOT pins.
- After reset these bits take the value selected by the BOOT pins. When booting from main Flash memory with BOOT pins set to 10 [(BOOT1,BOOT0) = (1,0)] this register takes the value 0x00.
- Other bits are used to swap FMC SDRAM Bank 1/2 with FMC Bank 3/4 and configure the Flash Bank 1/2 mapping

There are two possible FMC remap at address 0x0000 0000:

- FMC Bank 1 (NOR/PSRAM 1 and 2) remap:
  Only the first two regions of Bank 1 memory controller (Bank1 NOR/PSRAM 1 and NOR/PSRAM 2) can be remapped.
- FMC SDRAM Bank 1 remap.

In remap mode at address 0x0000 0000, the CPU can access the external memory via ICode bus instead of System bus which boosts up the performance.

Address offset: 0x00

Reset value: 0x0000 000X (X is the memory mode selected by the BOOT pins)

Note: Booting from NOR Flash memory or SDRAM is not allowed. The regions can only be mapped at 0x0000 0000 through software remap.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved				SWP_FMC		Res.	FB_MODE	Reserved					MEM_MODE[2:0]		
				rw	rw		rw						rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:10 **SWP_FMC**: FMC memory mapping swap

Set and cleared by software. These bits are used to swap the FMC SDRAM Bank 1/2 and FMC Bank 3/4 (SDRAM Bank 1/2 and NAND Bank 2/PCCARD Bank) in order to enable the code execution from SDRAM Banks without a physical remapping at 0x0000 0000 address.

00: No FMC memory mapping swap

01: SDRAM banks and NAND Bank 2/PCCARD mapping are swapped. SDRAM Bank 1 and 2 are mapped at NAND Bank 2 (0x8000 0000) and PCCARD Bank (0x9000 0000) address, respectively. NAND Bank 2 and PCCARD Bank are mapped at 0xC000 0000 and 0xD000 0000, respectively.

10: Reserved

11: Reserved

Bit 9 Reserved, must be kept at reset value.

Bit 8 **FB_MODE**: Flash Bank mode selection

Set and cleared by software. This bit controls the Flash Bank 1/2 mapping.
0: Flash Bank 1 is mapped at 0x0800 0000 (and aliased at 0x0000 0000) and Flash Bank 2 is mapped at 0x0810 0000 (and aliased at 0x0010 0000)
1: Flash Bank 2 is mapped at 0x0800 0000 (and aliased at 0x0000 0000) and Flash Bank 1 is mapped at 0x0810 0000 (and aliased at 0x0010 0000)

Bits 7:3 Reserved, must be kept at reset value.

Bits 2:0 **MEM_MODE:** Memory mapping selection

Set and cleared by software. This bit controls the memory internal mapping at address 0x0000 0000. After reset these bits take the value selected by the Boot pins (except for FMC).
000: Main Flash memory mapped at 0x0000 0000
001: System Flash memory mapped at 0x0000 0000
010: FMC Bank1 (NOR/PSRAM 1 and 2) mapped at 0x0000 0000
011: Embedded SRAM (SRAM1) mapped at 0x0000 0000
100: FMC/SDRAM Bank 1 mapped at 0x0000 0000
Other configurations are reserved

Note: Refer to Section 2.3: Memory map for details about the memory mapping at address 0x0000 0000.

9.3.2 SYSCFG peripheral mode configuration register (SYSCFG_PMC)

Address offset: 0x04

Reset value: 0x0000 0000

<table>
<tr><td>31</td><td>30</td><td>29</td><td>28</td><td>27</td><td>26</td><td>25</td><td>24</td><td>23</td><td>22</td><td>21</td><td>20</td><td>19</td><td>18</td><td>17</td><td>16</td></tr>
<tr><td colspan="8" rowspan="2">Reserved</td><td>MII_RMII_SEL</td><td colspan="4" rowspan="2">Reserved</td><td colspan="3">ADCxDC2</td></tr>
<tr><td>rw</td><td>rw</td><td>rw</td><td>rw</td></tr>
<tr><td>15</td><td>14</td><td>13</td><td>12</td><td>11</td><td>10</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
<tr><td colspan="16">Reserved</td></tr>
</table>

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **MII_RMII_SEL:** Ethernet PHY interface selection

Set and Cleared by software.These bits control the PHY interface for the Ethernet MAC.
0: MII interface is selected
1: RMII PHY interface is selected

Note: This configuration must be done while the MAC is under reset and before enabling the MAC clocks.

Bits 22:19 Reserved, must be kept at reset value.

Bits 18:16 **ADCxDC2**:

0: No effect.

1: Refer to AN4073 on how to use this bit.

Note: These bits can be set only if the following conditions are met:

- ADC clock higher or equal to 30 MHz.

- Only one ADCxDC2 bit must be selected if ADC conversions do not start at the same time and the sampling times differ.

- These bits must not be set when the ADCDC1 bit is set in PWR_CR register.

Bits 15:0 Reserved, must be kept at reset value.

9.3.3 SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 0 to 3)

These bits are written by software to select the source input for the EXTIx external interrupt.

0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin
1001: PJ[x] pin
1010: PK[x] pin

9.3.4 SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 4 to 7)

These bits are written by software to select the source input for the EXTIx external interrupt.

0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin
1001: PJ[x] pin
1010: PK[x] pin

9.3.5 SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 8 to 11)

These bits are written by software to select the source input for the EXTIx external interrupt.
0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin
1001: PJ[x] pin

Note: PK[11:8] are not used.

9.3.6 SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 12 to 15)

These bits are written by software to select the source input for the EXTIx external interrupt.
0000: PA[x] pin
0001: PB[x] pin
0010: PC[x] pin
0011: PD[x] pin
0100: PE[x] pin
0101: PF[x] pin
0110: PG[x] pin
0111: PH[x] pin
1000: PI[x] pin
1001: PJ[x] pin

Note: PK[15:12] are not used.

9.3.7 Compensation cell control register (SYSCFG_CMPCR)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Reserved							READY	Reserved							CMP_PD
							r								rw

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 **READY:** Compensation cell ready flag
0: I/O compensation cell not ready
1: O compensation cell ready

Bits 7:2 Reserved, must be kept at reset value.

Bit 0 **CMP_PD:** Compensation cell power-down
0: I/O compensation cell power-down mode
1: I/O compensation cell enabled

9.3.8 SYSCFG register maps for STM32F42xxx and STM32F43xxx

The following table gives the SYSCFG register map and the reset values.

Table 42. SYSCFG register map and reset values (STM32F42xxx and STM32F43xxx)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	SYSCFG_MEMRMP	Reserved																				SWP_FMC		Reserved	FB_MODE	Reserved					MEM_MODE		
	Reset value																					0	0		0						x	x	x
0x04	SYSCFG_PMC	Reserved								MII_RMII_SEL	Reserved				ADC3DC2	ADC2DC2	ADC1DC2	Reserved															
	Reset value									0					0	0	0																
0x08	SYSCFG_EXTICR1	Reserved																EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C	SYSCFG_EXTICR2	Reserved																EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x10	SYSCFG_EXTICR3	Reserved																EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x14	SYSCFG_EXTICR4	Reserved																EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	SYSCFG_CMPCR	Reserved																							READY	Reserved							CMP_PD
	Reset value																								0								0

Refer to *Section 2.3: Memory map* for the register boundary addresses.

10 DMA controller (DMA)

This section applies to the whole STM32F4xx family, unless otherwise specified.

10.1 DMA introduction

Direct memory access (DMA) is used in order to provide high-speed data transfer between peripherals and memory and between memory and memory. Data can be quickly moved by DMA without any CPU action. This keeps CPU resources free for other operations.

The DMA controller combines a powerful dual AHB master bus architecture with independent FIFO to optimize the bandwidth of the system, based on a complex bus matrix architecture.

The two DMA controllers have 16 streams in total (8 for each controller), each dedicated to managing memory access requests from one or more peripherals. Each stream can have up to 8 channels (requests) in total. And each has an arbiter for handling the priority between DMA requests.

10.2 DMA main features

The main DMA features are:

- Dual AHB master bus architecture, one dedicated to memory accesses and one dedicated to peripheral accesses
- AHB slave programming interface supporting only 32-bit accesses
- 8 streams for each DMA controller, up to 8 channels (requests) per stream
- Four-word depth 32 first-in, first-out memory buffers (FIFOs) per stream, that can be used in FIFO mode or direct mode:
  - FIFO mode: with threshold level software selectable between 1/4, 1/2 or 3/4 of the FIFO size
  - Direct mode

    Each DMA request immediately initiates a transfer from/to the memory. When it is configured in direct mode (FIFO disabled), to transfer data in memory-to-peripheral mode, the DMA preloads only one data from the memory to the internal